Biochemistry



Section B
Amino acid and protein

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Section B Amino acid and protein

- General introduction
- Amino acid structure (B1) 20 AA
- Acid and bases (B2)
- Protein structure (B3)
- Myoglobin and Hemoglobin (B4)
- Collagen (B5)
- Protein purification (B6, B7,B8)
- Protein sequencing (B9)



B0. General Introduction

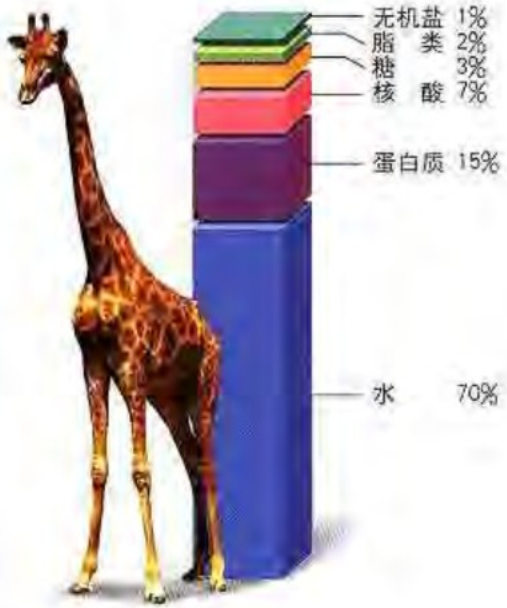


图2-9　生物体的一般化学组成　不同的细胞或组织，其分子组成可以有差异。

B0. General Introduction

Proteins have widely diverse forms and functions.

enzymes(酶), hormones(激素 , 荷尔蒙), antibodies(抗体), transporters(转运蛋白),

muscle(肌肉), lens protein of eyes(眼睛晶状体), spider webs(蜘蛛网), rhinoceros horn(犀牛角),

Antibiotics （抗生素）

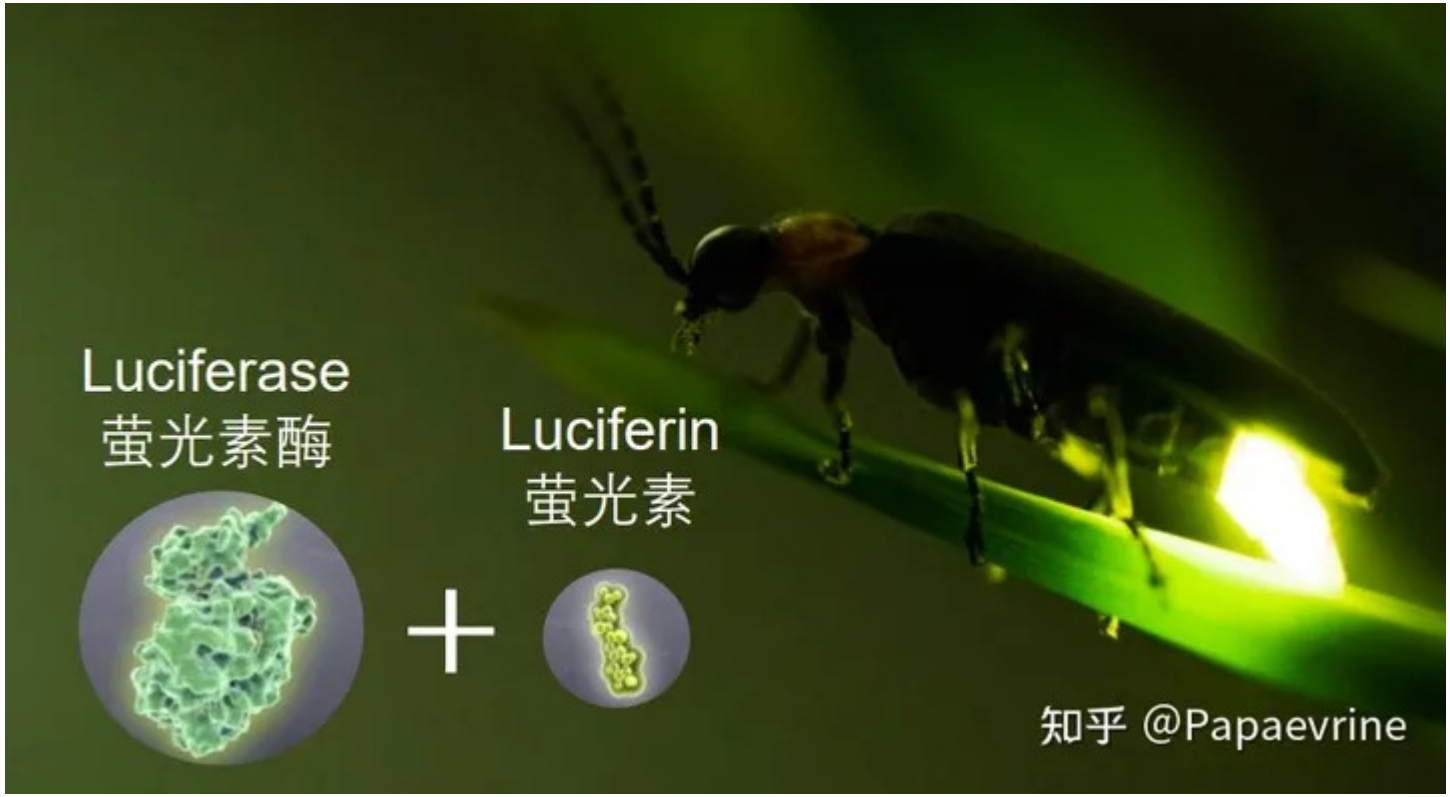
mushroom poisons(蘑菇毒素).

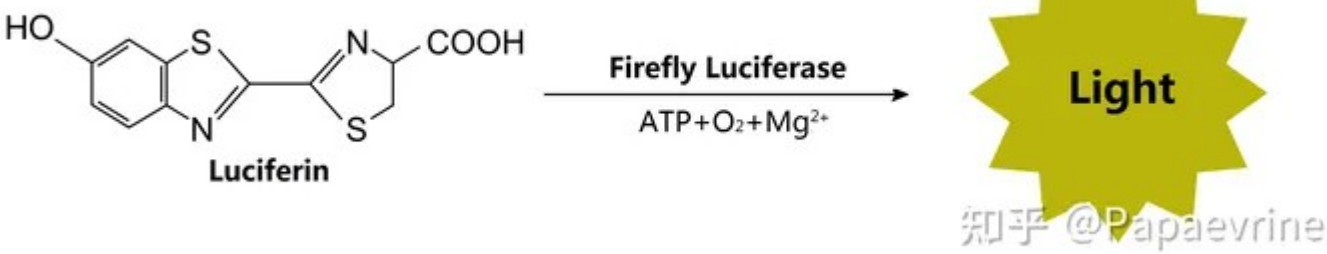
A few examples on protein diversity…



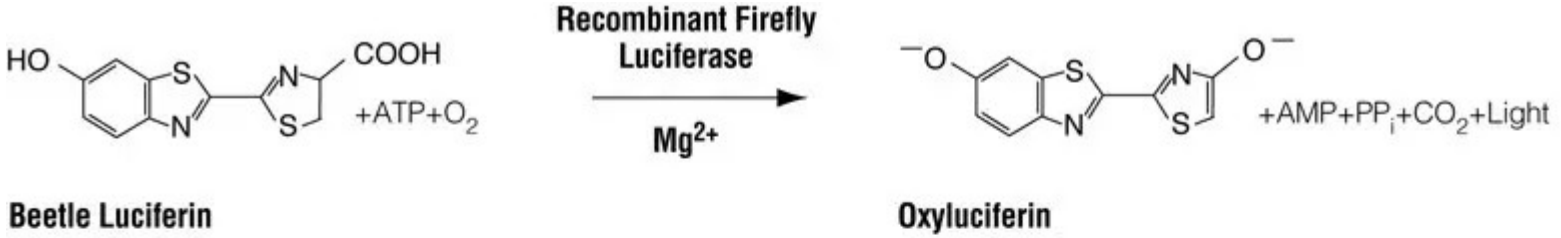
(a)

Fireflies emit light catalyzed by luciferase （萤光素酶） with ATP

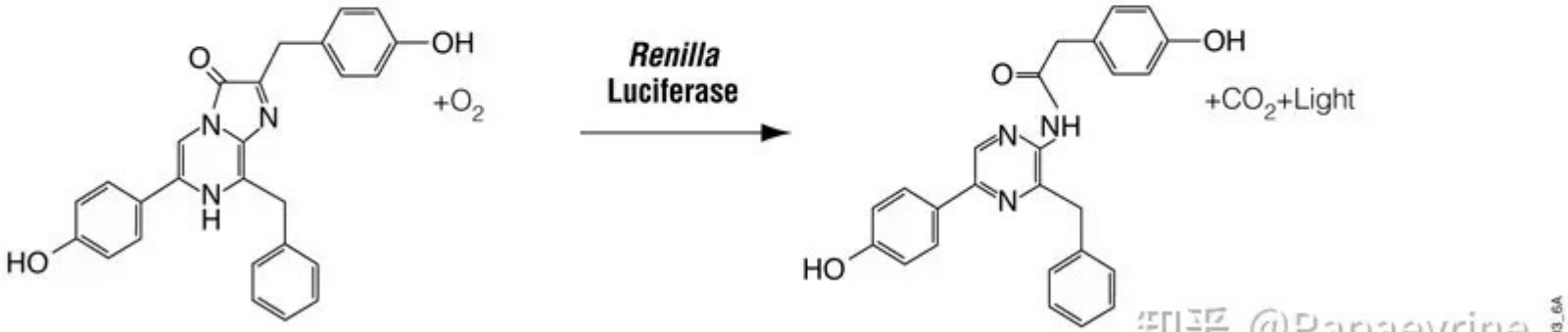

Luciferase
萤光素酶
Luciferin
萤光素
知乎 @Papaevrine


HO
S
N
COOH
N
S
Luciferin
Firefly Luciferase
ATP+O₂+Mg²⁺
Light
知乎 @Papaevrine





Beetle Luciferin **Oxyluciferin**



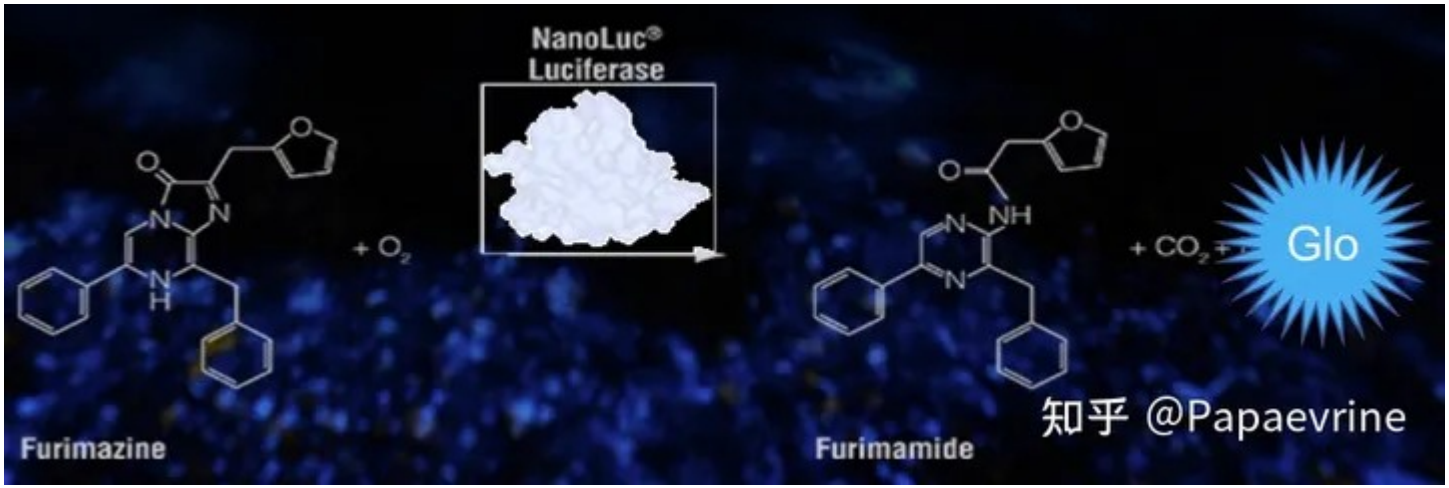
Coelenterazine **Coelenteramide**

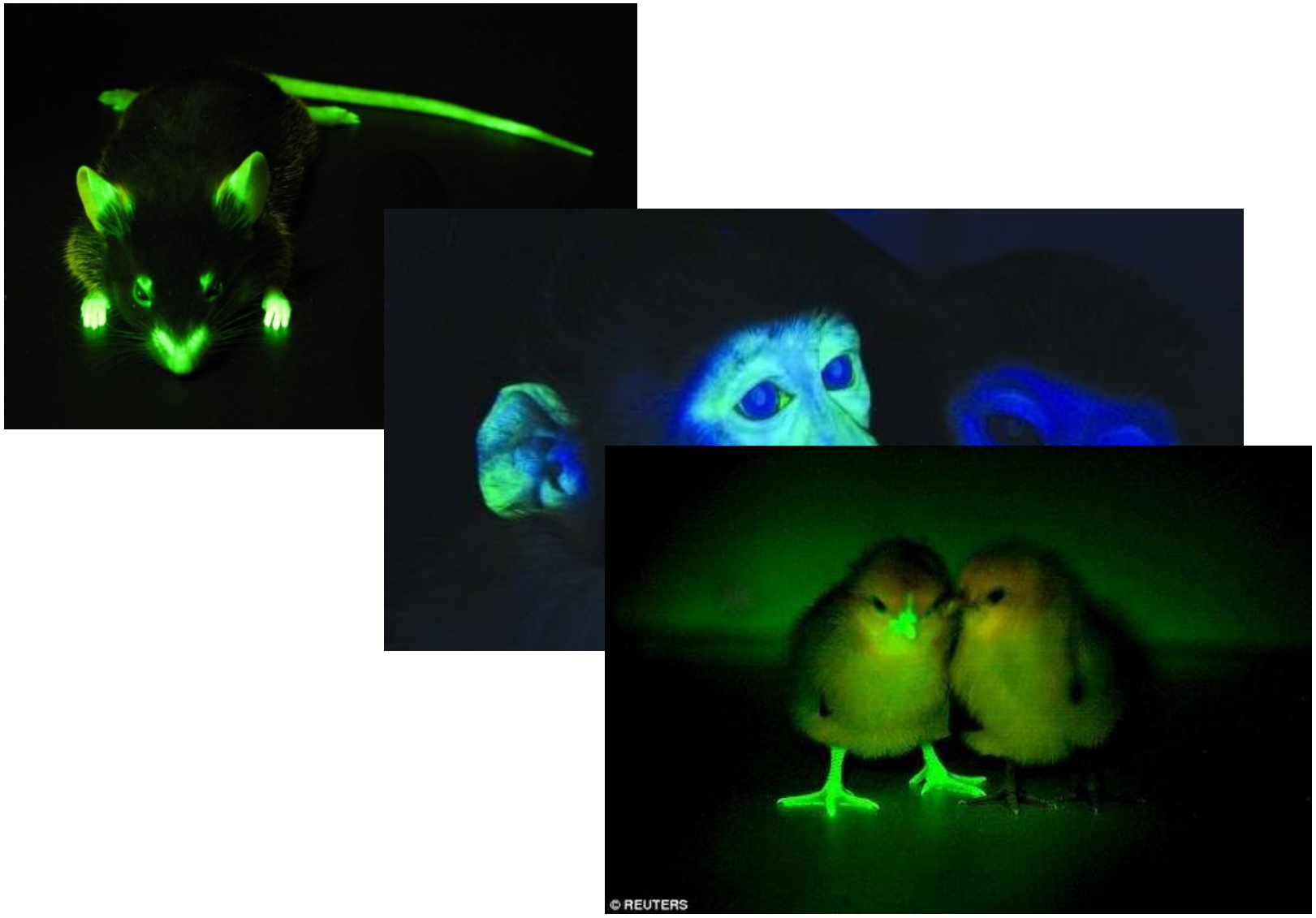
知乎 @Papaevrine

1399MA03_6A

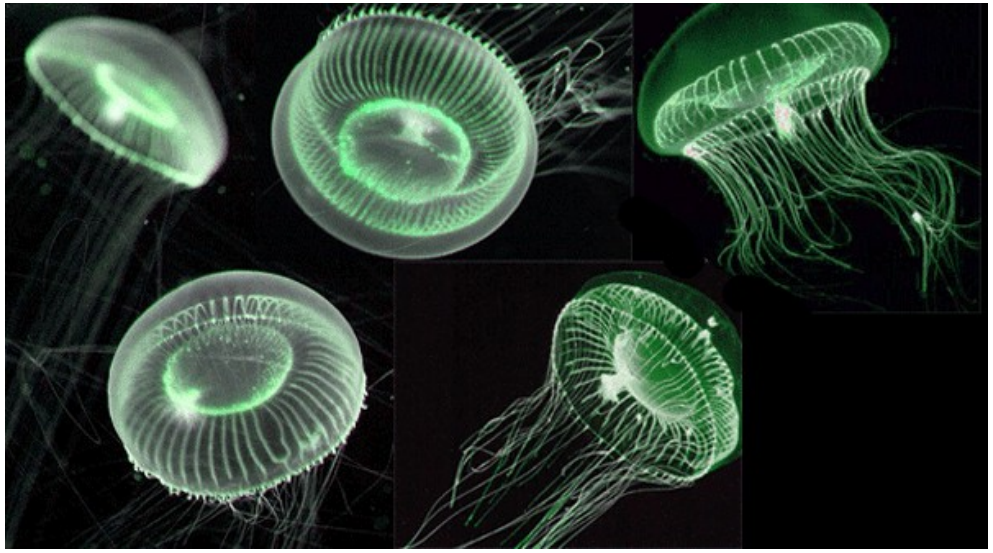
NanoLuc（Nluc）

源自深海细脚刺虾，是一个经过基因工程改造的小分子酶（19.1kDa， 171 个氨基酸， 513bp），是 Promega 的专利萤光素酶，结构简单，没有翻译后修饰或者二硫键。它使用一种新型底物— **Furimazine**，不依赖 ATP，光信号更亮。



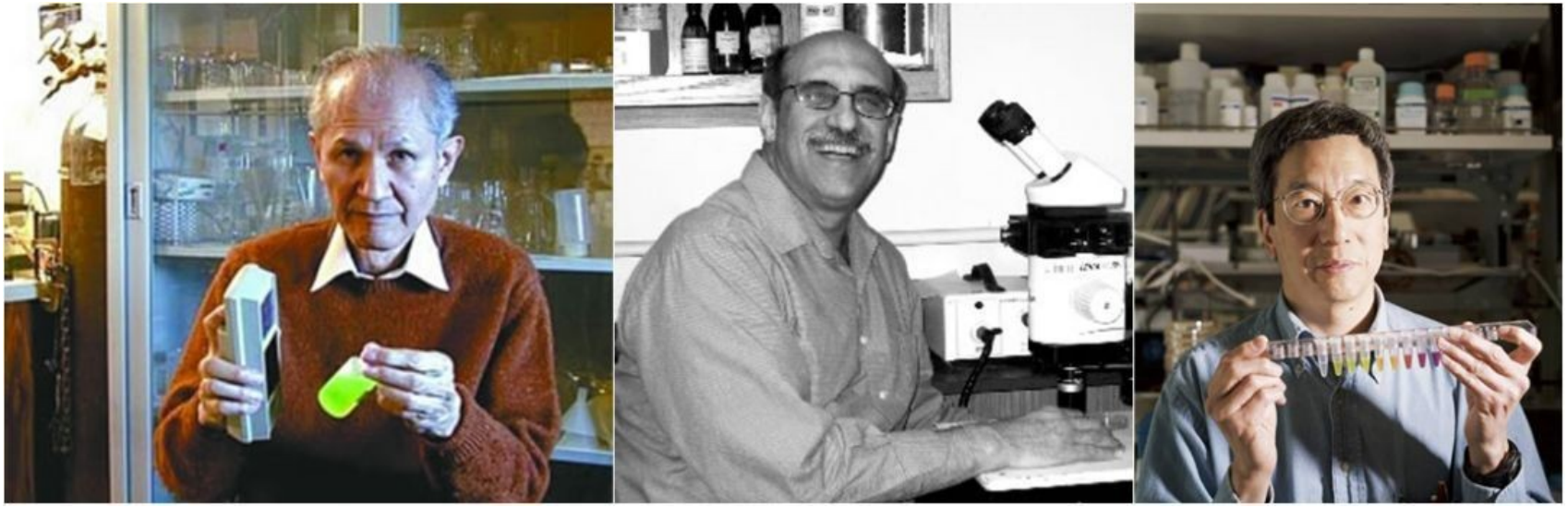


——百度图片



Aequorea

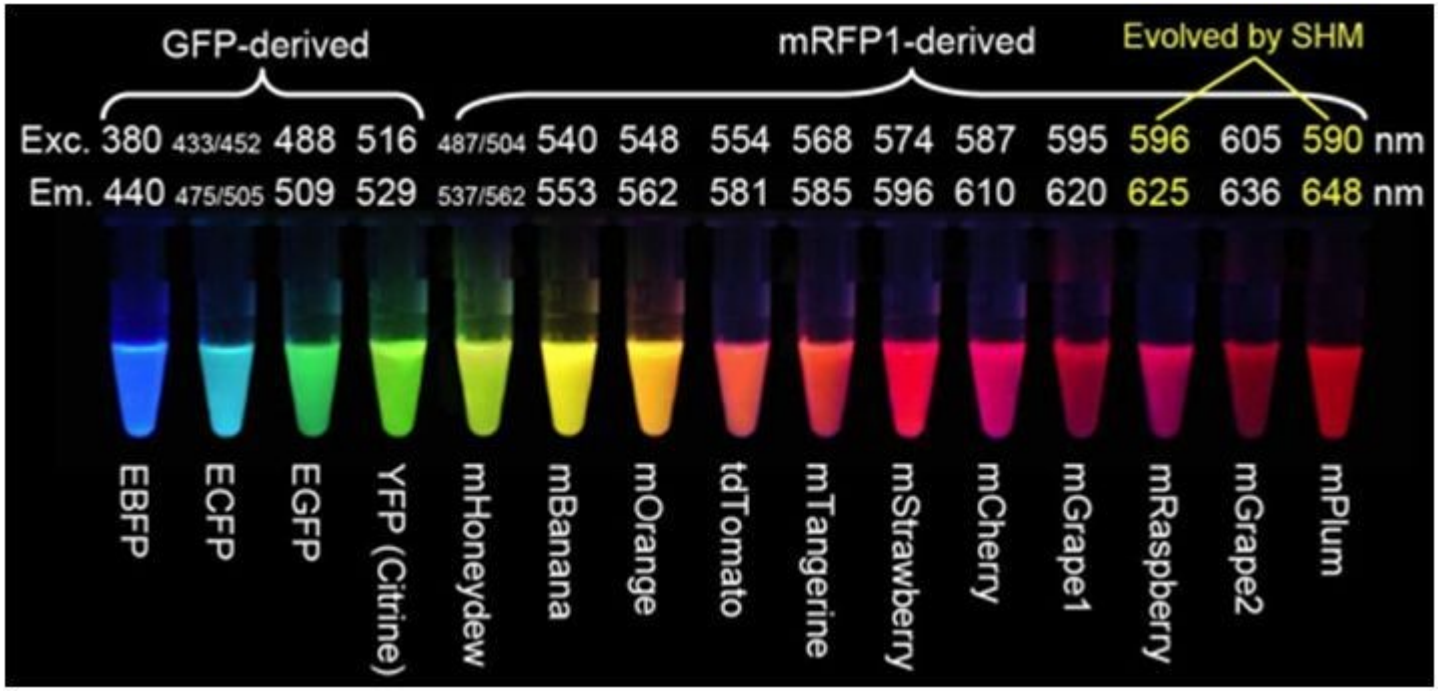
Green fluorescent protein
绿色荧光蛋白



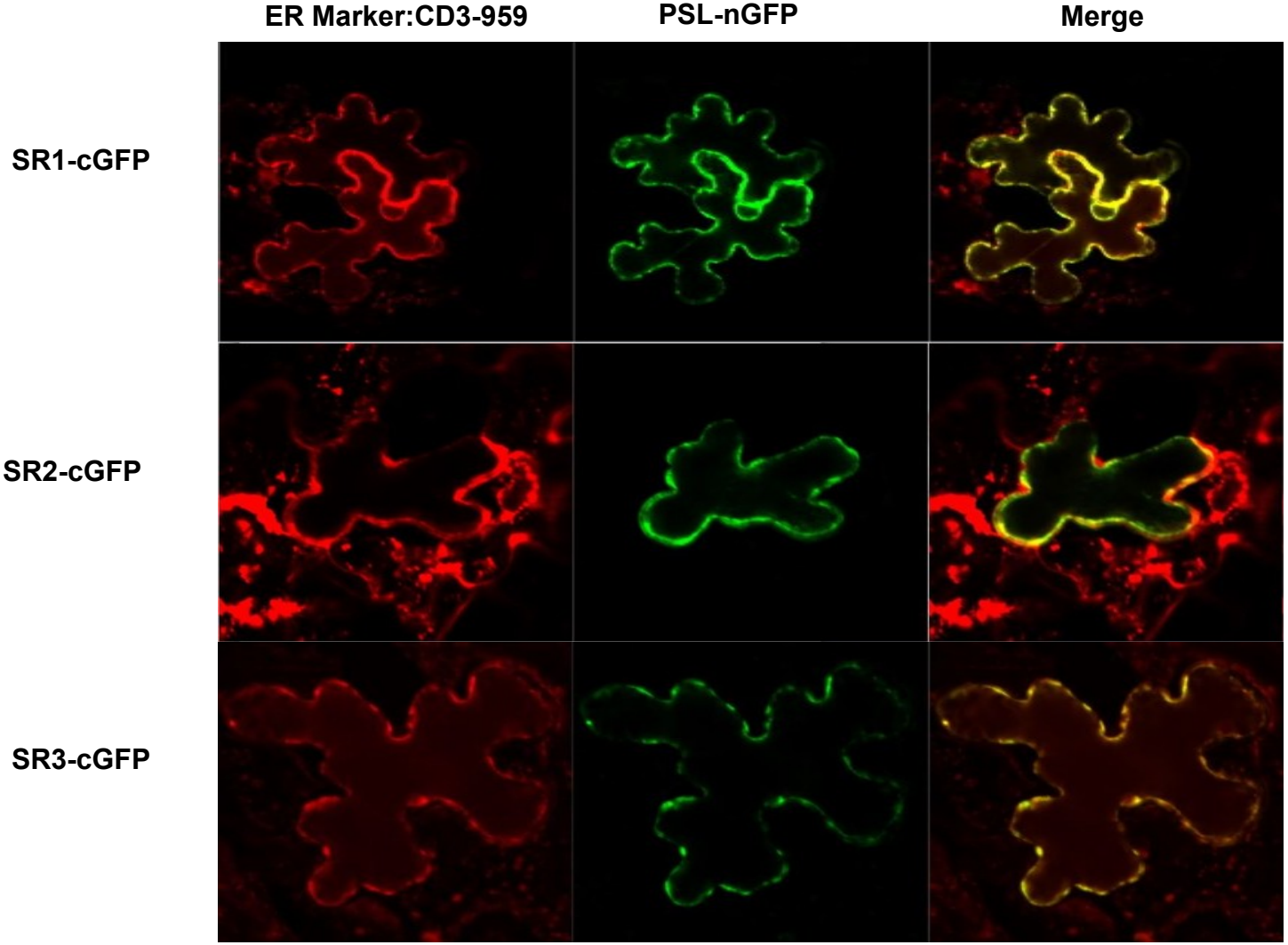
2008 年，瑞典皇家科学院将当年诺贝尔化学奖分别授予日本科学家下村修（左一）、美国科学家马丁·查尔菲（中）和美籍华裔科学家钱永健（左三）

钱永健演讲

https://www.bilibili.com/video/BV1f4411k755?is_story_h5=false&p=1&share_from=ugc&share_medium=android&share_plat=android&share_session_id=c4e40565-2c67-4466-8031-7021e3f37bdc&share_source=WEIXIN&share_tag=s_i&timestamp=166229522



钱永健实验室改造的一系列荧光蛋白变体及其被激发光照射后呈现出的颜色



路俊尧完成

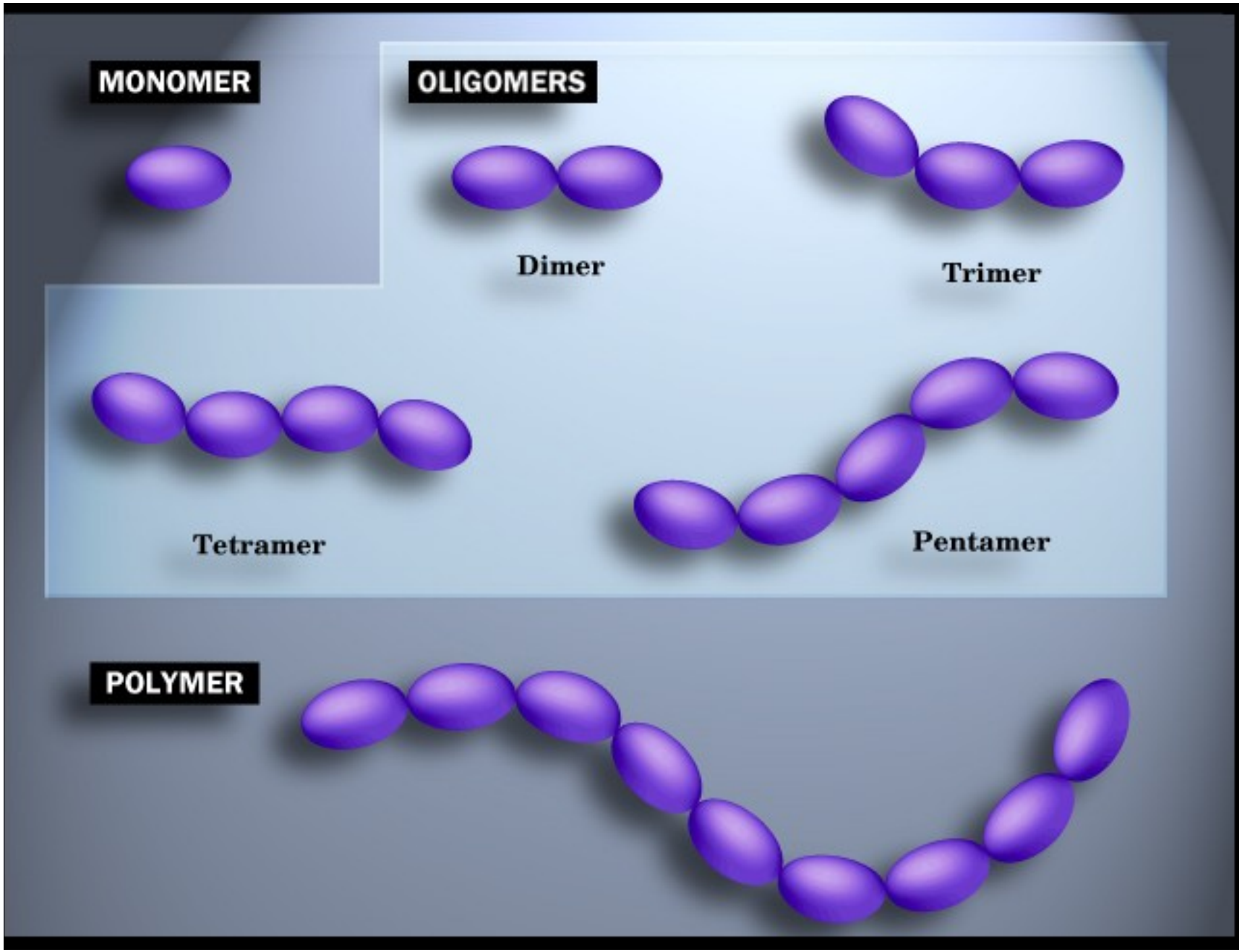
Biochemistry

B0. General Introduction

Proteins are polymers(多聚体) of amino acids.

20 amino acids (氨基酸) ➔ millions of proteins with different properties and activities.




MONOMER
OLIGOMERS
Dimer
Trimer
Tetramer
Pentamer
POLYMER

monomer —— 单体；

from Greek mono "one" and meros "part")

dimer —— 二聚体；

trimer —— 三聚体；

tetramer —— 四聚体；

pentamer —— 五聚体；

hexamer —— 六聚体；

heptamer —— 七聚体；

octamer —— 八聚体；

nonamer —— 九聚体；

decamer —— 十聚体；

Single (1) (also: singleton, sole, only, etc.)

Double (2) (also: pair, twice)

Triple (3) (also: triplet, treble, thrice, threesome, troika, trio)

Quadruple (4)

Quintuple or Pentuple (5)

Sextuple or Hextuple (6)

Septuple (7)

Octuple (8)

Nonuple (9)

Decuple (10)

Hendecuple or Undecuple (11)

Duodecuple (12)

Centuple (100)

Biological Monomers

What to Look For = What's Important:

Functional Groups: amino, carboxyl, carbonyls (both), alcohol, methyl, phosphate, sulfhydryl, and others.

Ionization state, or not.

Solubility

Covalent Bonds – single, double, triple.

Weak Bonds = H-bonds, Ionic bonds, hydrophobic interactions, van der Waals forces.

How Monomers are Polymerized

Proteins are polymers(多聚体) of amino acids.

20 amino acids (氨基酸) ➔ millions of proteins with different properties and activities.

Protein structures are studied at primary,

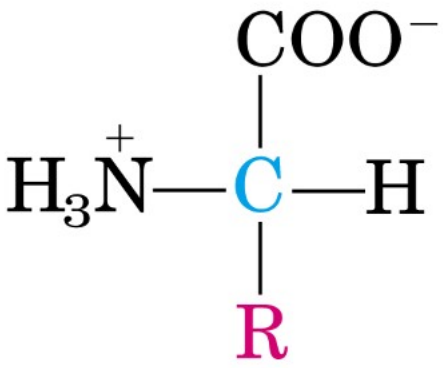
secondary,

tertiary

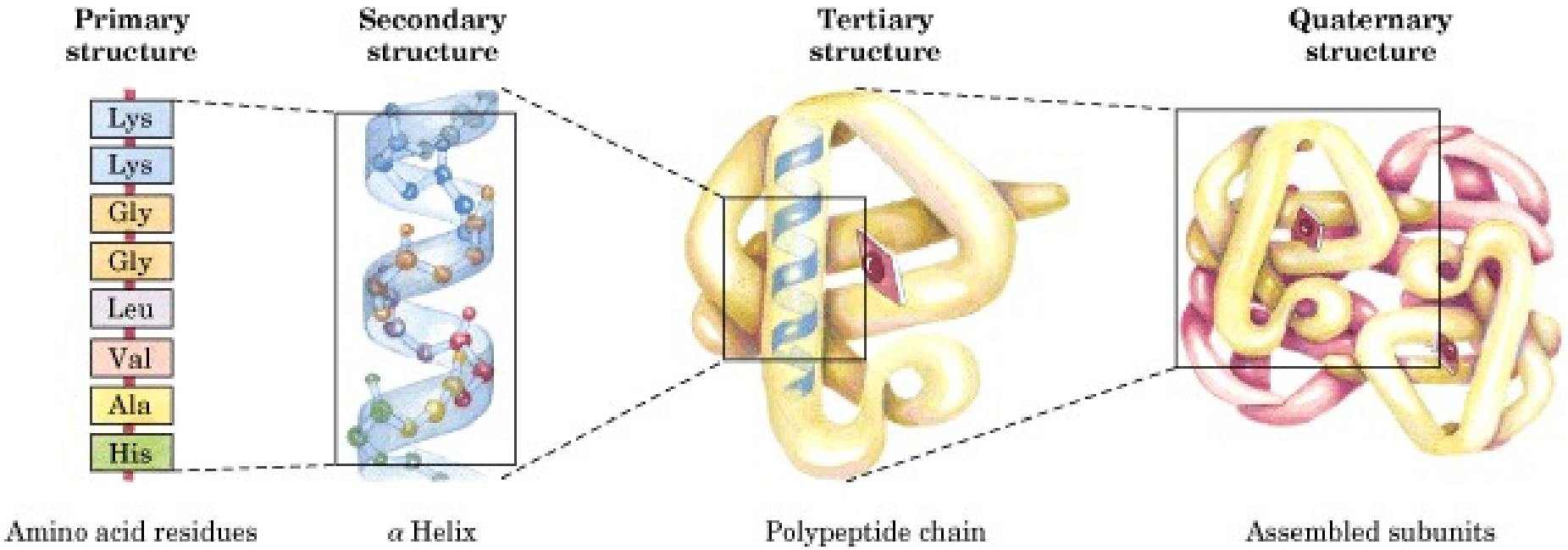
and quaternary levels.

高级构象




COO⁻
H₃N⁺—C—H
R

amino acid


Primary structure
Secondary structure
Tertiary structure
Quaternary structure
Lys
Lys
Gly
Gly
Leu
Val
Ala
His
Amino acid residues
α Helix
Polypeptide chain
Assembled subunits

Carbon Bonding

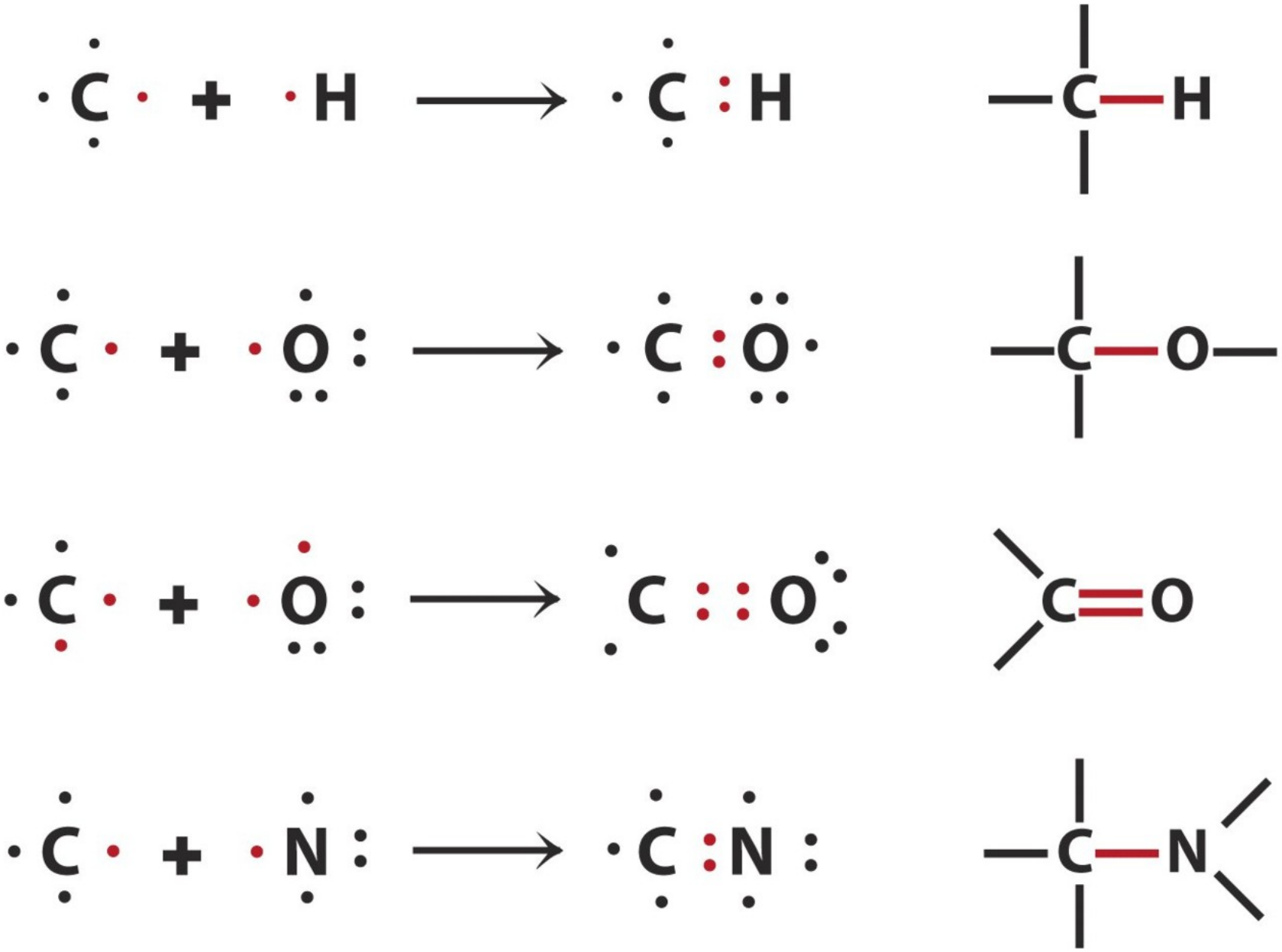


TABLE 1–1 Strengths of Bonds Common in Biomolecules

Type of bond	*Bond dissociation energy* (kJ/mol)*	*Type of bond*	*Bond dissociation energy (kJ/mol)*
Single bonds		**Double bonds**	
O—H	470	C=O	712
H—H	435	C=N	615
P—O	419	C=C	611
C—H	414	P=O	502
N—H	389		
C—O	352	**Triple bonds**	
C—C	348	C≡C	816
S—H	339	N≡N	930
C—N	293		
C—S	260		
N—O	222		
S—S	214		

*The greater the energy required for bond dissociation (breakage), the stronger the bond.

Biological molecules typically have several functional groups

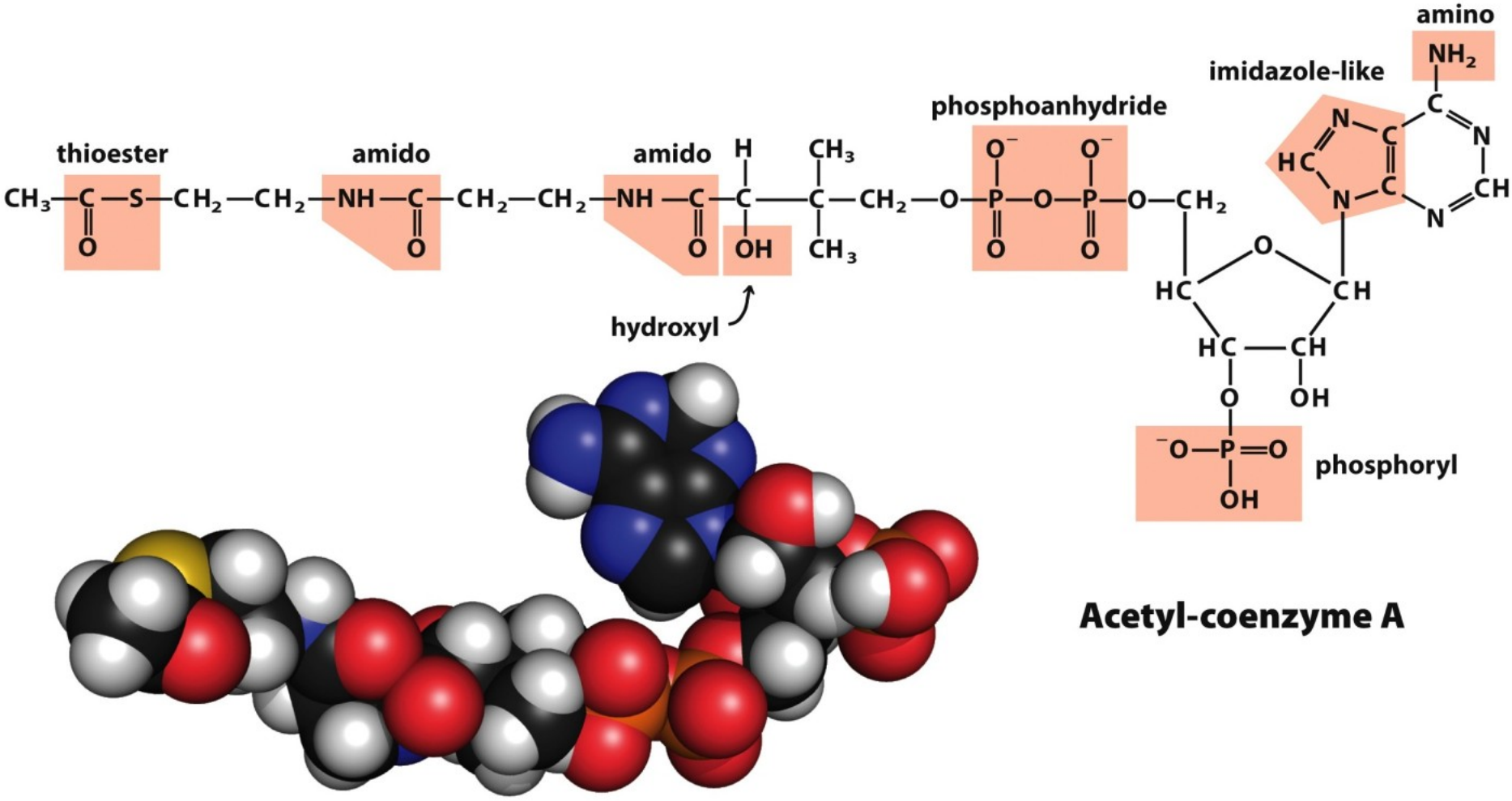
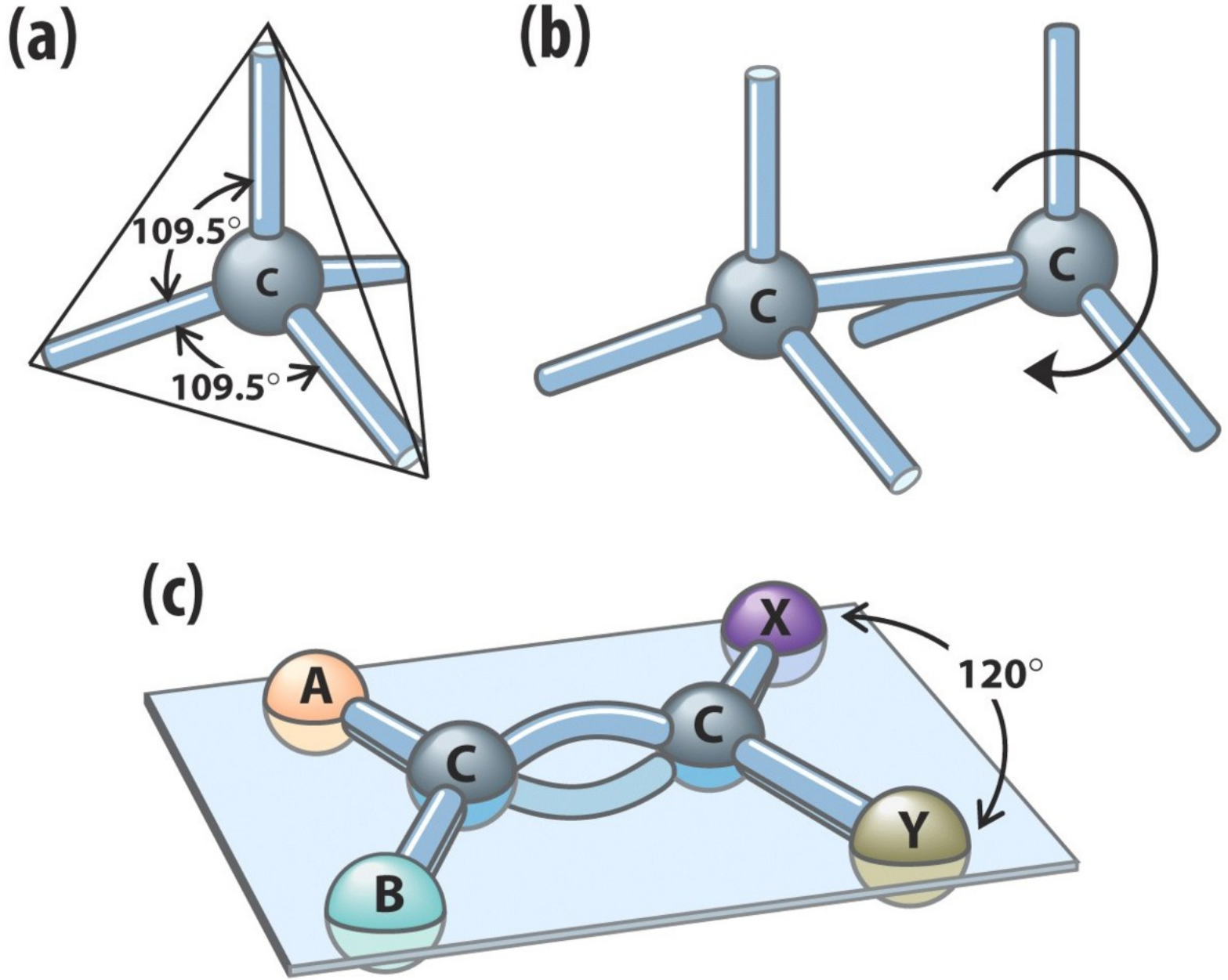
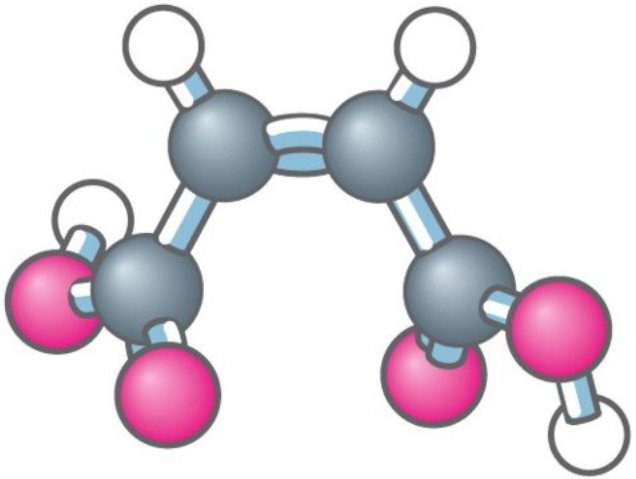


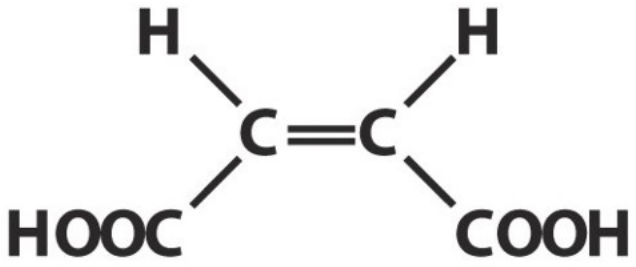
Figure 1-17
Lehninger Principles of Biochemistry, Sixth Edition
© 2013 W. H. Freeman and Company

Bond Angles and Rotation

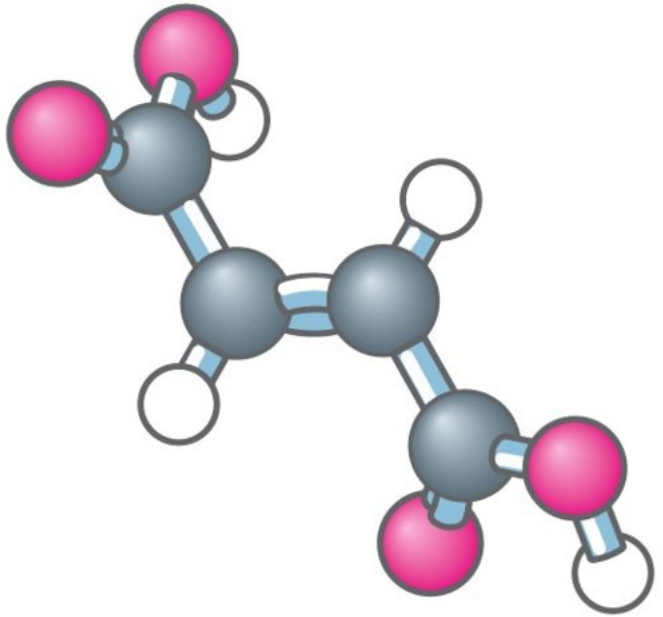


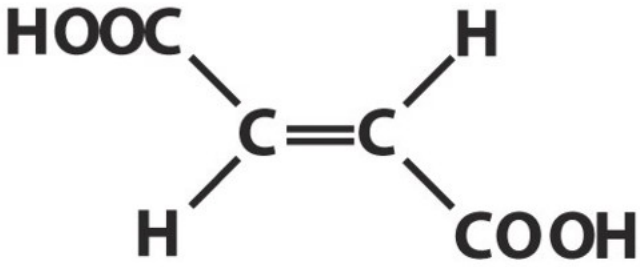
Cis and *Trans*





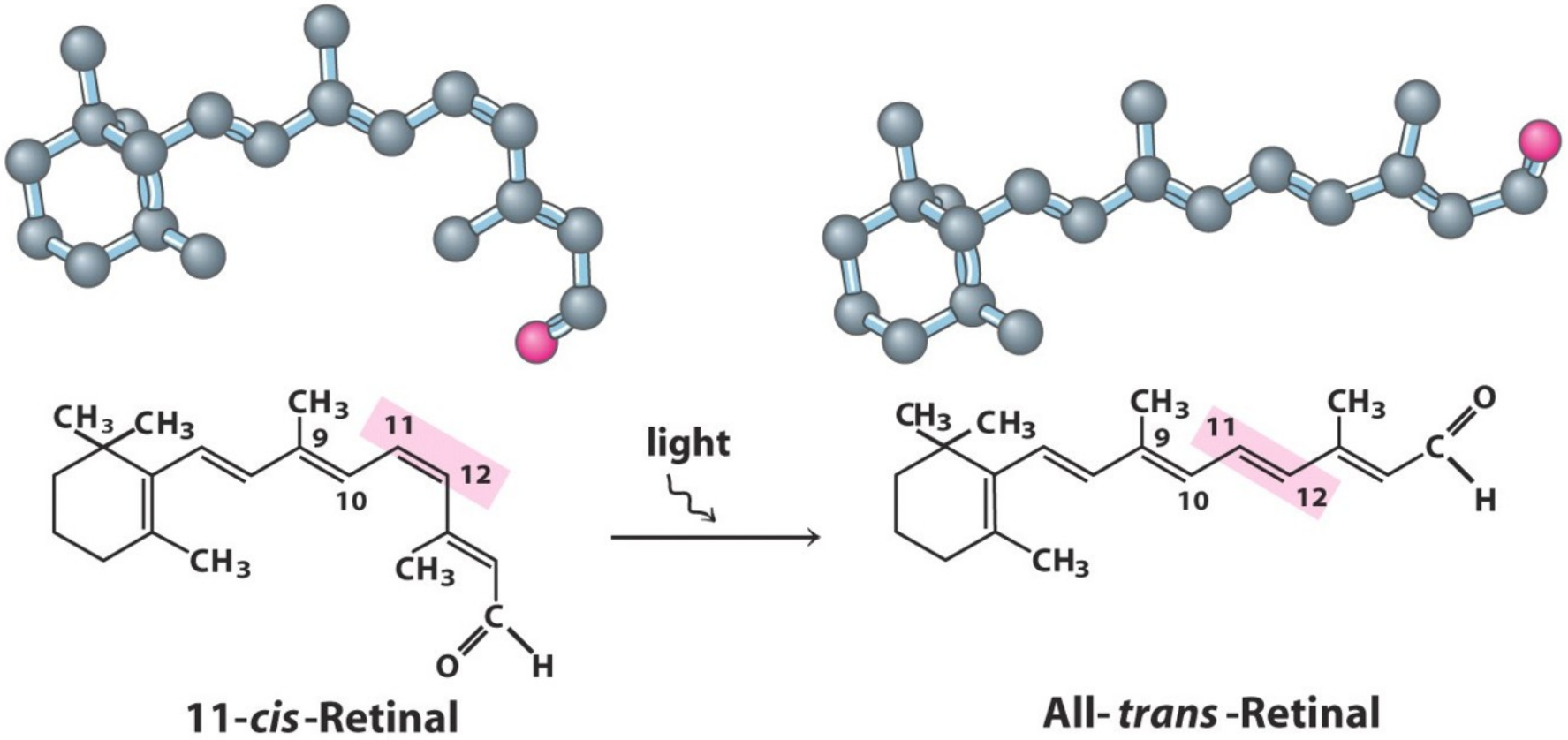
Maleic acid (cis)





Fumaric acid (trans)

Cis and Trans –Change



Summary of B0

- Proteins are polymers of 20 amino acids;
- Protein structures are studied at primary, secondary, tertiary and quaternary levels.
- Biomolecules are compounds of carbon with a variety of functional groups.



Section B1 Amino acid structure

1. All natural proteins were found to be built from the same set of 20 standard α-amino acids.

2. The 20 amino acids are usually grouped according to the properties (mainly polarity) of their R groups.

3. Nonstandard amino acids are found in certain proteins, generally as a result of post-translational modifications.

4. The amino acids ionize in aqueous solutions.

1. All natural proteins were found to be built from the same set of 20 standard α-amino acids

The earliest studies of proteins focused on the free amino acids derived from these proteins.

- **The 1st amino acid (asparagine) was discovered in 1806 from asparagus (a green vegetable).**
- **The last (threonine) was not identified until 1938!**
- **Glutamate from wheat gluten (sticky). Tyrosine from cheese .**

The names of the amino acids are often abbreviated, either to three letters or to a single letter.

- Glycine (Gly, G)　甘氨酸
- Alanine (Ala, A)　丙氨酸
- Valine (Val, V)　缬氨酸
- Leucine (Leu, L)　亮氨酸
- Isoleucine (Ile, I)　异亮氨酸
- Methionine(Met, M)　蛋氨酸
- Proline (Pro, P)　脯氨酸

Hydrophobic, aliphatic amino acids

Polar, uncharged amino acid

- Serine (Ser, S) 丝氨酸
- Threonine (Thr, T) 苏氨酸
- Cysteine (Cys, C) 半胱氨酸
- Asparagine (Asn, N) 天冬酰胺
- Glutamine (Gln, Q) 谷氨酰氨



Hydrophobic, aromatic amino acids

- Phenylalanine (Phe,F) 苯丙氨酸
- Tyrosine (Tyr,Y) 酪氨酸
- Tryptophan (Trp, W) 色氨酸



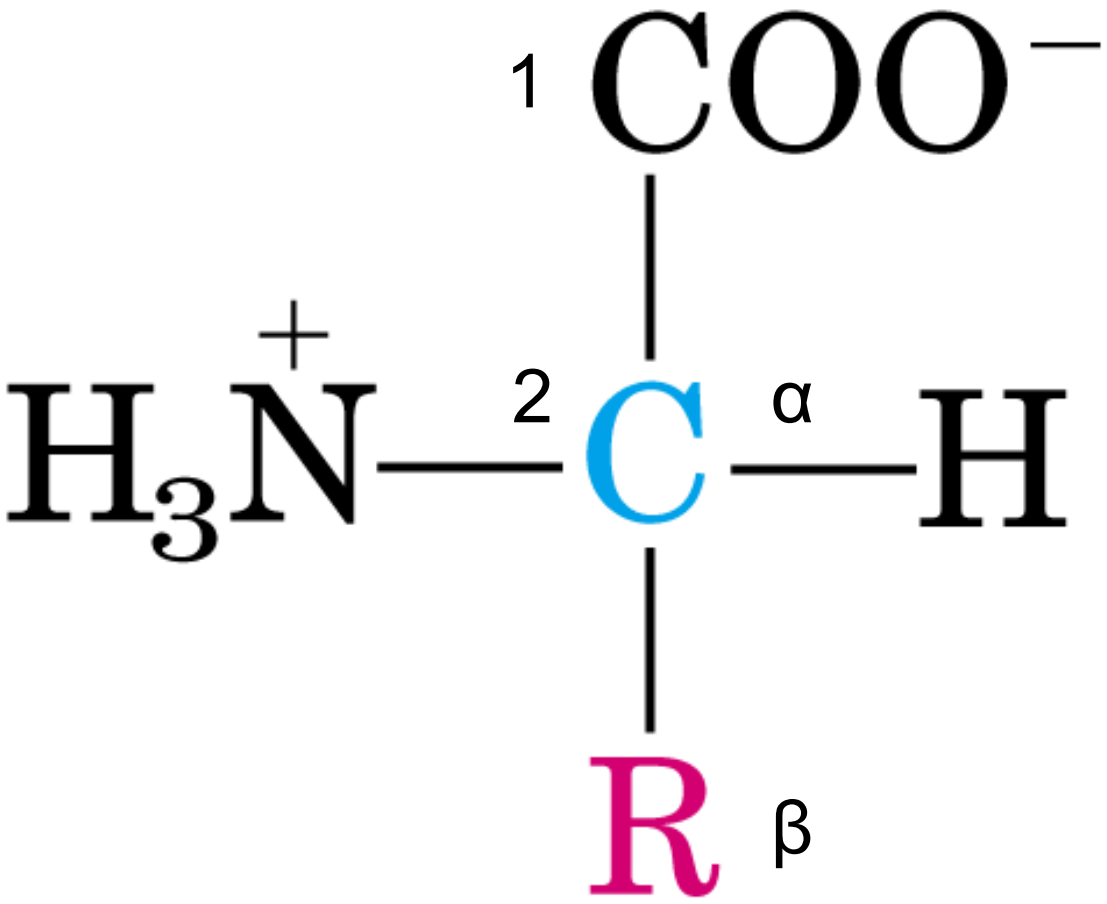
Polar, charged amino acid

- Arginine (Arg, R)　精氨酸
- Lysine (Lys, K)　赖氨酸
- Histidine (His, H)　组氨酸
- Aspartate (Asp, D)　天冬氨酸
- Glutamate (Glu, E)　谷氨酸

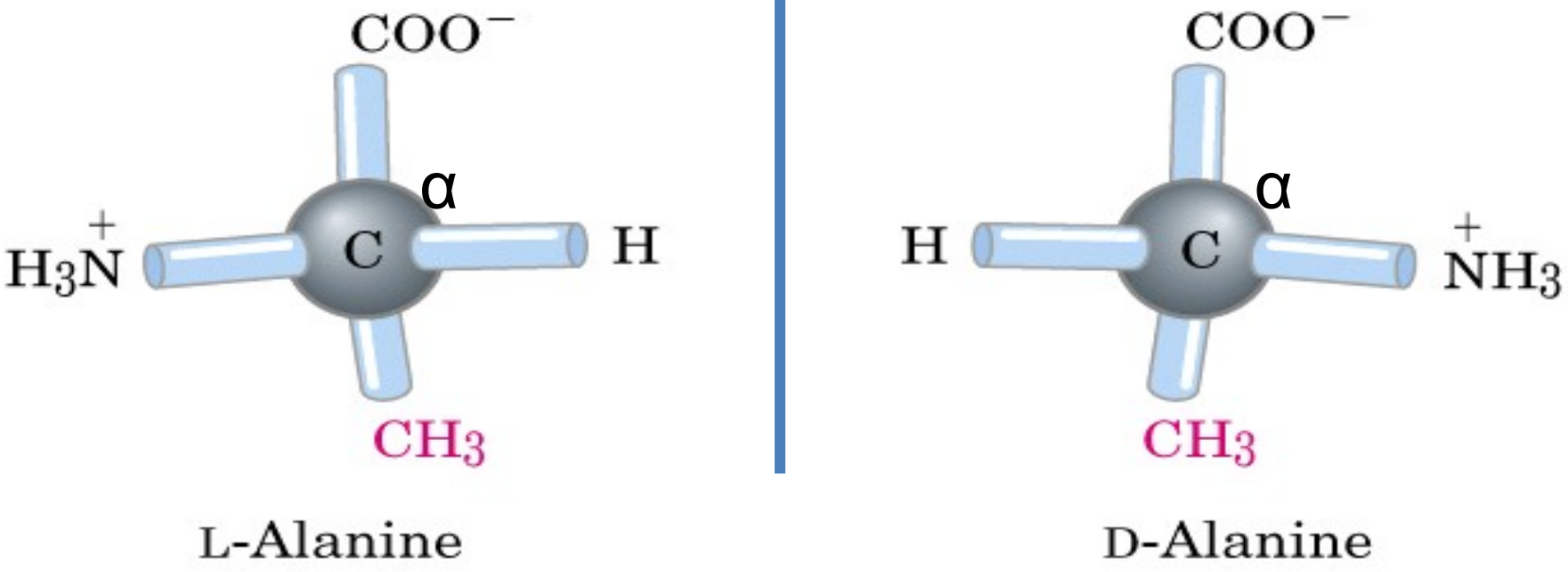
1.1 The 20 α-amino acids share common structural features.

- Each has a carboxyl group and an amino group (but one has an imino group in proline) bonded to the same carbon atom, designated as the α-carbon.

- **Each has a different side chain (or R group, R="Remainder of the molecule").**

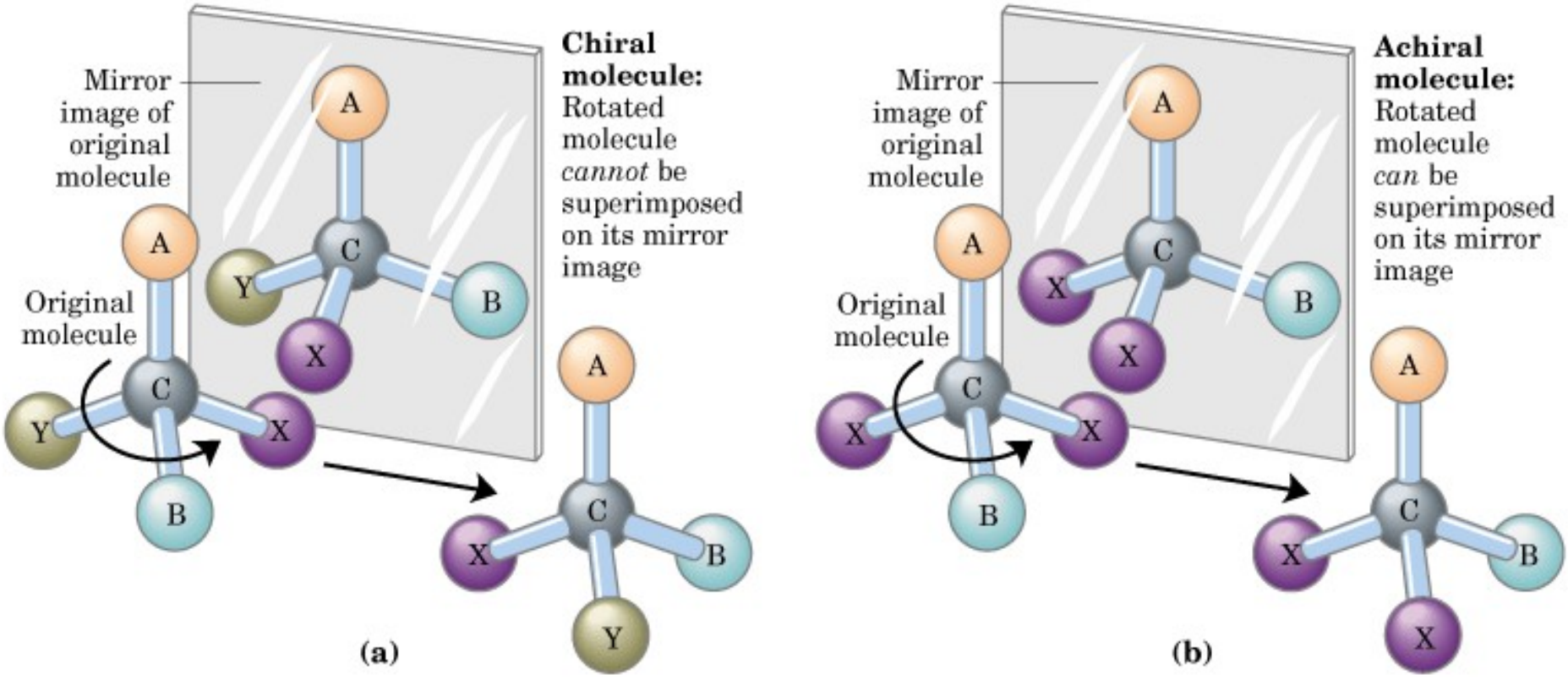


The α-carbons for 19 of them are asymmetric (or chiral), thus being able to have two stereoisomers. Glycine has no chirality.

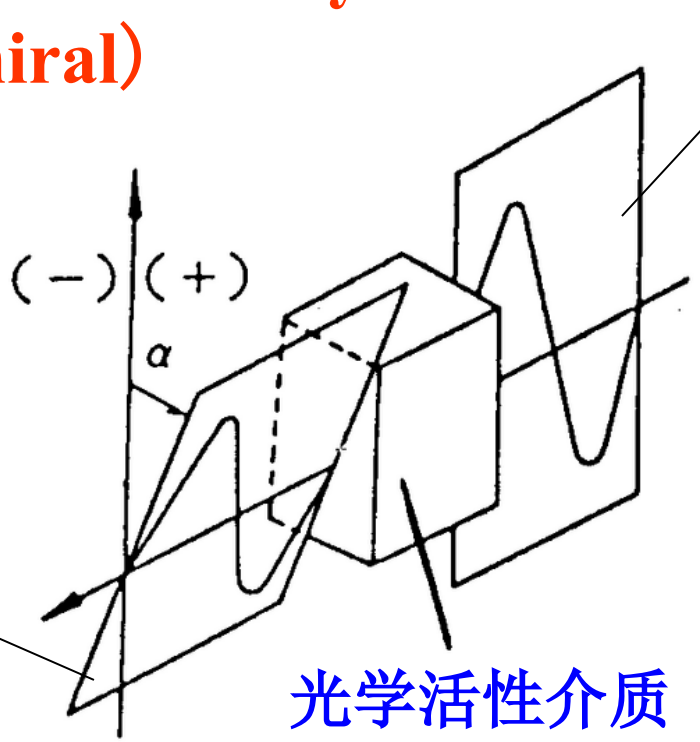


(a)

Enantiomers (The Cα atom is an asymmetric center or chiral center)



- 旋光性（ optical rotation ）
- 光学活性（ optical activity ）
- “手性” （chiral）



某些光学活性介质能使平面偏振光的偏振面发生旋转

About AA stereoisomers :

- **The correlation of structure (or configuration) with optical rotation is very complex and has not been successful to date! (i.e., the D- and L-signs do not tell anything about their optical rotation!)**
- **Only the L-amino acids have been found in proteins (D-isomers have been found only in small peptides of bacteria cell walls and in some peptide antibiotics).**

L-α-AA

1.2 Each amino acid is given a three-letter abbreviation and a one-letter symbol. They often use the first three letter and the first letter. When there is confusion, an alternative is used. These must be remembered.

- Glycine (Gly, G)
- Alanine (Ala, A)
- Valine (Val, V)
- Leucine (Leu, L)
- Isoleucine (**Ile**, I)
- Methionine(Met, M)
- Proline (Pro, P)

(total 7 AAs)

- Arginine (Arg, **R**)
- Lysine (Lys, **K**)
- Histidine (His, H)
- Aspartate (Asp, **D**)
- Glutamate (Glu, **E**)
- (total 5 AAs)

- Serine (Ser, S)
- Threonine (Thr, T)
- Cysteine (Cys, C)
- Asparagine (**Asn**, **N**)
- Glutamine (**Gln**, **Q**)
- (Total 5 AAs)

- Phenylalanine (Phe,**F**)
- Tyrosine (Tyr,**Y**)
- Tryptophan (**Trp**, **W**)
- (total 3 AAs)

1.3 All proteins in all species (from bacteria to human) are constructed from the same set of 20 amino acids.

- All proteins, no matter how different they are in structure and function, are made from the same set of 20 standard amino acids.
- This fundamental alphabet of the protein language is at least two billion years old.



2. The 20 amino acids are usually grouped according to the properties (mainly polarity) of their R groups

2.1 seven amino acids have nonpolar, aliphatic (hydrophobic) R groups.

- They are Gly, Ala, Val, Leu, Ile, Met and Pro.
- Gly has a hydrogen as its R group, having minimal steric hindrance.
- In protein structure Gly offers the most flexibility!
- Ala, Val, Leu, Ile and Met have hydrocarbon R groups, often involved in hydrophobic interactions.

- **Pro has an imino group, instead of an amino group, forming a five-membered ring structure, being rigid in conformation.**
- **Pro is often found in the bends of folded protein chains and often present on the surface of proteins. It offers the least flexibility.**

Gly, G, 甘氨酸

Ala, A, 丙氨酸

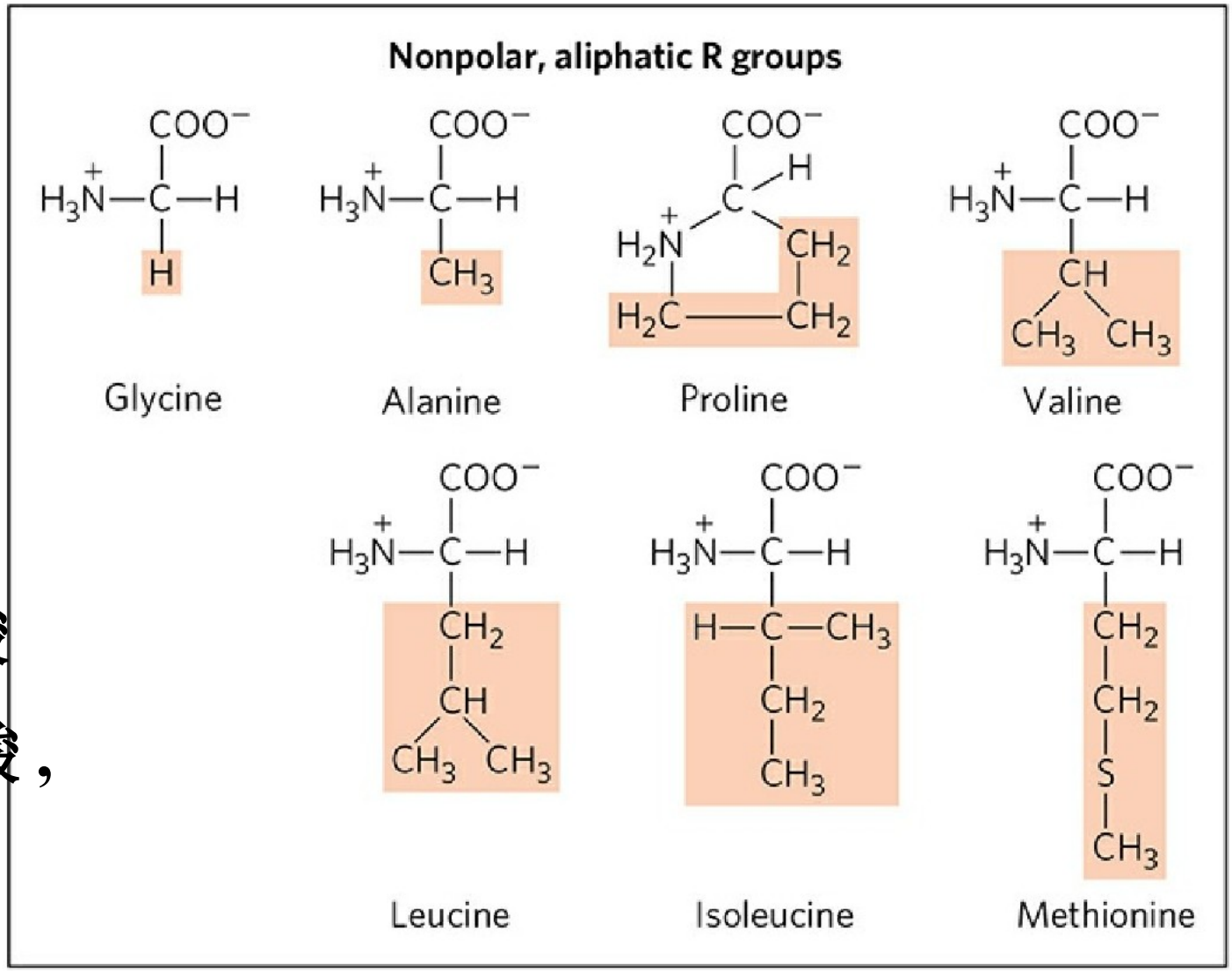
Val, V, 缬氨酸

Leu, L, 亮氨酸

Ile, I, 异亮氨酸

Met, M, 蛋氨酸 ,
甲硫氨酸

Pro, P, 脯氨酸



2.2 Ser, Thr, Asn, Gln, and Cys have polar, uncharged R groups.

- The R groups are more hydrophilic, due to the presence of hydroxyl groups, sulfur atoms, or amide groups.
- -SH group (sulfhydryl) of two Cys in proteins can be oxidized to form a covalent disulfide bond.
- Cys often participates in hydrophobic interactions.

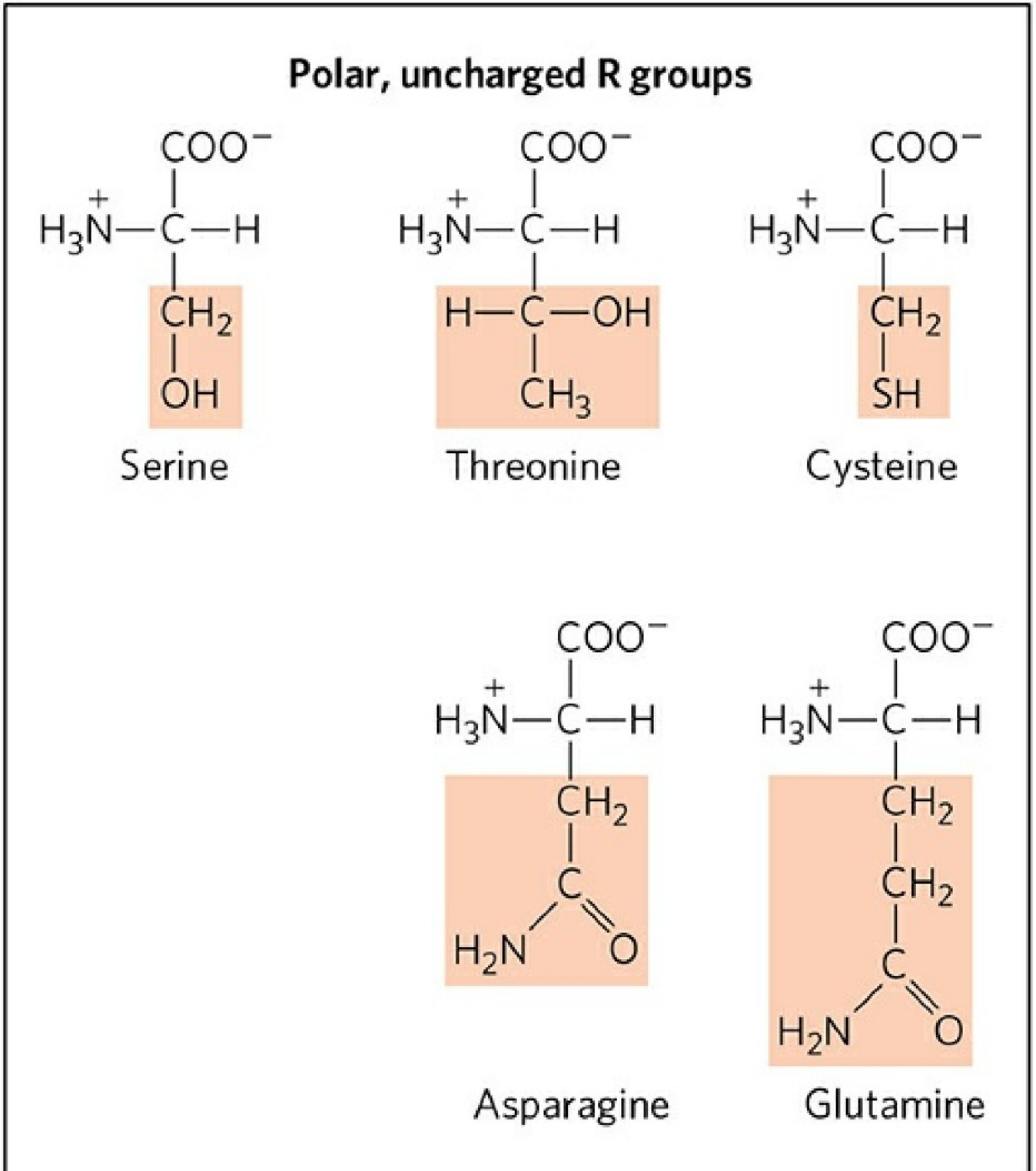
Ser, S, 丝氨酸

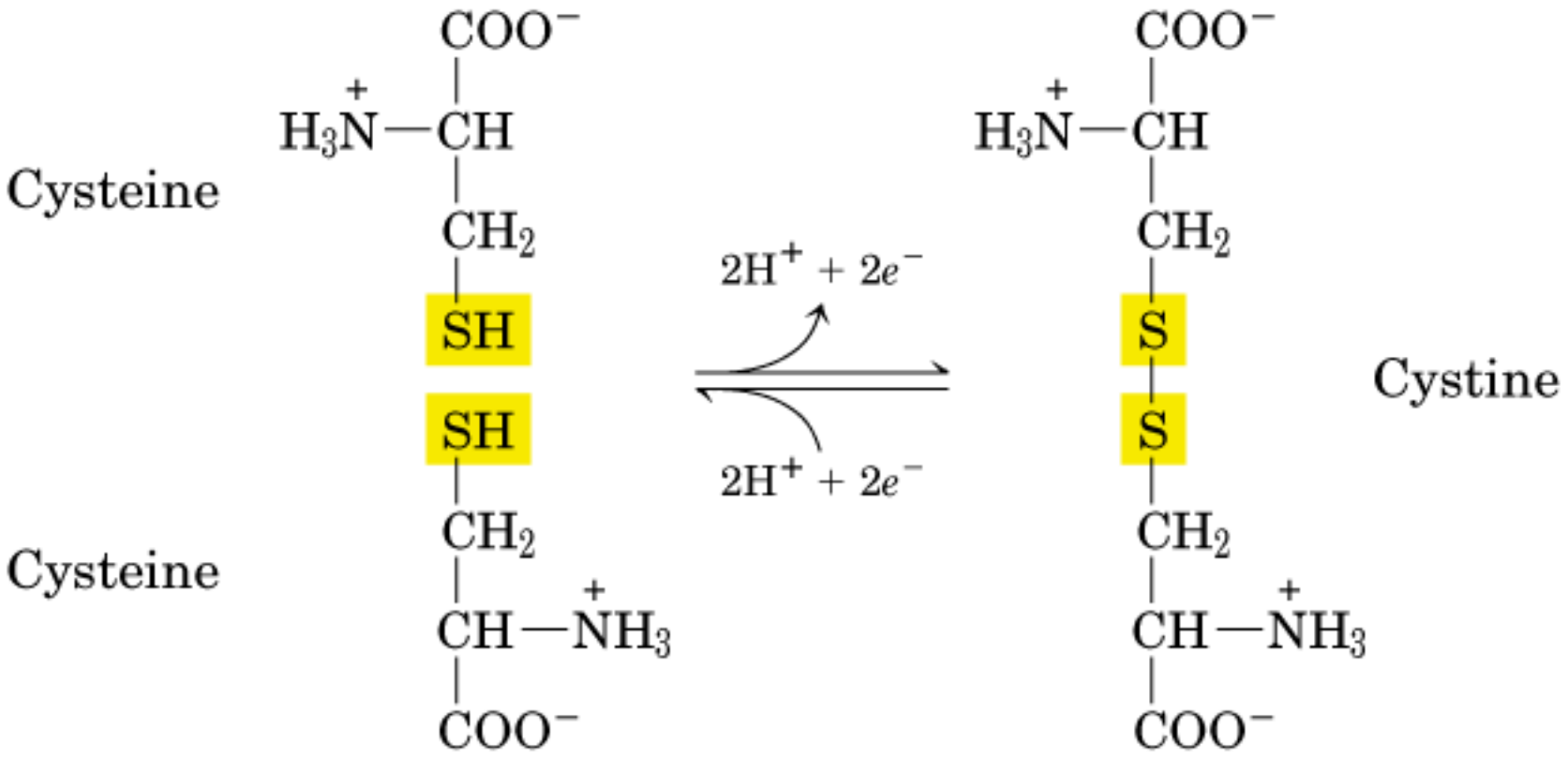
Thr, T, 苏氨酸

Cys, C, 半胱氨酸

Asn, N, 天冬酰胺

Gln, Q, 谷氨酰胺



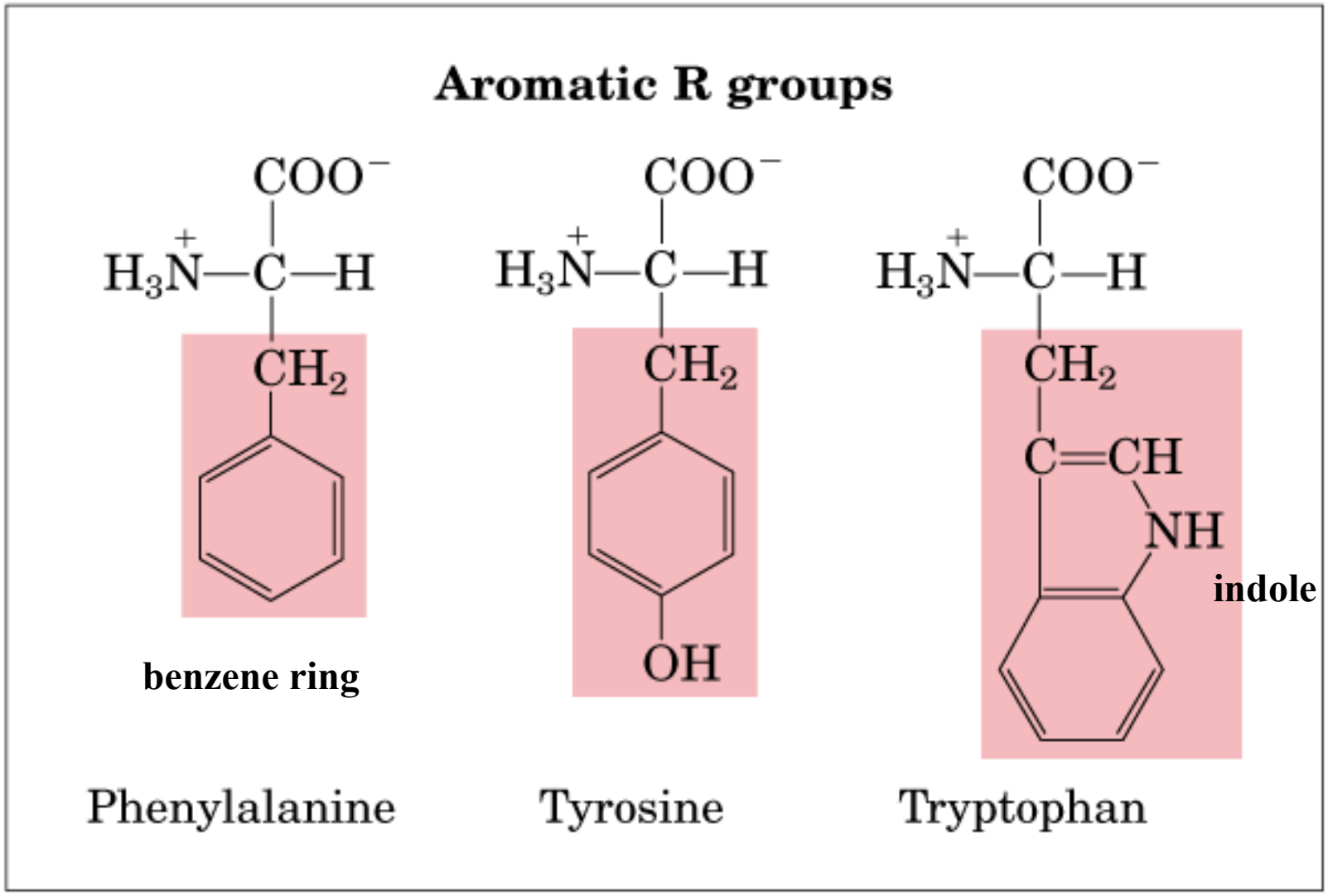


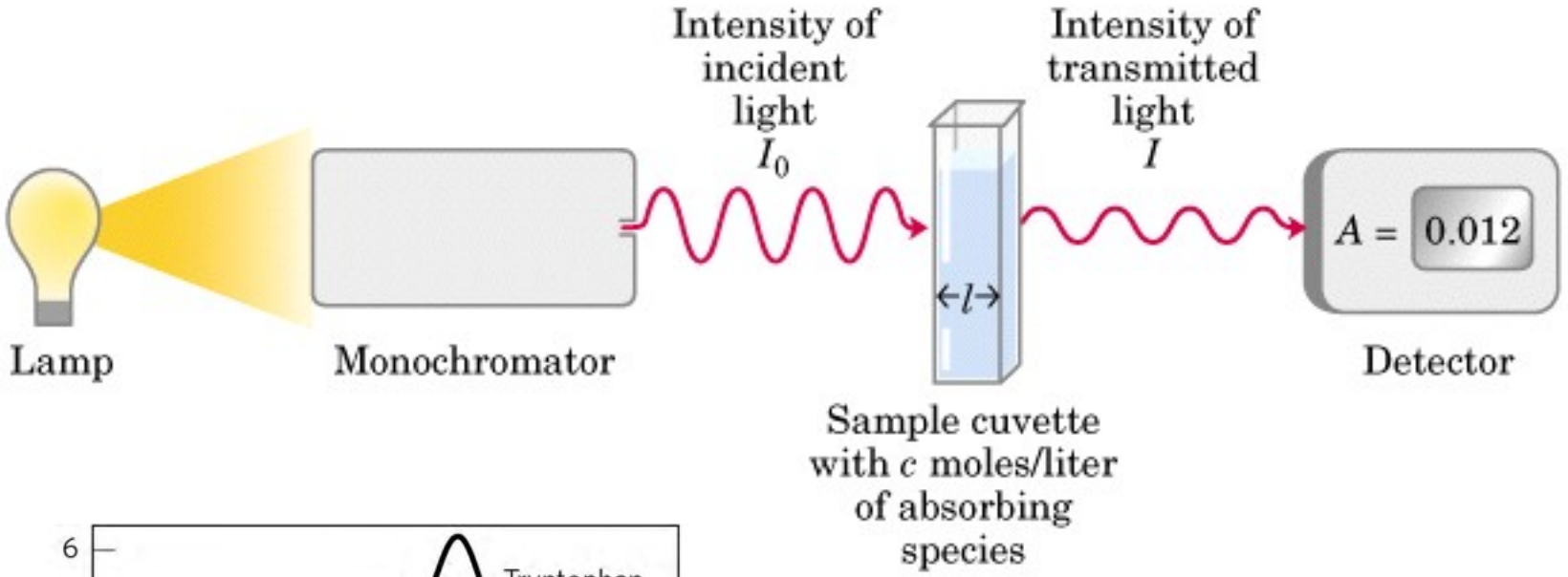
Cystine (胱氨酸) is a dimer of cysteine (半胱氨酸)

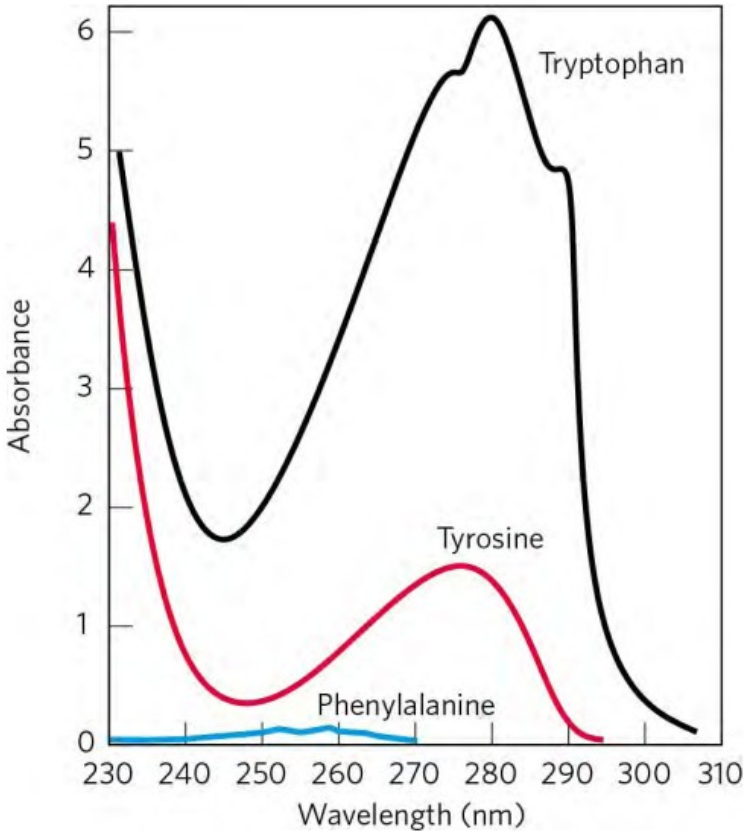
2.3 Phe, Tyr, and Trp have aromatic (芳香族) R groups

- **Phe and Tyr both have benzene rings.**
- **Tryptophan has an indole ring （吲哚） .**
- **All three participate in hydrophobic interactions.**
- **The -OH group in Tyr is an important functional group in proteins. (phosphorylation, hydrogen bond, etc)**
- **They are jointly responsible for the light absorption of proteins at 280 nm, a property used as a measure of the concentration of proteins.**

Phe, F, 苯丙氨酸 ; Tyr, Y, 酪氨酸 ; Trp, W, 色氨酸







Lambert-Beer's law:

$A = \text{Log } I_o/I = \varepsilon cl$

ε, extinction coefficient;

c, concentration;

l, optical length

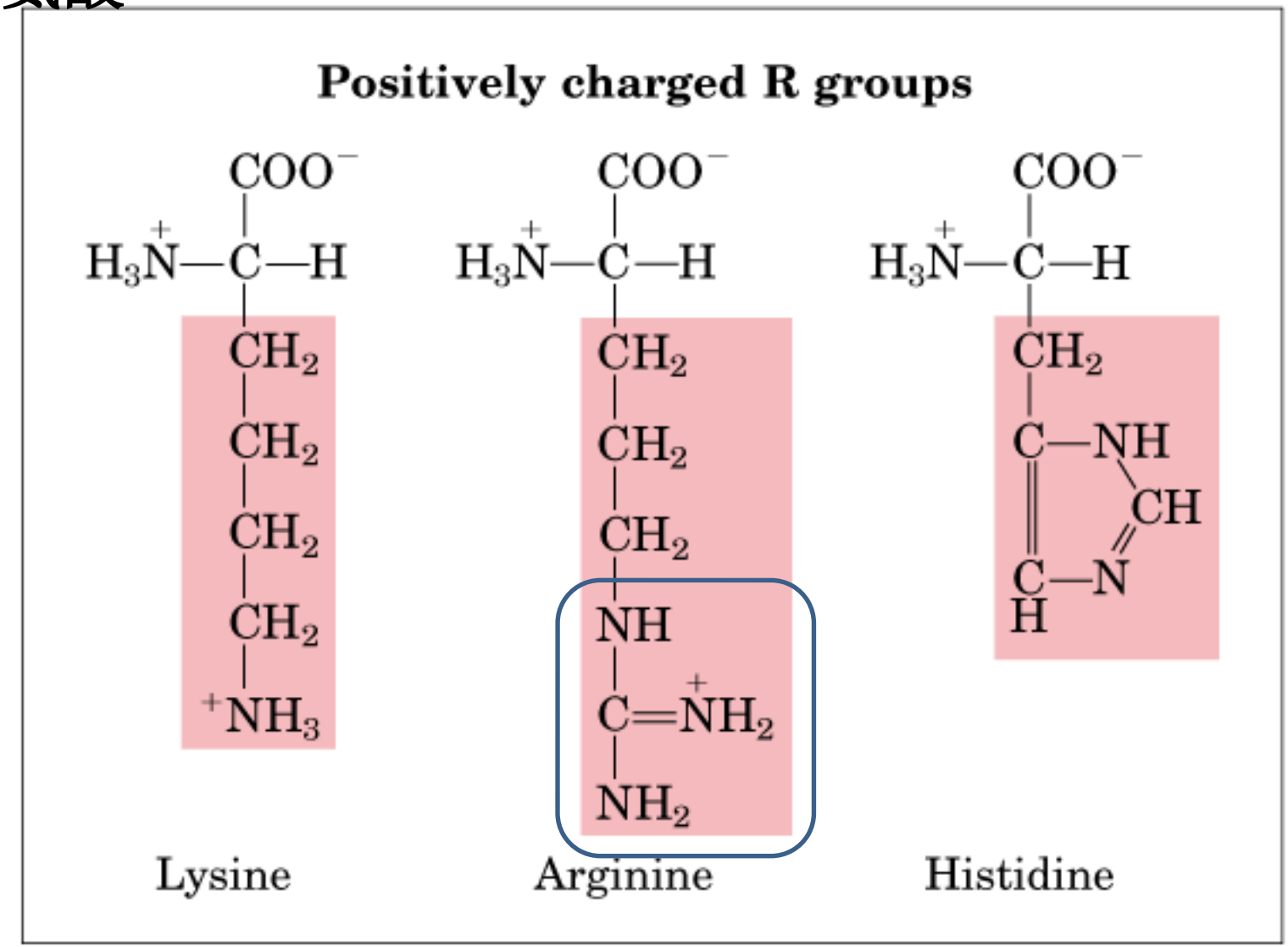
2.4 Positively charged (basic) and negatively charged (acidic) R groups

- **The most hydrophilic R groups are those either positively or negatively charged.**
- **Asp and Glu have carboxyl in their R groups. They have net negative charge at pH 7.0, thus usually named as aspartate and glutamate (conjugate base names, instead of aspartic acid and glutamic acid, un-ionized form).**

- **Arg, Lys, and His have positively charged R groups at pH 7.0.**
  - **Their R groups contain guanidino(胍基), amino, imidazole(咪唑基) groups respectively.**
  - **The side chain of His can be positively or uncharged depending on the local environment near pH 7.0**



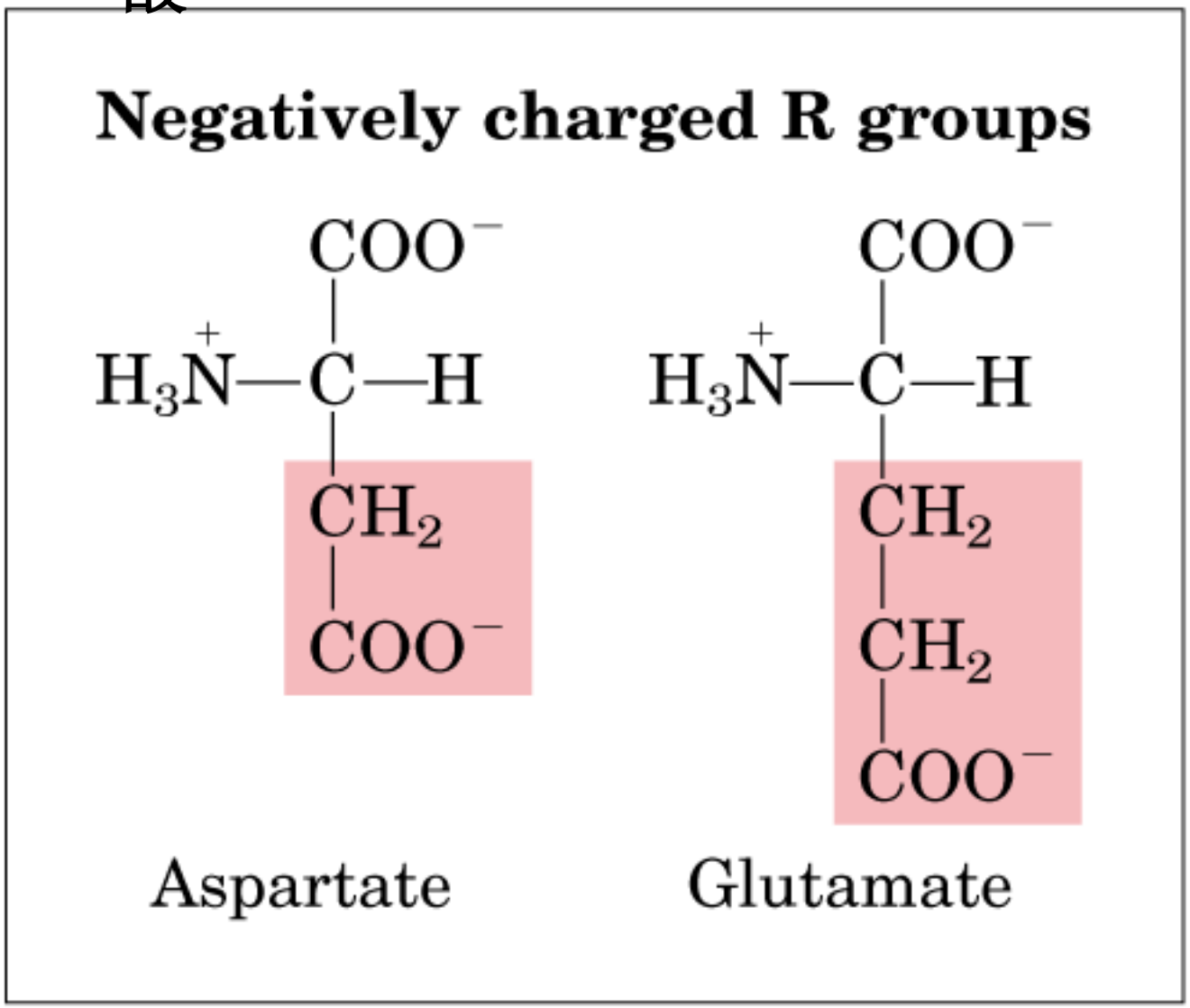
Lys, K, 赖氨酸 ; Arg, R, 精氨酸 ; His, H, 组氨酸

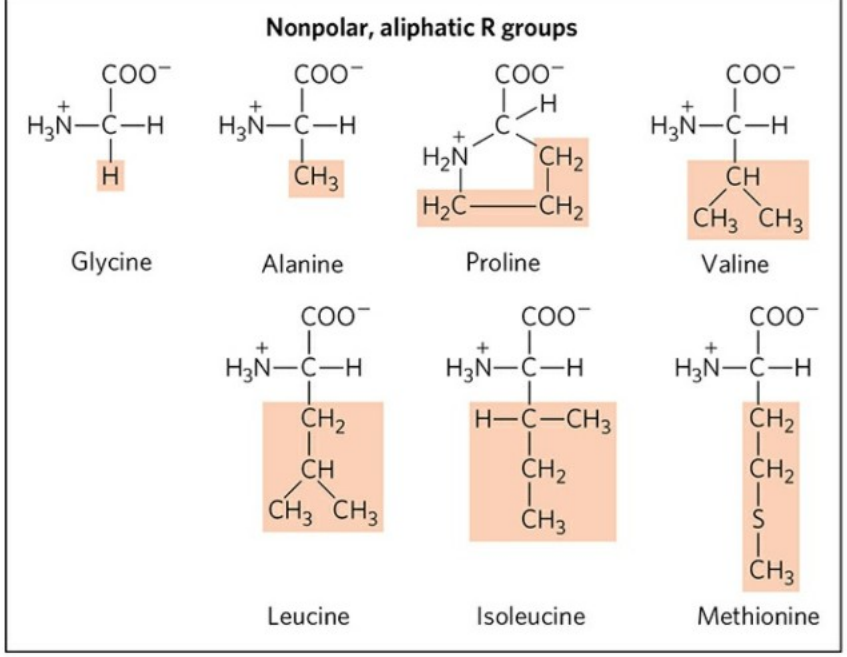


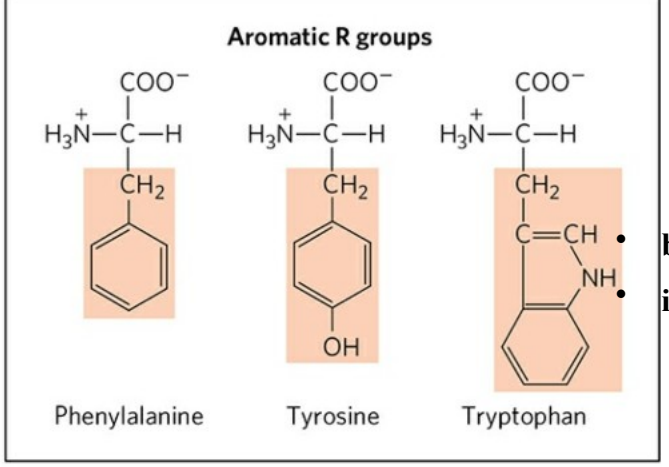
guanidino(胍基)

imidazole(咪唑基)

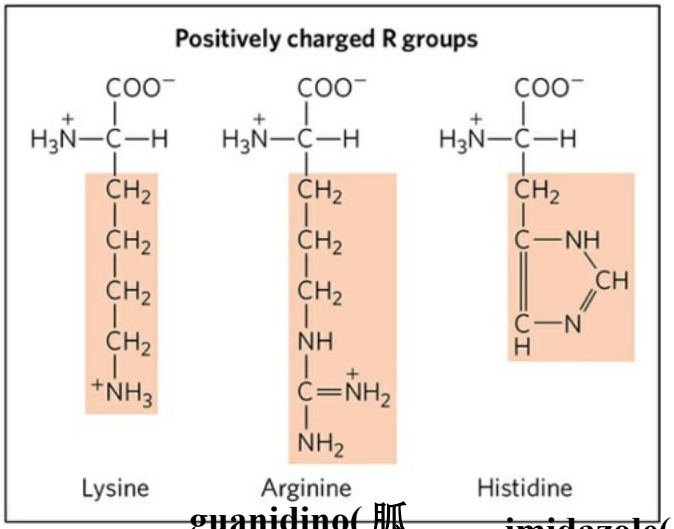
Asp, D, 天冬氨酸 ; Glu, E, 谷氨酸





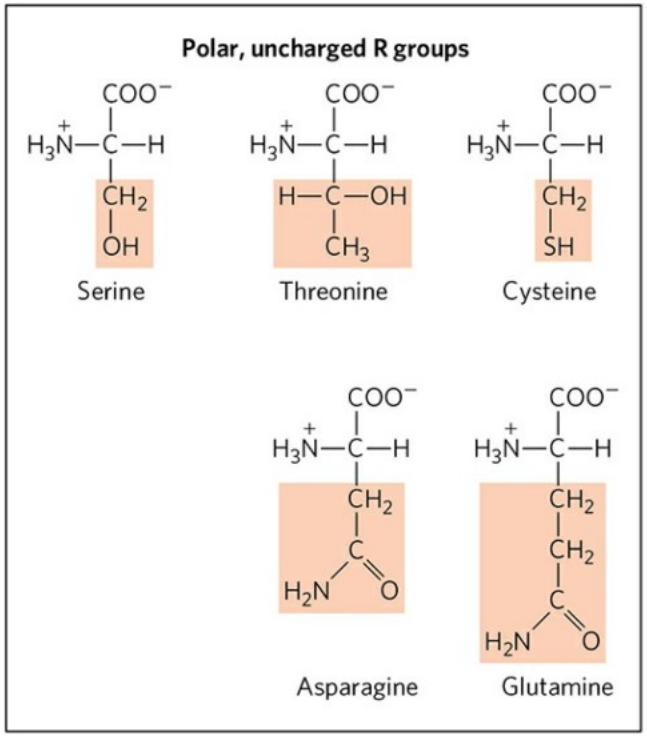


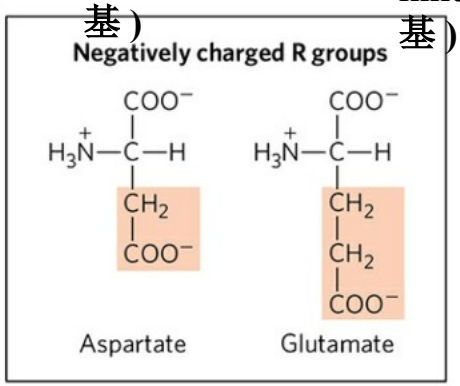
- **benzene rings.**
- **indole ring** （吲哚）



guanidino(胍基)

imidazole(咪唑基)

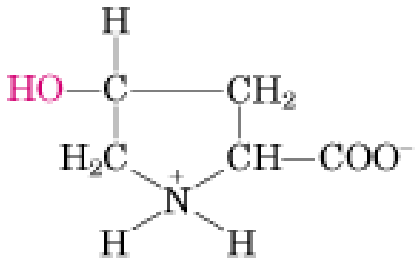




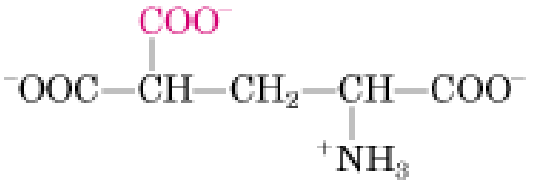
next

3. Nonstandard amino acids are found in certain proteins, generally as a result of post-translational modifications.

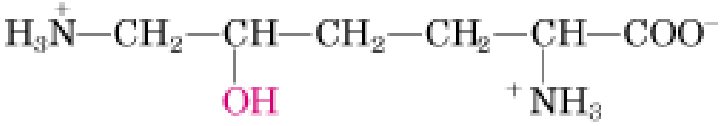
- **4-Hydroxyglutamate and 5-hydroxylysine in collagen(胶原).**
- **γ-carboxyglutamate is found in the blood-clotting prothrombin (凝血酶原 ,an enzyme).**
- **Many additional nonstandard amino acids are found in cells, but not in proteins (e.g., ornithine 鸟氨酸 and citrulline 瓜氨酸 , intermediates in amino acid metabolism).**



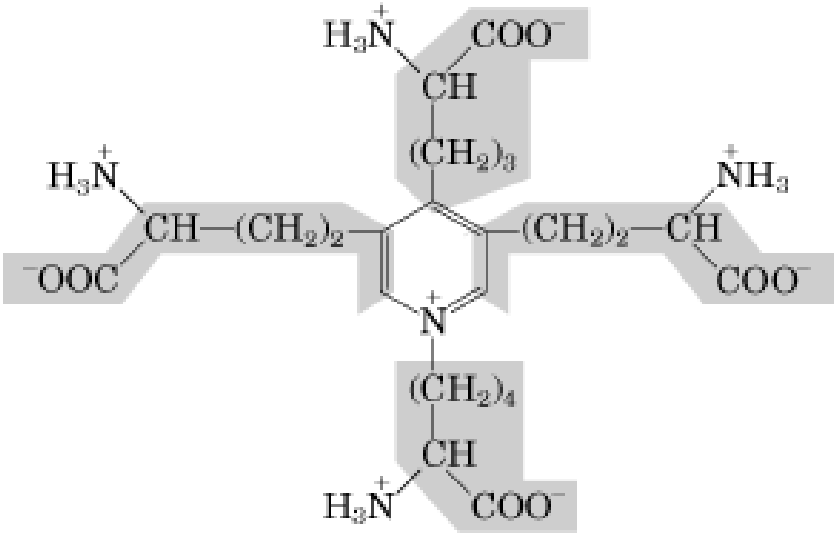
4-Hydroxyproline



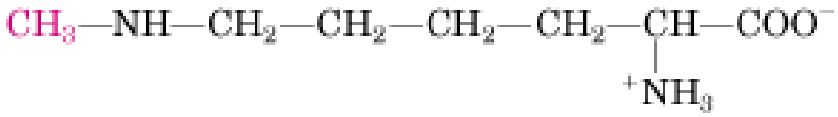
γ-Carboxyglutamate



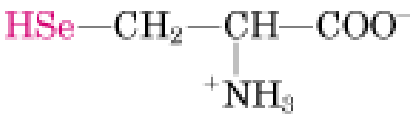
5-Hydroxylysine



Desmosine

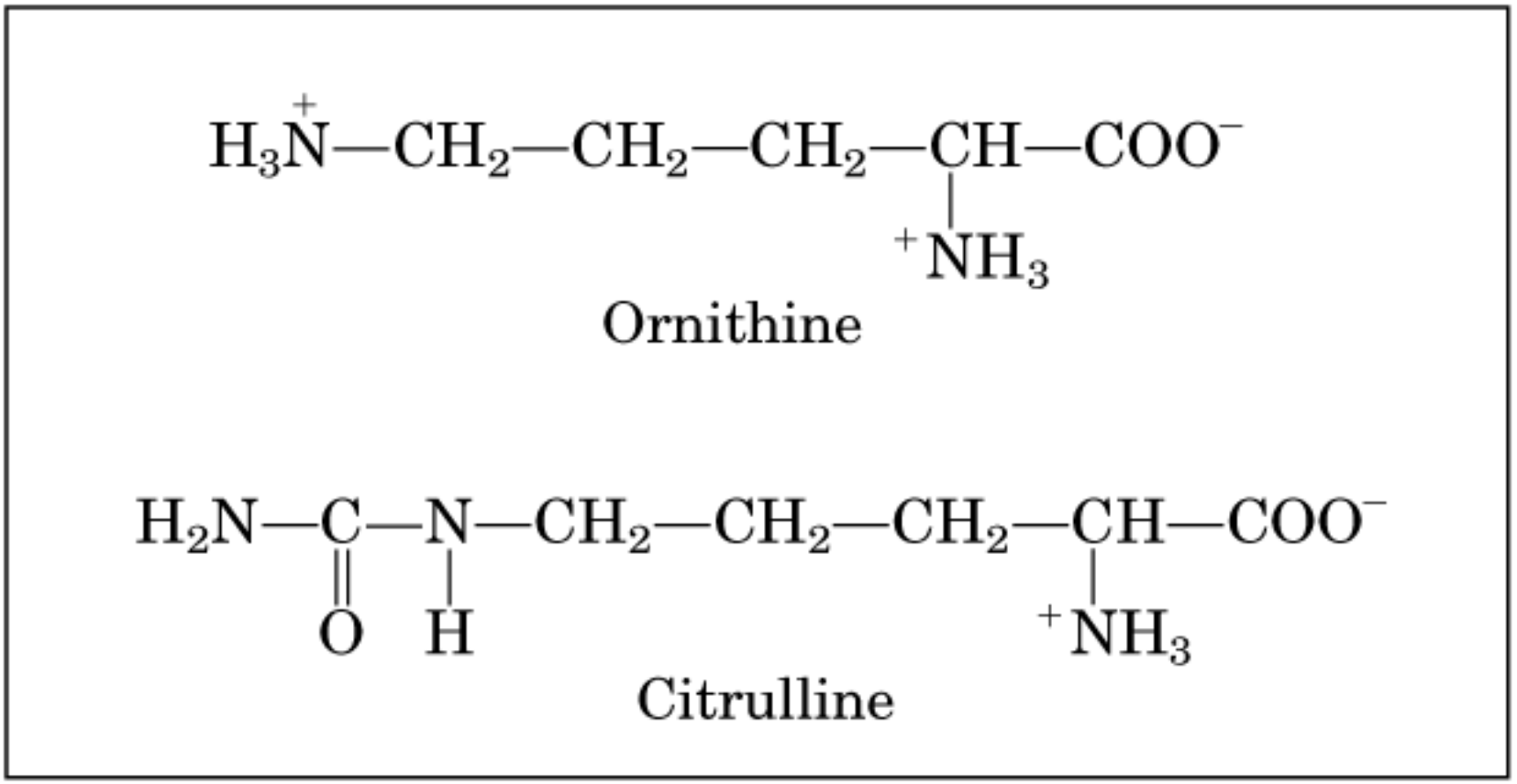


6-*N*-Methyllysine



Selenocysteine

(a)


$H_3\overset{+}{N}-CH_2-CH_2-CH_2-CH(^+NH_3)-COO^-$
Ornithine
$H_2N-C(=O)-NH-CH_2-CH_2-CH_2-CH(^+NH_3)-COO^-$
Citrulline

(b)

Two additional amino acids have been identified

Selenocysteine(硒代半胱氨酸): encoded by the RNA nucleotide triplet UGA found in Archaea, eubacteria and animals including mammals.

Pyrrolysine (吡咯赖氨酸): encoded by the RNA nucleotide triplet UAG, found in Archaea, and eubacteria.

Science (2002) vol. 296, 1409

Nature may yet surprise us with more directly encoded amino acids.

References and Notes

1. N. P. Kristensen, in *The Hierarchy of Life. Molecules and Morphology in Phylogenetic Analysis*, B. Fernholm, K. Bremer, H. Jörnvall, Eds. (Elsevier, Amsterdam, 1989), pp. 295–306.
2. _____, in *The Insects of Australia*, CSIRO, Ed. (Melbourne Univ. Press, Carlton, ed. 2, 1991), vol. 1, chap. 5.
3. A. H. Staniczek, *Zool. Anz.* **239**, 147 (2000).
4. In June 2001, J. Marshall (Natural History Museum, London) showed O.Z. a male insect from Tanzania, which had been submitted for an opinion 16 years ago. Shortly afterward O.Z. received from F. Kernegger a male Baltic-amber insect (subsequently described as *Raptophasma kerneggeri*), whose close similarity to the Tanzania specimen was immediately obvious. In July 2001, O.Z. discovered in the unsorted alcohol collection of Phasmatodea in the Museum für Naturkunde (Berlin) an adult female of a similar insect from Namibia.
5. According to one school of thought among contemporary systematists, the naming of higher taxa that only contain a single genus is "empty formalism"; we accept the logical merits of this stand. Pragmatically we believe, however, that any recognized genus should be assigned to a "family" and an "order," because these categories play an important role in how biologists communicate and how biological knowledge is systematized.
6. The description includes many characters currently known only from a single specimen, because genitalic characters can only be observed in a specimen of the respective sex, muscle characters can only be observed in the ethanol-preserved female, and only a few characters can be judged in the *Raptophasma* fossils.
7. O. Zompro, *Mitt. Geol.-Pal. Inst. Hamburg* **85**, 229 (2001).
8. A. Arillo, V. M. Ortuño, A. Nel, *Bull. Soc. Entomol. Fr.* **102**, 11 (1997).
9. W. C. Wheeler, M. Whiting, Q. D. Wheeler, J. M. Carpenter, *Cladistics* **17**, 113 (2001).

Pyrrolysine Encoded by UAG in Archaea: Charging of a UAG-Decoding Specialized tRNA

Gayathri Srinivasan, Carey M. James, Joseph A. Krzycki*

Pyrrolysine is a lysine derivative encoded by the UAG codon in methylamine methyltransferase genes of *Methanosarcina barkeri*. Near a methyltransferase gene cluster is the *pylT* gene, which encodes an unusual transfer RNA (tRNA) with a CUA anticodon. The adjacent *pylS* gene encodes a class II aminoacyl-tRNA synthetase that charges the *pylT*-derived tRNA with lysine but is not closely related to known lysyl-tRNA synthetases. Homologs of *pylS* and *pylT* are found in a Gram-positive bacterium. Charging a $tRNA_{CUA}$ with lysine is a likely first step in translating UAG amber codons as pyrrolysine in certain methanogens. Our results indicate that pyrrolysine is the 22nd genetically encoded natural amino acid.

In *Methanosarcina* species, specific methyltransferases initiate methanogenesis and carbon assimilation from substrates such as trimethylamine (TMA), dimethylamine (DMA), or monomethylamine (MMA). The highly expressed, nonhomologous genes encoding these methyltransferases have in-frame UAG (amber) codons that do not stop translation during synthesis of the full-length

pool from MMA-grown *M. barkeri* MS revealed an RNA of the size expected for the $tRNA_{CUA}$ product of *pylT* (Fig. 1C). The predicted secondary structure of $tRNA_{CUA}$ has unusual properties compared with typical tRNAs (*8*). Even though the structure has the expected sizes for the acceptor, D, and T stems and D, T, and anticodon loops, the anticodon stem could

DOI: 10.1126/science.1069588

Summary of B1 Amino acid structure

1. All natural proteins were found to be built from the same set of 20 standard L-α-amino acids.

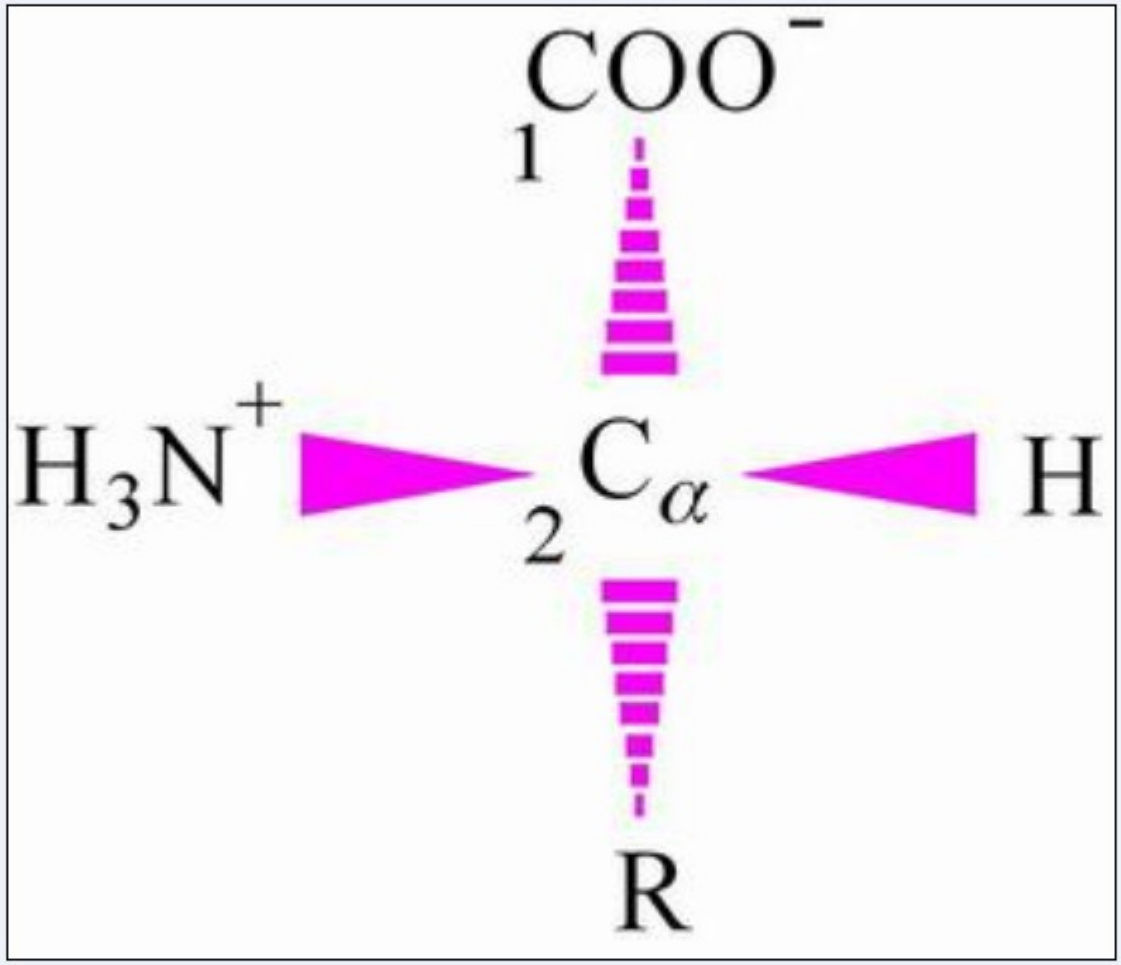
2. The 20 amino acids are usually grouped according to the properties (mainly polarity) of their R groups. How many groups? And what they are?

3. Nonstandard amino acids are found in certain proteins, generally as a result of post-translational modifications.

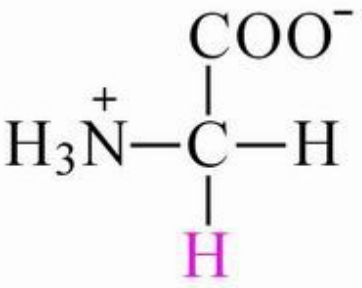
Summary of B1

- Standard amino acid /Nonstandard amino acid
- **L- α-amino acids**
- **nonpolar, aliphatic** amino acid
- **polar, uncharged** amino acid
- **Aromatic** amino acid
- **Positively charged (basic)**
- **and negatively charged (acidic)** amino acid
- Essential amino acid
- Nonessential amino acid

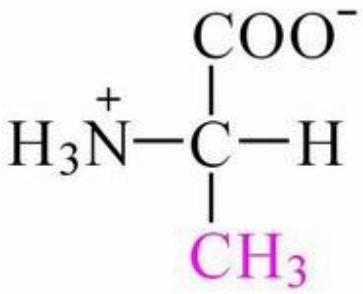



COO⁻
1
H3N+
Cα
2
H
R

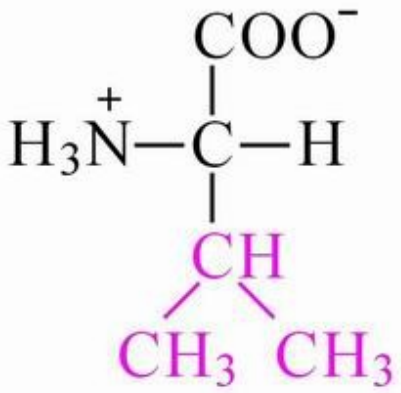
非极性、脂肪族侧链



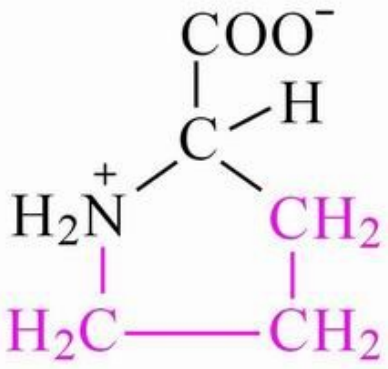
甘氨酸



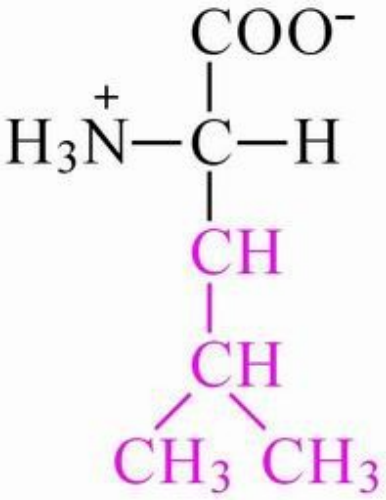
丙氨酸



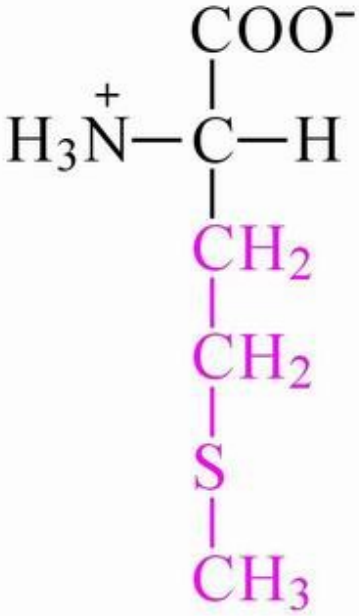
缬氨酸



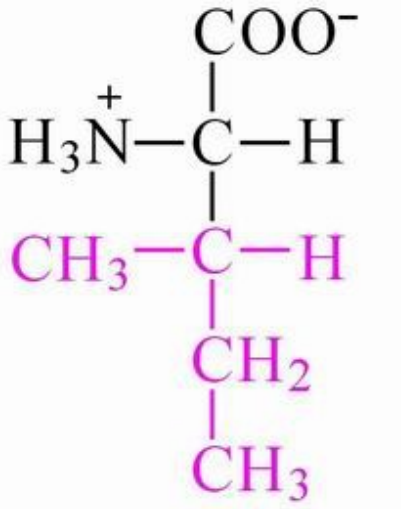
脯氨酸



亮氨酸



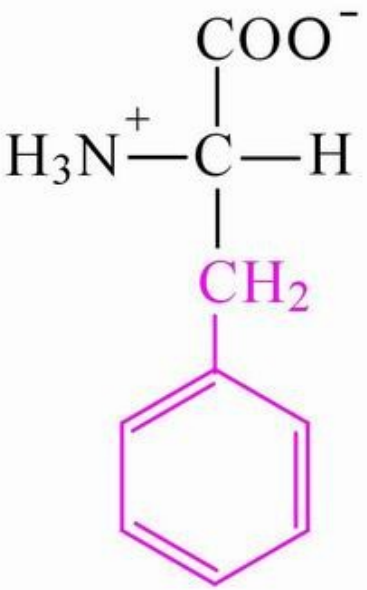
甲硫氨酸



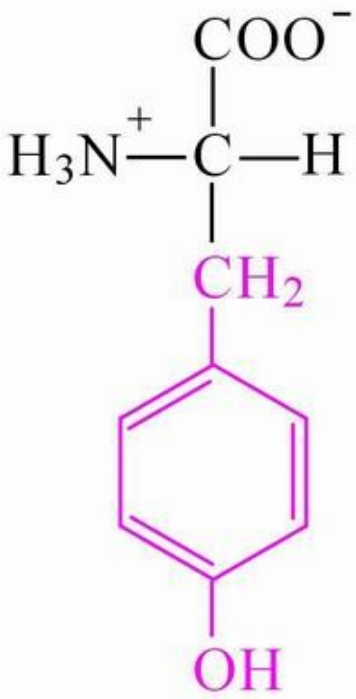
异亮氨酸

蛋氨酸

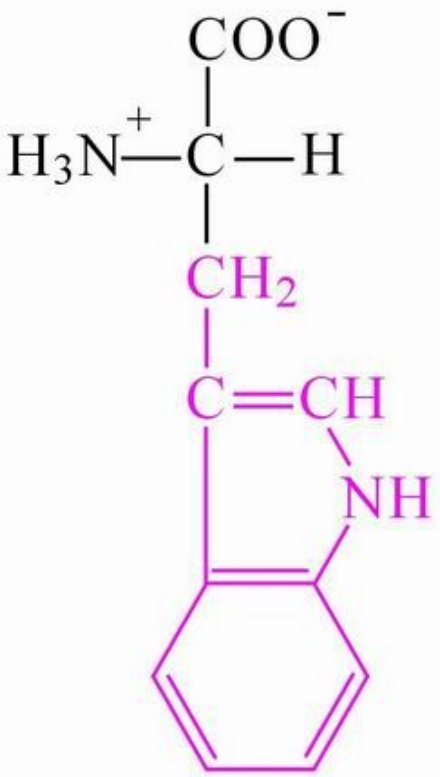
非极性、芳香族侧链



苯丙氨酸

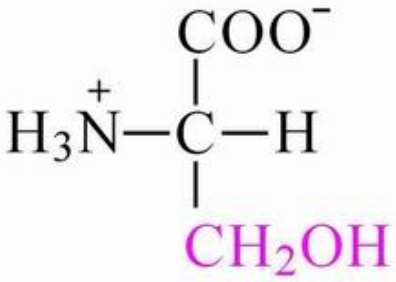


酪氨酸

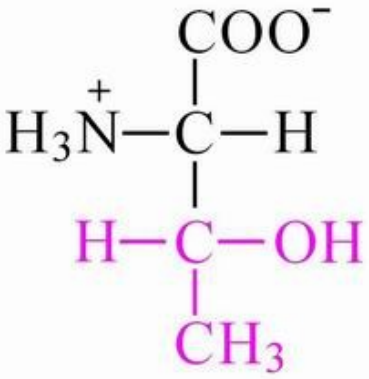


色氨酸

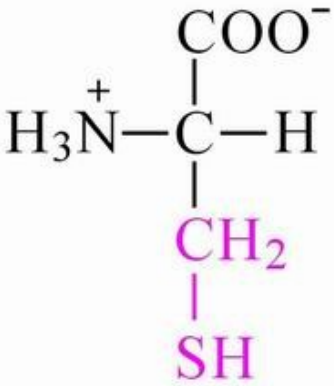
极性、不带电侧链



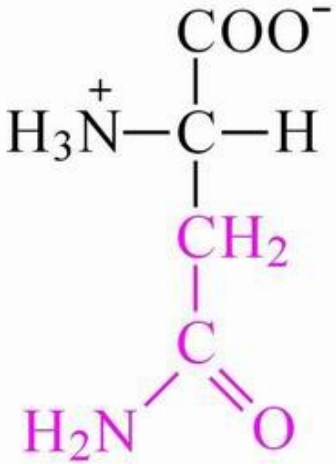
丝氨酸



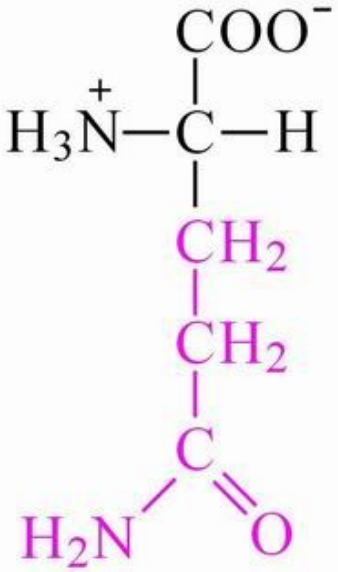
苏氨酸



半胱氨酸

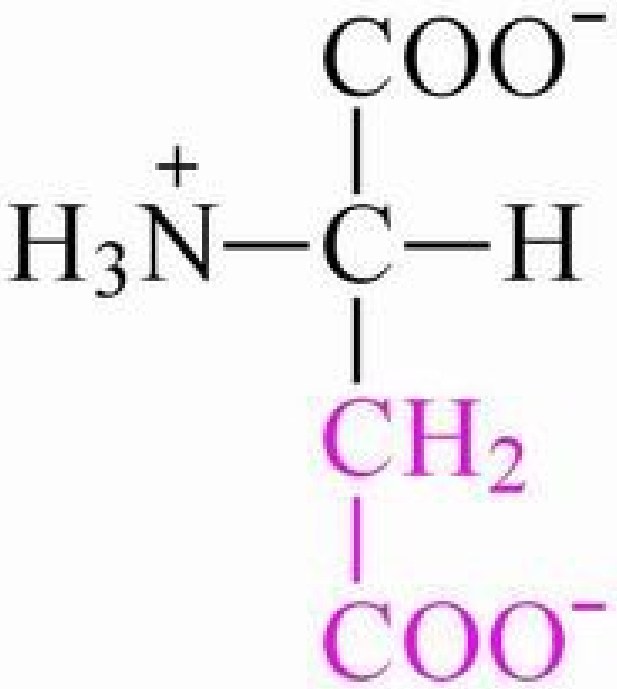


天冬酰胺

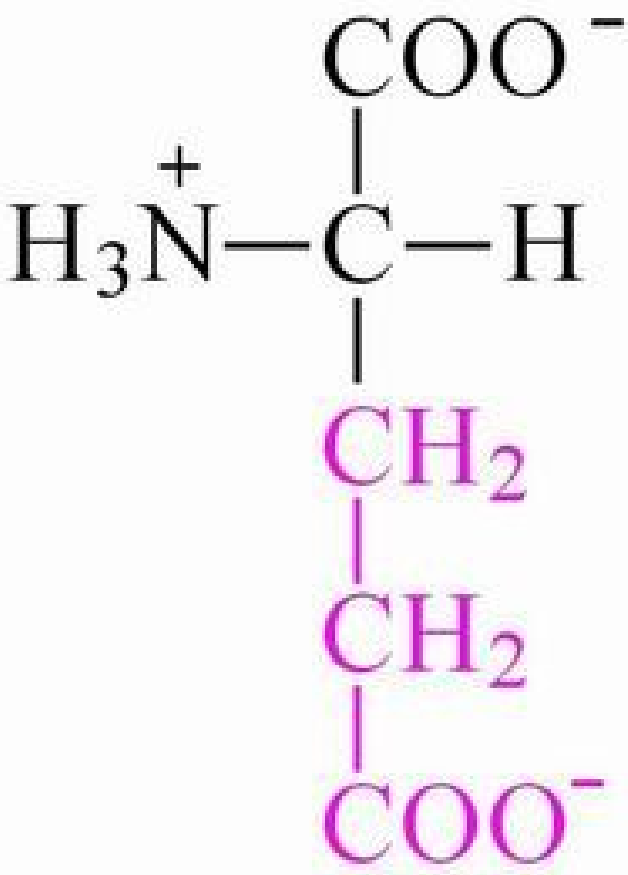


谷氨酰胺

带负电侧链

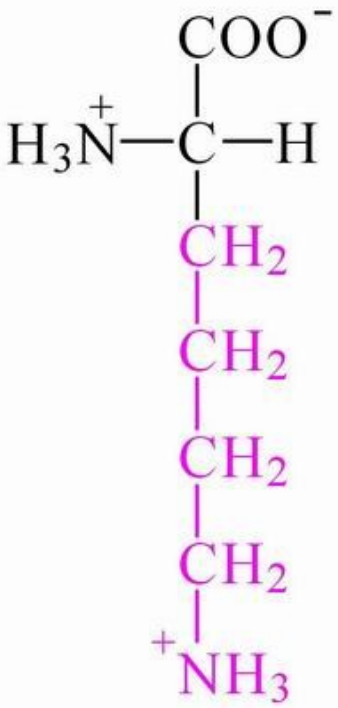


天冬氨酸

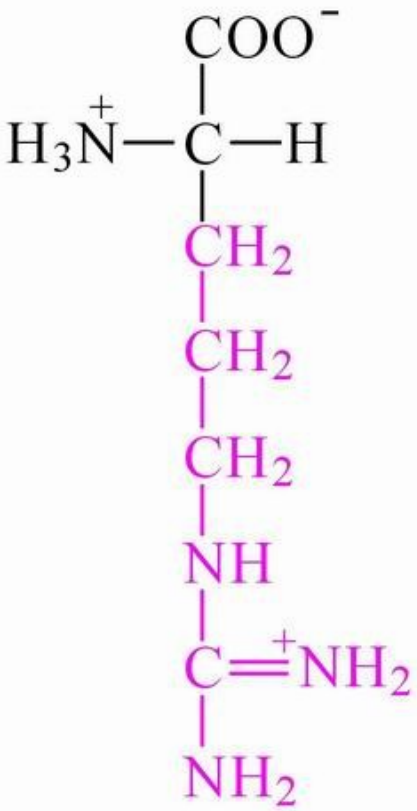


谷氨酸

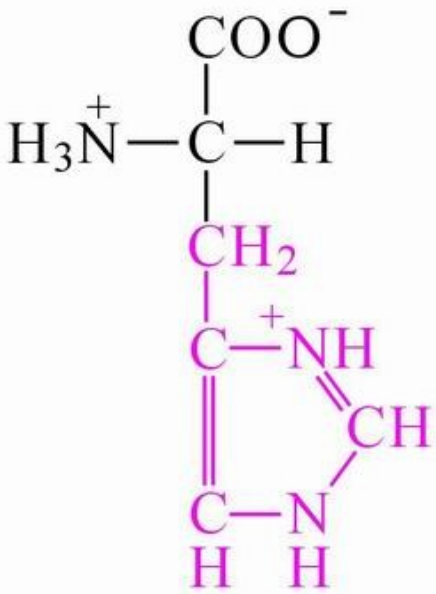
带正电侧链



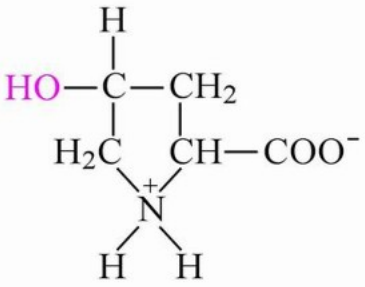
赖氨酸



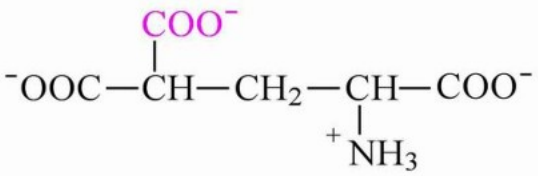
精氨酸



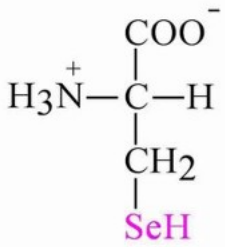
组氨酸



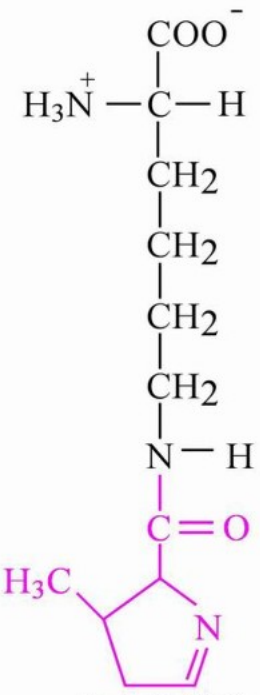
4-羟脯氨酸



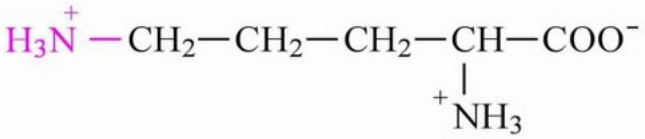
γ-羧基谷氨酸



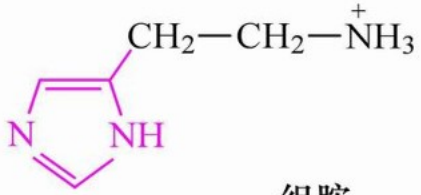
硒代半胱氨酸



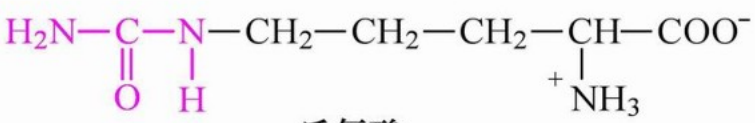
吡咯赖氨酸



鸟氨酸



组胺



瓜氨酸



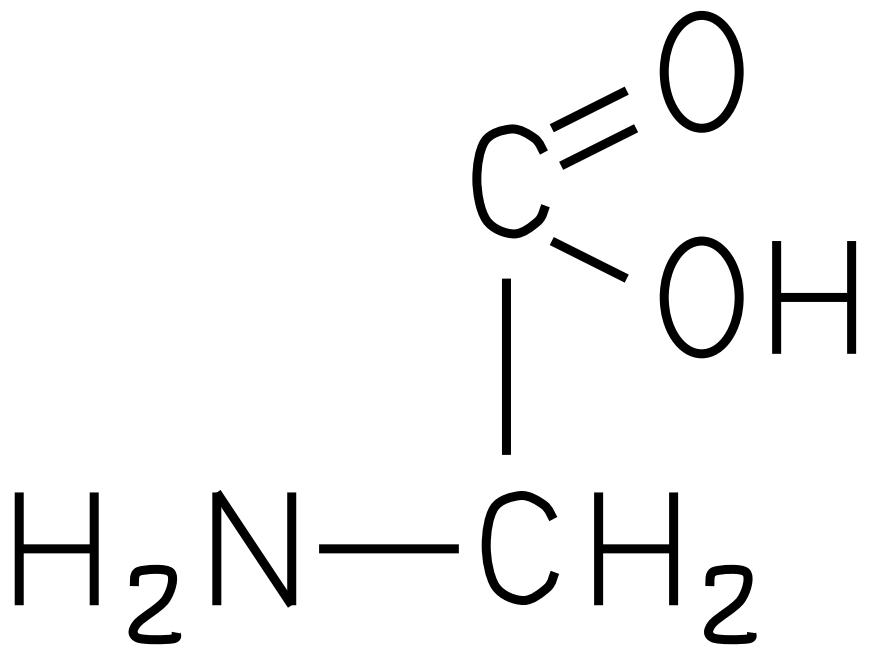
γ-氨基丁酸

氨基酸的其他分类方法

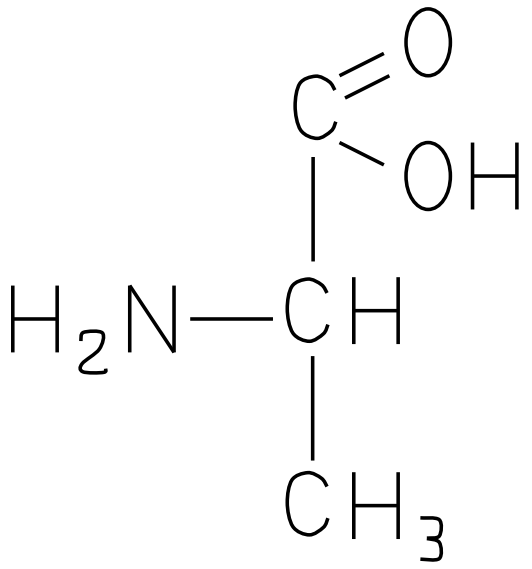
- **2C 甘**
- **3C 丙 丝 半 色 酪 苯 组**
- **4C 天 天 苏**
- **5C 谷 谷 脯 缬 蛋**
- **6C 亮 异 赖 精**

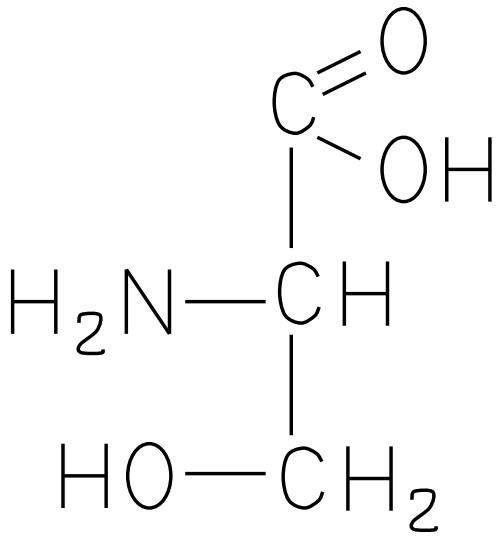


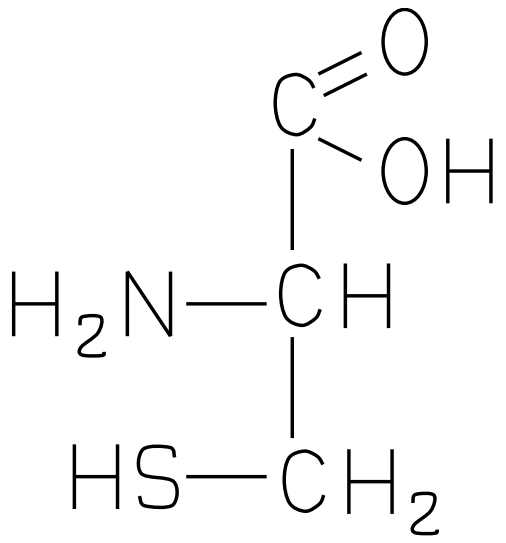
2C



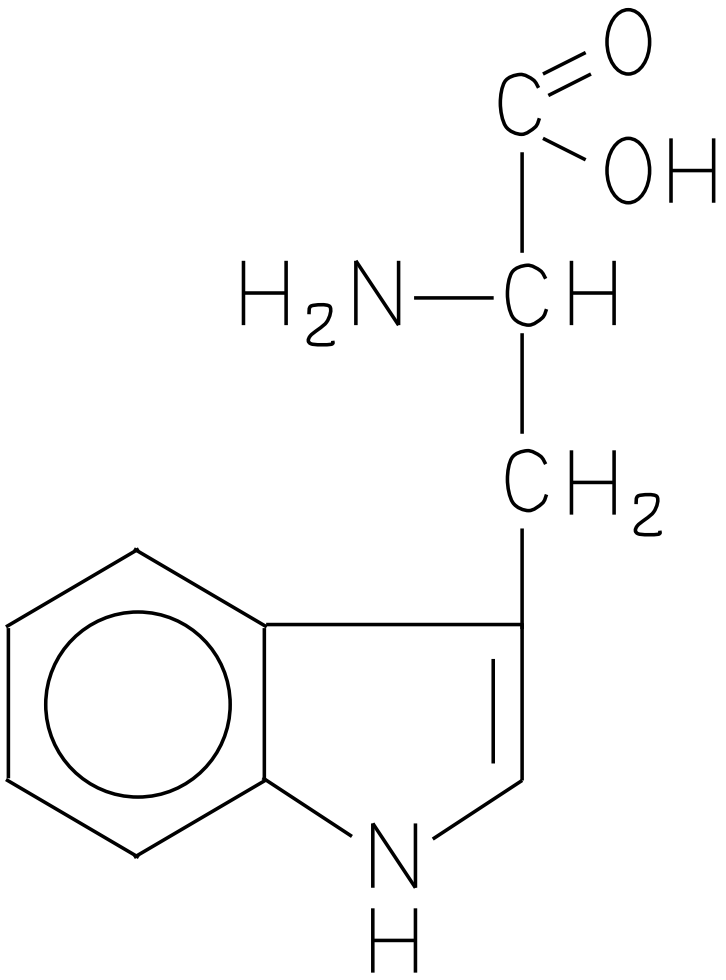
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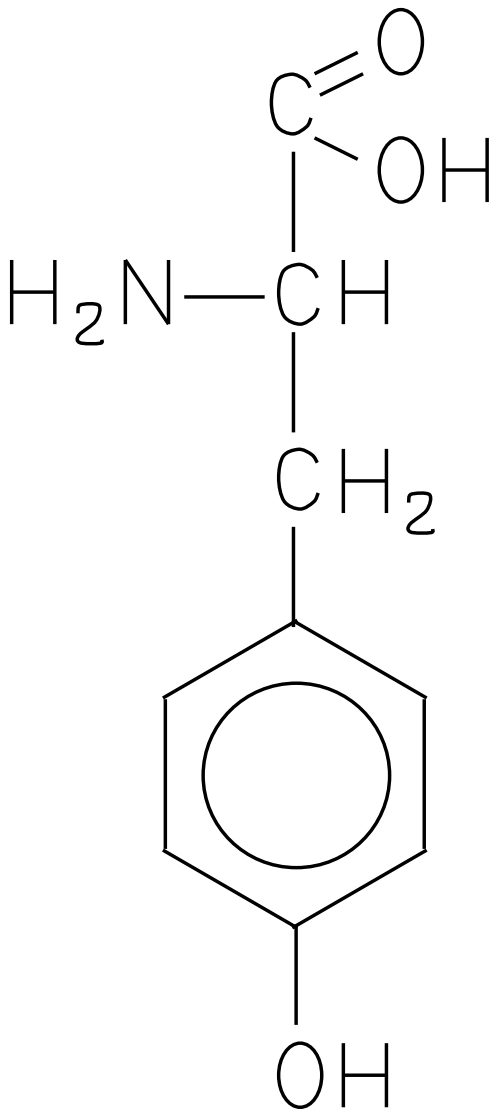




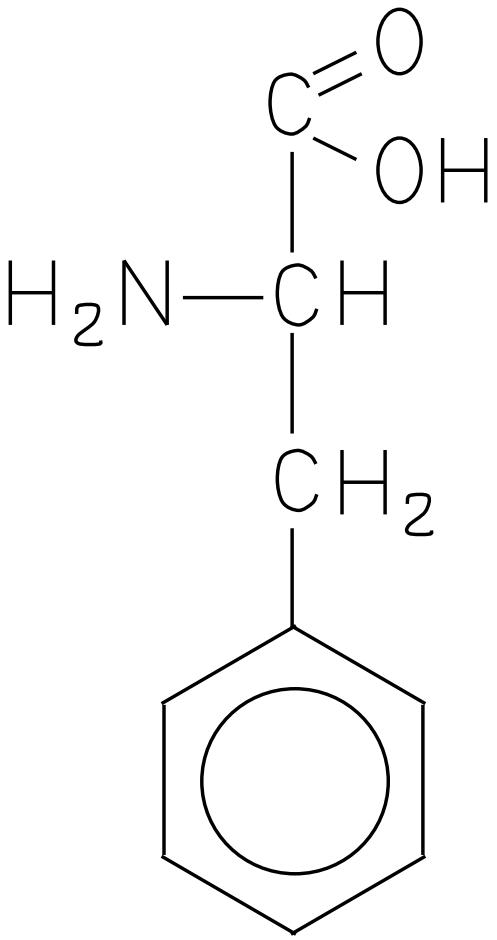


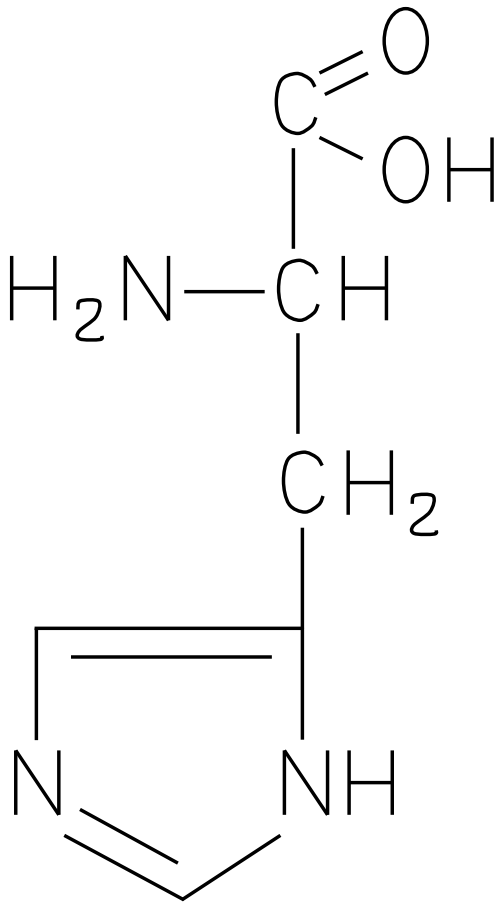
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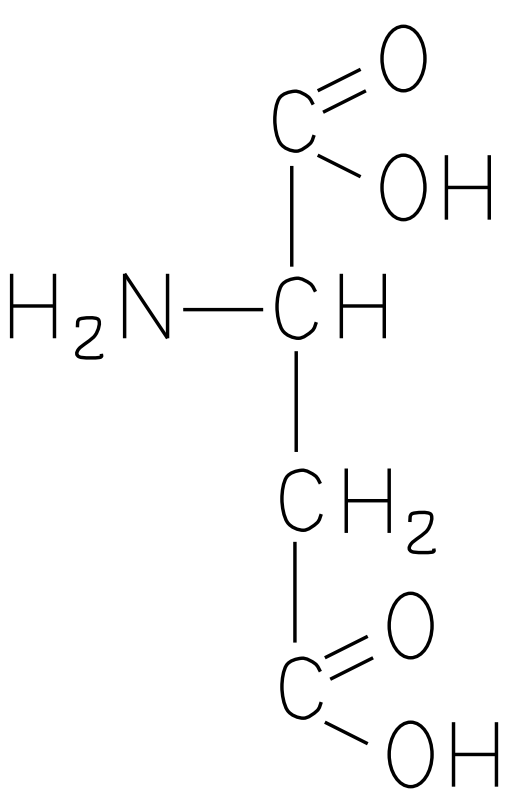


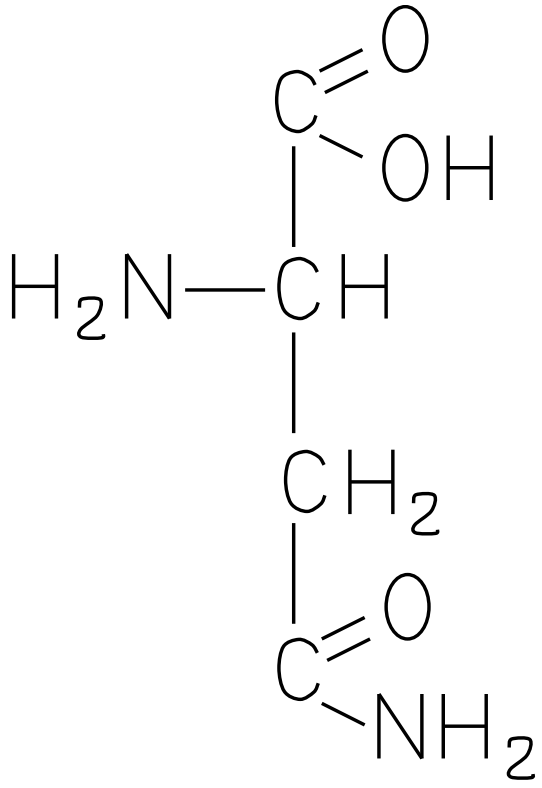
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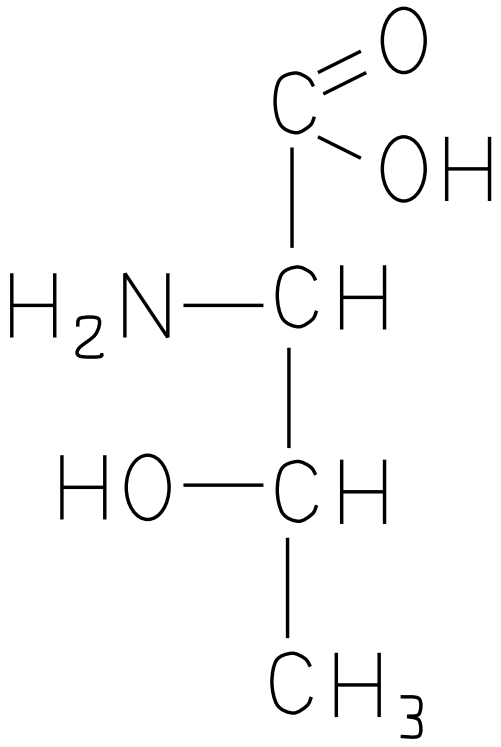




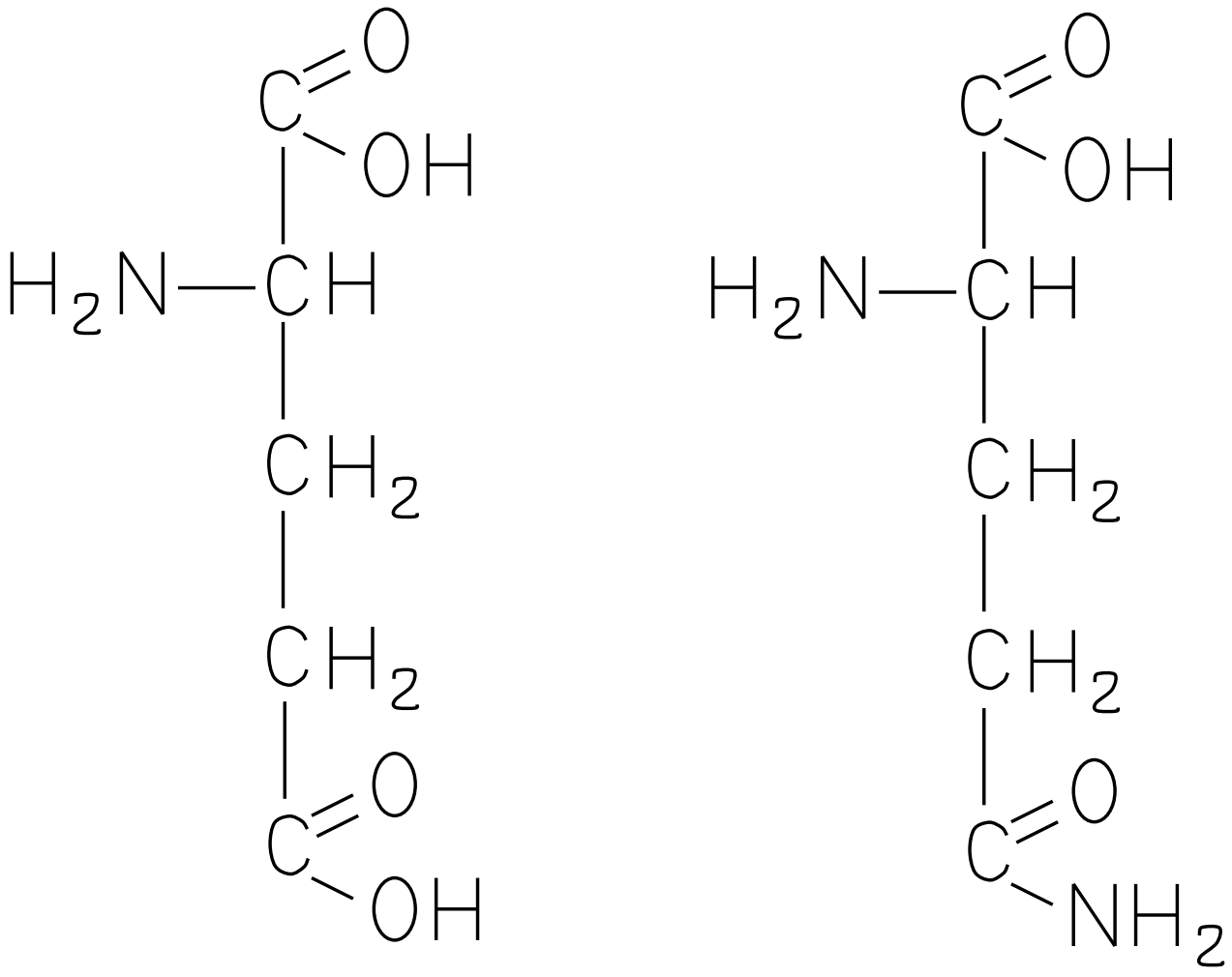
4C



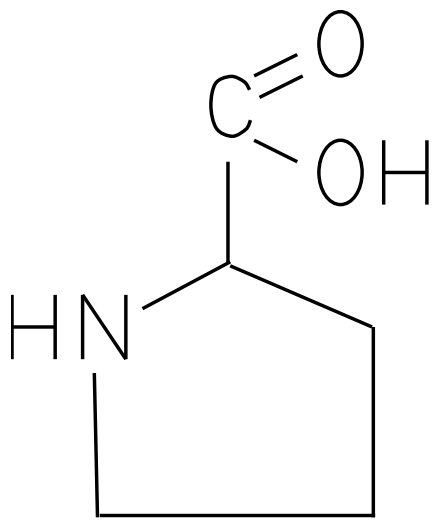


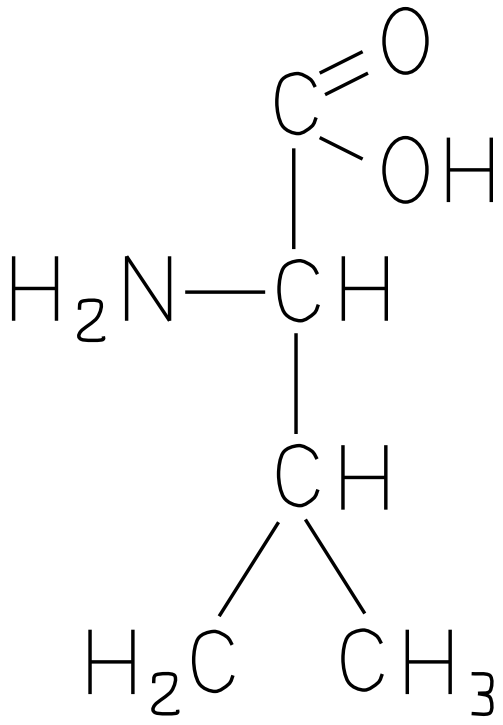


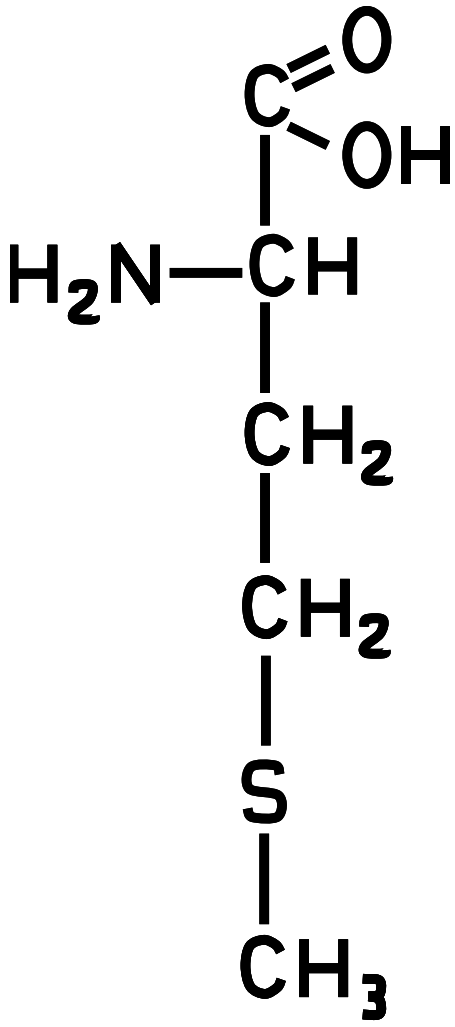
5C



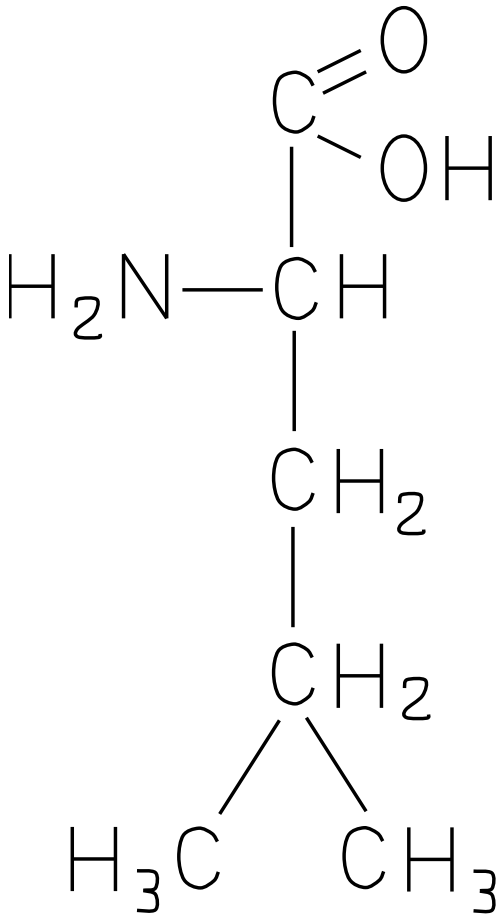
5C

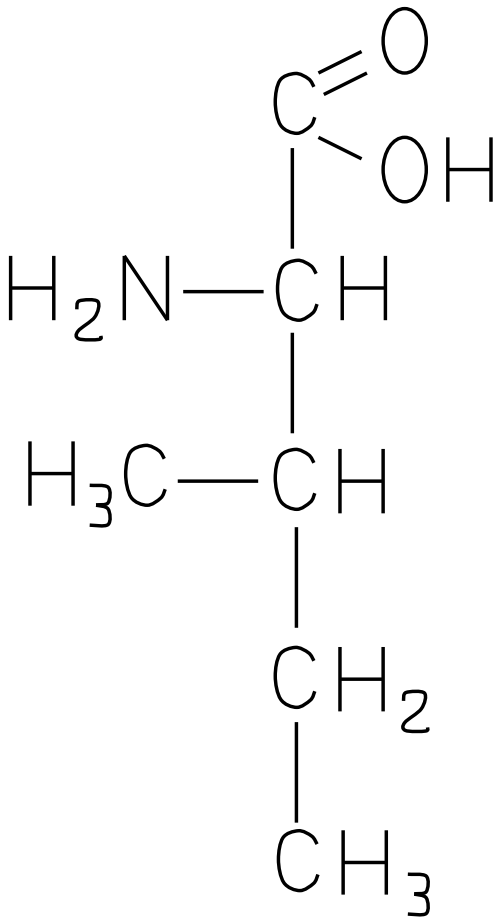




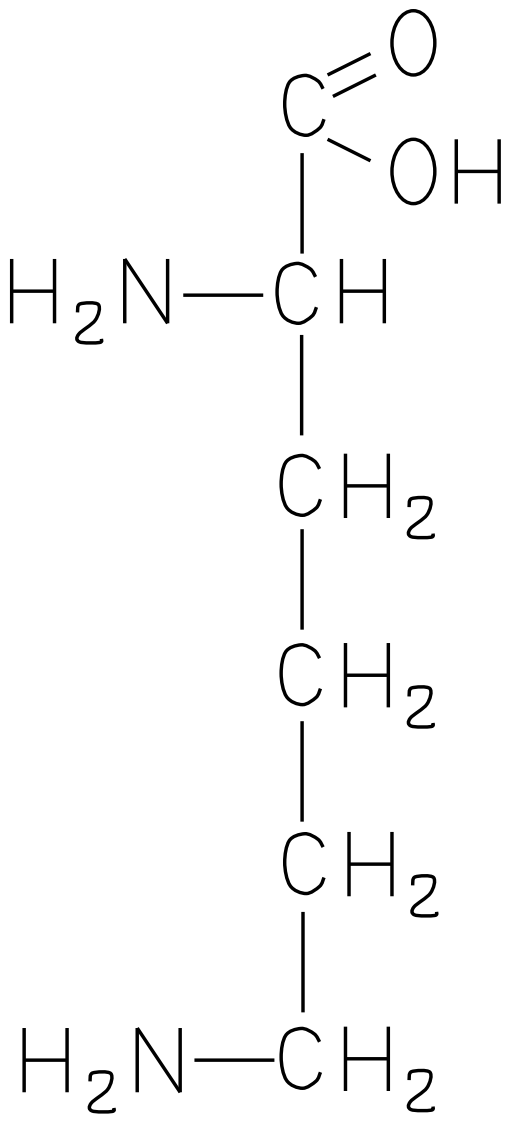


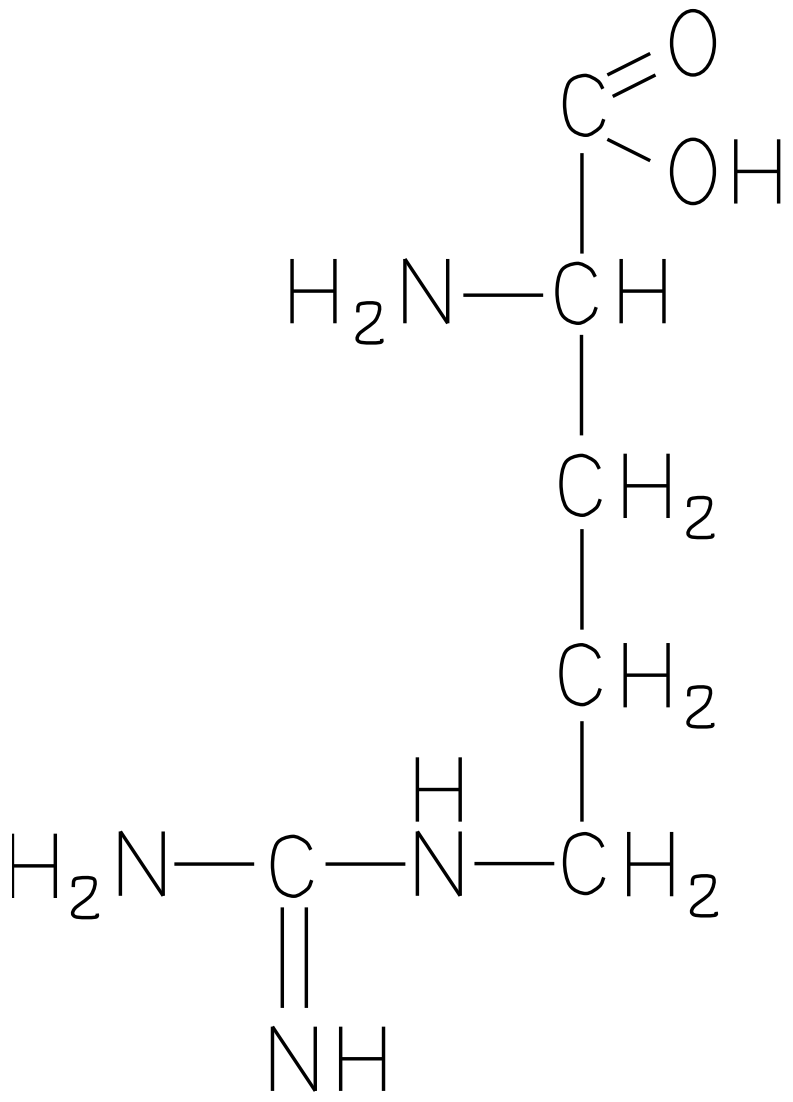
6C





6C

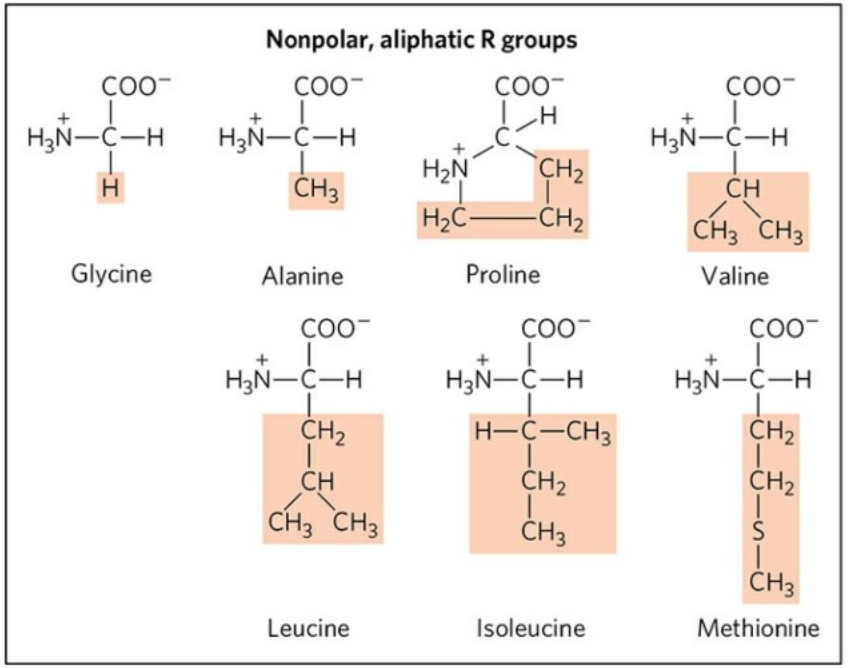


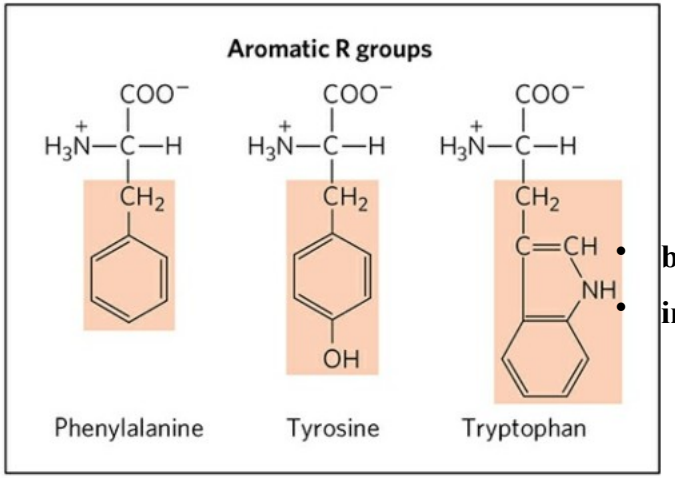


B2 Acids and Bases

- Acids, bases and pH
- Buffers
- Ionization of amino acids

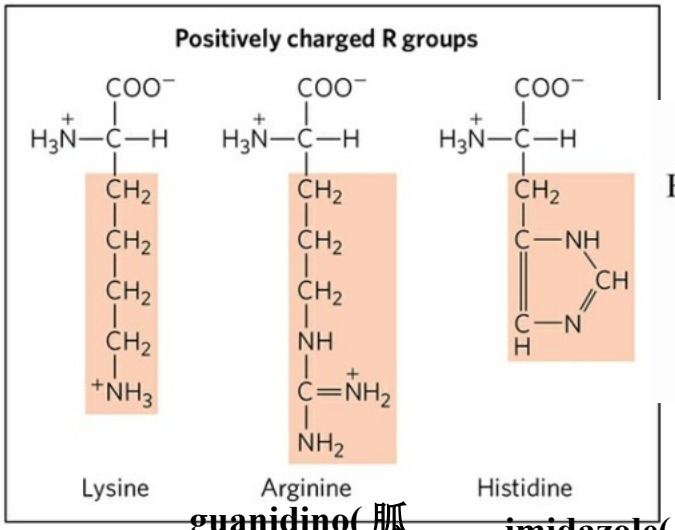


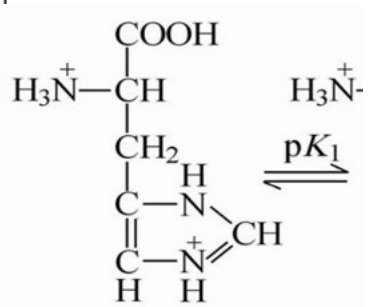




• **benzene rings.**

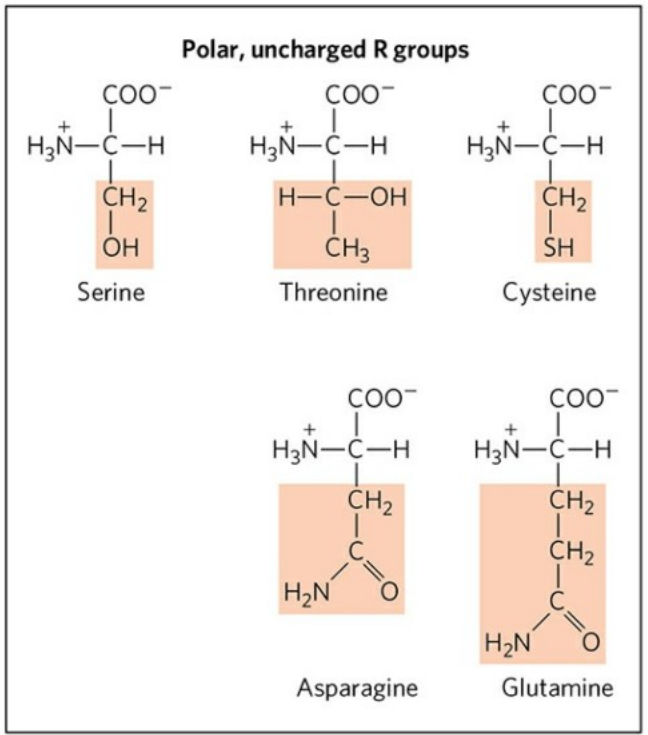
• **indole ring** （吲哚）

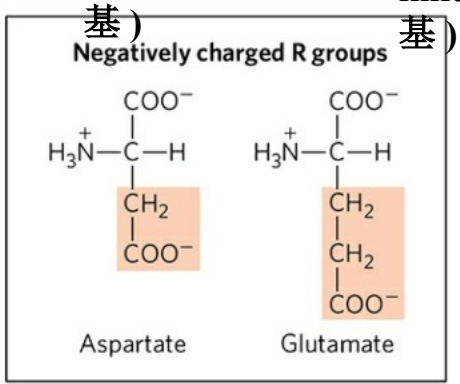




guanidino(胍基)

imidazole(咪唑基)





1 Weak acids (proton don
(proton acceptors) do not
dissolved in water.



双牛曳犁 (汉画像石) 河南南阳

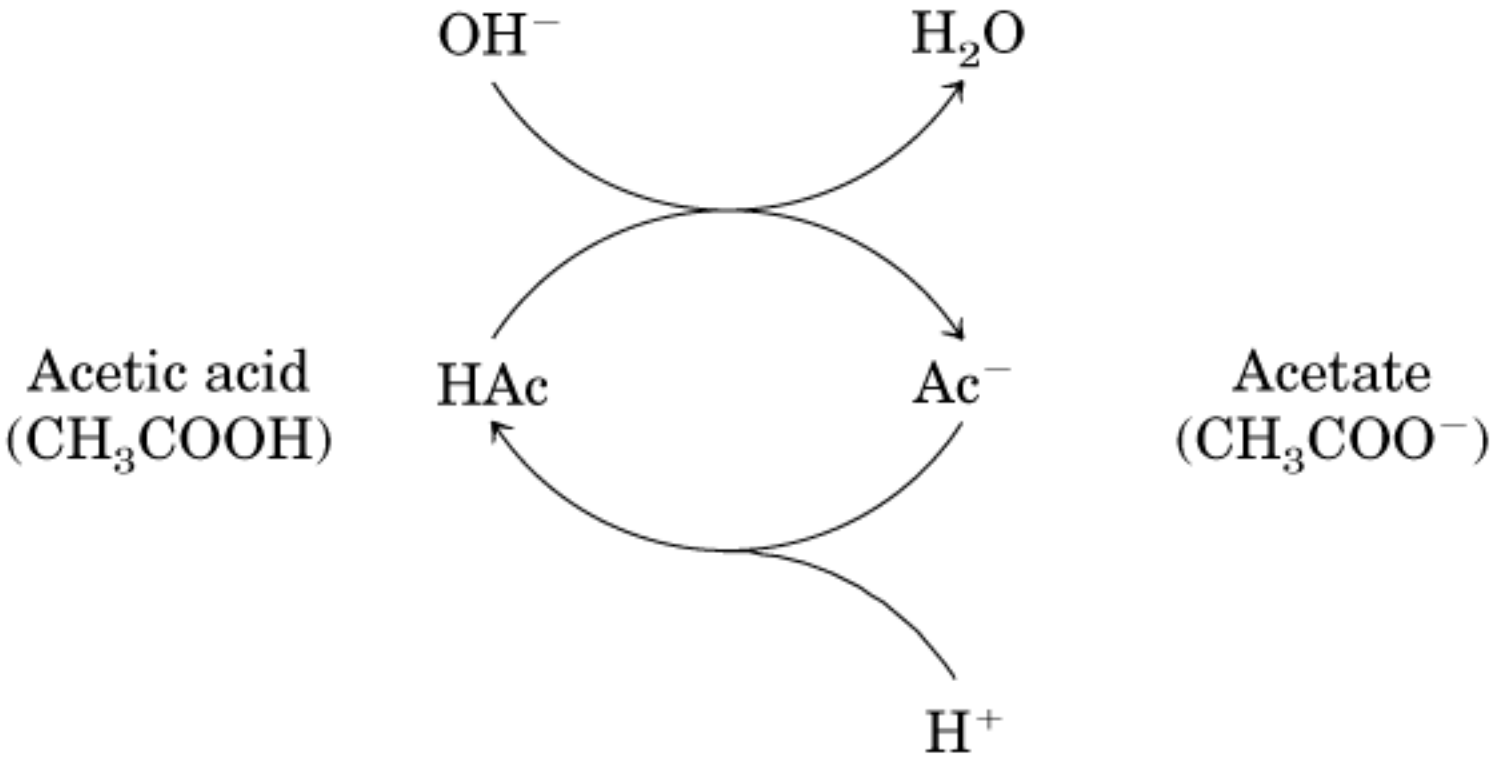
1.1 strong acids (e.g.
sulfuric and nitric acids) a
NaOH and KOH) ionizes
dissolved in water.

1.2 A proton donor and its corresponding proton acceptor make up a conjugate acid-base pair.

1.3 The characteristic tendency of each weak acid for losing its proton is reflected by its dissociation constant Ka (stronger acids have larger Ka value).

1.4 For convenience, Ka is converted to pKa (the negative logarithm).

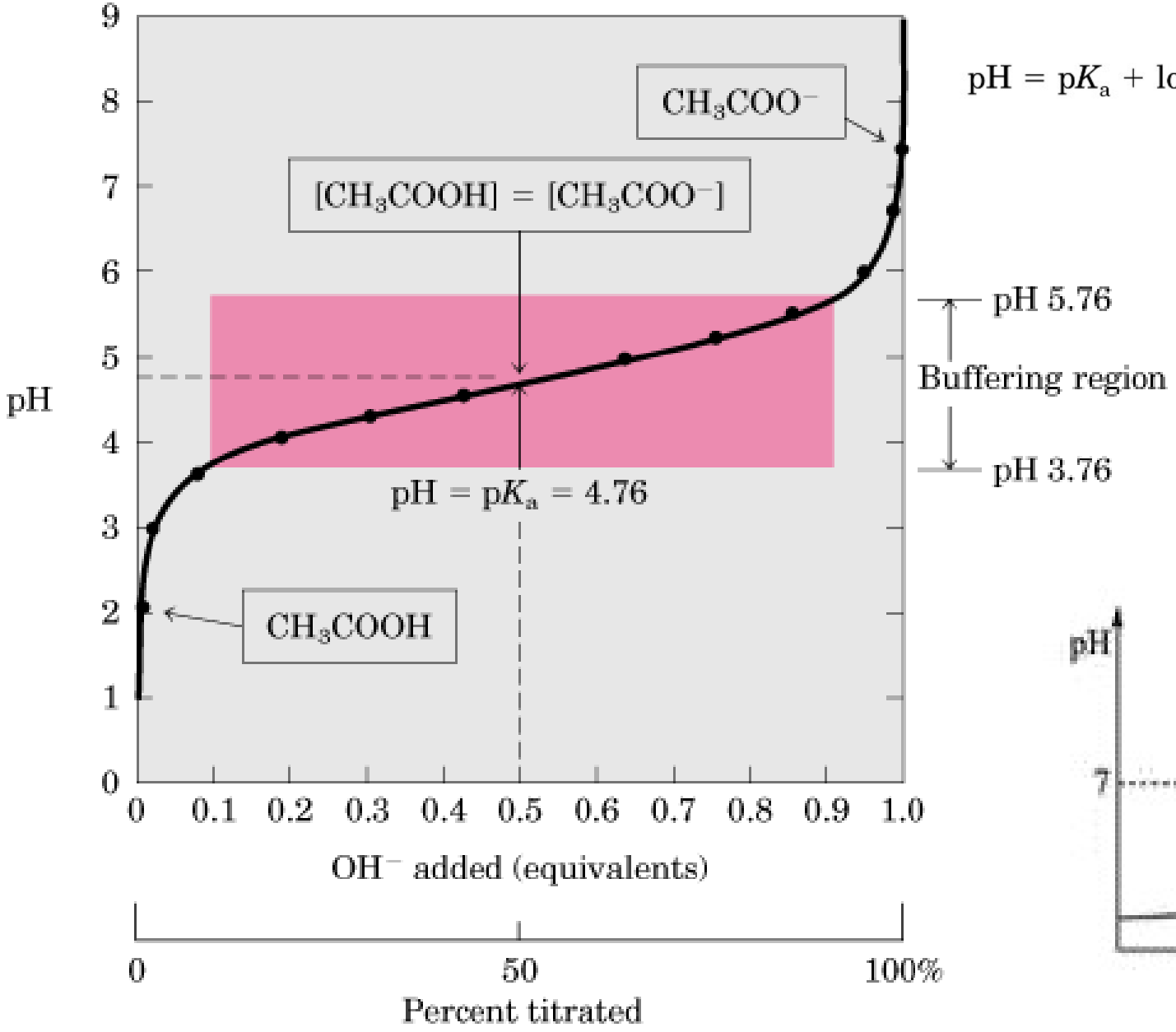
$$K_w = [H^+][OH^-]$$

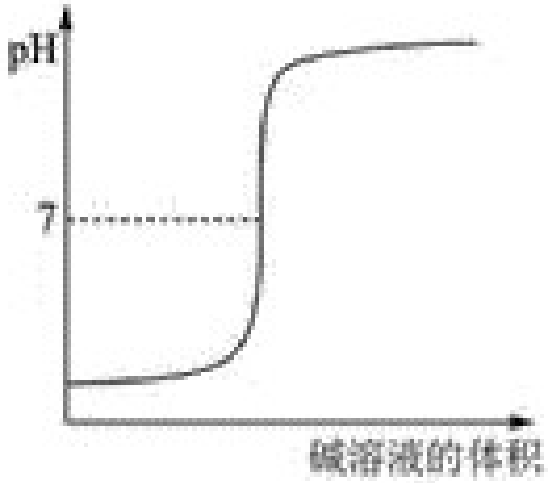


$$K_a = \frac{[H^+][Ac^-]}{[HAc]}$$

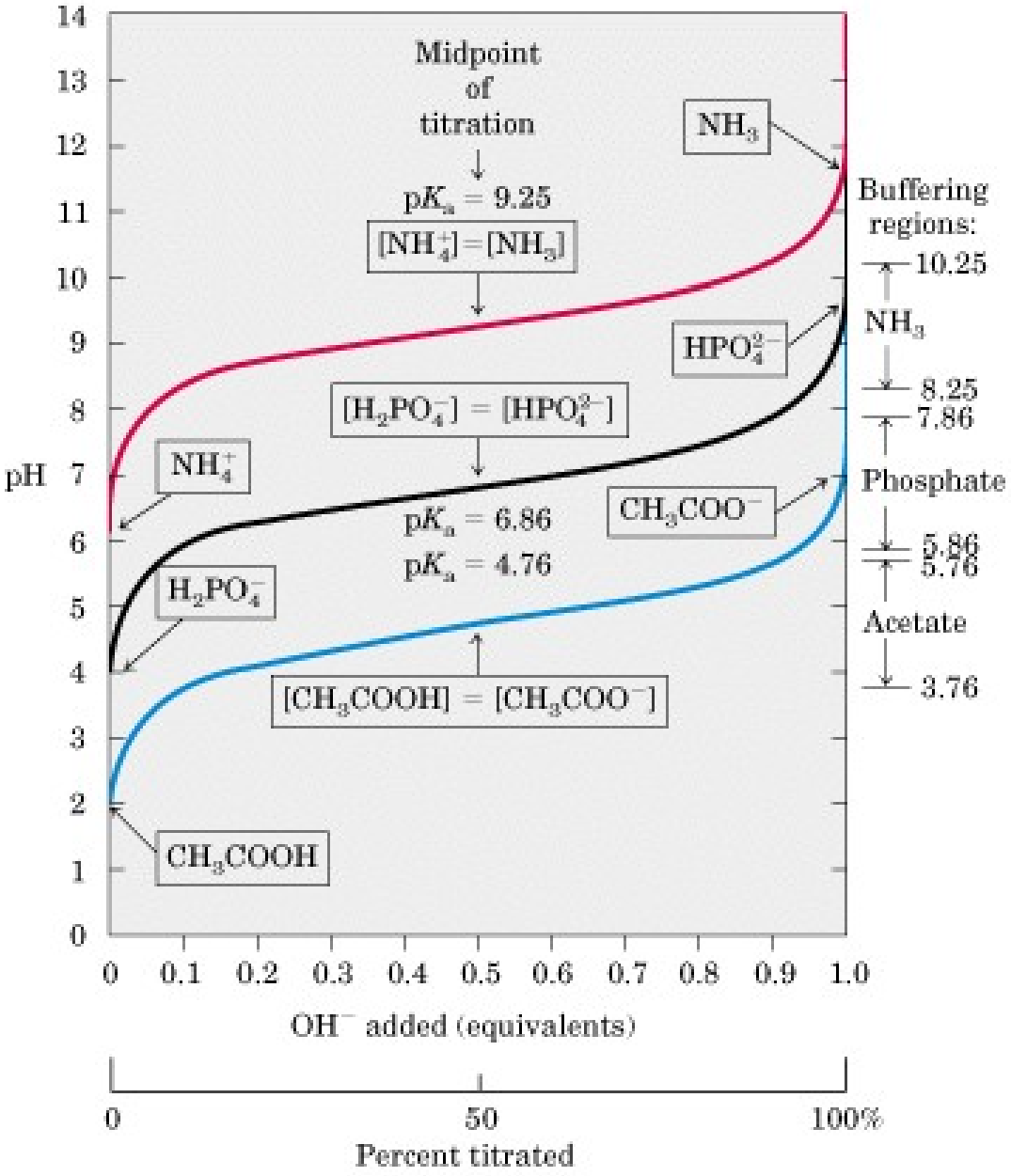
$$\mathrm{pH} = \mathrm{p}K_\mathrm{a} + \log \frac{[\mathrm{A}^-]}{[\mathrm{HA}]}$$

Henderson-Hasselbalch equation


pH = pKa + log [A⁻]/[HA]
CH3COO⁻
[CH3COOH] = [CH3COO⁻]
pH 5.76
Buffering region
pH 3.76
pH = pKa = 4.76
CH3COOH
pH
0 1 2 3 4 5 6 7 8 9
0 0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9 1.0
OH⁻ added (equivalents)
0 50 100%
Percent titrated


pH
7
碱溶液的体积

盐酸滴定曲线


Midpoint of titration
$pK_a = 9.25$
$[NH_4^+] = [NH_3]$
NH_3
HPO_4^{2-}
$[H_2PO_4^-] = [HPO_4^{2-}]$
$pK_a = 6.86$
$pK_a = 4.76$
NH_4^+
CH_3COO^-
$H_2PO_4^-$
$[CH_3COOH] = [CH_3COO^-]$
CH_3COOH
pH
OH⁻ added (equivalents)
Percent titrated
Buffering regions:
10.25
NH_3
8.25
7.86
Phosphate
5.86
5.76
Acetate
3.76

Monoprotic acids

Acetic acid
($K_a = 1.74 \times 10^{-5}$ M)

Ammonium
($K_a = 5.62 \times 10^{-10}$ M)

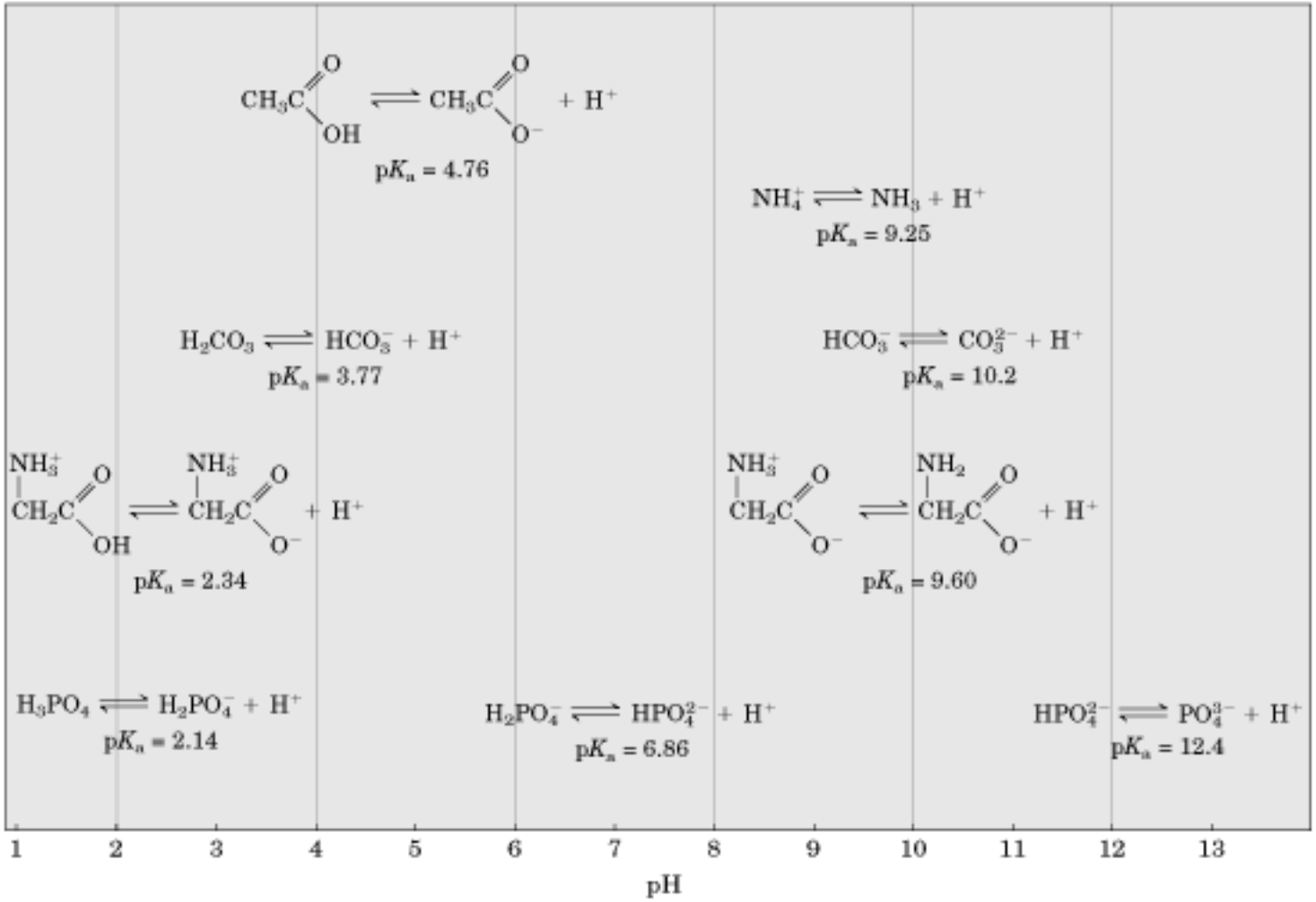
Diprotic acids

Carbonic acid
($K_a = 1.70 \times 10^{-4}$ M);
Bicarbonate
($K_a = 6.31 \times 10^{-11}$ M)

Glycine, carboxyl
($K_a = 4.57 \times 10^{-3}$ M);
Glycine, amino
($K_a = 2.51 \times 10^{-10}$ M)

Triprotic acids

Phosphoric acid
($K_a = 7.25 \times 10^{-3}$ M);
Dihydrogen phosphate
($K_a = 1.38 \times 10^{-7}$ M);
Monohydrogen phosphate
($K_a = 3.98 \times 10^{-13}$ M)



2 The titration curves of weak acids can be fitted by the Henderson-Hasselbach equation

2.1 The amount of weak acids in a solution can be determined by titrating with a strong base of known concentration .

2.2 Titration curves are made from plotting the pH of the solution against the amount of strong base added to neutralize the weak acid (until [proton donor]~0, [proton acceptor]=initial weak acid concentration, that is, all weak acid molecules are ionized, no more buffering effect, pH increases rapidly).

2.3 Titration curves of weak acids have nearly identical shapes (reflecting the same law behind the phenomenon)

2.4 The Henderson-Hasselbalch equation fits the titration curves of all weak acids!

$$\mathrm{pH} = \mathrm{p}K_\mathrm{a} + \log\frac{[\mathrm{A}^-]}{[\mathrm{HA}]}$$

pH = pKa + log[proton acceptor]/[proton donor]

2.5 The pH value at which the conjugate acid-base pair is at equimolar concentration equals to the pKa value of the weak acid. The plateau of the curve gives the pKa.

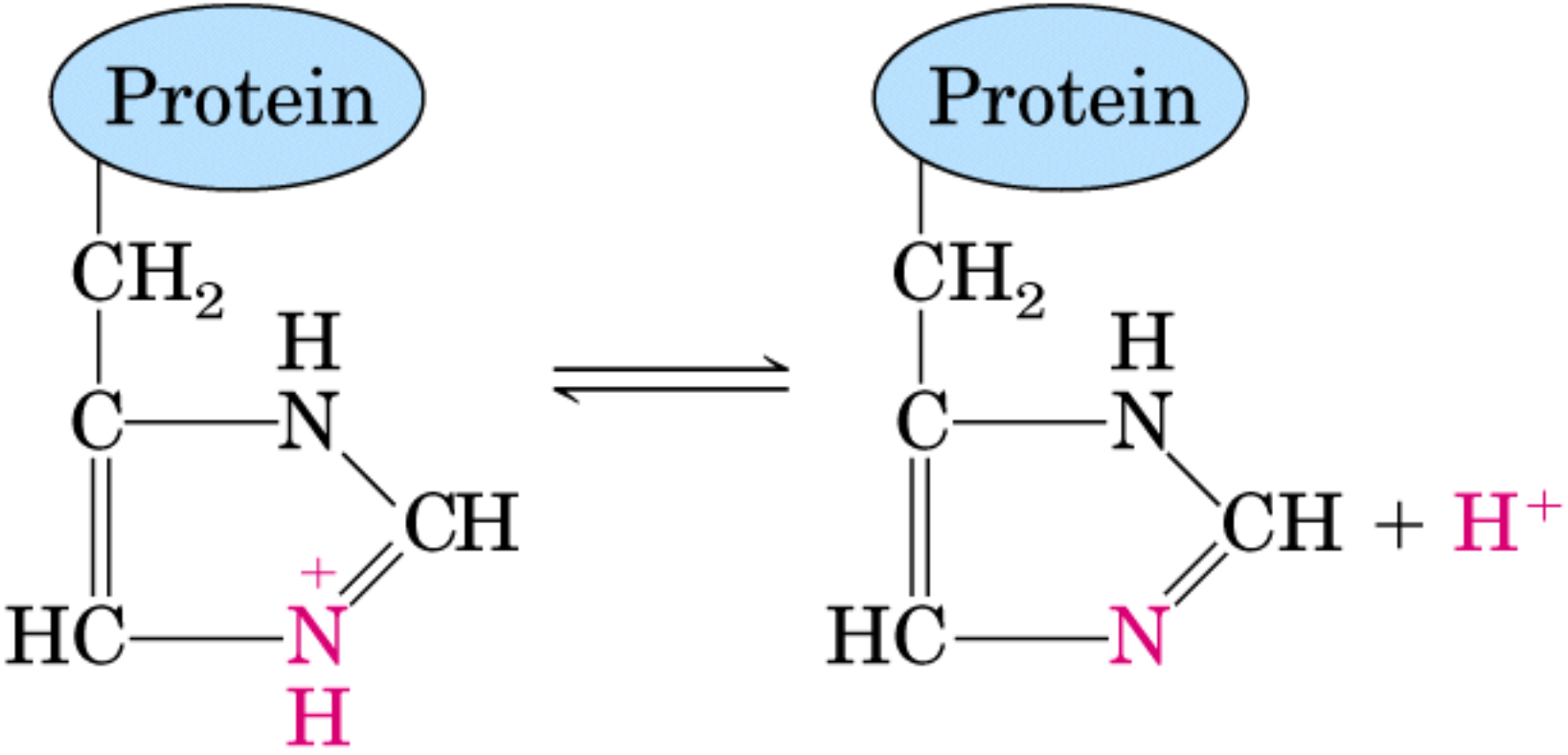
3 The titration curves of weak acids indicate that a conjugate acid-base pair can act as a buffer (resisting to pH changes of the system).

3.1 There is a relatively flat zone on all titration curves of weak acids extending about 1 pH units on either side of the pKa values.

3.2 pH changes little in the flat zone when H^+ or OH^- are added to the system .

3.3 pH values in biological systems are usually strictly kept constant (near pH 7.0) by buffering pairs of conjugate acid-bases.

(e.g., the phosphate buffer $H_2PO_4^-/HPO_4^{-2}$ in cytoplasm and the bicarbonate buffer H_2CO_3/HCO_3^- in blood, better with CO_2 gas to dissolved H_2CO_3 conversion).



Histidine **imidazole(咪唑基)**

4. The amino acids ionize in aqueous solutions.

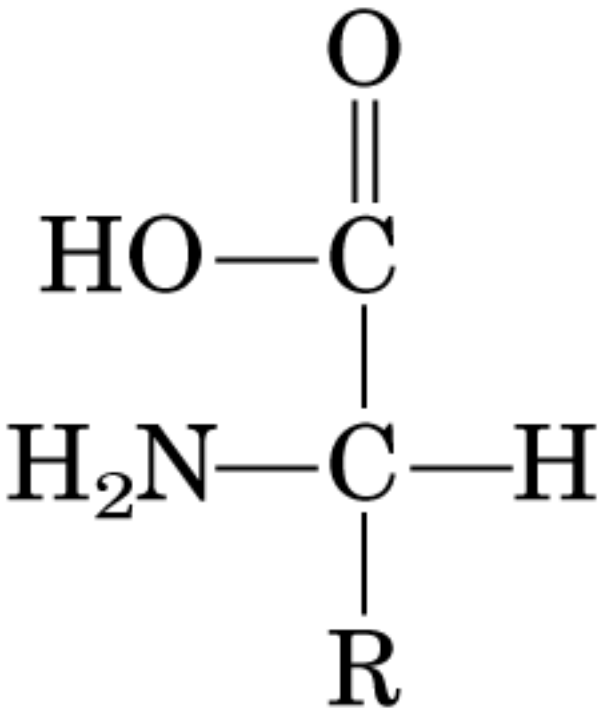
4.1 Crystalline amino acids (in neutral aqueous solutions) have melting points much higher than those of other organic molecules of similar size.

4.2 The amino acids ionize to various states depending on pH values.

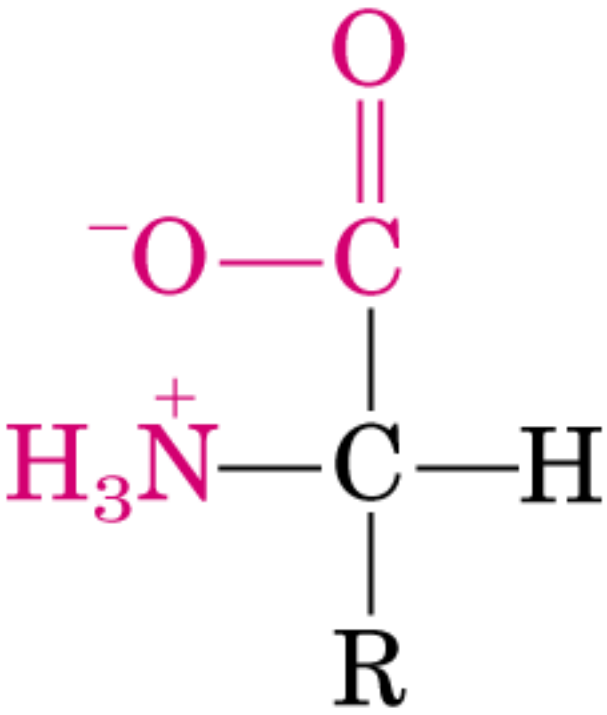
4.3 The amino acids (of neutral side chains) exist predominantly as dipolar ions, known as zwitterions (German for "hybrid ions").

4.4 Amino acids can act both as acids and bases. The zwitterion form of amino acids are ampholytes. Amino acids can be diprotic and triprotic acids.





Nonionic form



Zwitterionic form

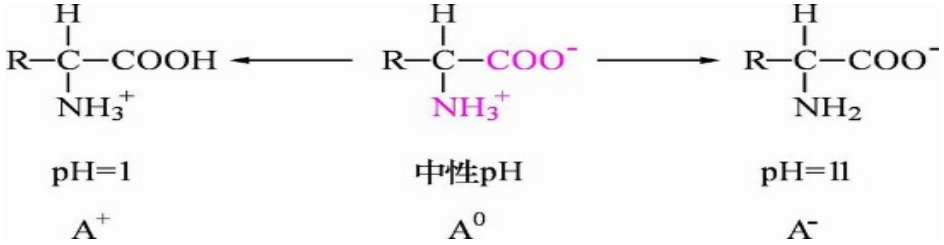
4.5 Monoamino monocarboxylic α-amino acids

(e.g., Gly, Ser, Phe with no ionizable R groups)

all have similar two-stage titration curves.

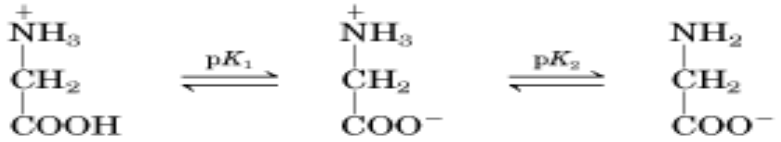
- The first stage reflects the deprotonation of the α-COOH group (pK_1).
- The second stage reflects the deprotonation of the α-NH_3^+ group (pK_2).

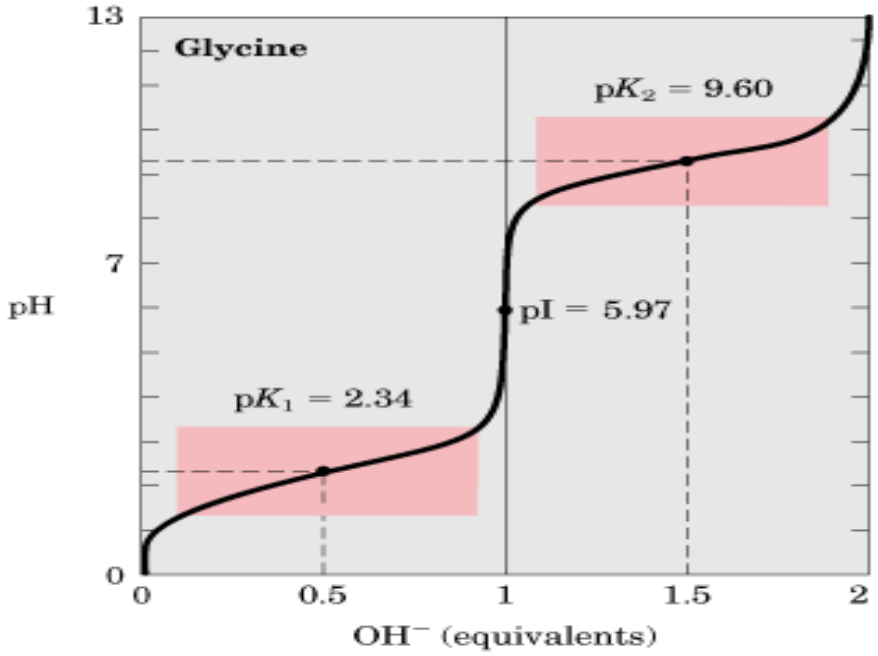
Amino acids, being both weak acids and bases, have characteristic titration curves and pKa values



Ampholyte

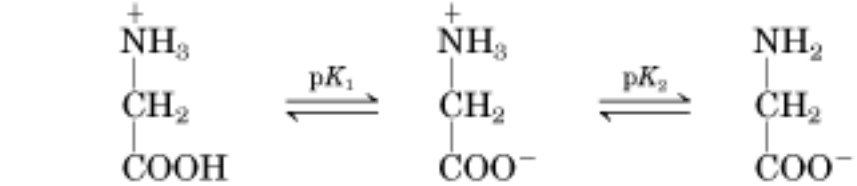
zwitterion

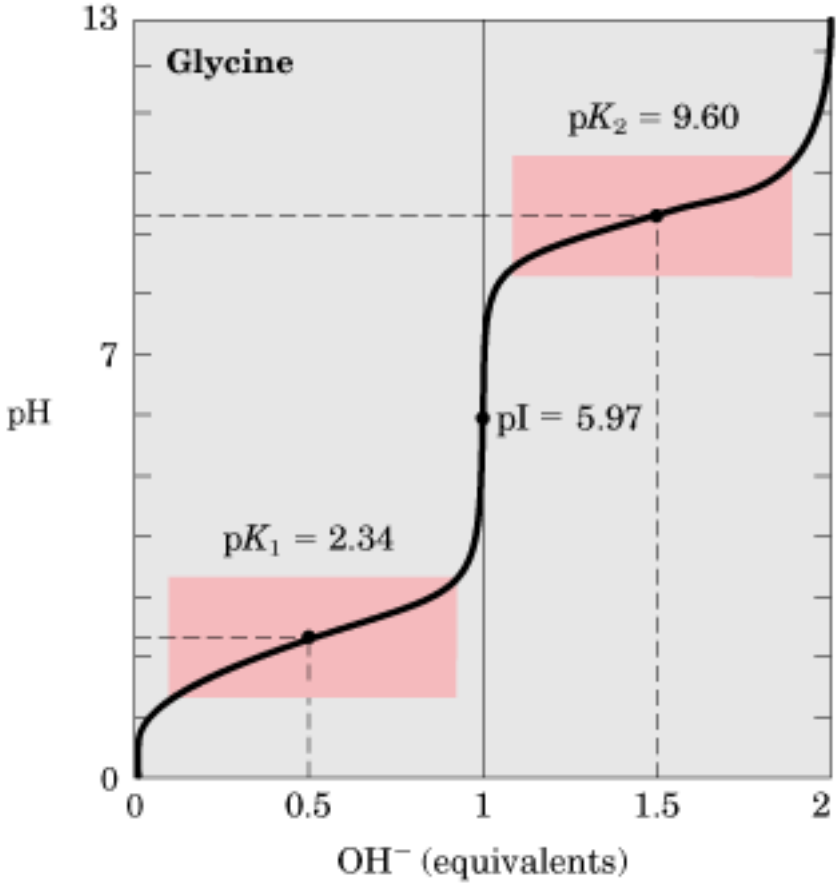




- **The pKa value of the α-COOH is more than 2 units smaller than that of acetic acid (pKa of 4.76), that is, a stronger (weak) acid.**
- **These amino acids have two buffering power regions.**




NH3+
CH2
COOH
pK1
NH3+
CH2
COO−
pK2
NH2
CH2
COO−


Glycine
pK2 = 9.60
pI = 5.97
pK1 = 2.34
pH
13
7
0
0
0.5
1
1.5
2
OH− (equivalents)

$$pI = (pK_1 + pK_2)/2$$

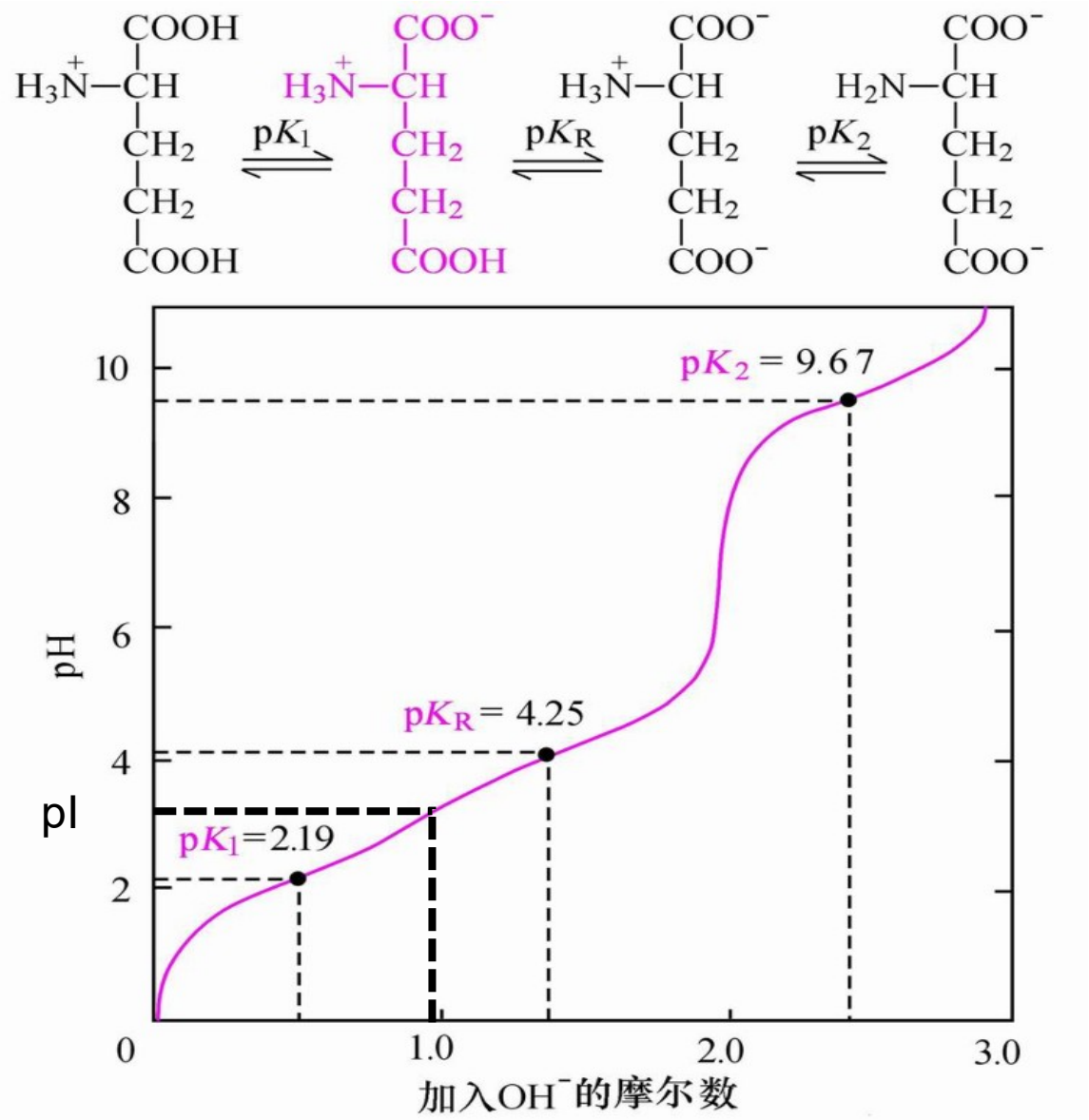
There is a specific pH (designated pI) at which an amino acid has equal positive and negative charge.

- **An amino acid does not move in an electric field at its pI, called isoelectric point.**
- **The pI of monoamino monocarboxylic amino acids reflects a status at which the α-COOH group is fully deprotonated, but the α-NH_3^+ group has not yet started deprotonating**

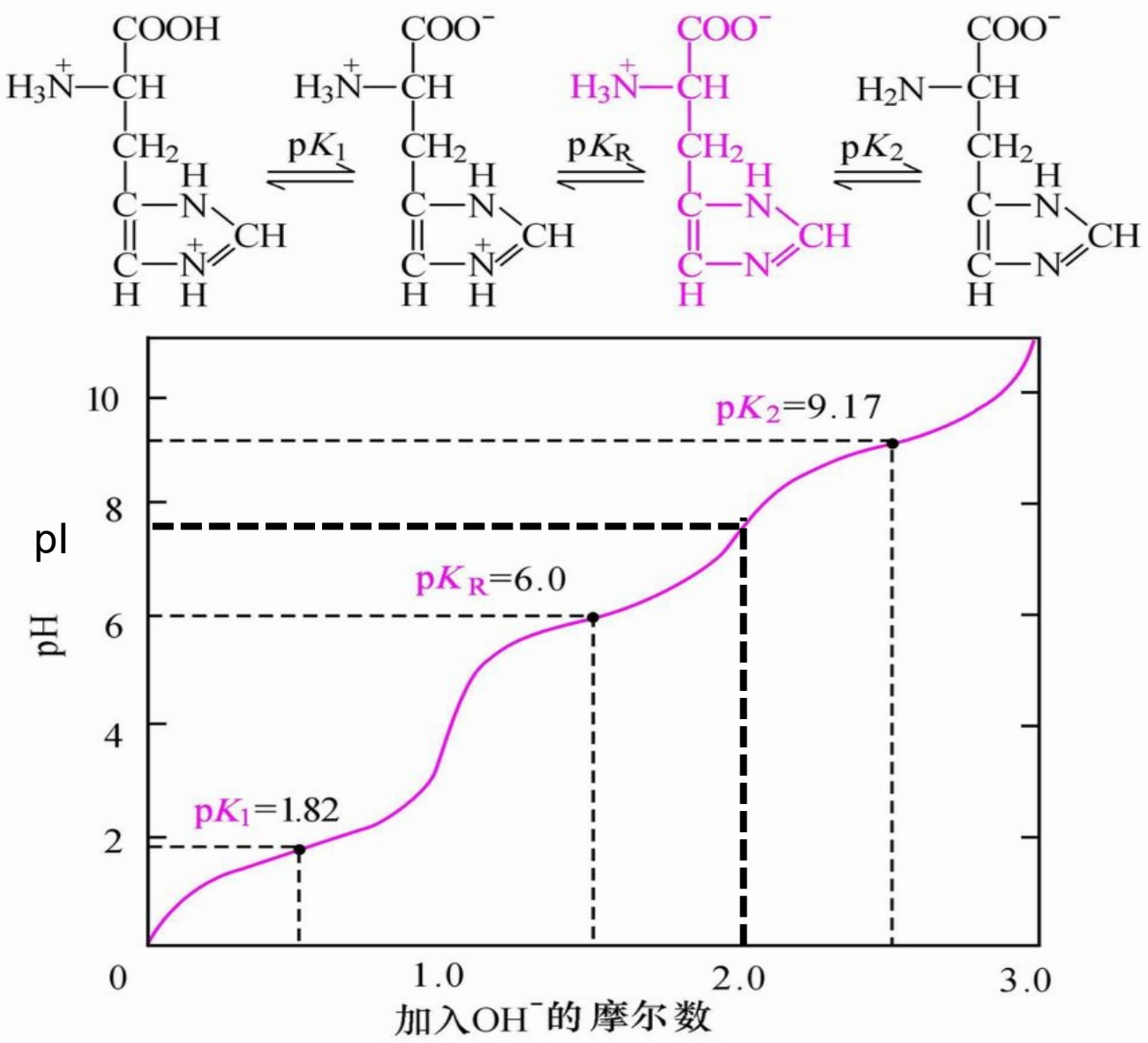
$$pI = (pK_1 + pK_2)/2$$

4.6 Acidic and basic amino acids have three-stage titration curves. The additional stage is for the ionizable group on the side chains (pK_R).

An acidic amino acid pI=(pK1+pK$_R$)/2



A basic amino acid pI=(pK$_R$+pK2)/2



The pI point of an acidic amino acid reflects a status at which the α-COOH is fully deprotonated, but the side chain -COOH and the a-NH_3^+ group have not yet started deprotonating

$$pI = (pK_1+pK_R)/2$$

The pI point of a basic amino acid reflects a status at which the α-COOH and the side chain NH_3^+ have fully deprotonated, but the α- NH_3^+ group have not yet deprotonated.

$$pI = (pK_R+pK_2)/2$$

氨基酸的重要反应：

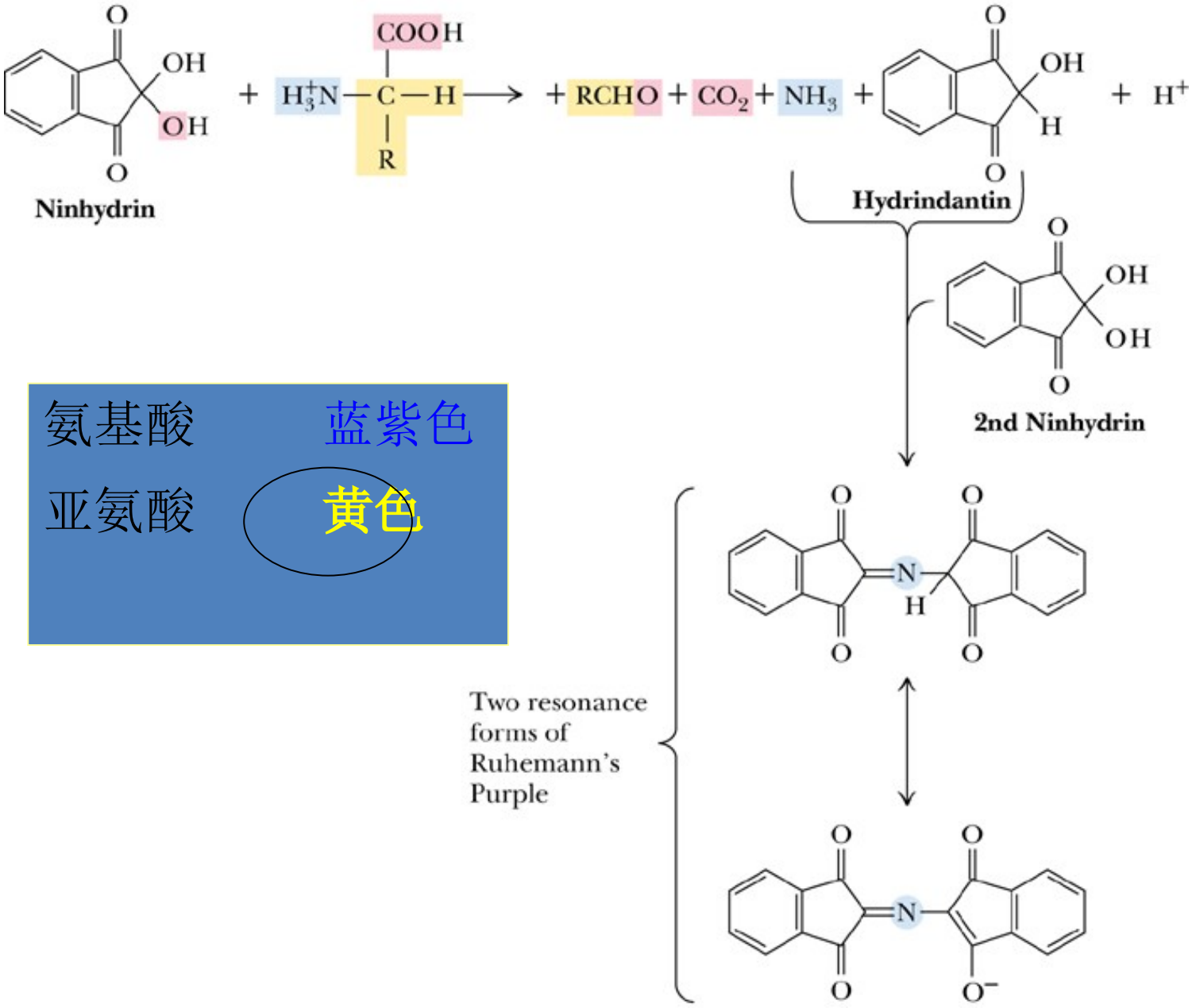
- 与茚三酮反应
- Sanger 反应
- Edman 反应



Ninhydrin Reaction

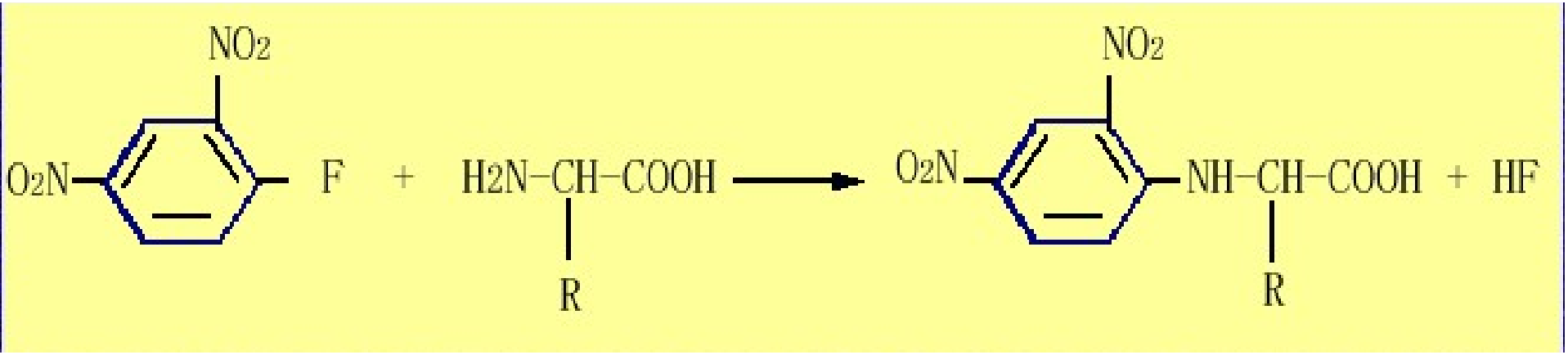


The amino acids are detected by colorimetric reaction with ninhydrin (茚三酮) .


Ninhydrin
COOH
H3+N—C—H
R
RCHO
CO2
NH3
Hydrindantin
2nd Ninhydrin
H+
氨基酸 蓝紫色
亚氨酸 黄色
Two resonance forms of Ruhemann's Purple

- Edman degradation and Sanger reaction

Sanger reaction

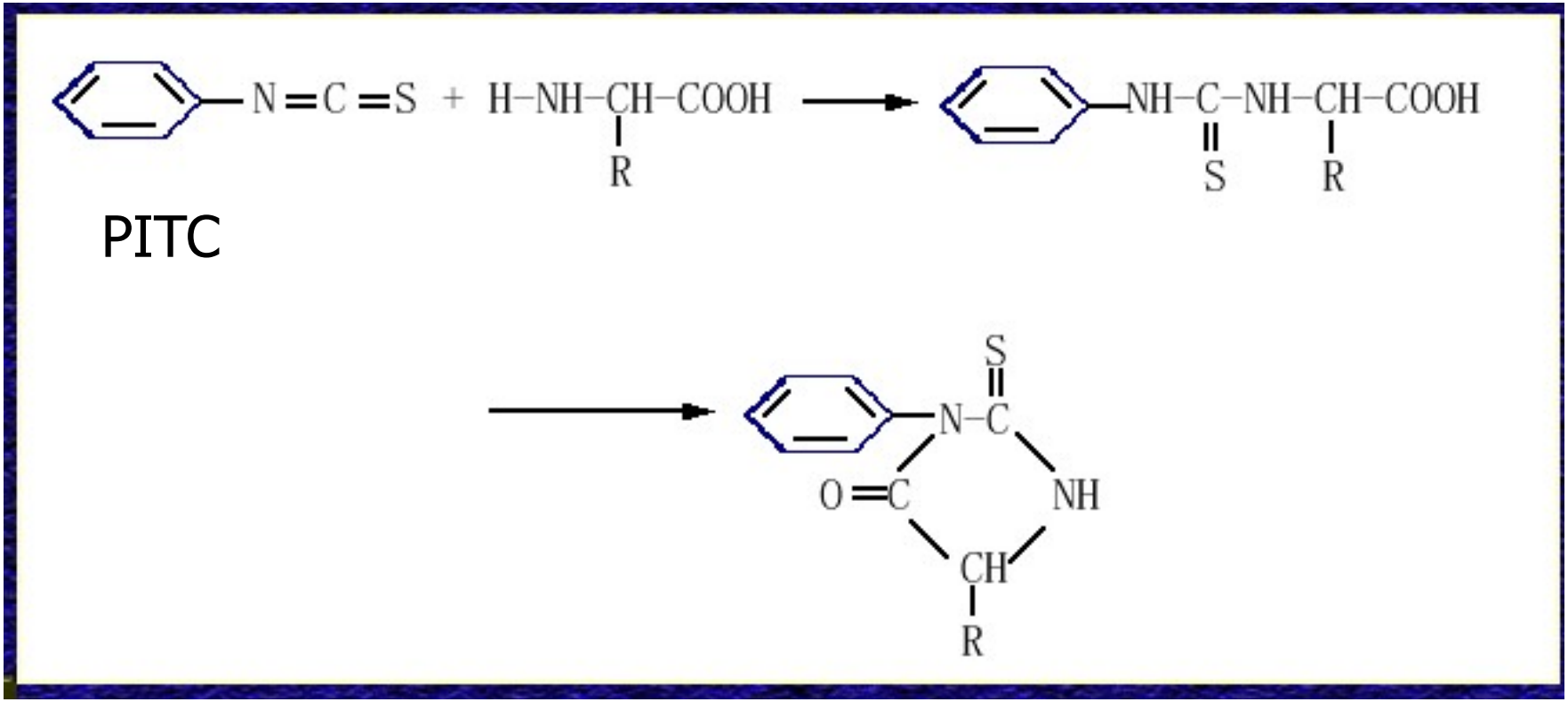


DNFB or FDNB

二硝基氟苯

（ sanger reagent ）

Edman reaction



The amino acid sequence of a protein can be determined by Edman degradation which sequentially removes one residue at a time from the N terminus.

Summary of B2 Acids and Bases

1. **Weak acids (proton donors) and weak bases (proton acceptors) do not ionize completely when dissolved in water.**

Titration curves, Henderson-Hasselbach equation, buffering region

$$\mathrm{pH} = \mathrm{p}K_a + \log\frac{[\mathrm{A}^-]}{[\mathrm{HA}]}$$

2. α-amino acids all have similar two-stage titration curves.

- **the deprotonation of : the α-COOH group (pK_1);**
  (R-COOH, pK_R)
  (R-NH_3^+, pK_R)
  α-NH_3^+ group (pK_2).

pI?

3. Three reactions： ninhydrin, Sanger reaction, Edman reaction.

B3 Protein structure

- Peptide
- The Three-dimensional Structure of Proteins



1. Amino acids covalently join one another to form peptides

1.1. The α-carboxyl group of one amino acid joins with the α-amino group of another amino acid by a peptide bond (actually an amide bond)

1.1.1 This is a condensation reaction where a water molecule is liberated or eliminated.

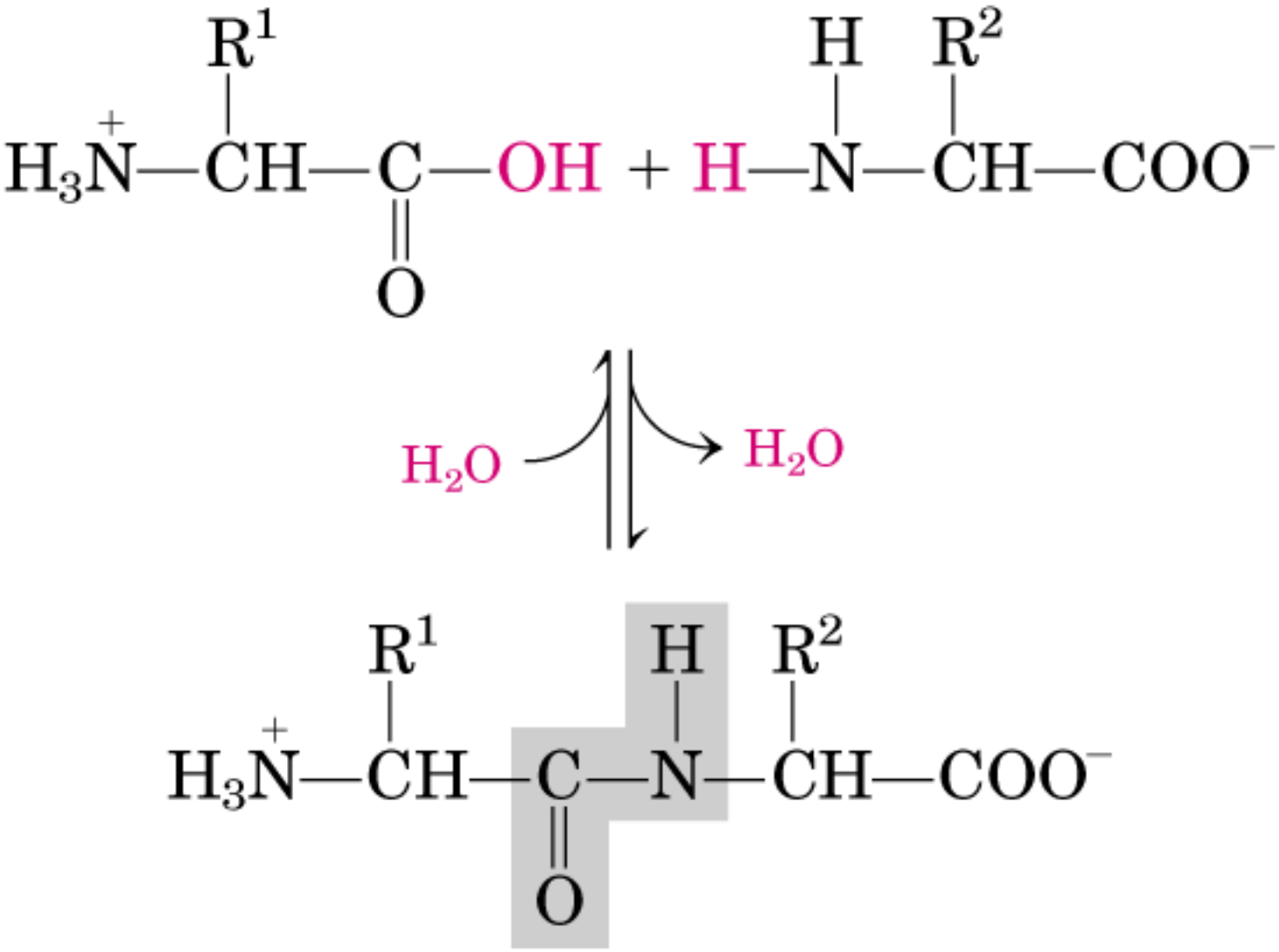
1.1.2 The condensation reaction can occur repeatedly to form oligopeptides (with less than 50 aa), polypeptides (btw 50-100 aa), and proteins (longer).

1. Amino acids covalently join one another to form peptides

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1.1.2 The condensation reaction can occur repeatedly to form oligopeptides (with less than 50 aa), polypeptides (btw 50-100 aa), and proteins (longer).


R1
H3N+—CH—C—OH + H—N—CH—COO−
H
R2
O
H2O
H2O
R1
H
R2
H3N+—CH—C—N—CH—COO−
O

1.2. The peptide chain is directional.

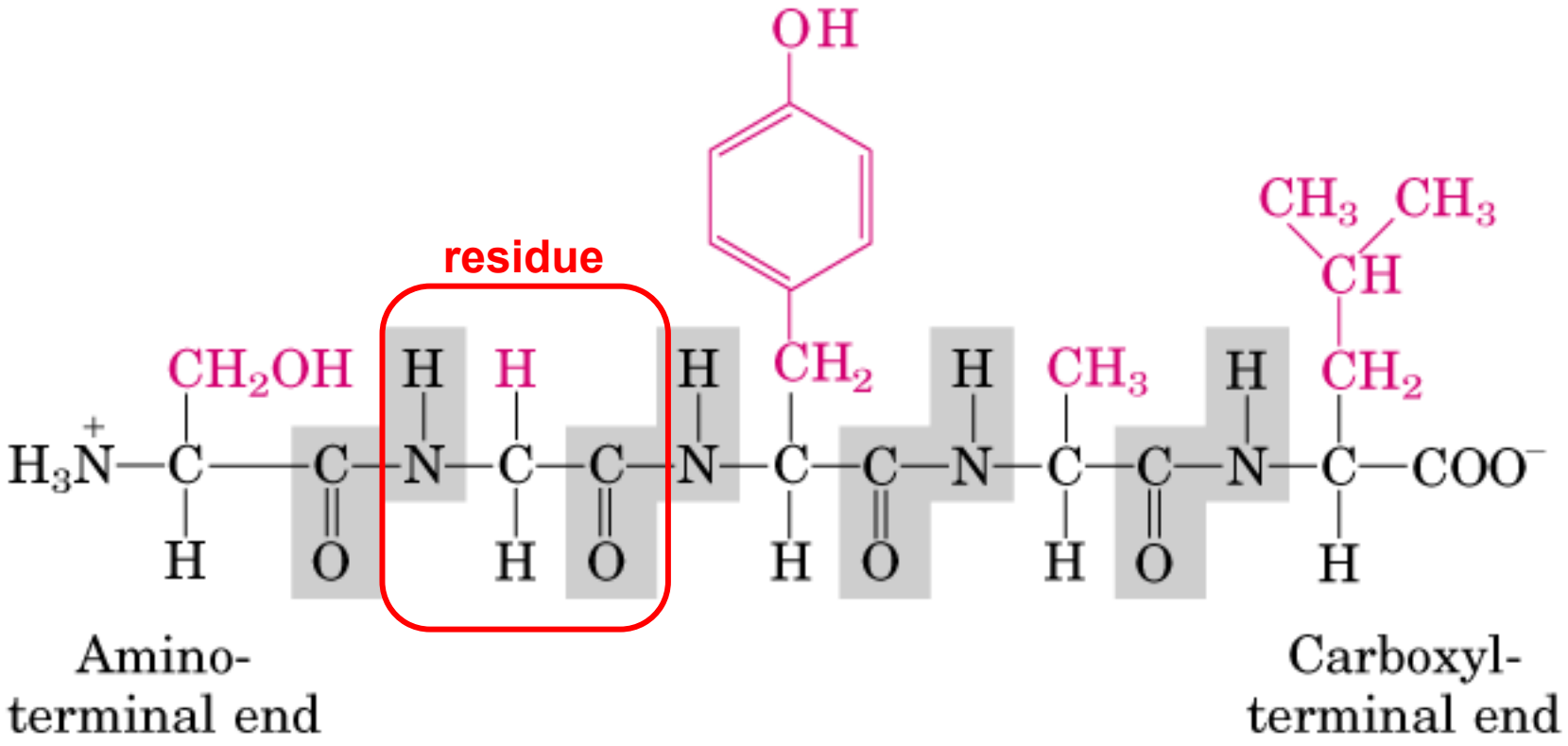
1.2.1 An amino acid unit in a peptide chain is called a residue.

1.2.2 The end having a free α-amino group is called amino-terminal or N-terminal.

1.2.3 The end having a free α-carboxyl group is called carboxyl-terminal or C-terminal.

1.2.4 By convention, the N-terminal is taken as the beginning of the peptide chain, and put at the left (C-terminal at the right).

Biosynthesis starts from the N-terminal.



N-terminus **C-terminus**

1.2.5 The peptide chain consist of a regularly repeating main chain (or backbone) and the variable side chains of the residues.

1.2.6 Amino acid residues have an order or sequence on a peptide.

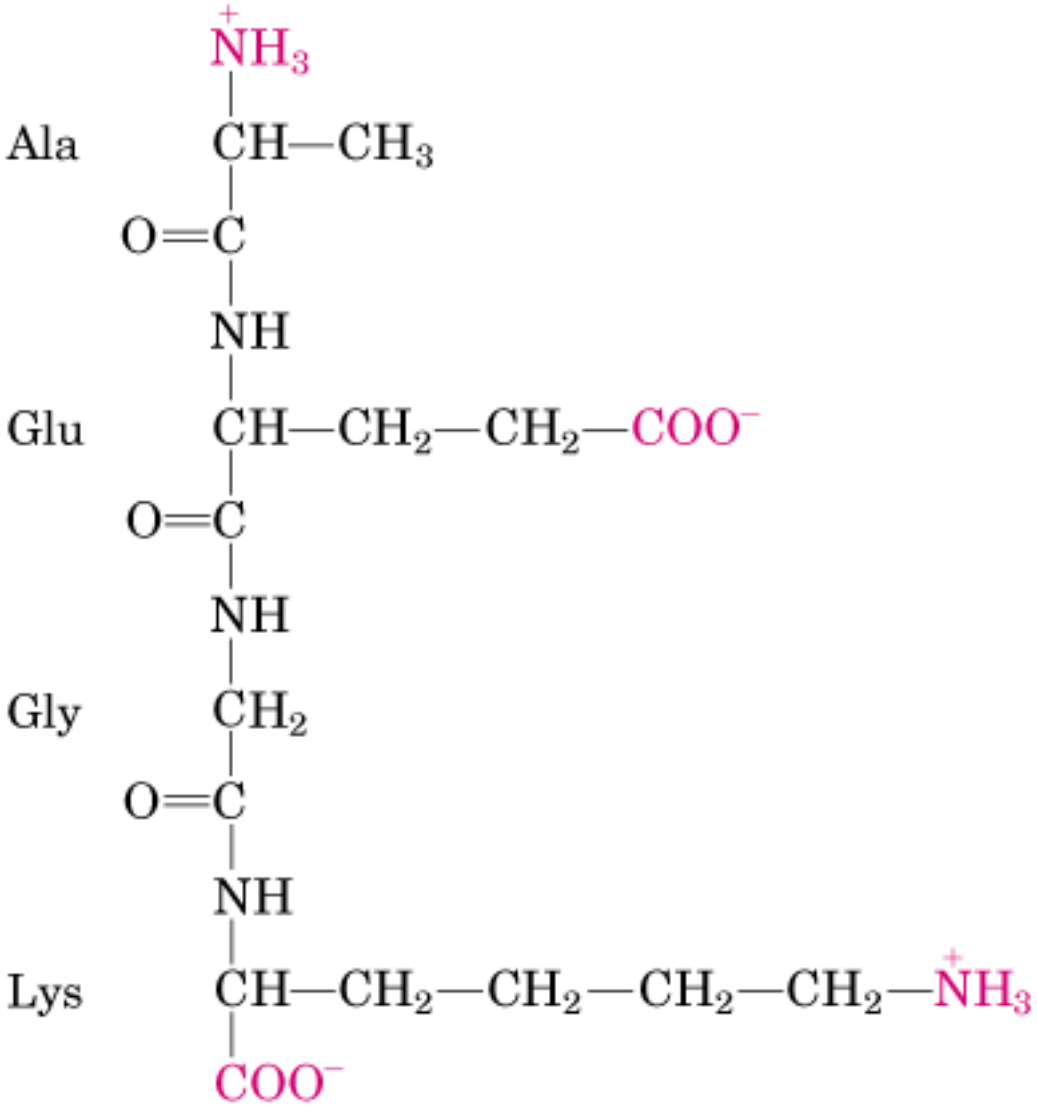
1.2.7 The 20 amino acids are analogous to the 26 letters in English; the number of different peptides made of them is unlimited.

1.3. The size of a peptide can be described by its total number of residues (e.g., a pentapeptide, an octapeptide) or relative molecular mass (molecular weight).

1.3.1 The mean molecular weight of an amino acid residue in a peptide is ~110 Dalton.

1.3.2 Most natural polypeptide chains contain between 50 and 2000 amino acid residues, thus having relative molecular mass between 5500 daltons and 220,000 daltons, or 5.5 kDa and 220 kDa, respectively.

1.4. Each peptide has a characteristic titration curve and an isoelectric point (pI).



This tetrapeptide has one free α-amino group, one free α-carboxyl group, plus two ionizable R groups.

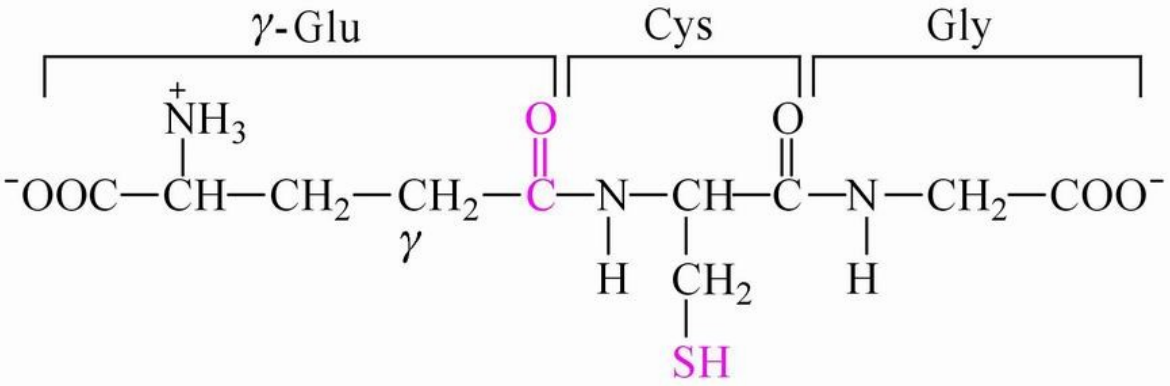
2. Many short peptides have important biological activities

2.1 Glutathione (GSH), present in plants, animals, and some bacteria, often at high levels, can be thought of as a redox buffer.

2.2 Some short peptides, such as neuropeptides （神经肽）, act as neurotransmitters（神经传递素）, neurohormones（神经激素）, and neuromodulators.

Glutathione

谷胱甘肽(GSH)
还原型



Glu-Cys-Gly
SH

GSH

Glu-Cys-Gly
S
S
GSSG
Glu-Cys-Gly

2.3 Some short peptides act as antibiotics

Gramicidin(短杆菌肽) A (15-residue) is a well studied peptide antibiotic. Its structure has been determined. It contains alternating L- and D-amino acid residues. It is not synthesized on ribosomes!

2.4 Many short peptides are used as defensive poisons.

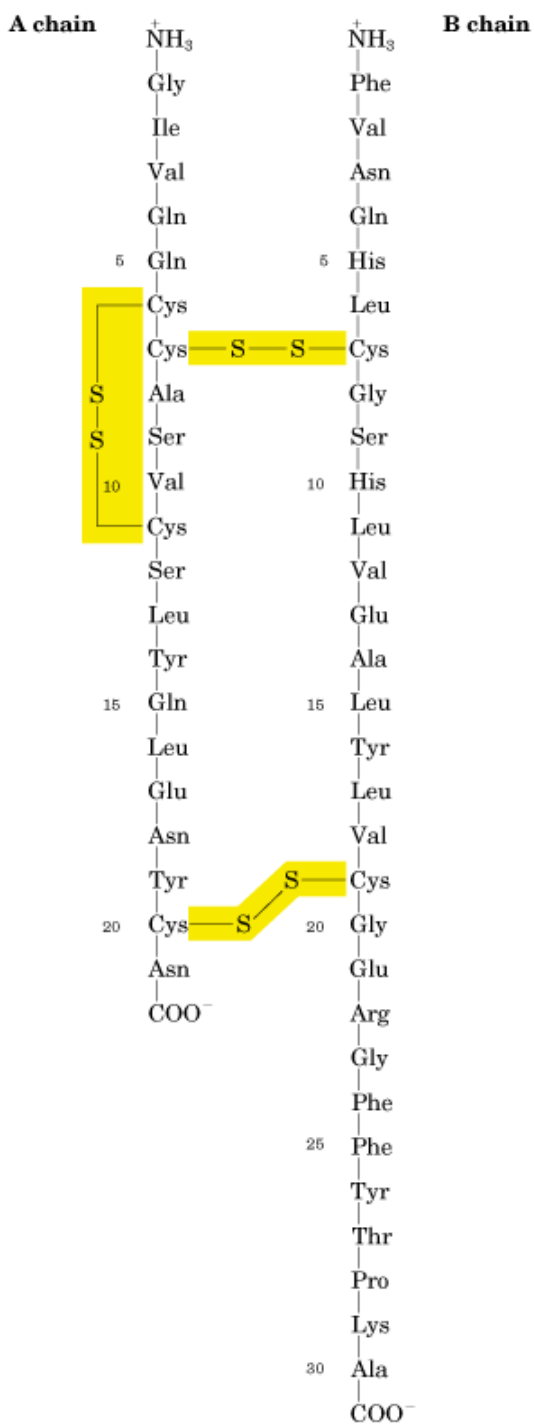
2.3.1 α-Amanitin (鹅膏蕈碱) (8-residue, circular) in mushroom is an extremely toxic peptide (inhibiting RNA polymerases II and III at picomolar levels!)

2.3.2 Very toxic short peptides are also found in snake venom, spider.

2.5 Many vertebrate hormones are small polypeptides.

Insulin (胰岛素)(51-residue) is produced by the pancreas and acts to lower blood glucose level.

Glucagon (胰高血糖素) (29-residue) is also produced by the pancreas and acts to increase the blood glucose level.



Insulin (胰岛素)(51-residue)

3. Proteins in general

3.1 While proteins and polypeptides are normally used interchangeably, polypeptides generally have MW of less than 10,000; and also proteins may have multiple polypeptide subunits.

table 5–2

Molecular Data on Some Proteins

	Molecular weight	Number of residues	Number of polypeptide chains
Cytochrome *c* (human)	13,000	104	1
Ribonuclease A (bovine pancreas)	13,700	124	1
Lysozyme (egg white)	13,930	129	1
Myoglobin (equine heart)	16,890	153	1
Chymotrypsin (bovine pancreas)	21,600	241	3
Chymotrypsinogen (bovine)	22,000	245	1
Hemoglobin (human)	64,500	574	4
Serum albumin (human)	68,500	609	1
Hexokinase (yeast)	102,000	972	2
RNA polymerase (*E. coli*)	450,000	4,158	5
Apolipoprotein B (human)	513,000	4,536	1
Glutamine synthetase (*E. coli*)	619,000	5,628	12
Titin (human)	2,993,000	26,926	1



3.2 Proteins have characteristic AA compositions

table 5–3

Amino Acid Composition of Two Proteins*

Amino acid	Number of residues per molecule of protein: Bovine cytochrome *c*	Bovine chymotrypsinogen
Ala	6	22
Arg	2	4
Asn	5	15
Asp	3	8
Cys	2	10
Gln	3	10
Glu	9	5
Gly	14	23
His	3	2
Ile	6	10
Leu	6	19
Lys	18	14
Met	2	2
Phe	4	6
Pro	4	9
Ser	1	28
Thr	8	23
Trp	1	8
Tyr	4	4
Val	3	23
Total	104	245

*Note that standard procedures for the acid hydrolysis of proteins convert Asn and Gln to Asp and Glu, respectively. In addition, Trp is destroyed. Special procedures must be employed to determine the amounts of these amino acids.

3.3 Some proteins contain chemical groups other than AA, which are called the prosthetic groups（辅基）, and these proteins are referred as conjugated proteins while the amino acid part alone is called apoprotein.

Simple protein

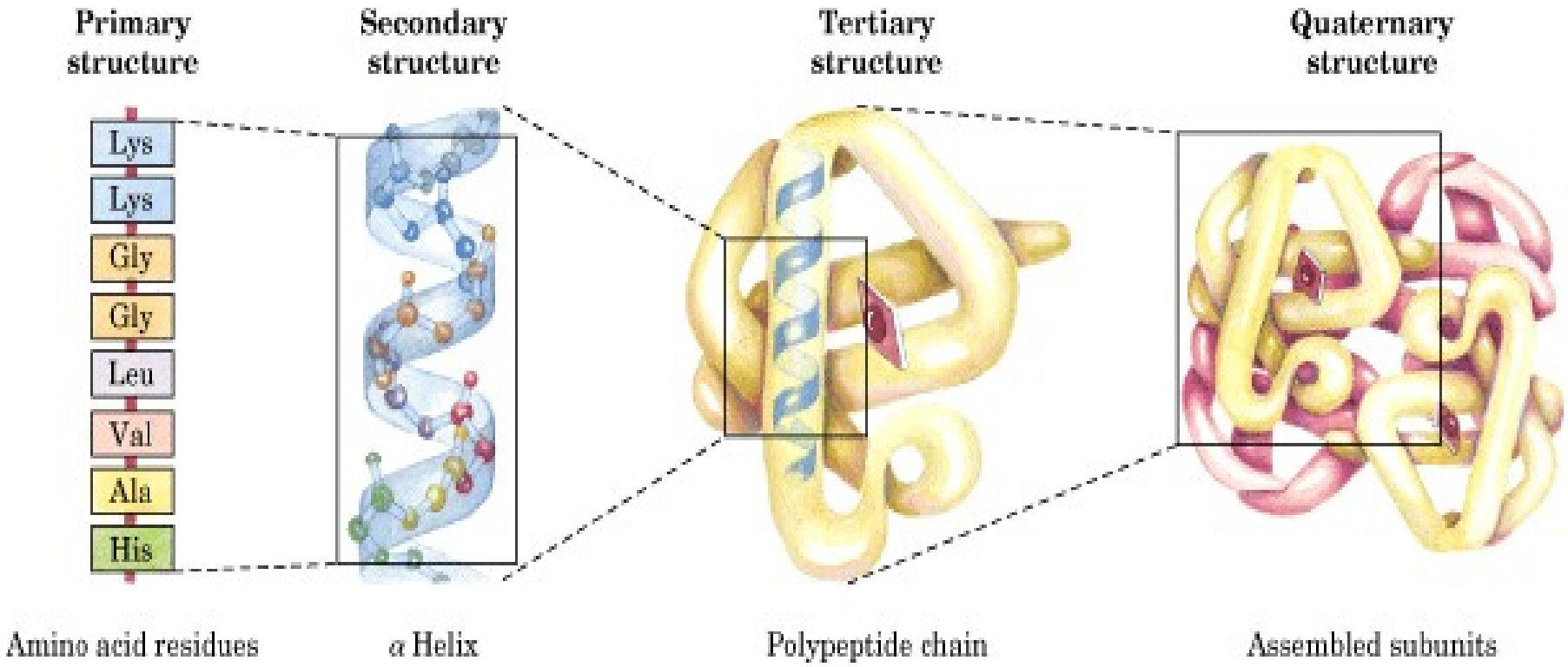
Conjugated protein （结合蛋白）

table 5–4

Conjugated Proteins

Class	Prosthetic group(s)	Example
Lipoproteins	Lipids	β_1-Lipoprotein of blood
Glycoproteins	Carbohydrates	Immunoglobulin G
Phosphoproteins	Phosphate groups	Casein of milk
Hemoproteins	Heme (iron porphyrin)	Hemoglobin
Flavoproteins	Flavin nucleotides	Succinate dehydrogenase
Metalloproteins	Iron	Ferritin
	Zinc	Alcohol dehydrogenase
	Calcium	Calmodulin
	Molybdenum	Dinitrogenase
	Copper	Plastocyanin

3.4 Protein structure is studied at different levels.



Protein structure

General studies of the peptide bond

Primary structure

Secondary structure

Tertiary structure

Quaternary structure

Protein stability



1.3 Protein structures have conventionally been understood at four different levels.

1.3.1 The primary structure is the amino acid sequence (including the locations of disulfide bonds).

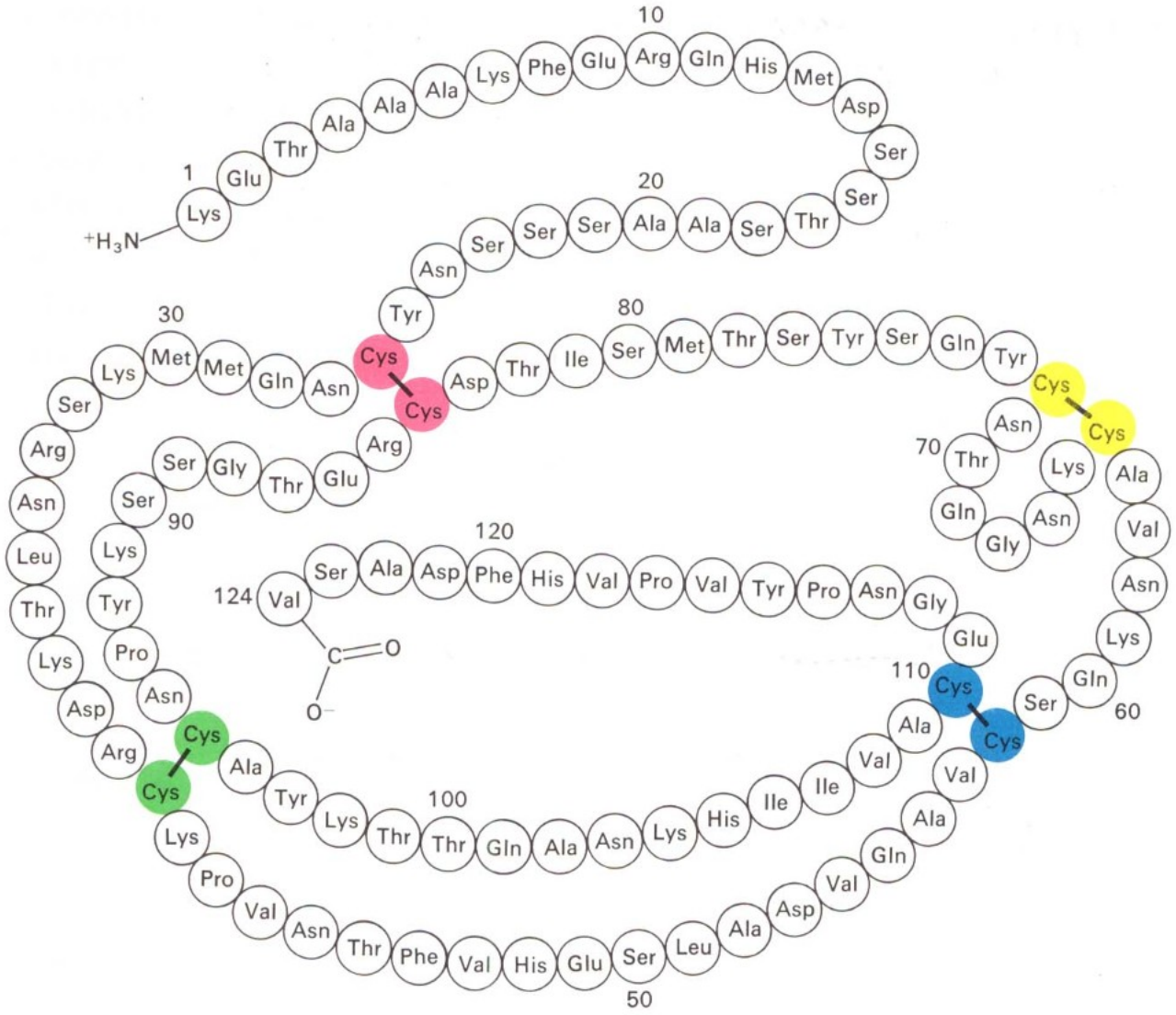
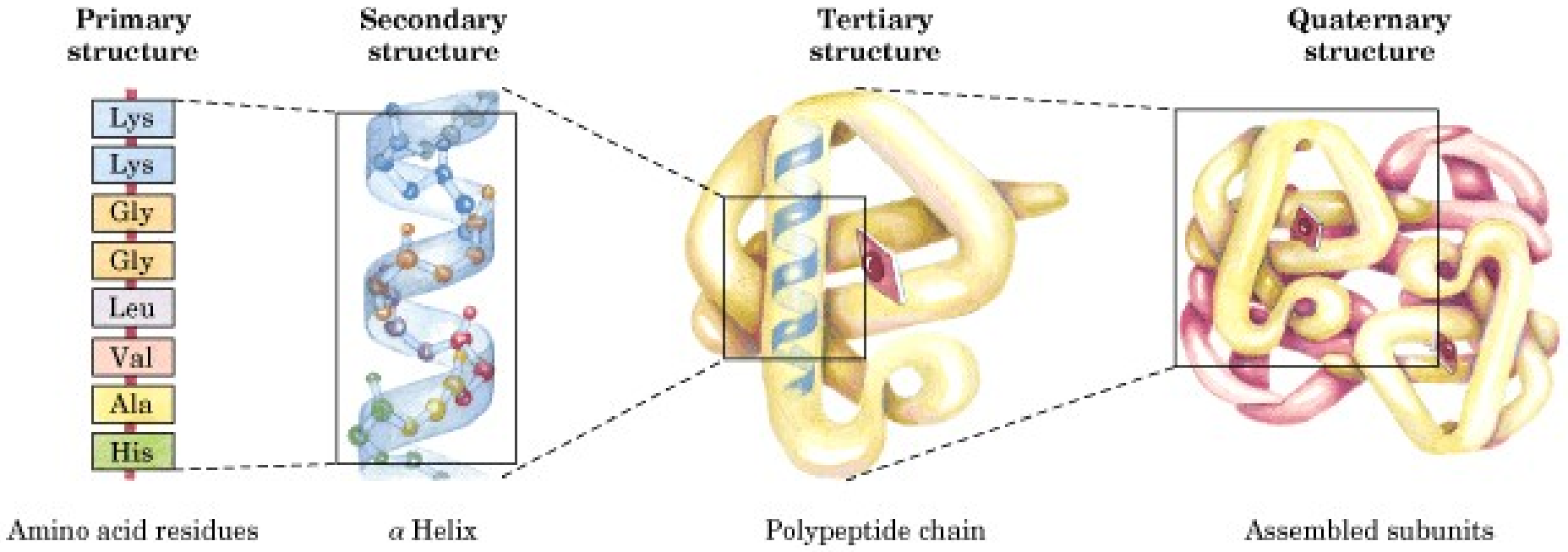


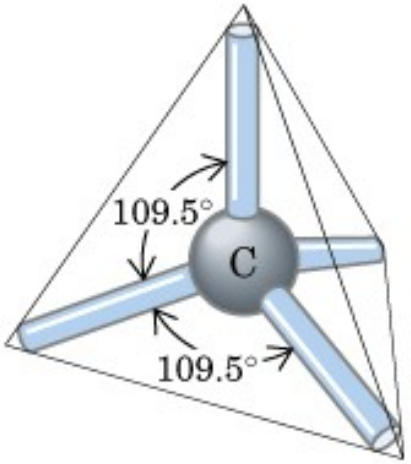
Figure 2-43
Amino acid sequence of bovine ribonuclease. The four disulfide bonds are shown in color. [After D. G. Smyth, W. H. Stein, and S. Moore. *J. Biol. Chem.* 238(1963):277.]

1.3.2 The secondary structure refers to the regular, recurring arrangements of adjacent residues resulting mainly from hydrogen bonding between backbone groups, with α-helices and β-pleated sheets as the two most common ones.

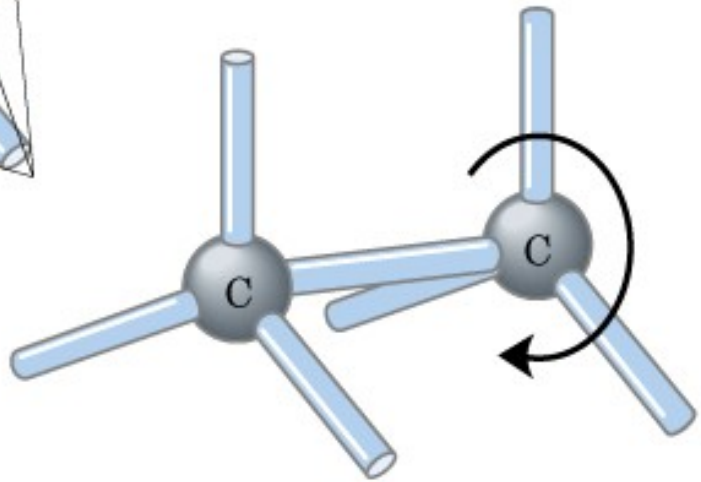
1.3.3 The tertiary structure refers to the spatial relationship among all amino acid residues in a polypeptide chain, that is, the complete three-dimensional structure.

1.3.4 The quaternary structure refers to the spatial arrangements of each subunit in a multisubunit protein, including nature of their contact.

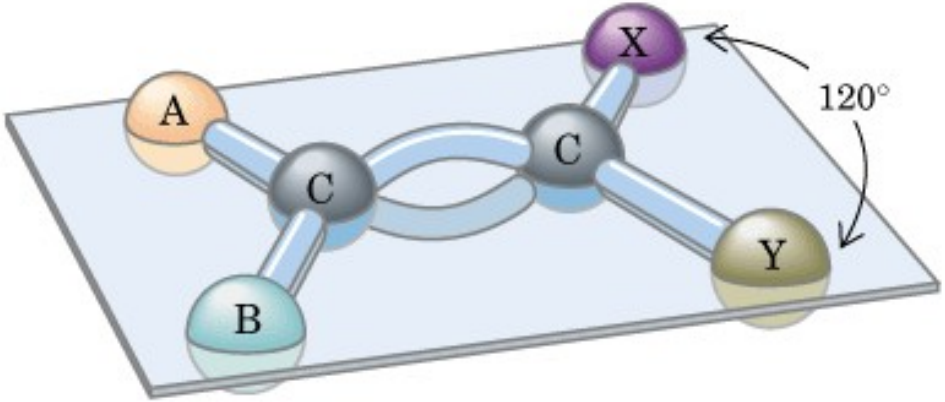

Primary structure
Secondary structure
Tertiary structure
Quaternary structure
Lys
Lys
Gly
Gly
Leu
Val
Ala
His
Amino acid residues
α Helix
Polypeptide chain
Assembled subunits



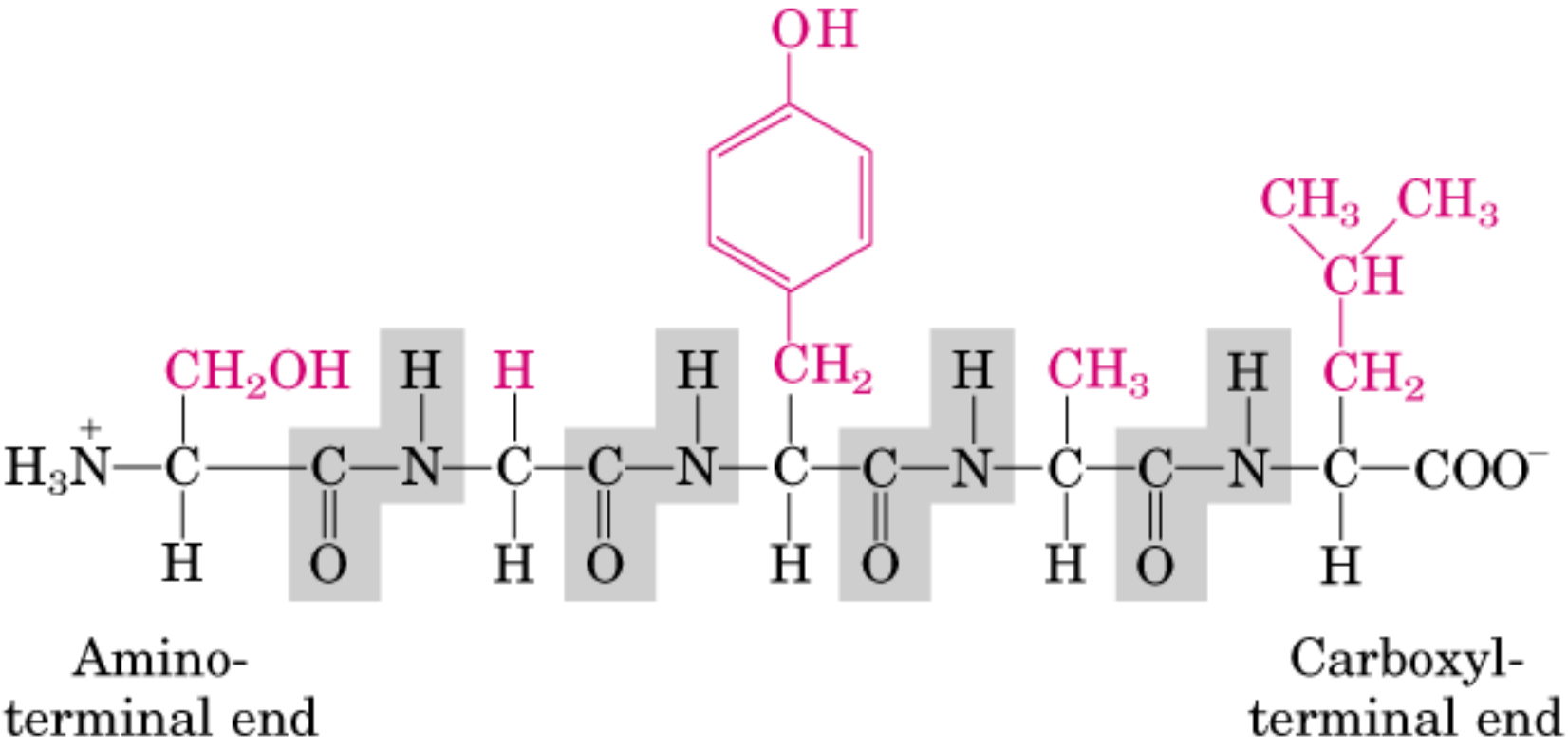
(a)



(b)



(c)


OH
CH3 CH3
CH
CH2OH
H
H
H
CH2
H
CH3
H
CH2
H3N+
C
C
N
C
C
N
C
C
N
C
C
N
C
COO−
H
O
H
O
H
O
H
O
H
Amino-
terminal end
Carboxyl-
terminal end

1. General studies of the peptide bond

1.1 The peptide (O=C-N-H) bond was found to be shorter than the C-N bond in a simple amine and atoms attached are coplanar.

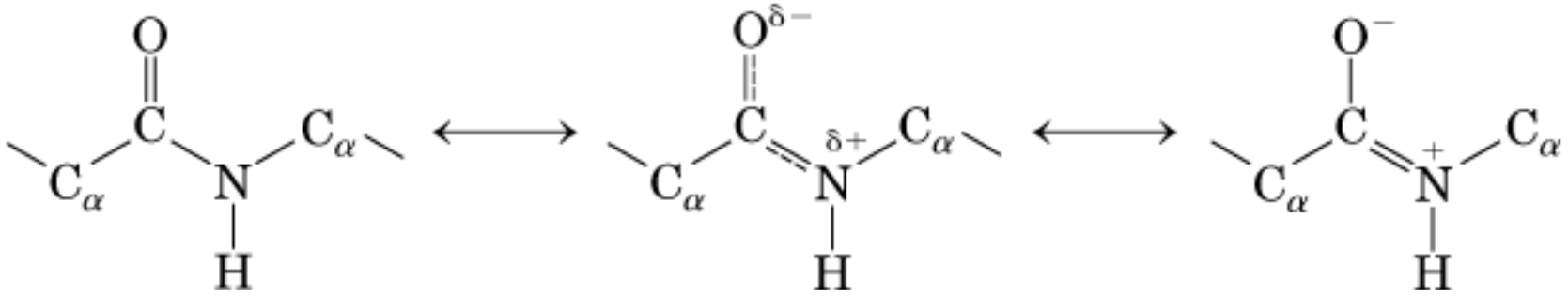
1.1.1 This was revealed by X-ray diffraction studies of amino acids and of simple dipeptides and tripeptides.

1.1.2 The peptide (amide) bond was found to be about 1.32 Å(angstrom) (C-N single bond, 1.49; C=N double bond, 1.27), thus having partial double bond feature (should be rigid and unable to rotate freely).

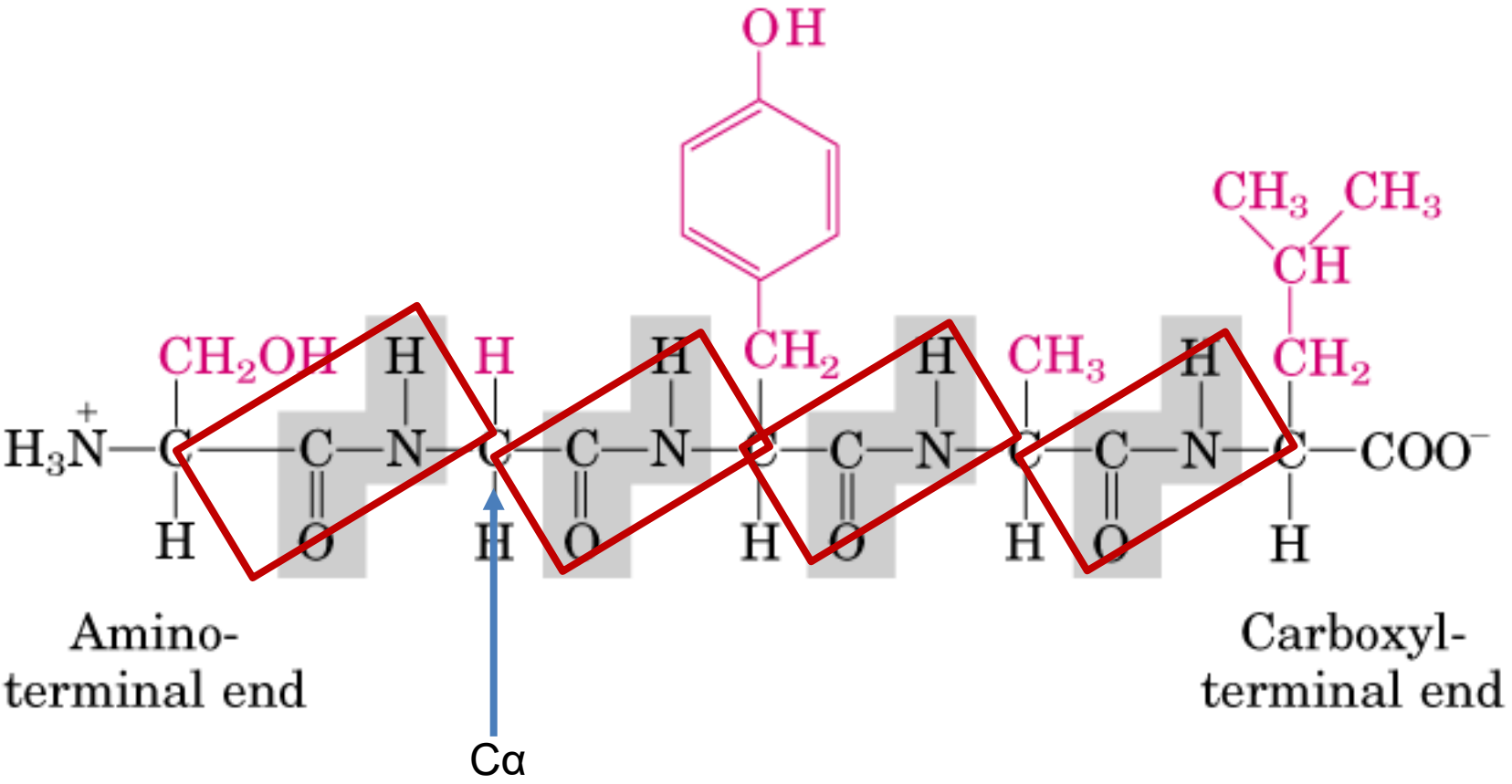
1.1.3 The partial double bond feature is a result of partial sharing (resonance) of electrons between the carbonyl oxygen and amide nitrogen.

1.1.4 The atoms attached to the peptide bond are coplanar with the oxygen and hydrogen atom in trans positions.

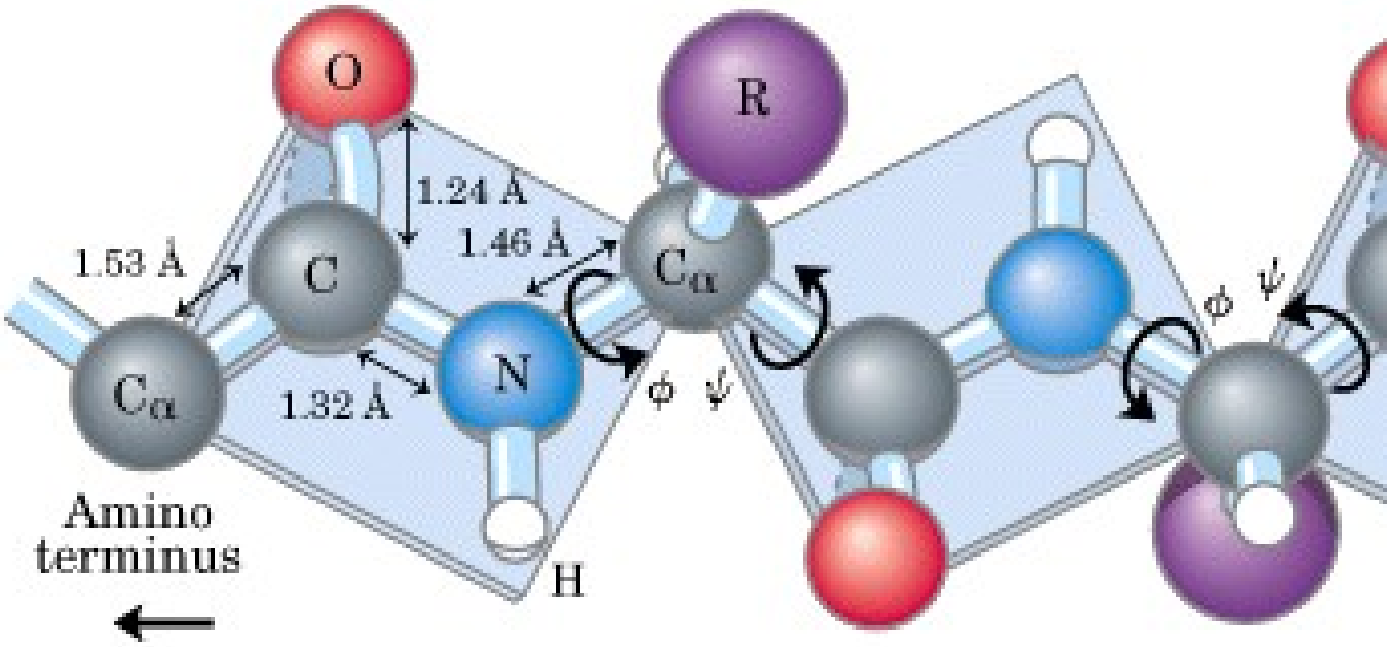
The carbonyl oxygen has a partial negative charge and the amide nitrogen a partial positive charge, setting up a small electric dipole. Virtually all peptide bonds in proteins occur in this trans configuration; an exception is noted in Figure 6–8b.



(a)


OH
CH3 CH3
CH
CH2OH
H
H
CH2
CH3
CH2
H3N+
COO−
Amino-
terminal end
Cα
Carboxyl-
terminal end

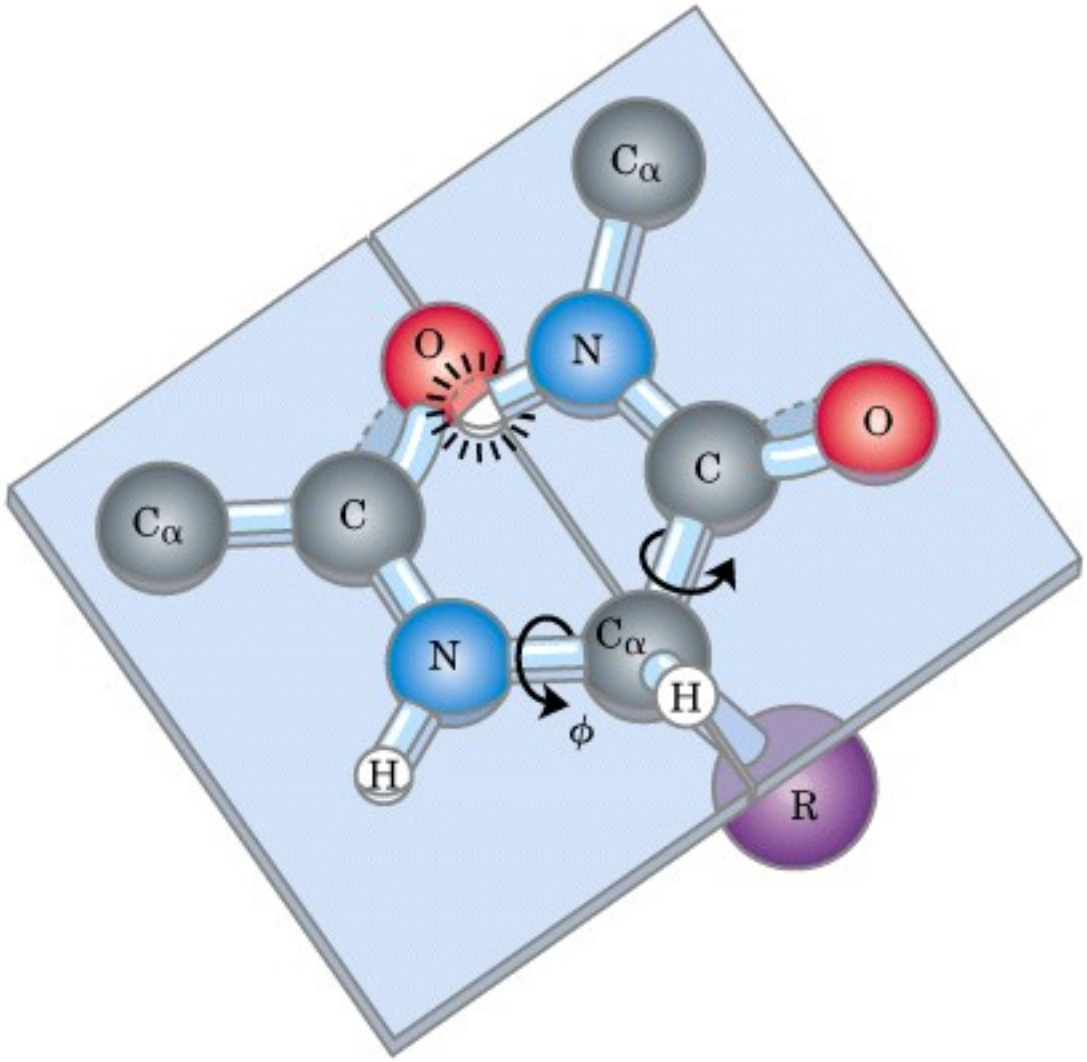
1.2 The peptide bond is rigid and planar



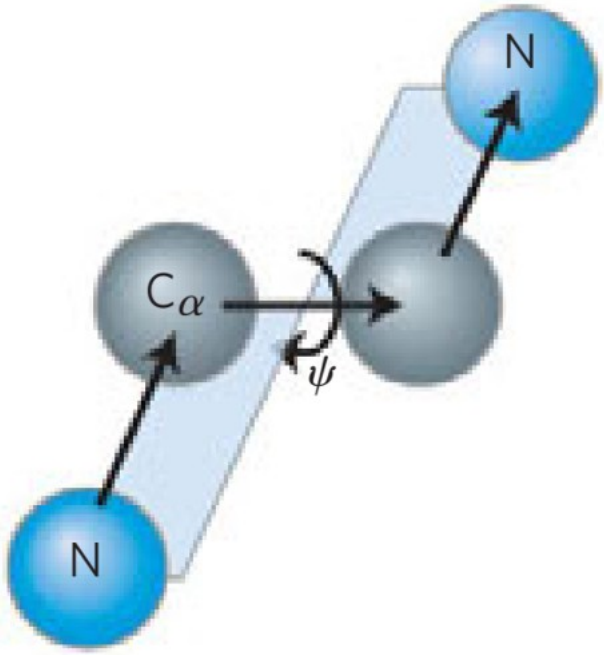
Της πλαναρ πεπτιδε

Dihedral angles : Φ and Ψ

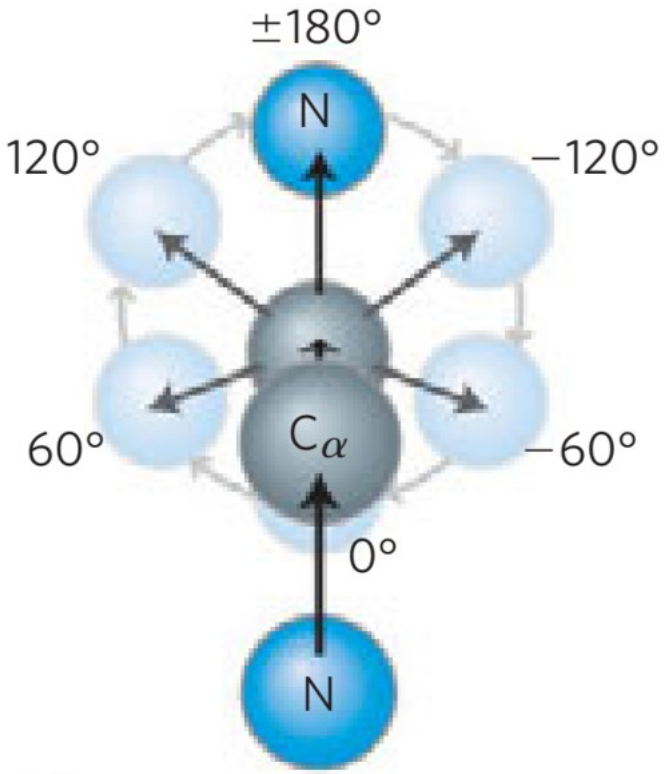
φηεδραλ



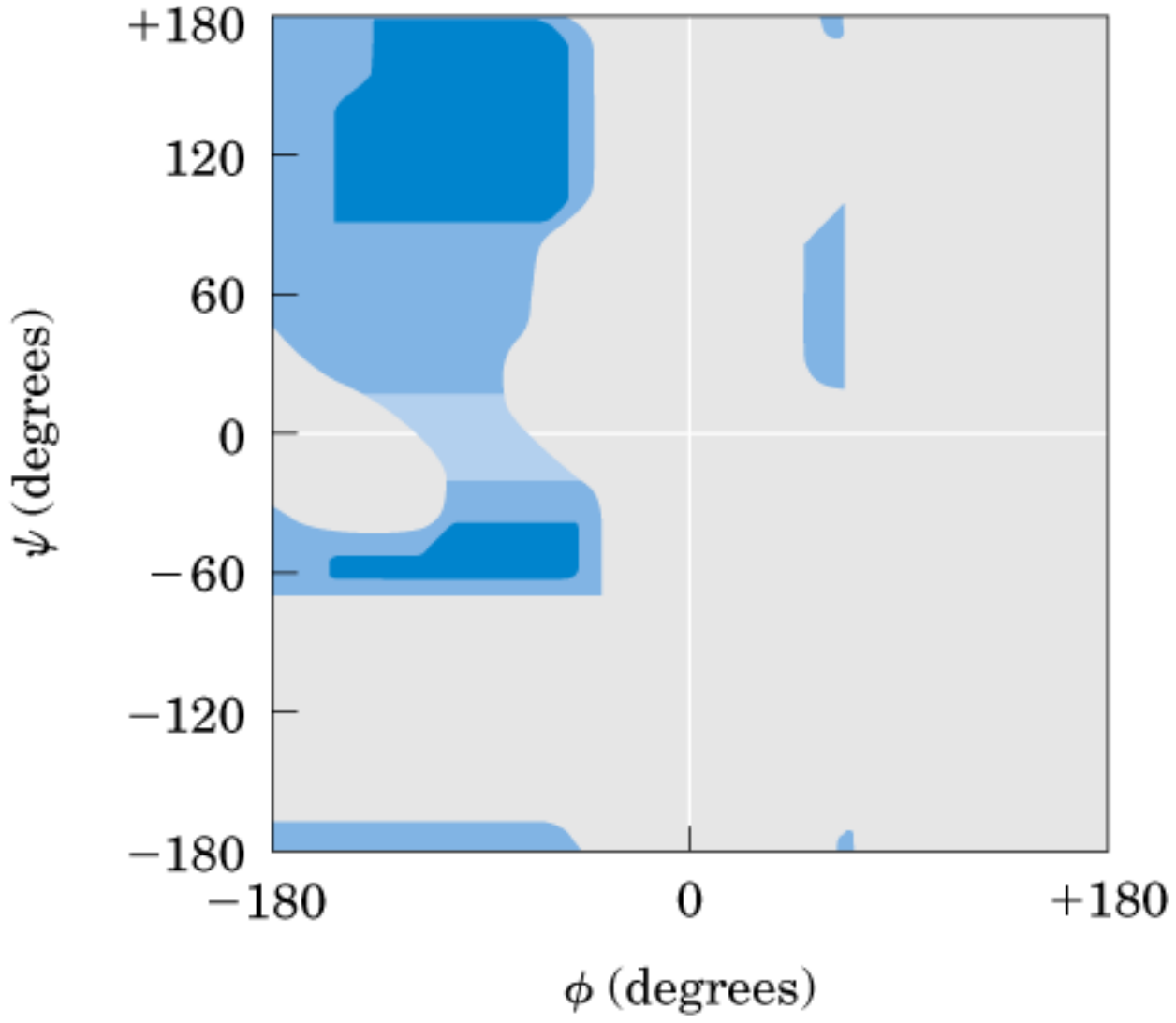
(c)

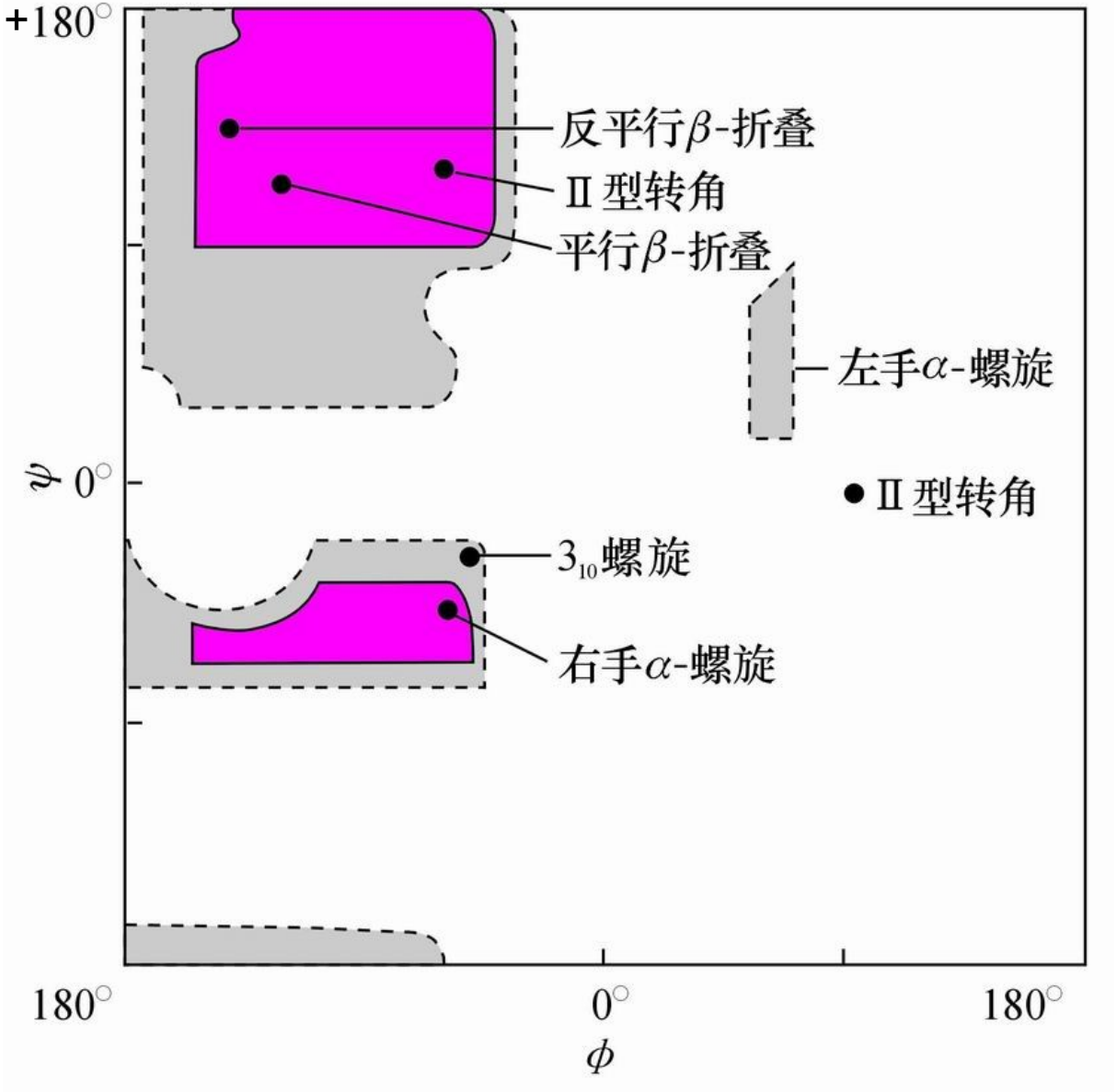


(c)



(d)


+180
120
60
0
−60
−120
−180
ψ (degrees)
−180
0
+180
ϕ (degrees)


+180°
反平行β-折叠
Ⅱ型转角
平行β-折叠
左手α-螺旋
ψ 0°
● Ⅱ型转角
3₁₀螺旋
右手α-螺旋
180°
0°
180°
φ

-

+

2. Protein Secondary Structure

Secondary structure refers to the regular folding of regions of the polypeptide chain or local conformation of some part of the polypeptide.

2.1 Single bonds other than the peptide bond in the backbone chain are free to rotate.

α-helix

β-pleated sheet

β -turn

2.2 The simplest arrangement of the polypeptide chain was proposed to be a helical structure called α-helix (Pauling and Corey, 1954 化学奖)

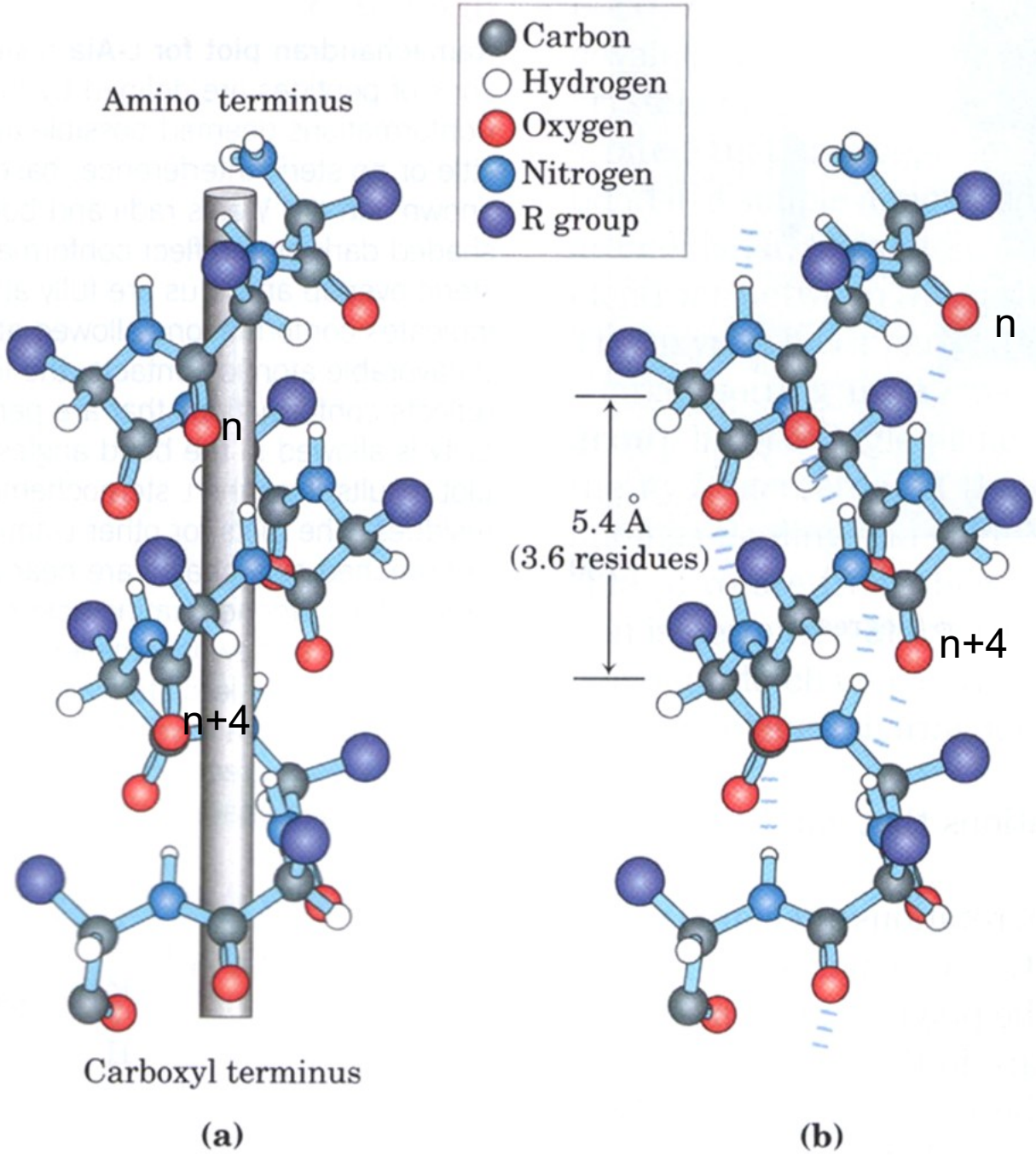
2.2.1 The polypeptide backbone is tightly wound around the long axis (rodlike).

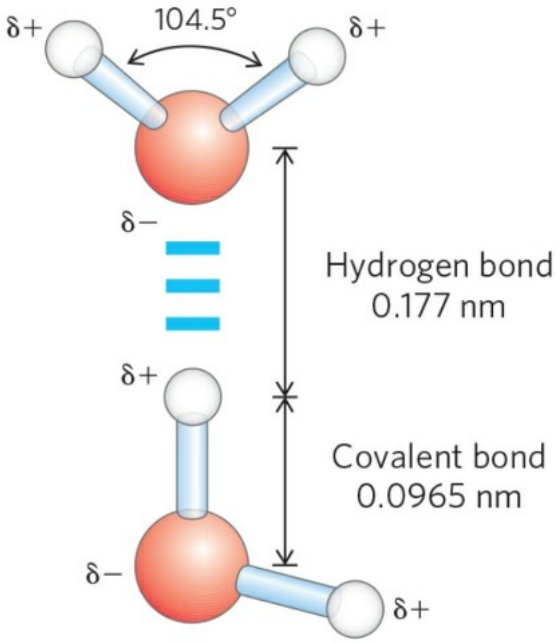
2.2.2 R groups protrude outward from the helical backbone.

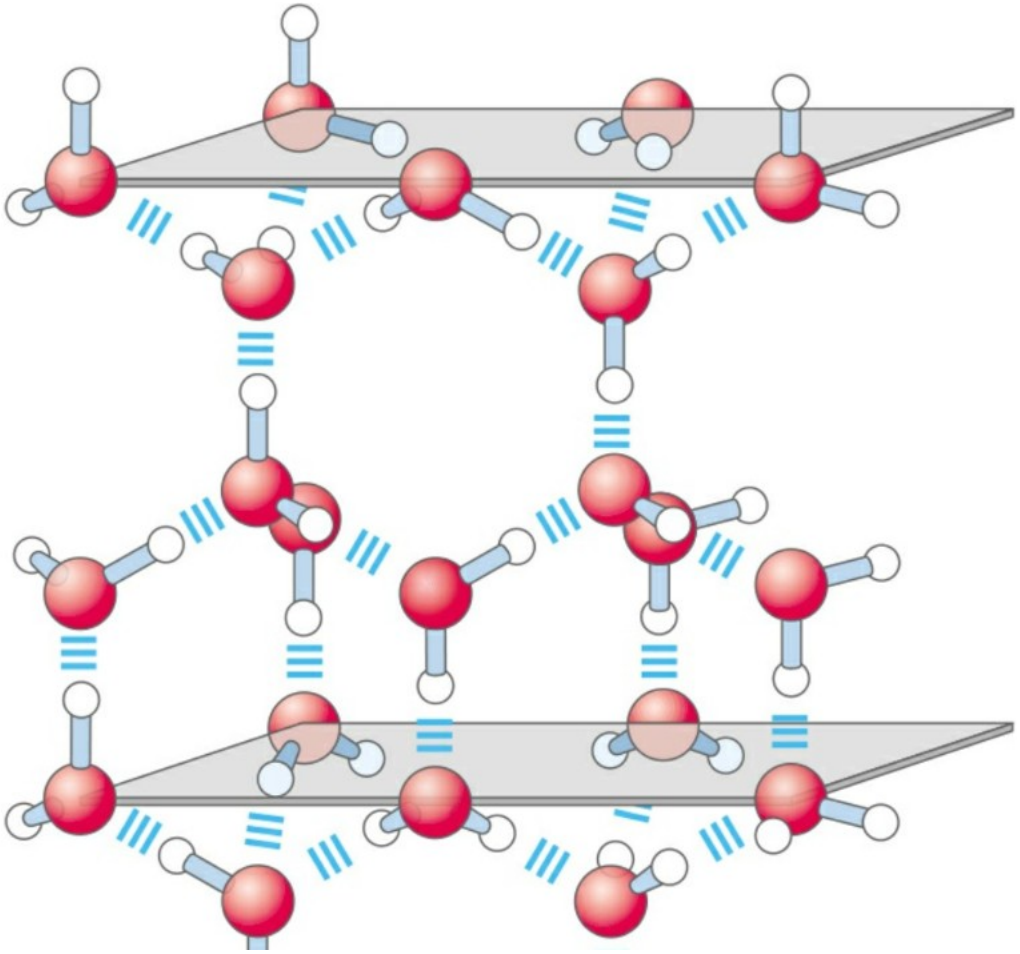
2.2.3 A single turn of the helix (corresponding to the repeating unit in α-keratin) extends about 5.4 Angstroms, including 3.6 residues (each residue arises 1.5 Å and rotate 100 degrees about the helix axis).

2.2.4 The model made optimal use of internal hydrogen bonding for structure stabilization.

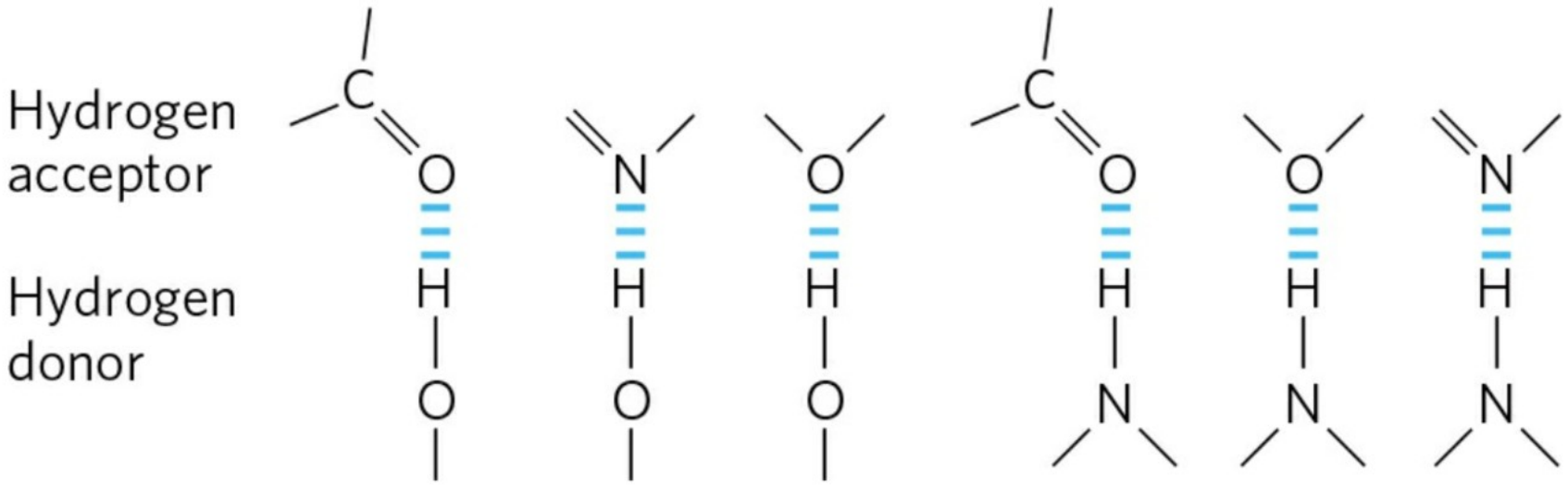
2.2.5 Each carbonyl oxygen (n) of the residue is hydrogen bonded to the hydrogen on the amino group of (n+4) amino acid away.


Carbon
Hydrogen
Oxygen
Nitrogen
R group
Amino terminus
n
n+4
Carboxyl terminus
(a)
5.4 Å
(3.6 residues)
n
n+4
(b)

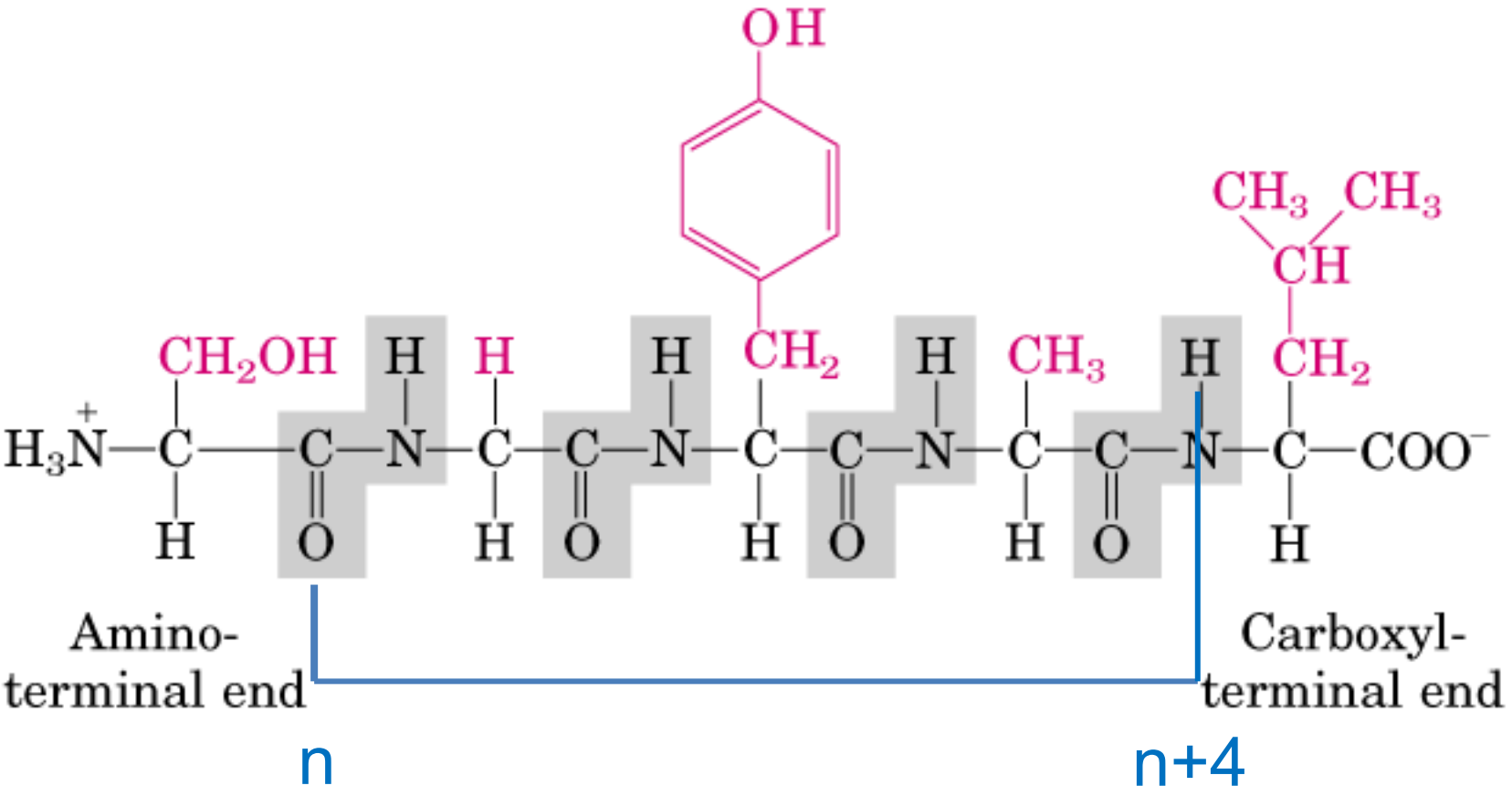


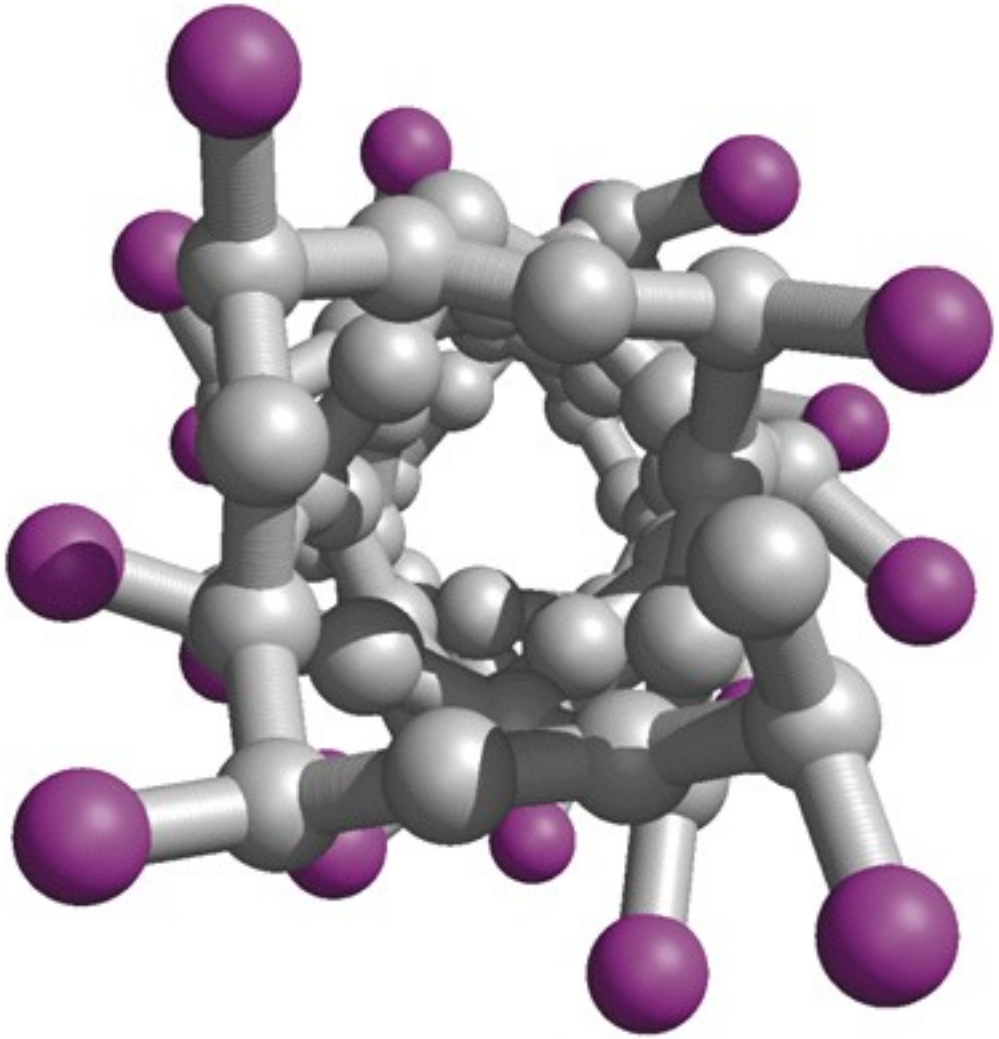


Hydrogen bonding in ice

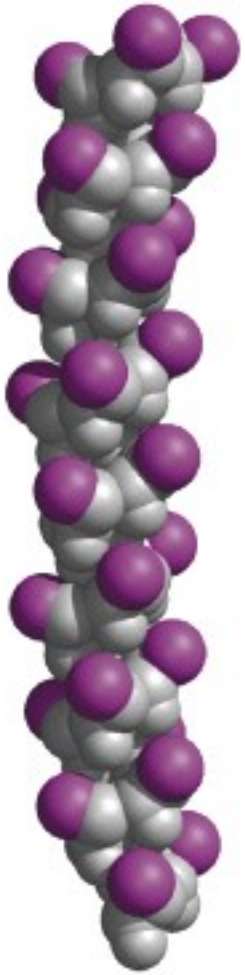


Common hydrogen bonds in biological systems

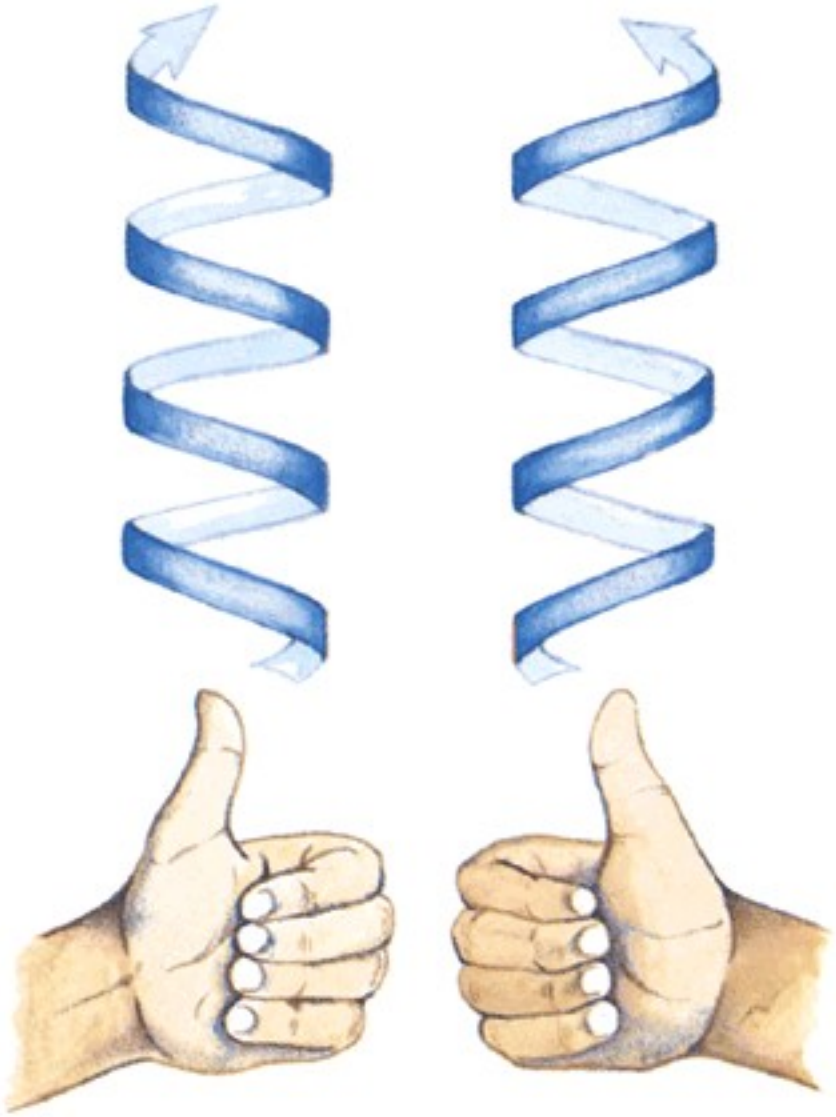

OH
CH3 CH3
CH
CH2OH
H
H
H
CH2
H
CH3
H
CH2
H3N+
C
C
N
C
C
N
C
C
N
C
C
N
C
COO−
H
O
H
O
H
O
H
O
H
Amino-
terminal end
Carboxyl-
terminal end
n
n+4



(c)



(d)



2.3 β-pleated sheet (or β conformation) was proposed to be the more extended conformation of the polypeptide chain.

2.3.1 The conformation is formed when two or more almost fully extended polypeptide chains are brought together side by side.

2.3.2 Regular hydrogen bonds are formed between the carbonyl oxygen and amide hydrogen between adjacent chains (look like a zipper).

2.3.3 The axial distance between the adjacent amino acid residues is ~3.5 Angstroms.

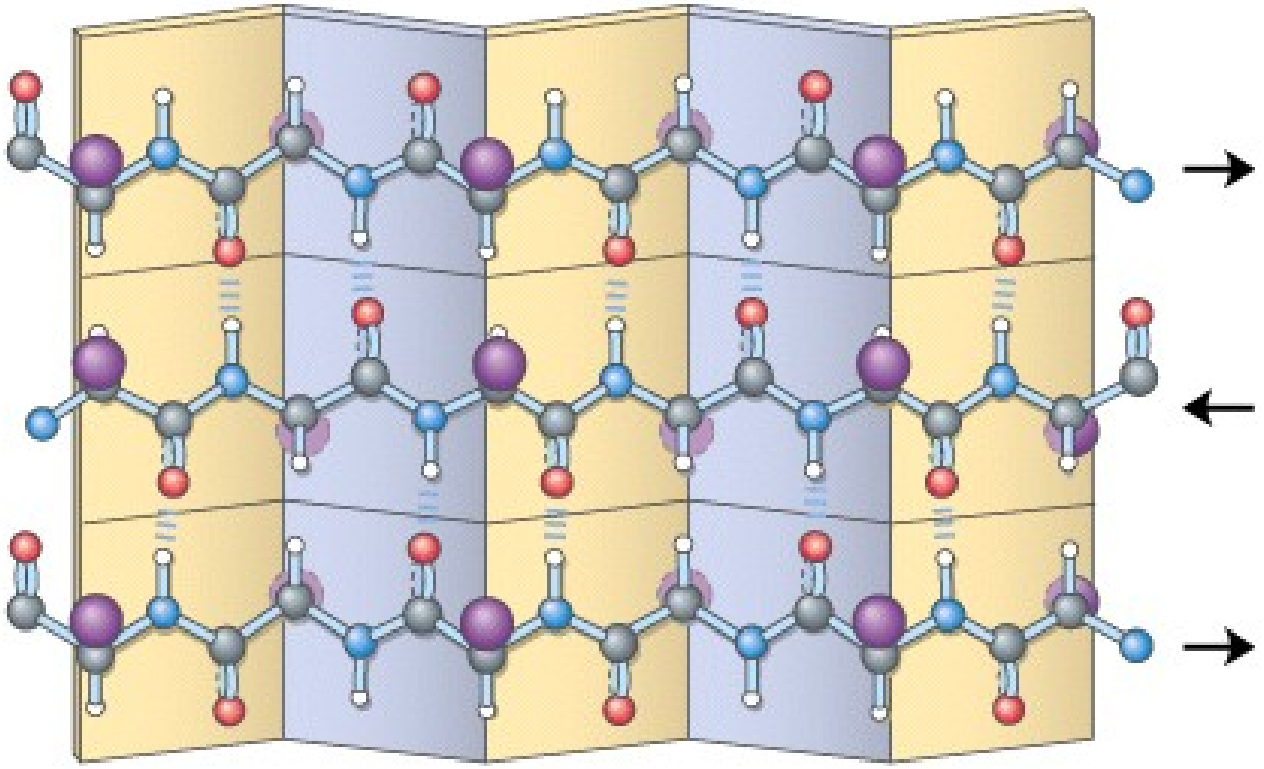
2.3.4 The planes of the peptide bonds arrange as pleated sheets.

2.3.5 The R groups of adjacent residues protrude in opposite directions.

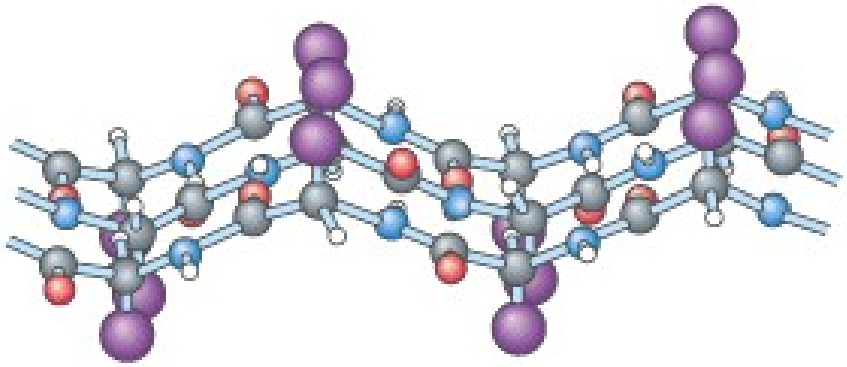
2.3.6 The adjacent polypeptide chains can be either parallel (the same direction) or antiparallel (the opposite direction).

(a) Antiparallel

Top view

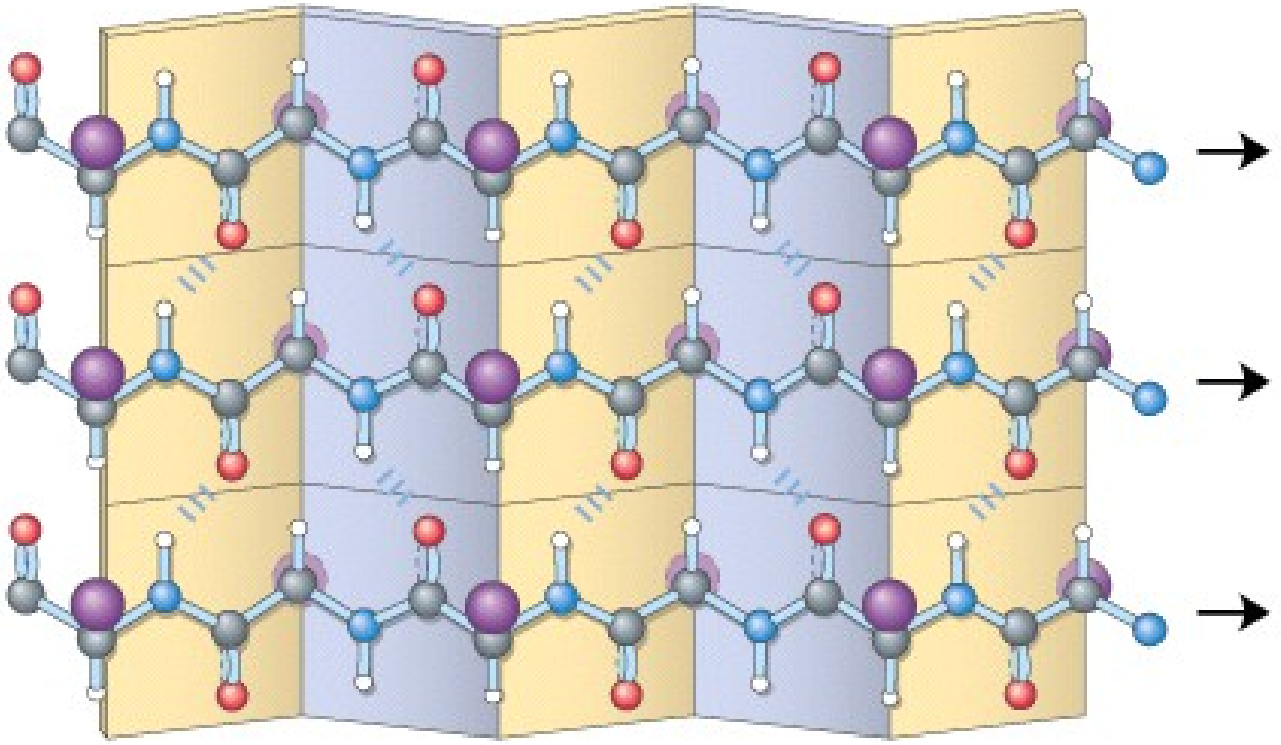


Side view

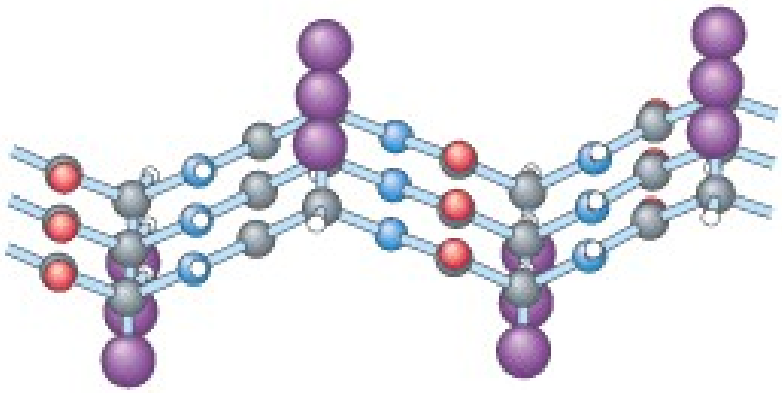


(b) Parallel

Top view

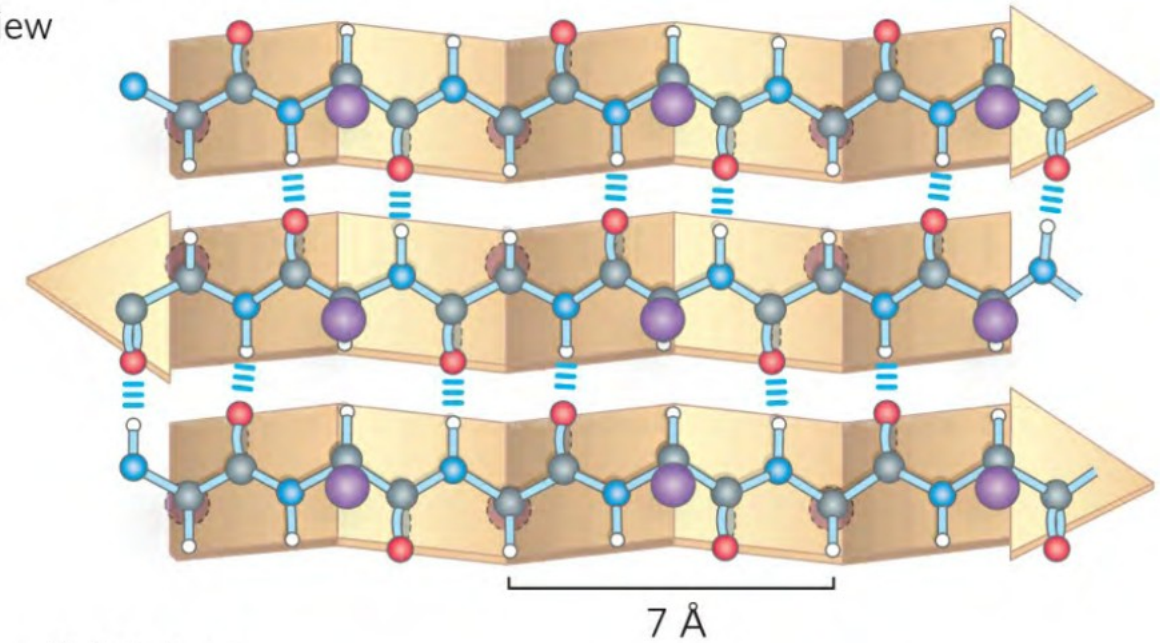


Side view



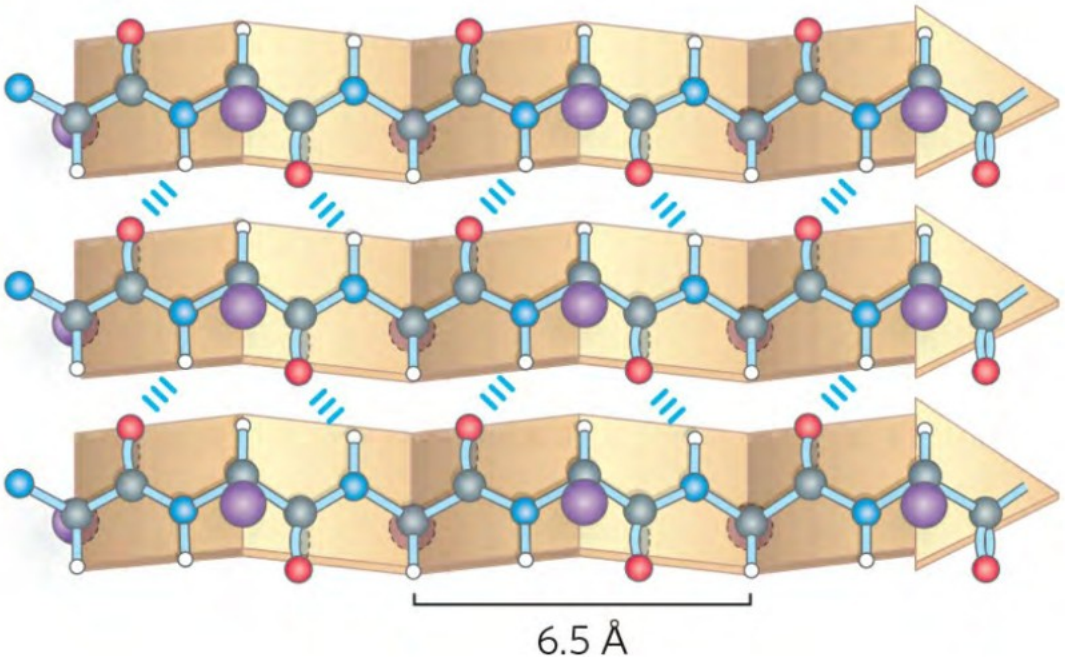
(b) Antiparallel β sheet

Top view

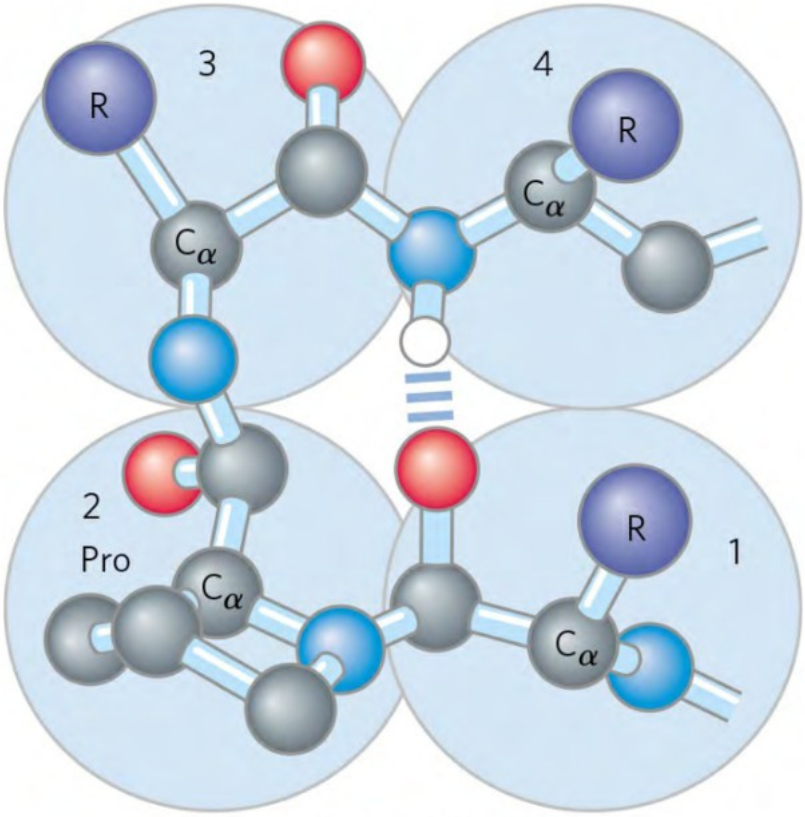


(c) Parallel β sheet

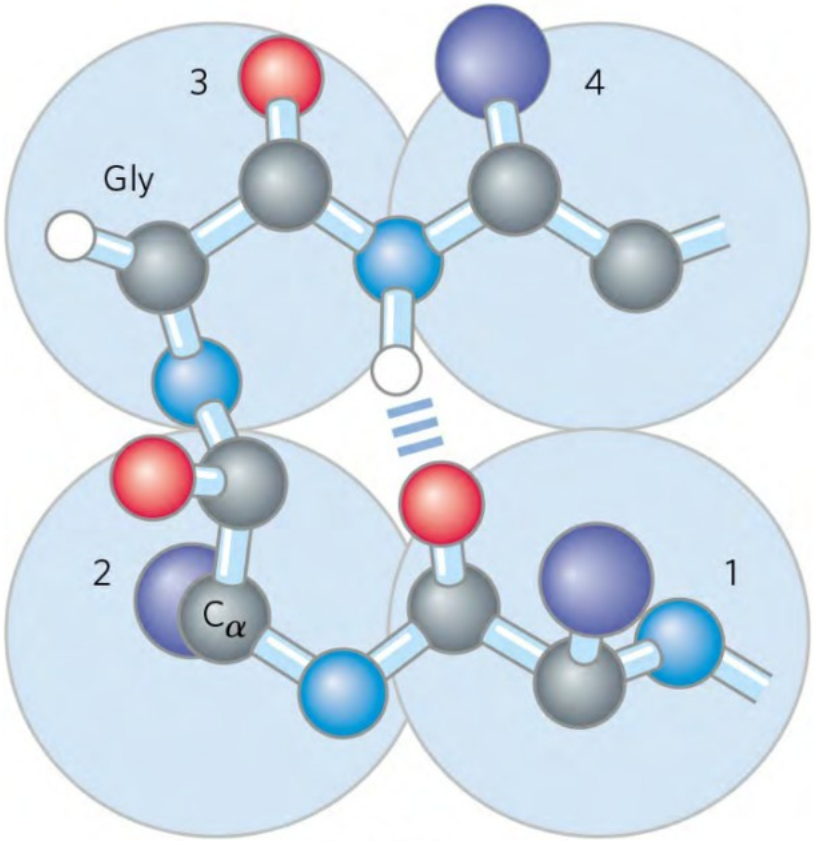
Top view



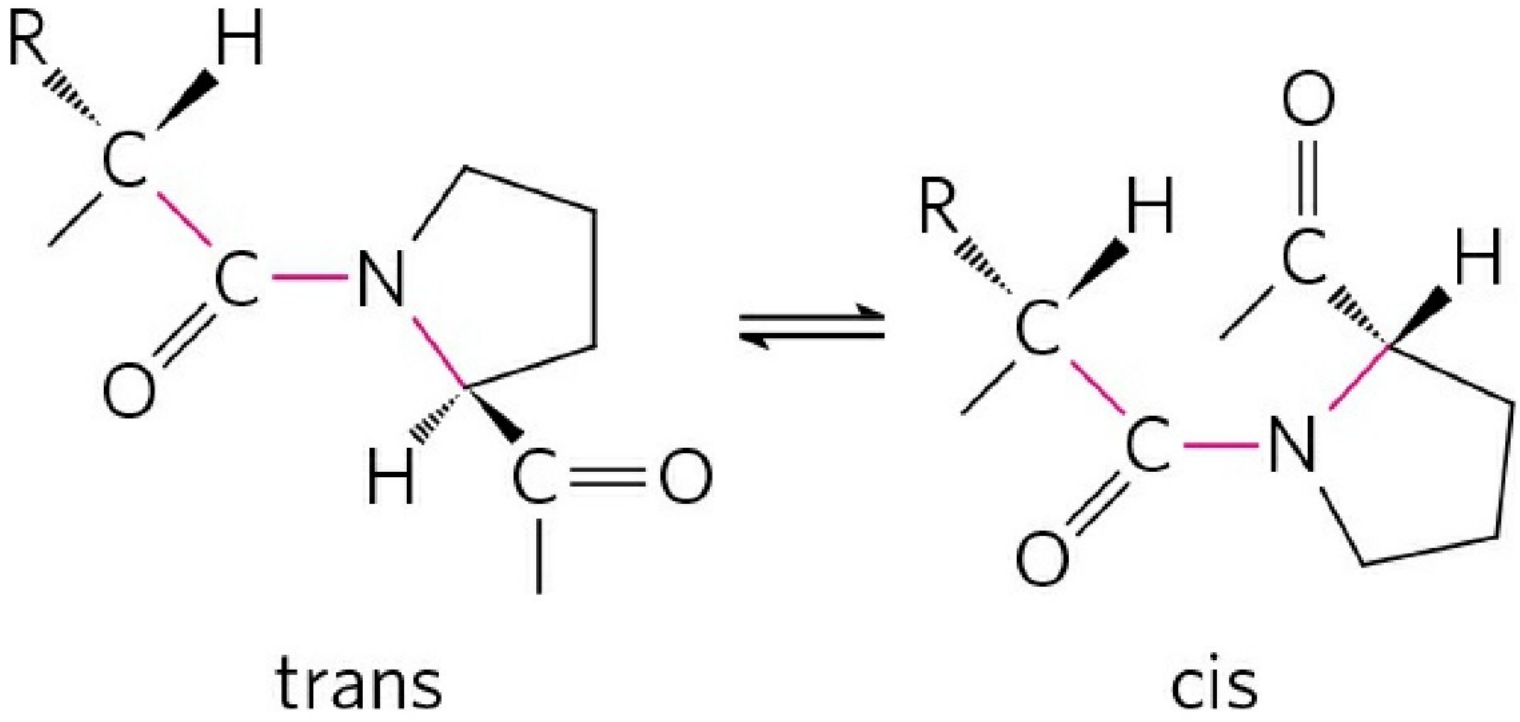
2.4 β turn (β -bend or hairpin 发卡 turn) is also a common secondary structure found where a polypeptide chain abruptly reverses its direction.



Type I β turn



Type II β turn



proline

2.4.1 It often connects the ends of two adjacent segments of an antiparallel β-pleated sheet.

2.4.2 It is a tight turn of ~180 degrees involving four amino acid residues.

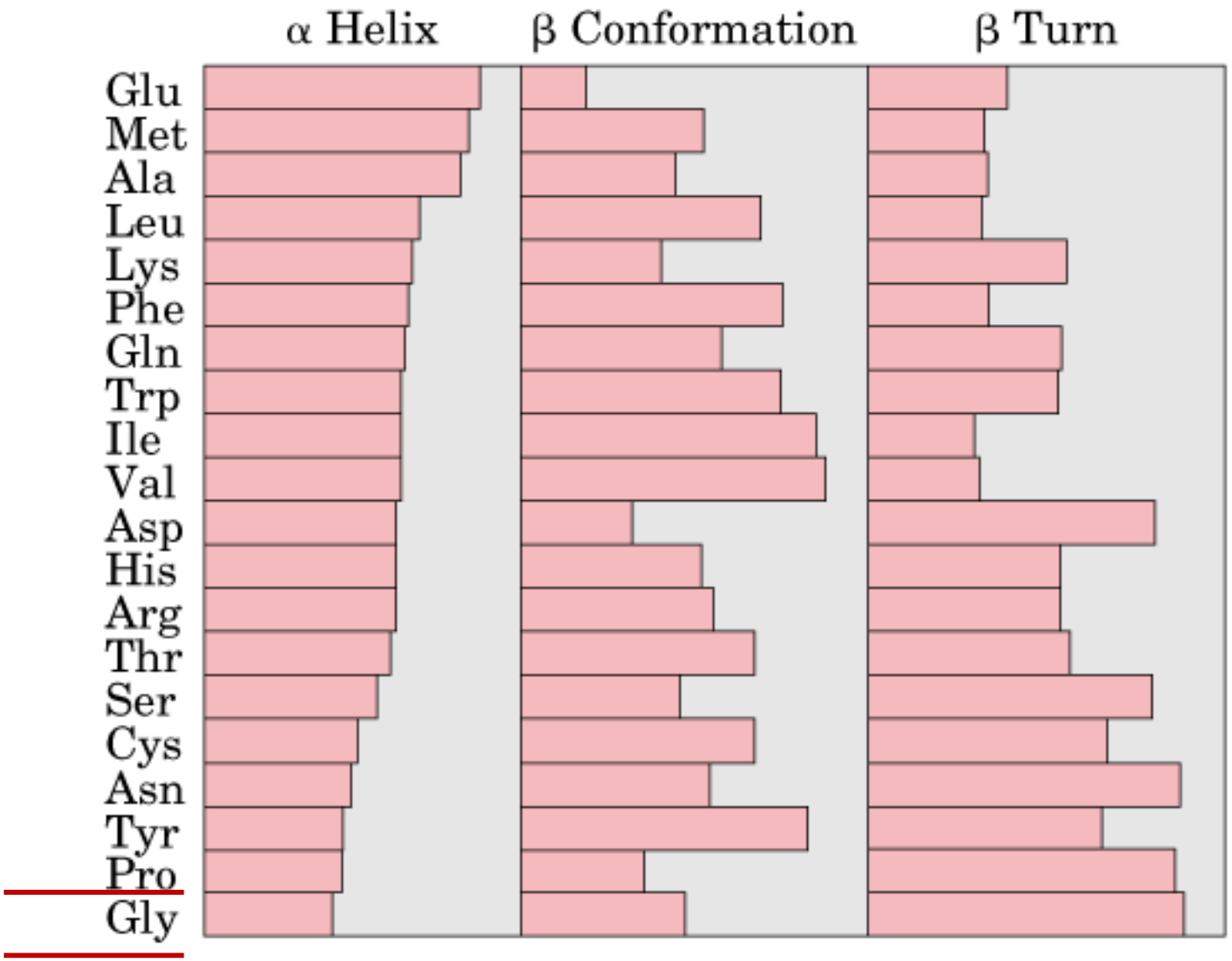
2.4.3 The essence of the structure is the hydrogen bonding between the C=O group of residue n and the NH group of the residue n+3.

2.4.4 Gly and Pro are often found in β turns.

2.4.5 β turns are often found near the surface of a protein.

2.5 Some amino acid residues are accommodated in the different types of secondary structure better than others.

2.6 Regions of polypeptide chain that are not in a regular secondary structure are said to have a coil or loop conformation.



Relative probabilities that a given amino acid will occur in the three common types of secondary structure.

table 6–2

Approximate Amounts of α Helix and β Conformation in Some Single-Chain Proteins*

Protein (total residues)	Residues (%)	
	α Helix	β Conformation
Chymotrypsin (247)	14	45
Ribonuclease (124)	26	35
Carboxypeptidase (307)	38	17
Cytochrome *c* (104)	39	0
Lysozyme (129)	40	12
Myoglobin (153)	78	0

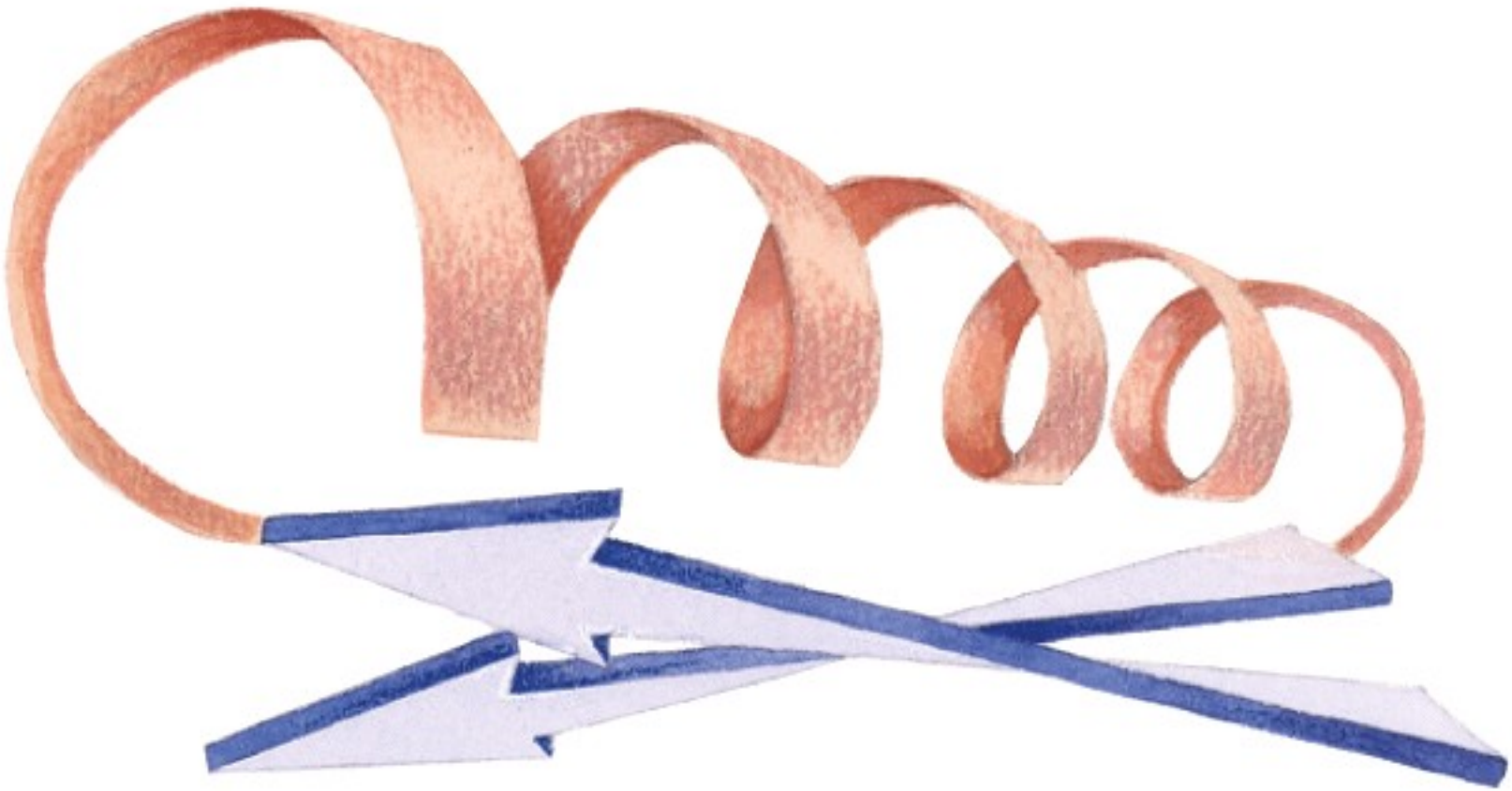
Source: Data from Cantor, C.R. & Schimmel, P.R. (1980) *Biophysical Chemistry,* Part I: *The Conformation of Biological Macromolecules,* p. 100, W.H. Freeman and Company, New York.

*Portions of the polypeptide chains that are not accounted for by α helix or β conformation consist of bends and irregularly coiled or extended stretches. Segments of α helix and β conformation sometimes deviate slightly from their normal dimensions and geometry.

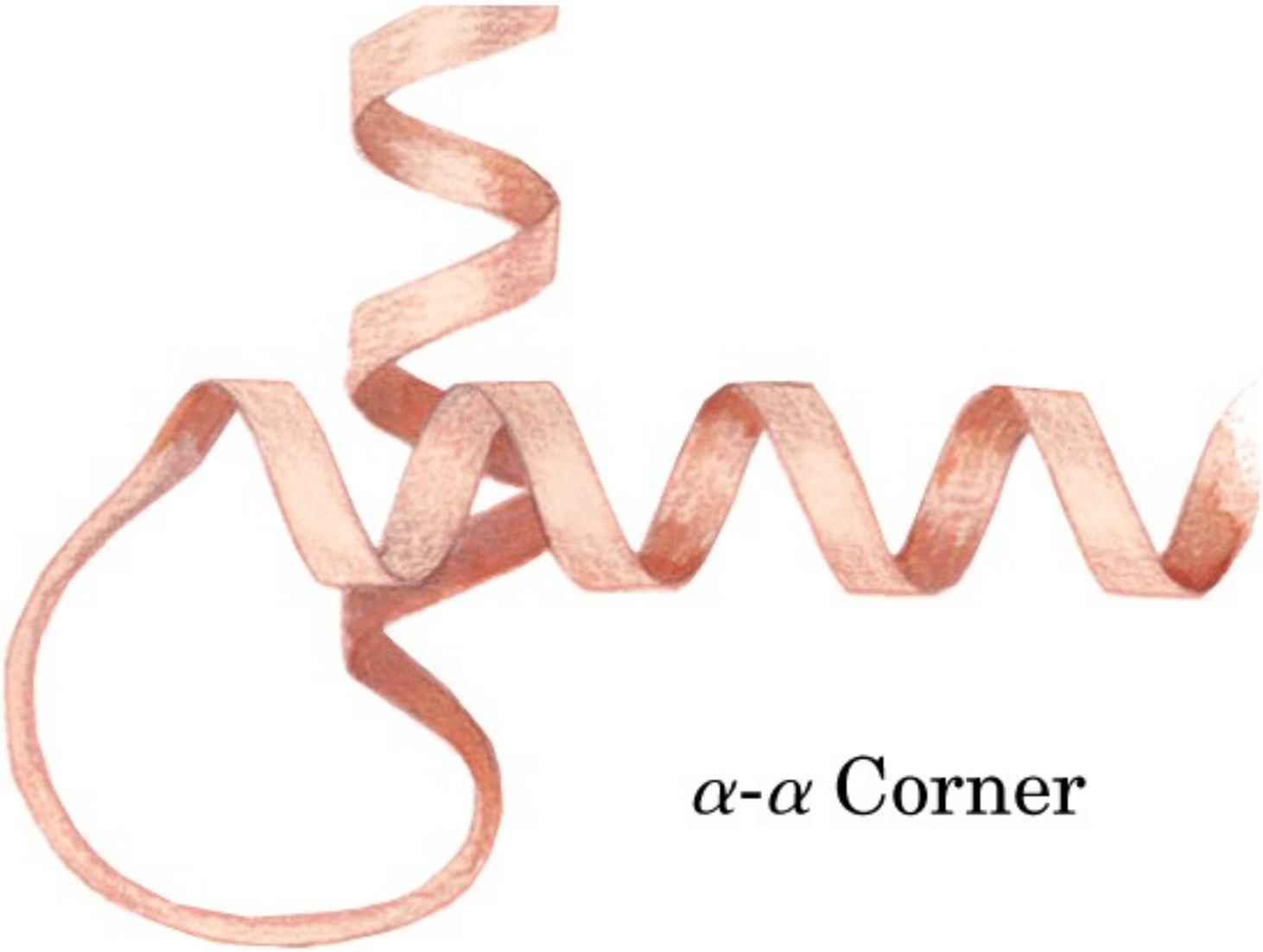
2.7 Supersecondary structures and domains

2.7.1 The supersecondary structures, also called motifs or simply folds, refers to clusters of secondary structures that repeatedly appear.

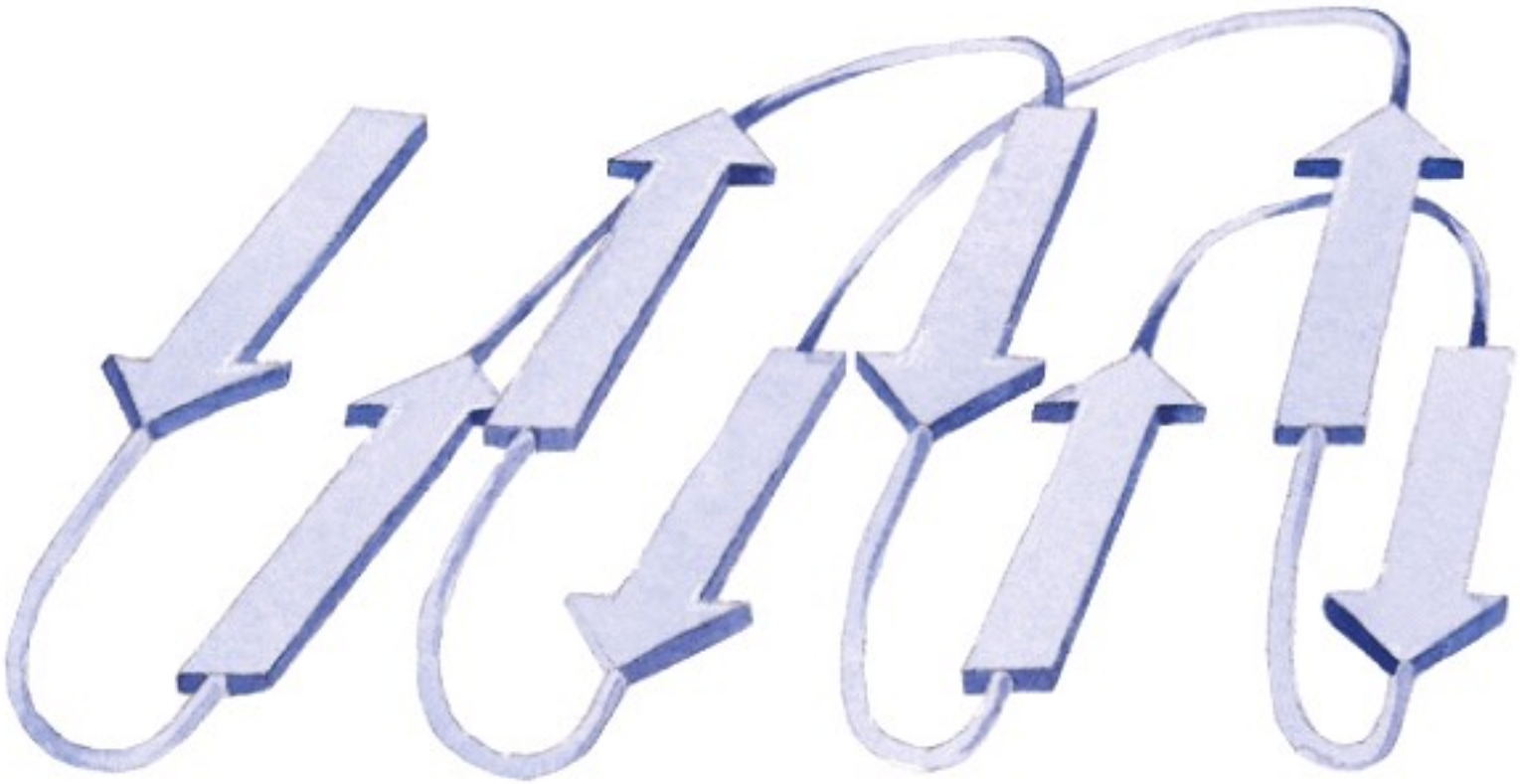
2.7.2 A compact region (usually including less than 200~400 residues) that is a distinct structural unit within a larger polypeptide chain is called a domain.



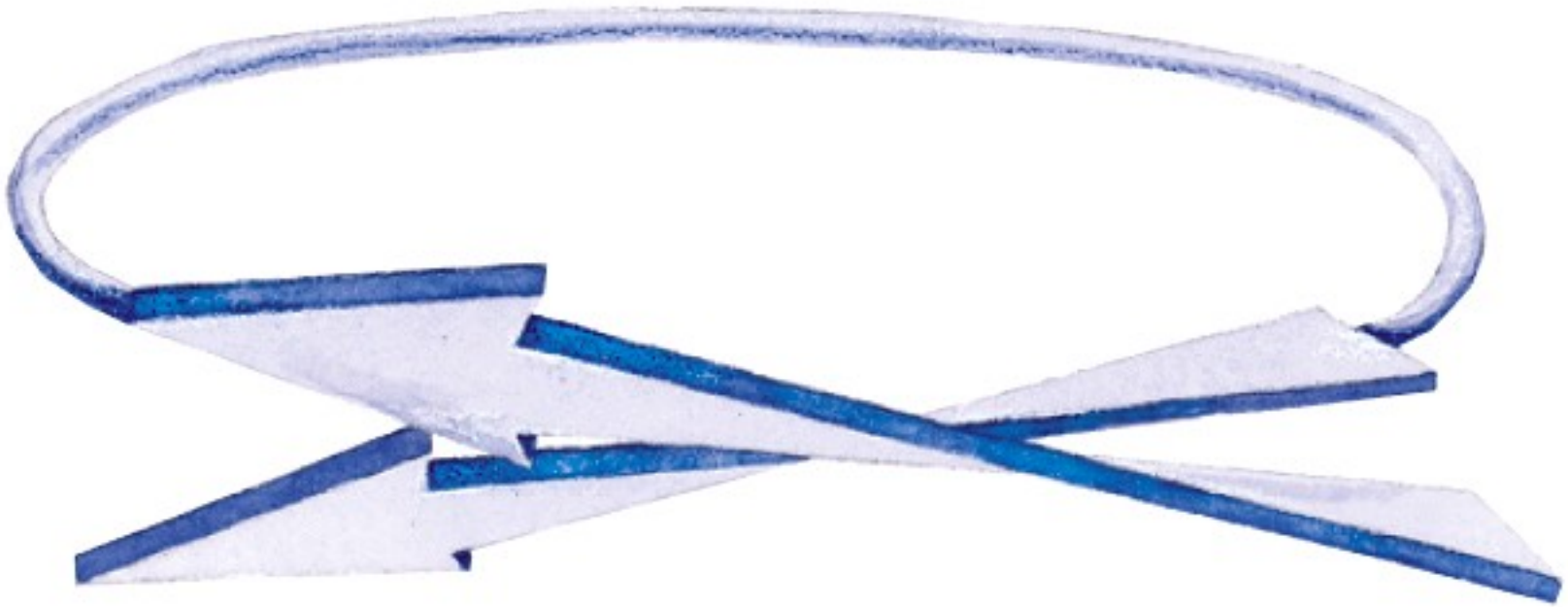
(a) β-α-β Loop



α-α Corner



(b) Typical connections in an all-β motif

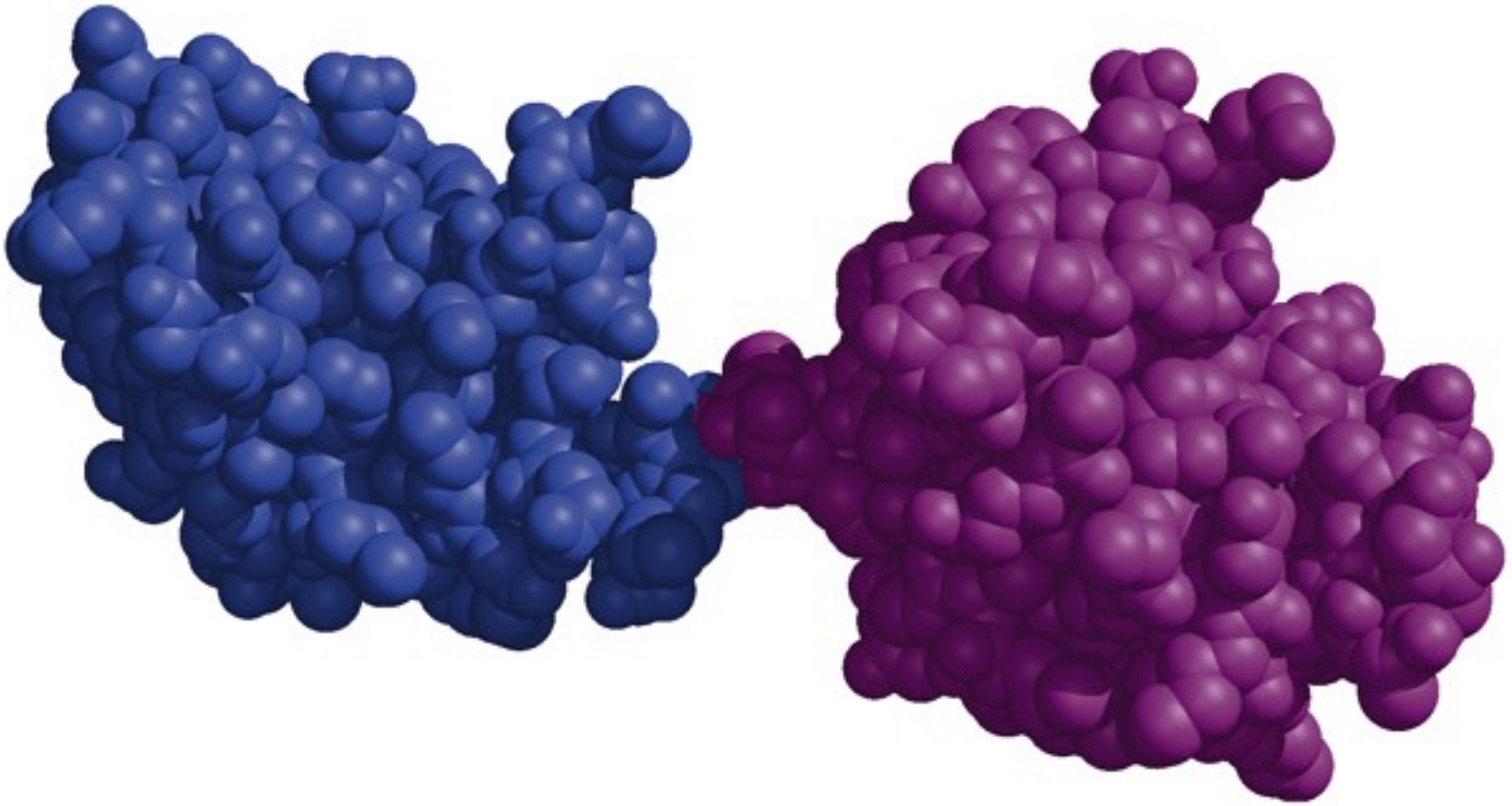


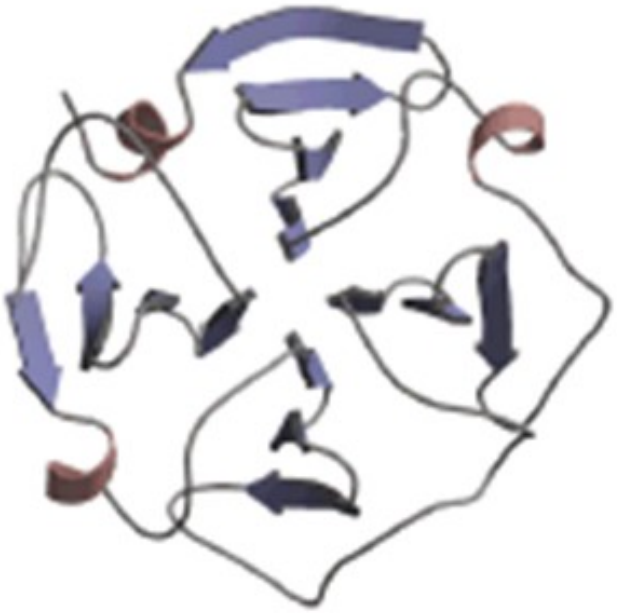
(c) Right-handed connection between β strands



Left-handed connection
between β strands
(very rare)

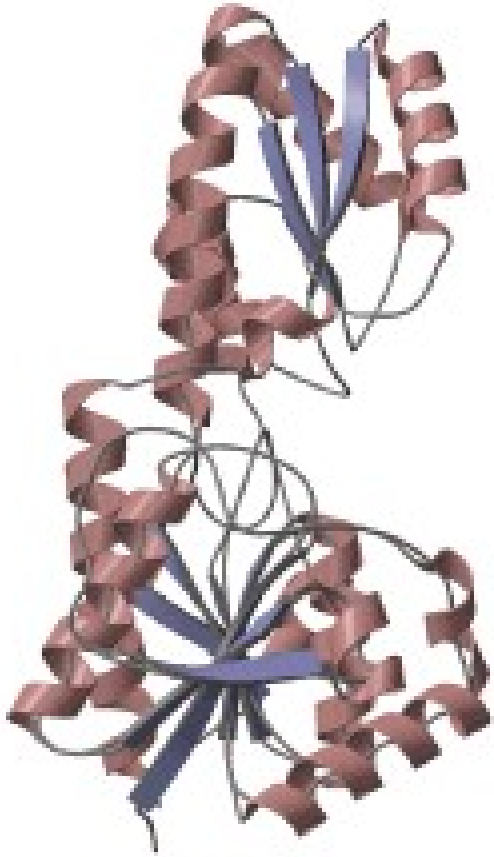
Structural domains in the polypeptide troponin (肌钙蛋白) C, two separate calcium-binding domains





1pex
Four-bladed β propeller
Hemopexin-like domain
Hemopexin-like domain
Collagenase-3 (MMP-13), carboxyl-terminal domain
Human (*Homo sapiens*)

motif



1pfk
Phosphofructokinase
Phosphofructokinase
Phosphofructokinase
Phosphofructokinase
Escherichia coli

domain

3. Protein Tertiary Structure---

the three-dimentional arrangement of the amino acid in the polypeptide chain.

All α



1a06
Serum albumin
Serum albumin
Serum albumin
Serum albumin
Human (*Homo sapiens*)



1bcf
Ferritin-like
Ferritin-like
Ferritin
Bacterioferritin (cytochrome b_1)
Escherichia coli



1gai
α/α toroid
Glycosyltransferases of the superhelical fold
Glucoamylase
Glucoamylase
Aspergillus awamori, variant x100



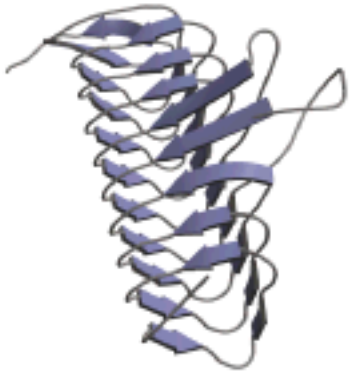
1enh
DNA-binding 3-helical bundle
Homeodomain-like
Homeodomain
engrailed Homeodomain
Drosophila melanogaster

Key
PDB Identifier
Fold
Superfamily
Family
Protein
Species

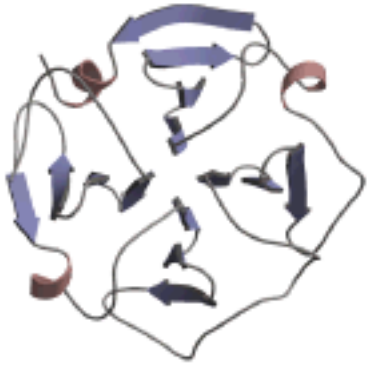
All β



1hoe
α-Amylase inhibitor
α-Amylase inhibitor
α-Amylase inhibitor
HOE-467A
Streptomyces tendae 4158



1lxe
Single-stranded left-handed β helix
Trimeric LpxA-like enzymes
UDP *N*-acetylglucosamine acyltransferase
UDP *N*-acetylglucosamine acyltransferase
Escherichia coli



1pex
Four-bladed β propeller
Hemopexin-like domain
Hemopexin-like domain
Collagenase-3 (MMP-13), carboxyl-terminal domain
Human (*Homo sapiens*)

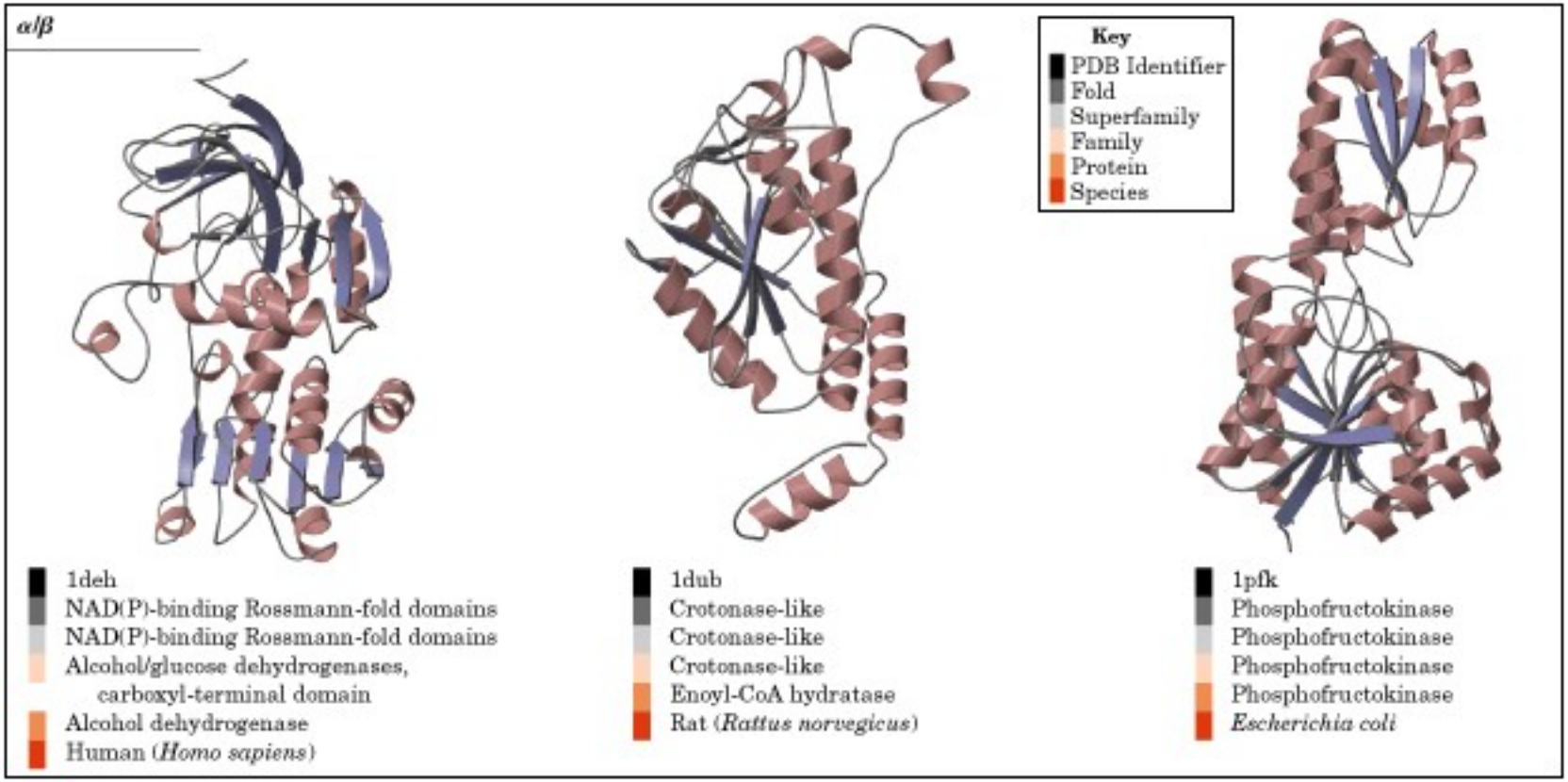


1jpc
β-Prism II
α-D-Mannose-specific plant lectins
α-D-Mannose-specific plant lectins
Lectin (agglutinin)
Snowdrop (*Galanthus nivalis*)

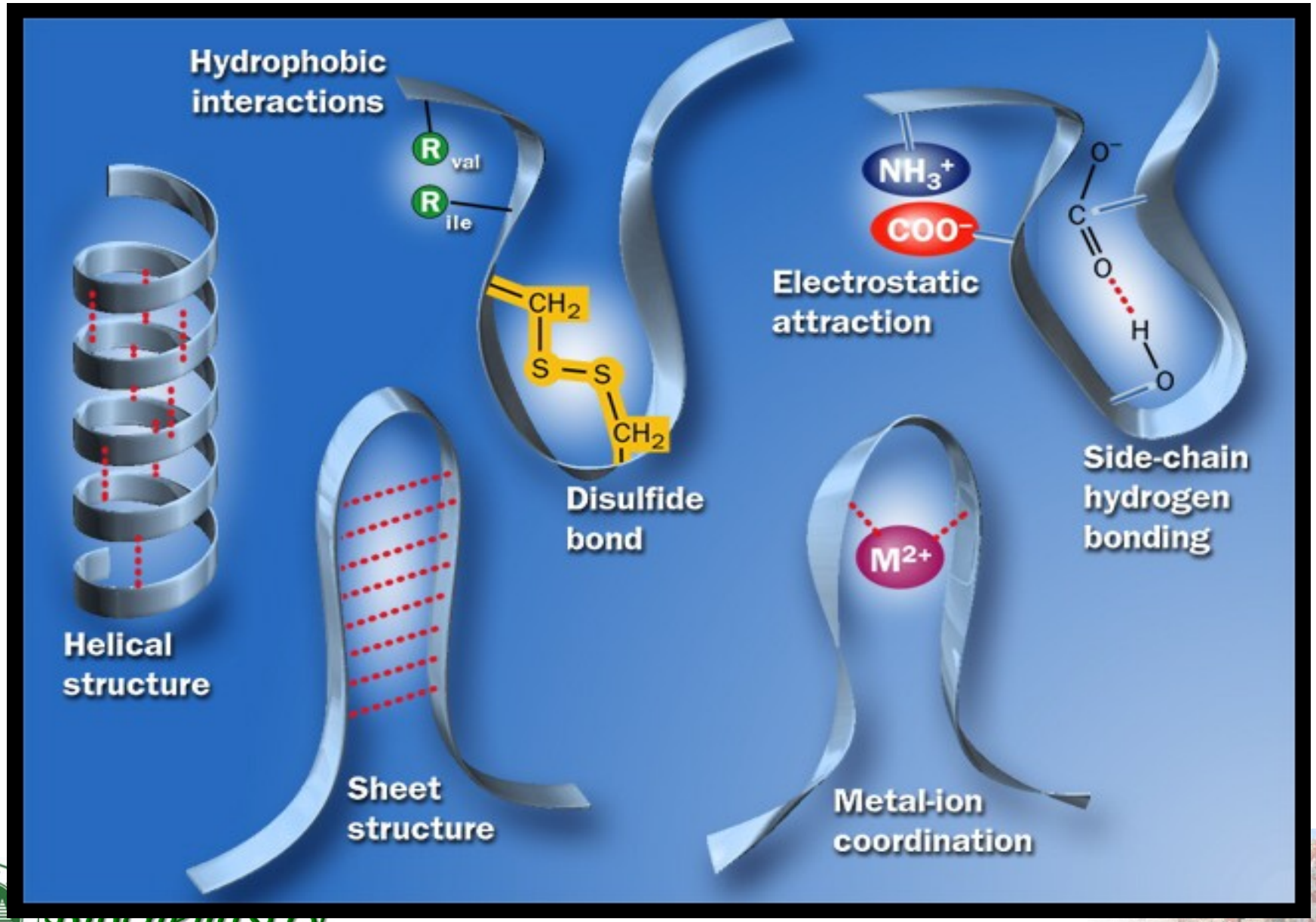


1cd8
Immunoglobulin-like β sandwich
Immunoglobulin
Antibody variable domain-like
CD8
Human (*Homo sapiens*)

Key
PDB Identifier
Fold
Superfamily
Family
Protein
Species

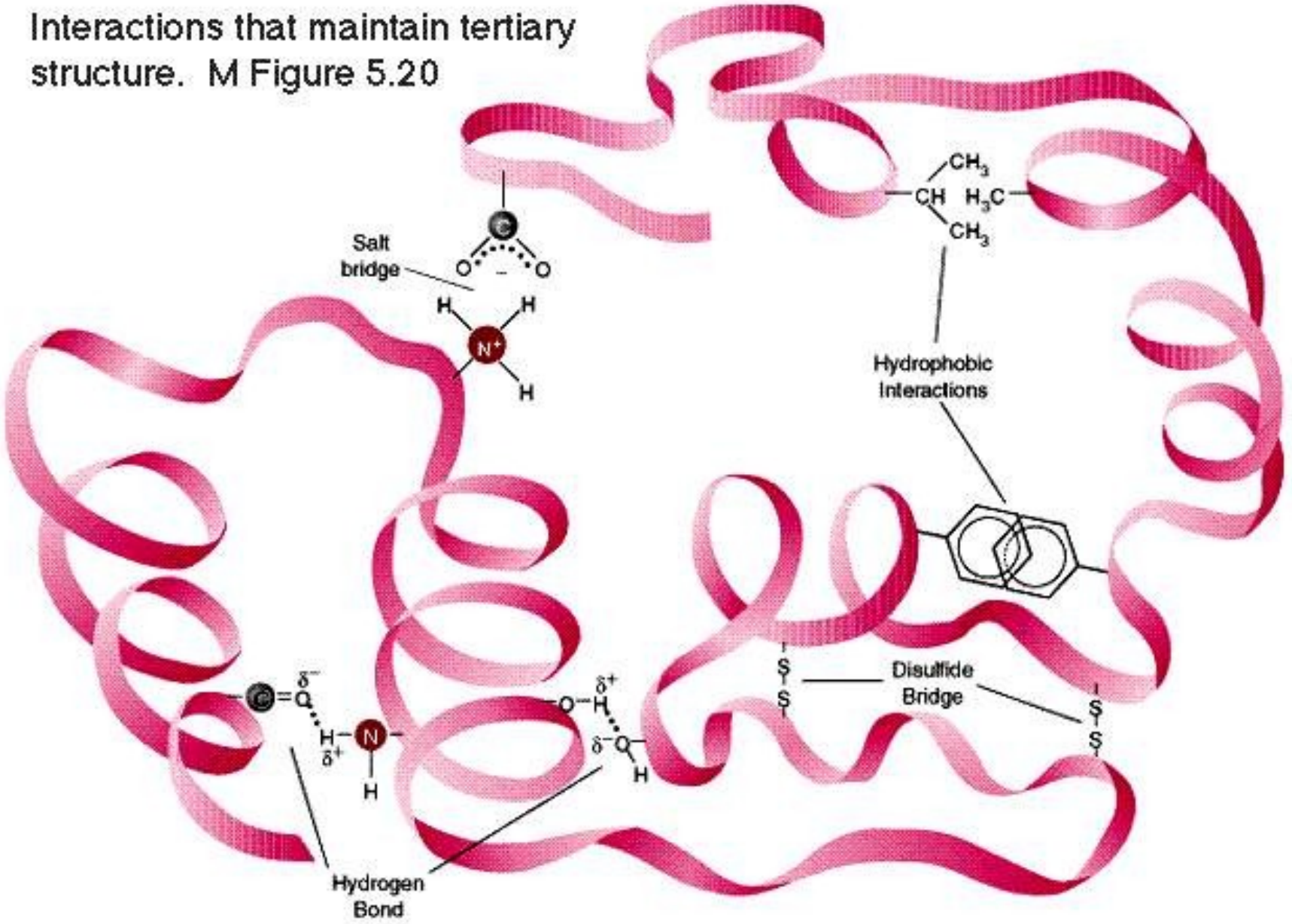

α/β
Key
PDB Identifier
Fold
Superfamily
Family
Protein
Species
1deh
NAD(P)-binding Rossmann-fold domains
NAD(P)-binding Rossmann-fold domains
Alcohol/glucose dehydrogenases, carboxyl-terminal domain
Alcohol dehydrogenase
Human (Homo sapiens)
1dub
Crotonase-like
Crotonase-like
Crotonase-like
Enoyl-CoA hydratase
Rat (Rattus norvegicus)
1pfk
Phosphofructokinase
Phosphofructokinase
Phosphofructokinase
Phosphofructokinase
Escherichia coli

Interactions that maintain protein structure



Biochemistry by McKee/McKee, ©1996 McGraw-Hill Higher Education

Interactions that maintain tertiary structure. M Figure 5.20



The polypeptide chain folds spontaneously so that the majority of its hydrophobic side-chains are buried in the interior, and the majority of its polar, charged side-chain are on the surface.

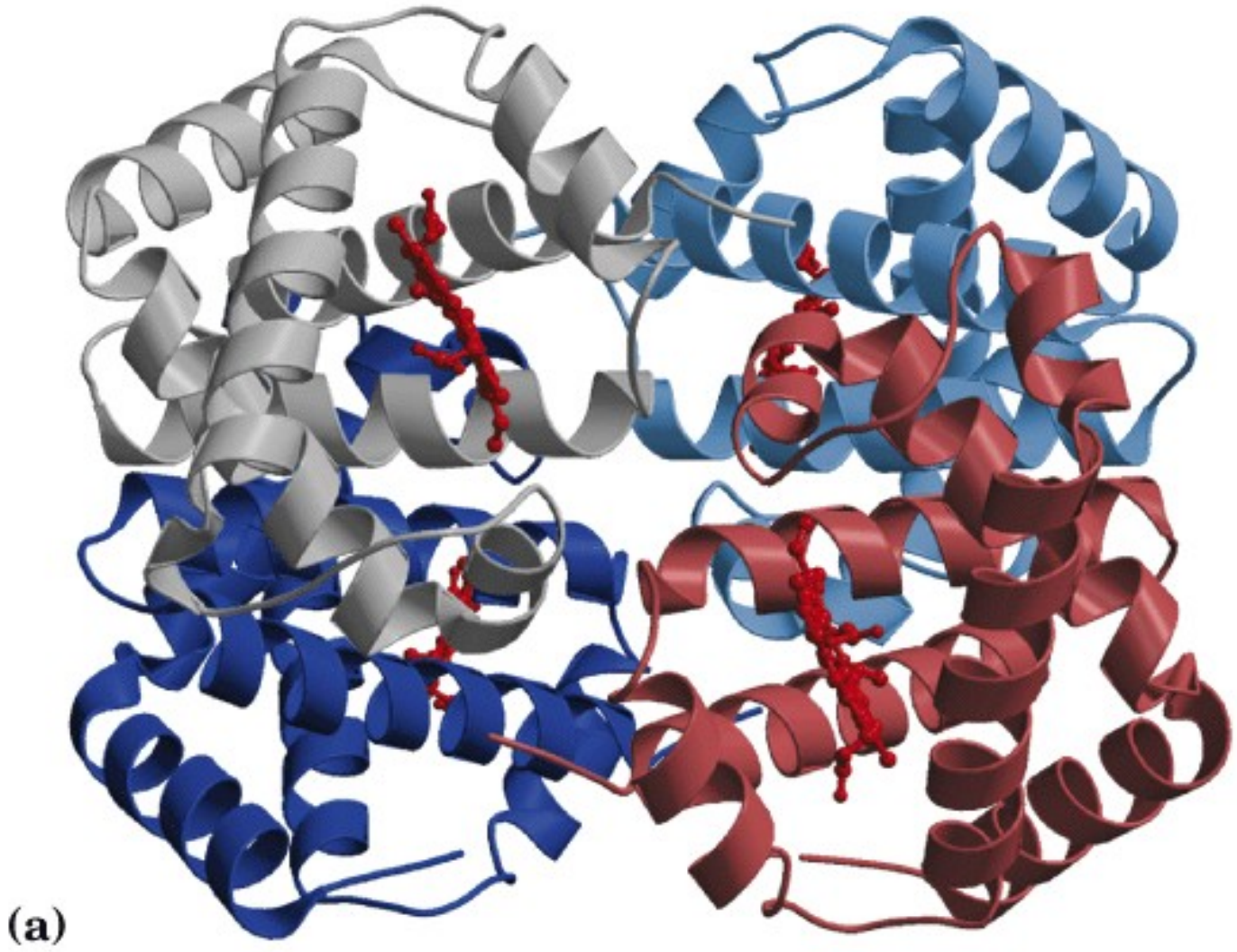
The native conformation is maintained not only by hydrophobic interactions, but also by salt bridges, multiple weak van der Waals interactions, hydrogen bonding and the covalent disulfide bonds.

3.1 Under physiological conditions of solvent and temperature, each protein folds spontaneously into one three-dimensional conformation, called the native conformation.

3.2 This conformation is usually the most stable, and predominate among the innumerable theoretically possible ones.

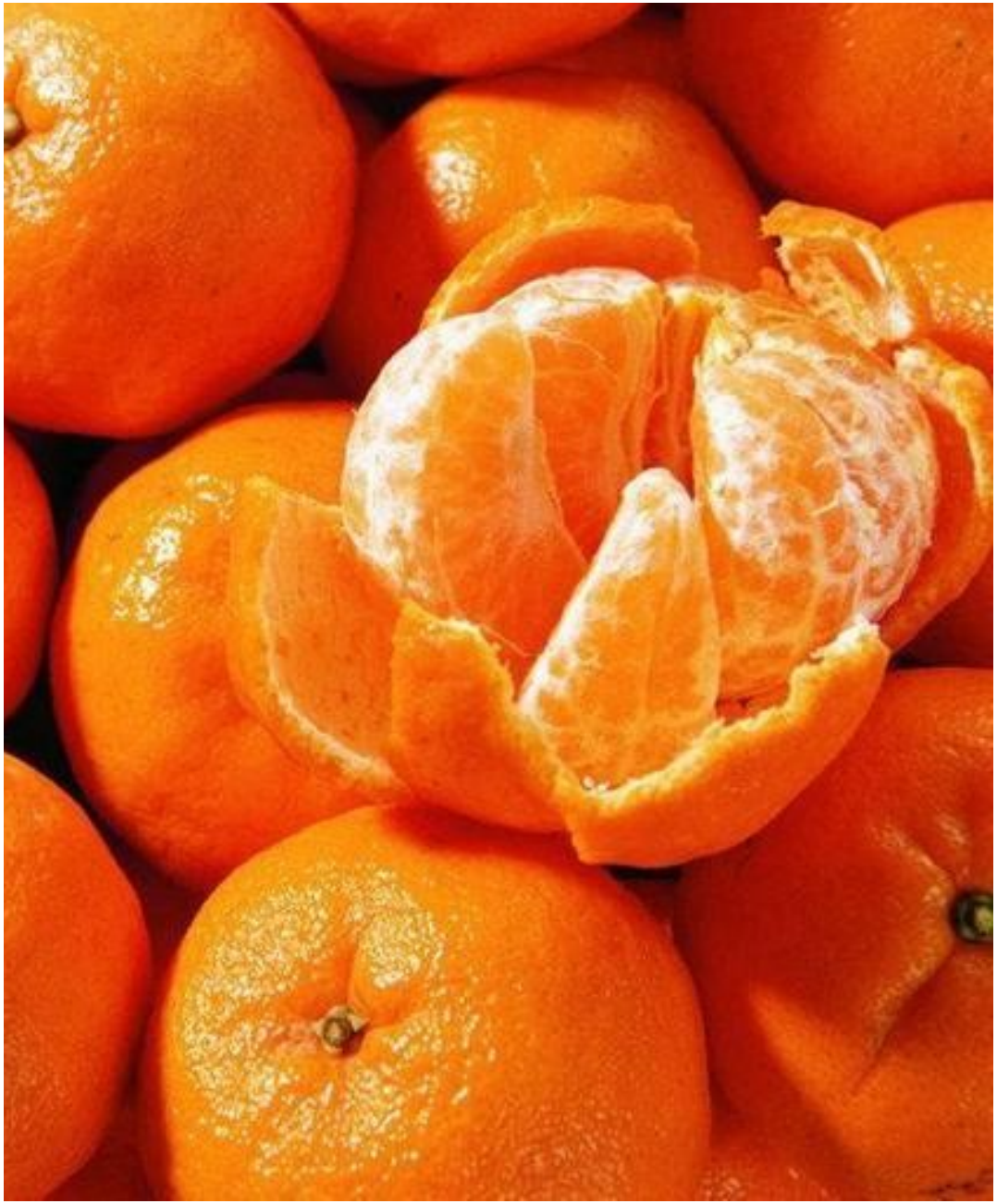
3.3 Usually only the native conformation is functional.

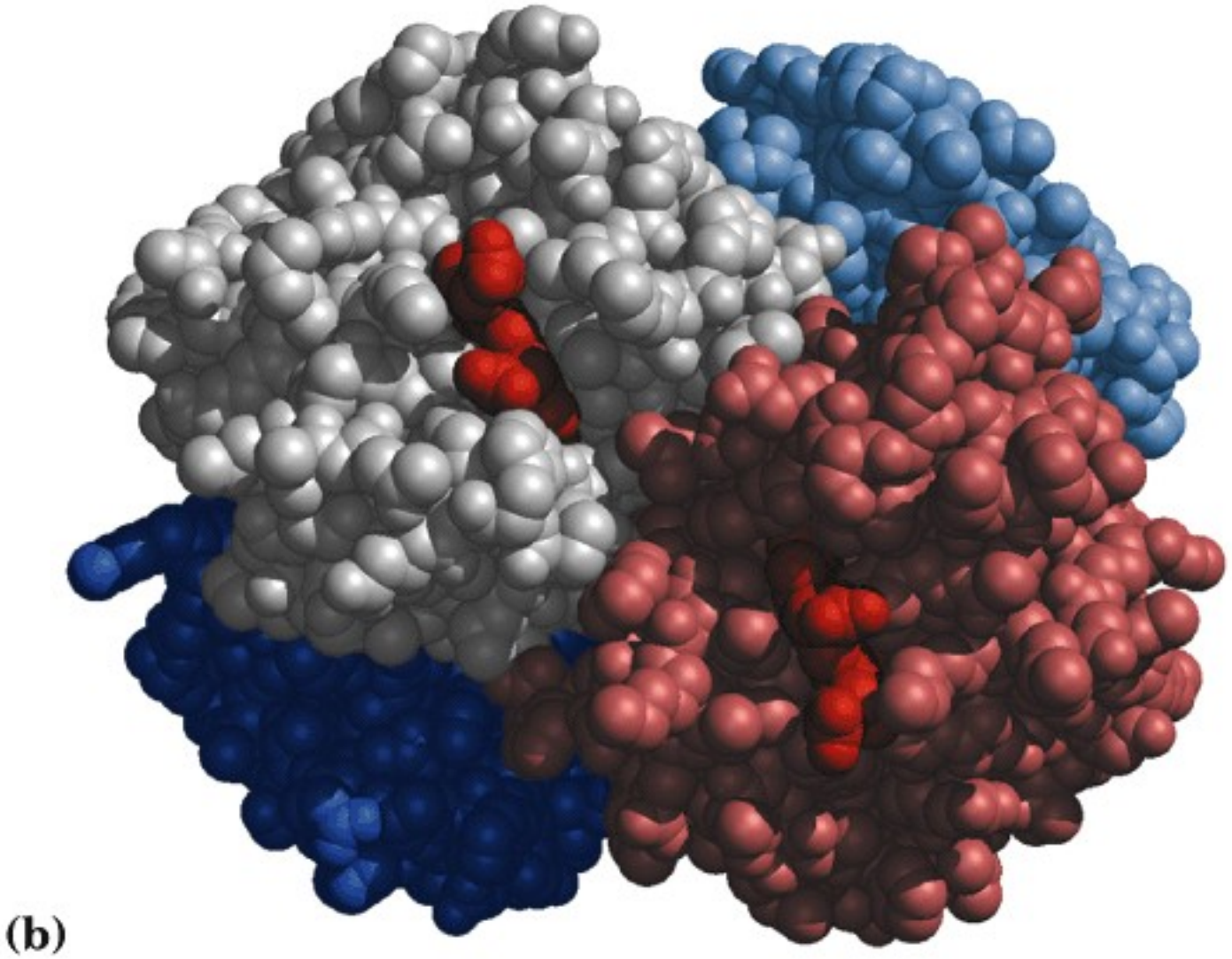
4. Quaternary structure: macromolecular assembly



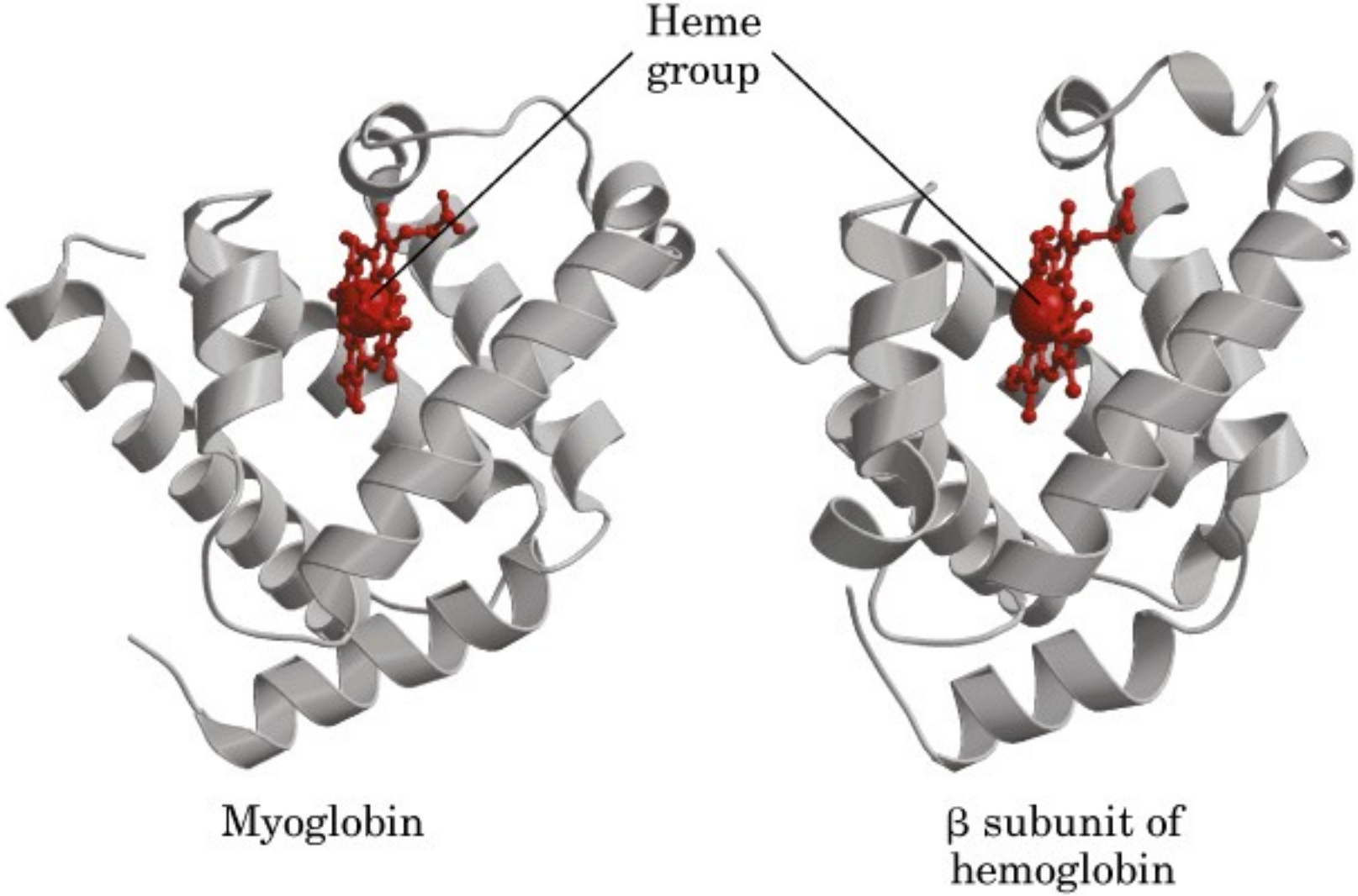
hemoglobin


昵图网 www.nipic.com
BY: maomaodou
NO:20120130210022515000





Hemoglobin


Heme
group
Myoglobin
β subunit of
hemoglobin

If a protein is made up of more than one polypeptide chain it is said to have quaternary structure. This refers to the spatial arrangement of the polypeptide subunits and the nature of the interactions between them.

What's the relationship between protein structure and function?

5. Proteins denature under stress

5.1 Loss of tertiary structure (native conformation) is accompanied by loss of function.

5.1.1 Proteins are relatively easy to lose their tertiary structures due to their marginal stability maintained by noncovalent interactions.

5.1.2 The process of total loss or randomization of three-dimensional structure of proteins is called denaturation.

5.1.3 Protein denaturation results from a change in the solvent environment that is sufficiently large to upset the forces that keep the protein structure intact.

5.1.4 Many means can cause protein to denature:

Heating: unbalance the compensating enthalpic and entropic contributions, thermal motion causes melting.

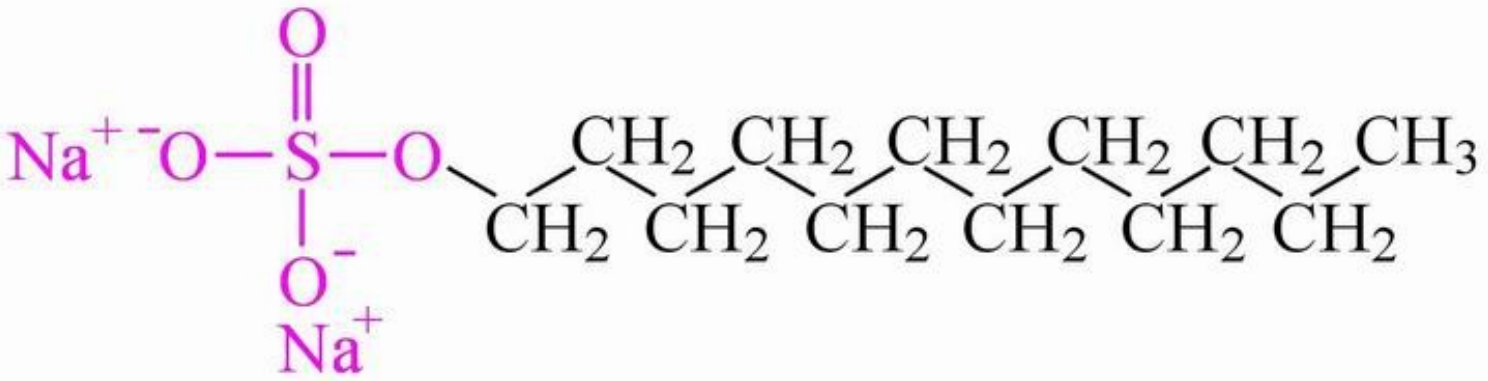
Extreme pH: upset the balance of a protein's charge interactions.

Miscible organic solvents (alcohol and acetone): solvent polarity and hydrogen bonding.

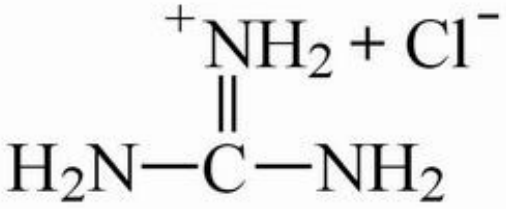
Solutes (urea, guanidine): provide alternative hydrogen bonding.

Detergents (SDS): introducing their hydrophobic tails into the protein's interior.

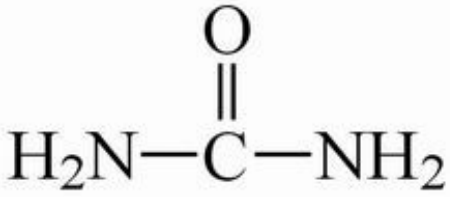
5.1.5 The mechanism of many of these denaturing processes are fully understood.



SDS



盐酸胍



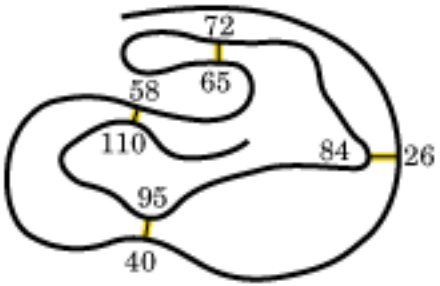
尿素

5.2 Denaturation of some proteins is reversible.

5.2.1 Some denatured globular proteins will regain their native structure and their biological activity once returned to conditions in which the native conformation is stable. This process is called renaturation.

5.2.2 The denaturation and renaturation phenomena were originally observed on ribonuclease A by chance by Christian Anfinsen (1960s).

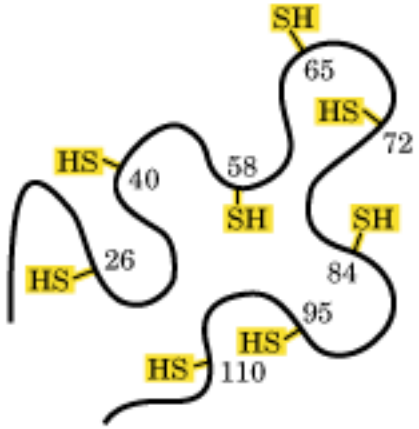
5.2.3 Ribonuclease A became reduced and randomly coiled (denatured) in 8 M urea plus β-mercaptoethanol, with a loss of the enzymatic activity.



Native state; catalytically active.

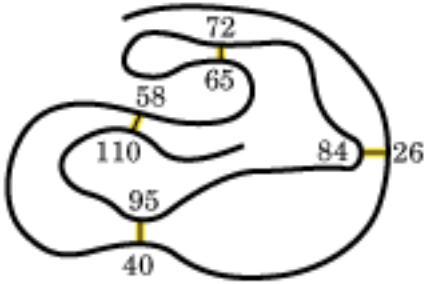
addition of urea and mercapto-ethanol

8 M urea plus β-mercaptoethanol



Unfolded state; inactive. Disulfide cross-links reduced to yield Cys residues.

removal of urea and mercapto-ethanol



Native, catalytically active state. Disulfide cross-links correctly re-formed.

5.2.4 When urea and β-mercaptoethanol were removed, the enzymatic activity was slowly regained until full recovery under stable conditions.

5.2.5 All the physical and chemical properties of the refolded enzyme were virtually identical with those of the native enzyme.

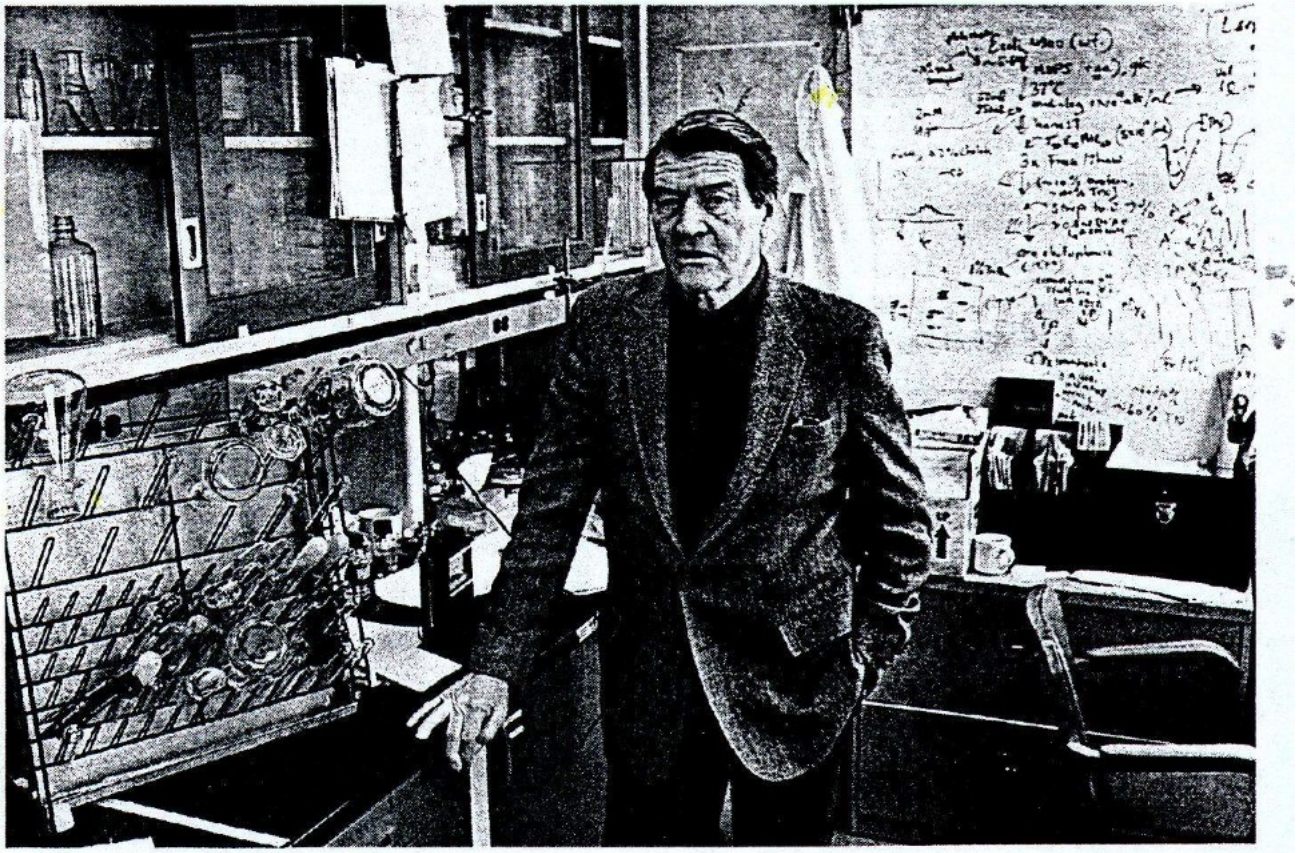
5.2.6 Conclusion: the information needed to specify the complex tertiary structure of ribonuclease A is all contained in its amino acid sequence.

The 3-D structure of a protein is determined by its amino acid sequence; The function of protein depends on its structure.

（一级结构决定高级结构，高级结构决定功能）

5.2.7 Subsequent studies have established the generality of this central principle of molecular biology: sequence specifies conformation.

Nobel Prize in Chemistry in 1972 to Anfinsen.



Nobel Prize Winner Christian Anfinsen, to whom the NIH conference was dedicated, in his laboratory at Johns Hopkins University in 1987.



Anfinsen Christian B.

1972 获诺贝尔奖

Anfinsen studied how the enzyme ribonuclease breaks down the ribonucleic acid (RNA) present in food. Anfinsen was able to ascertain how the ribonuclease molecule folds to form the characteristic three-dimensional structure that is compatible with its function.

5.3 The tertiary structures of proteins are not rigid.

5.3.1 Many studies have found that globular proteins have certain amount of flexibility in their backbones and undergo short-range internal fluctuations.

5.3.2 Many proteins undergo small conformational changes in the course of their biological function (e.g., O_2-bound hemoglobin differs from O_2-free hemoglobin, substrate binding to enzymes often causes conformational changes).

6.Protein folding

Although proteins can fold spontaneously *in intro* without the presence of accessory proteins, this process can take minutes to days. *In vivo* this process requires only a few minutes because the cells contain accessory proteins which assist the polypeptides to fold to their native conformation. There are three main classes of protein folding accessory proteins.



Folding accessory proteins

1. Protein disulfide isomerases (PDI)

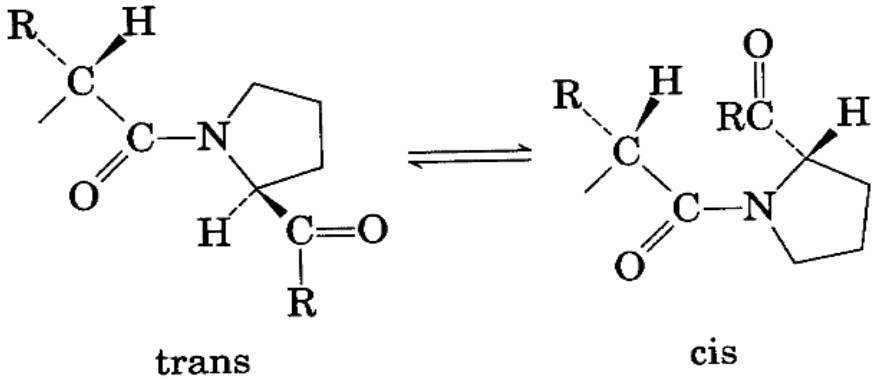
2. Peptidyl prolyl Cis-Trans isomerases (PPI)

3. Molecular chaperones

Accessory Proteins

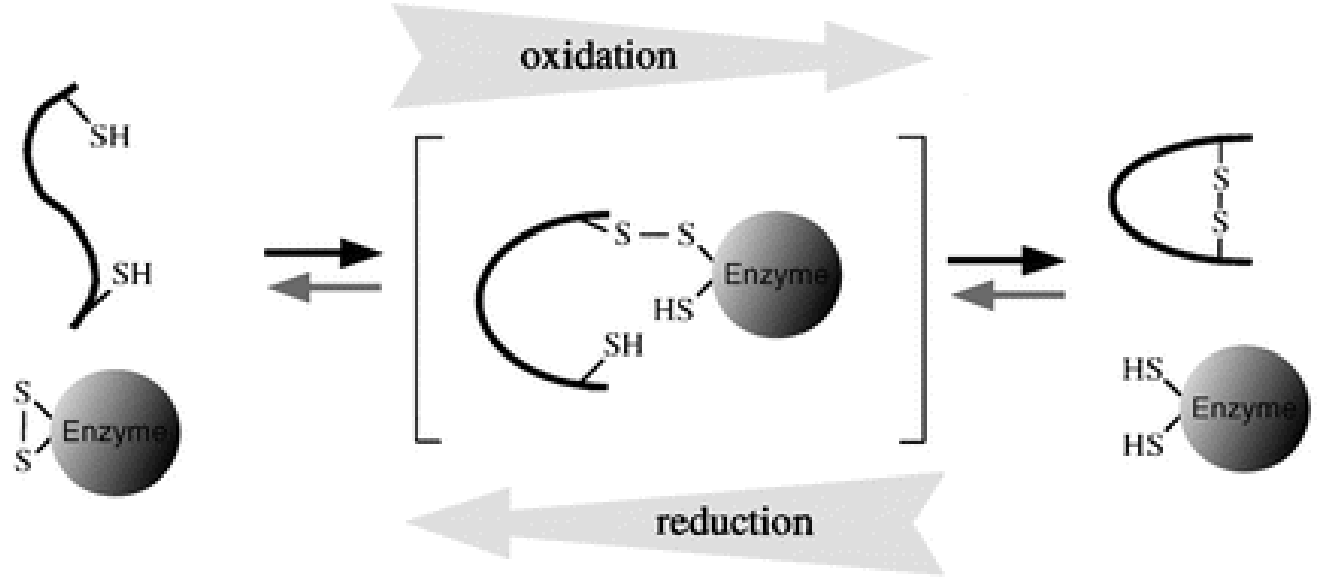
- **Molecular Chaperones**
- **Enzymes (Foldases)**
  - **PDI**
  - **PPI**

$$CySH + CySH \rightleftharpoons CyS\text{–}CyS$$

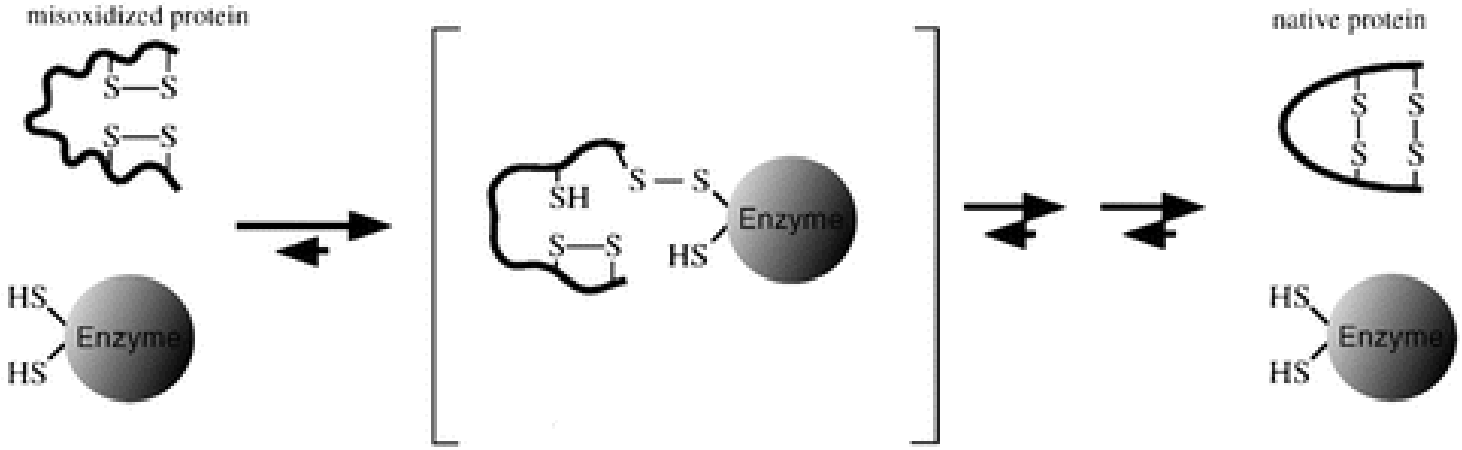


PDI

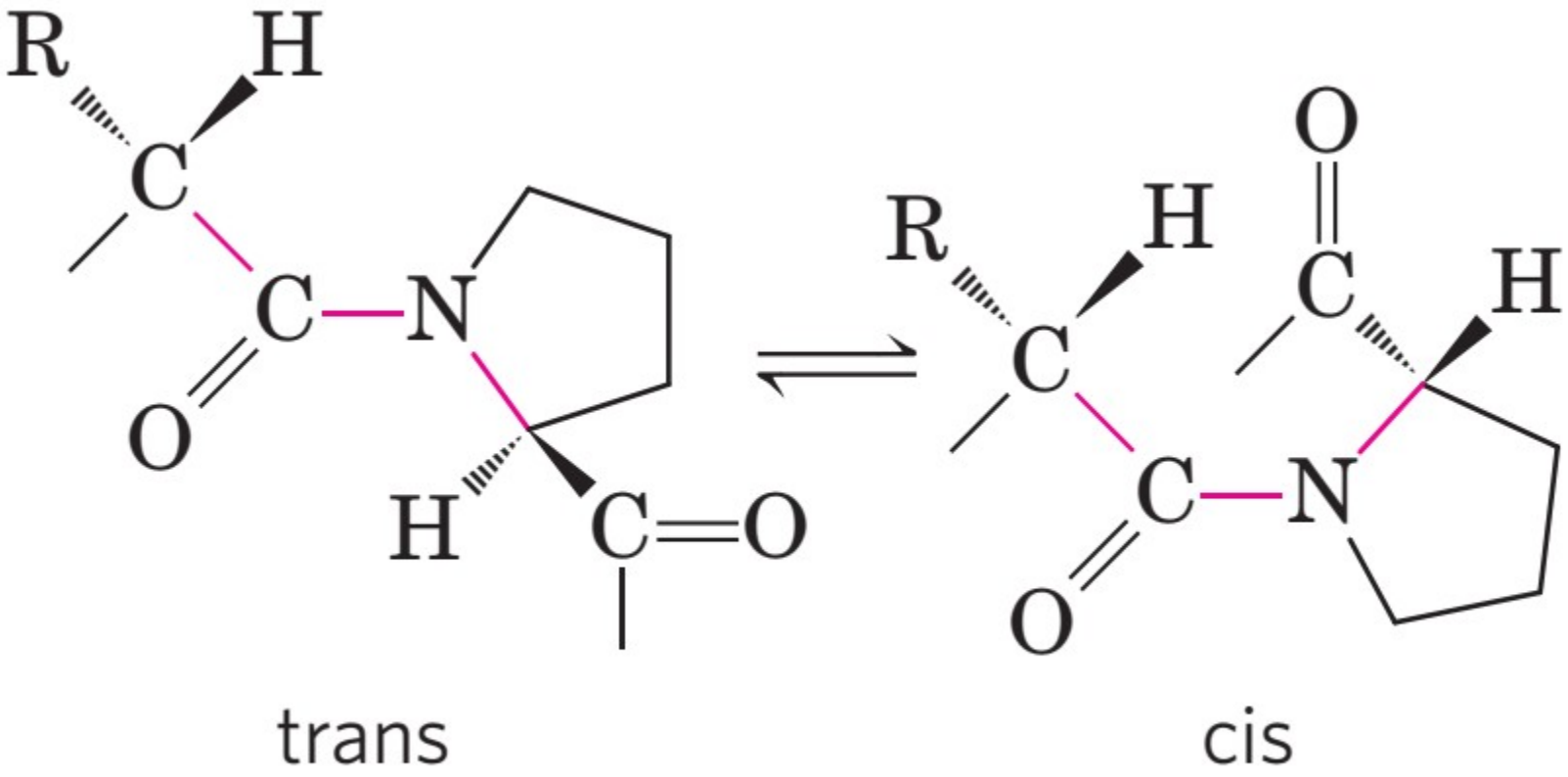
Disulfide Bond Formation/Reduction



Disulfide Bond Isomerization



PPI



Proline isomers

Protein disulfide isomerase (PDI) that catalyzes the interchange or shuffling of disulfide bonds until the bonds of native conformation are formed.

Peptide (Peptidyl) prolyl cis-trans isomerase (PPI) catalyzes the interconversion of the cis and trans isomers of proline peptide bonds.

- **Molecular Chaperones appear to be essential for the correct folding of certain polypeptide chains *in vivo,* for their assembly into oligomers, and for preventing improper folding and aggregation of proteins that may otherwise occur as internal hydrophobic regions are exposed to one another.**





'The *in vivo* folding problem', or 'Out of the ribosome, into the jungle'

7. Three-dimensional structures of proteins can be determined by several methods

7.1 X-Ray Diffraction (X-Ray crystallography)

7.2 Nuclear Magnetic Resonance (NMR)(获得 5 次诺贝尔奖)

7.3 cryo-electron microscopy(冷冻电镜 ）（ The Nobel Prize In Chemistry 2017 ）

7.4 Many proteins have since been determined, such as cytochrome c, lysozyme and ribonuclease A.

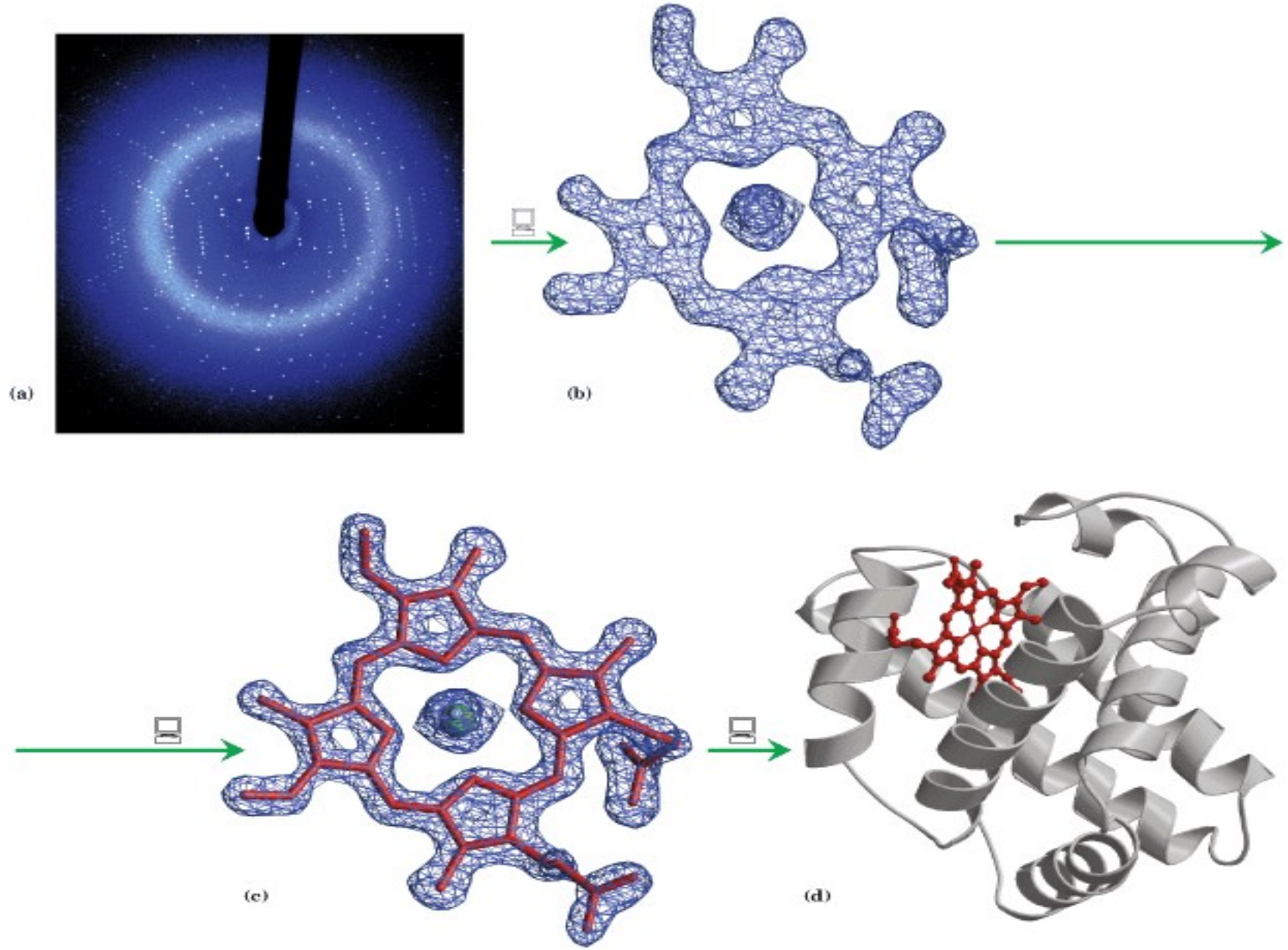
7.5 Some common characteristics of proteins

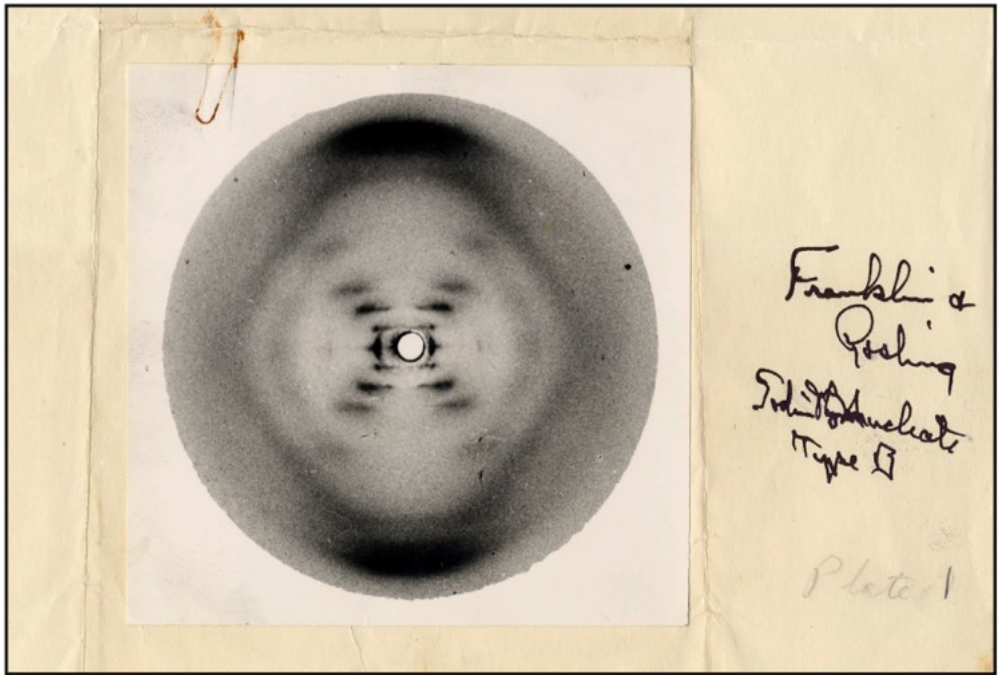
All the water soluble globular proteins have a hydrophobic core and a mainly hydrophilic outer surface.

Proteins are stabilized mainly by noncovalent interactions (sometimes by disulfide bonds).

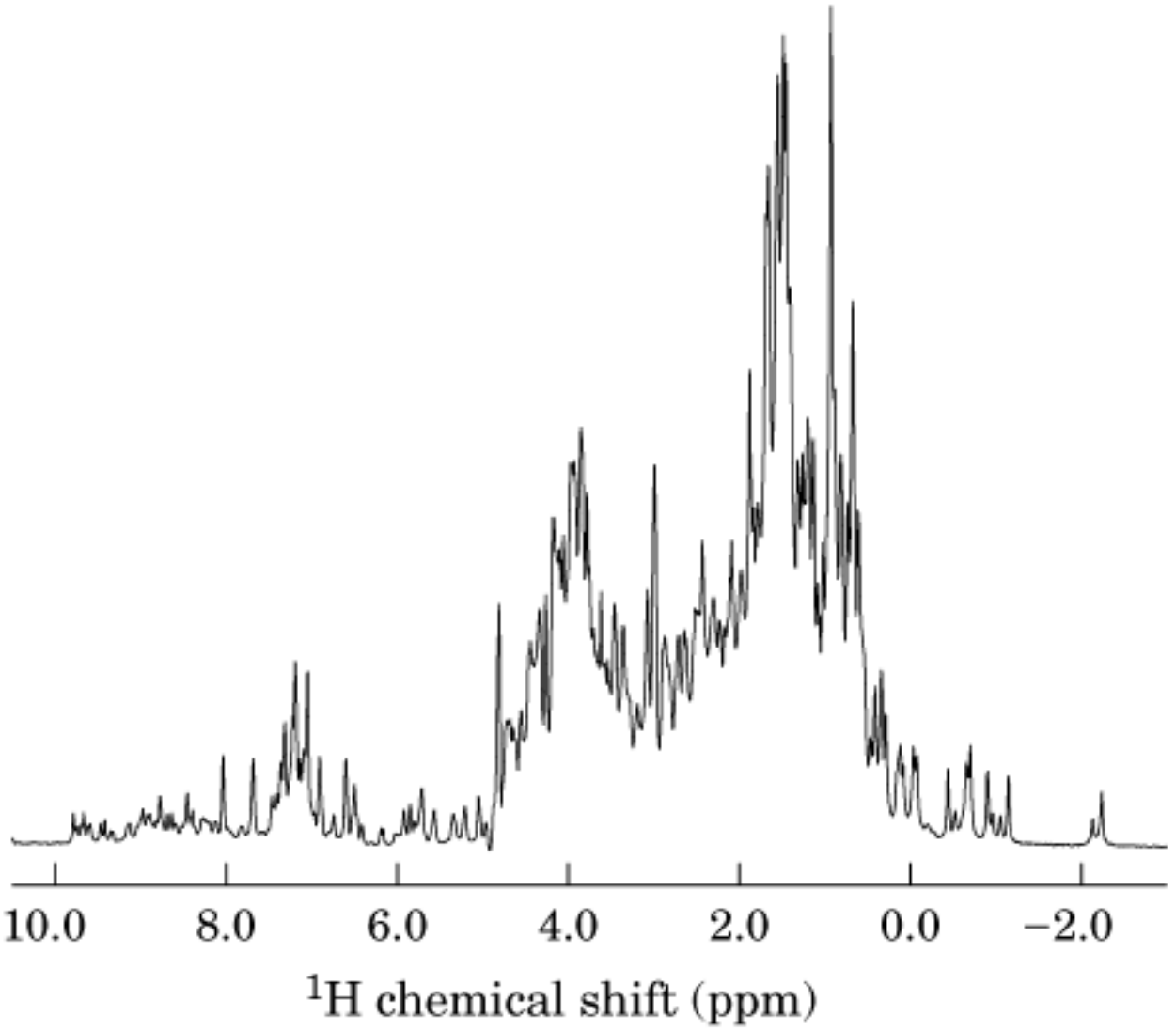
Each protein has a unique structure to perform its unique function.

X-Ray Diffraction

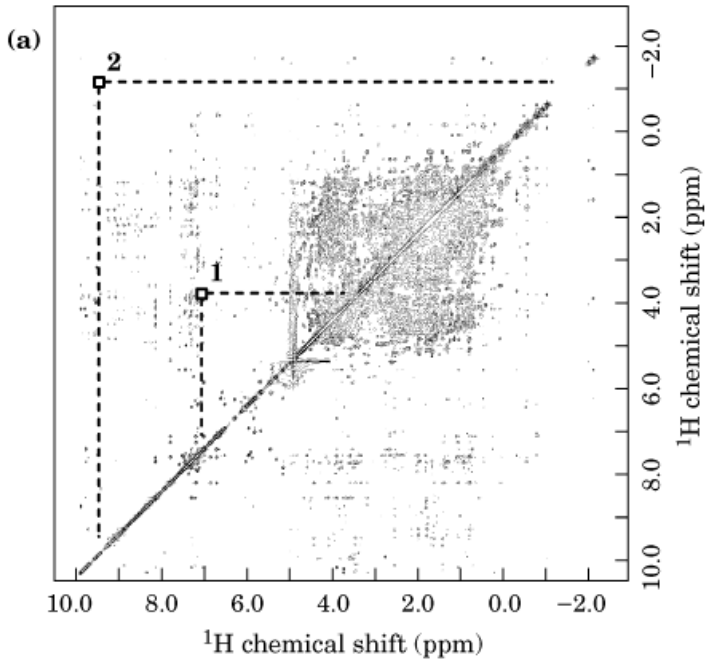



Franklin & Gosling
Type B
Plate 1

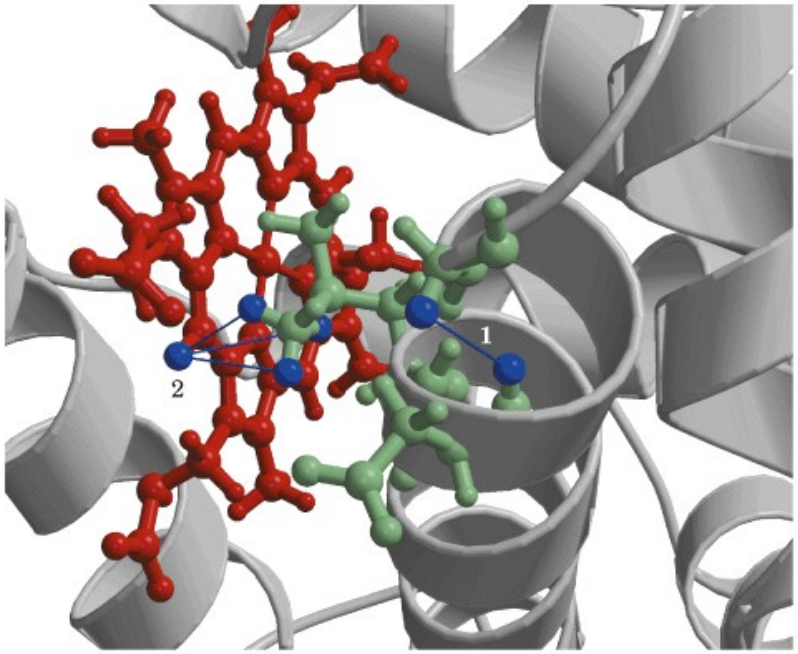
One-dimensional NMR



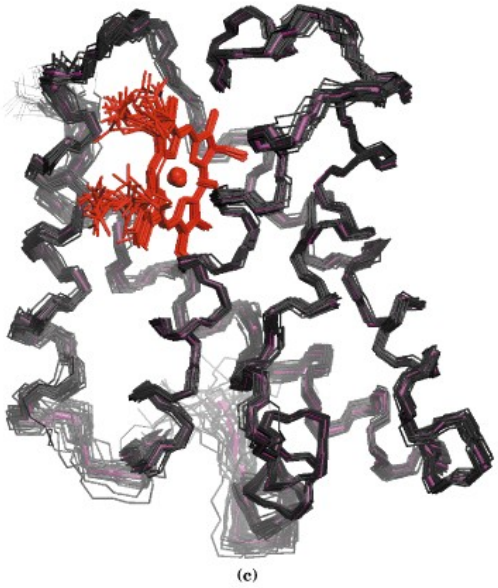
Two-dimensional NMR

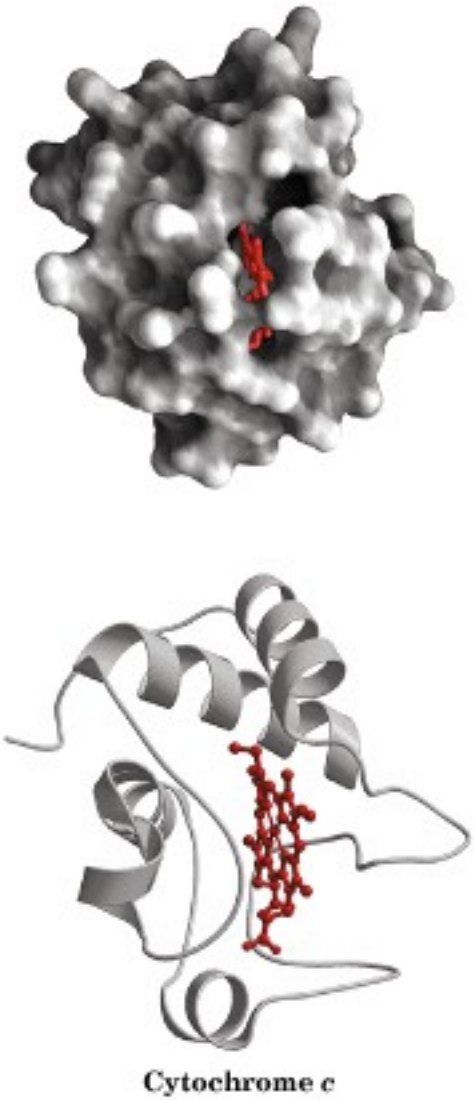


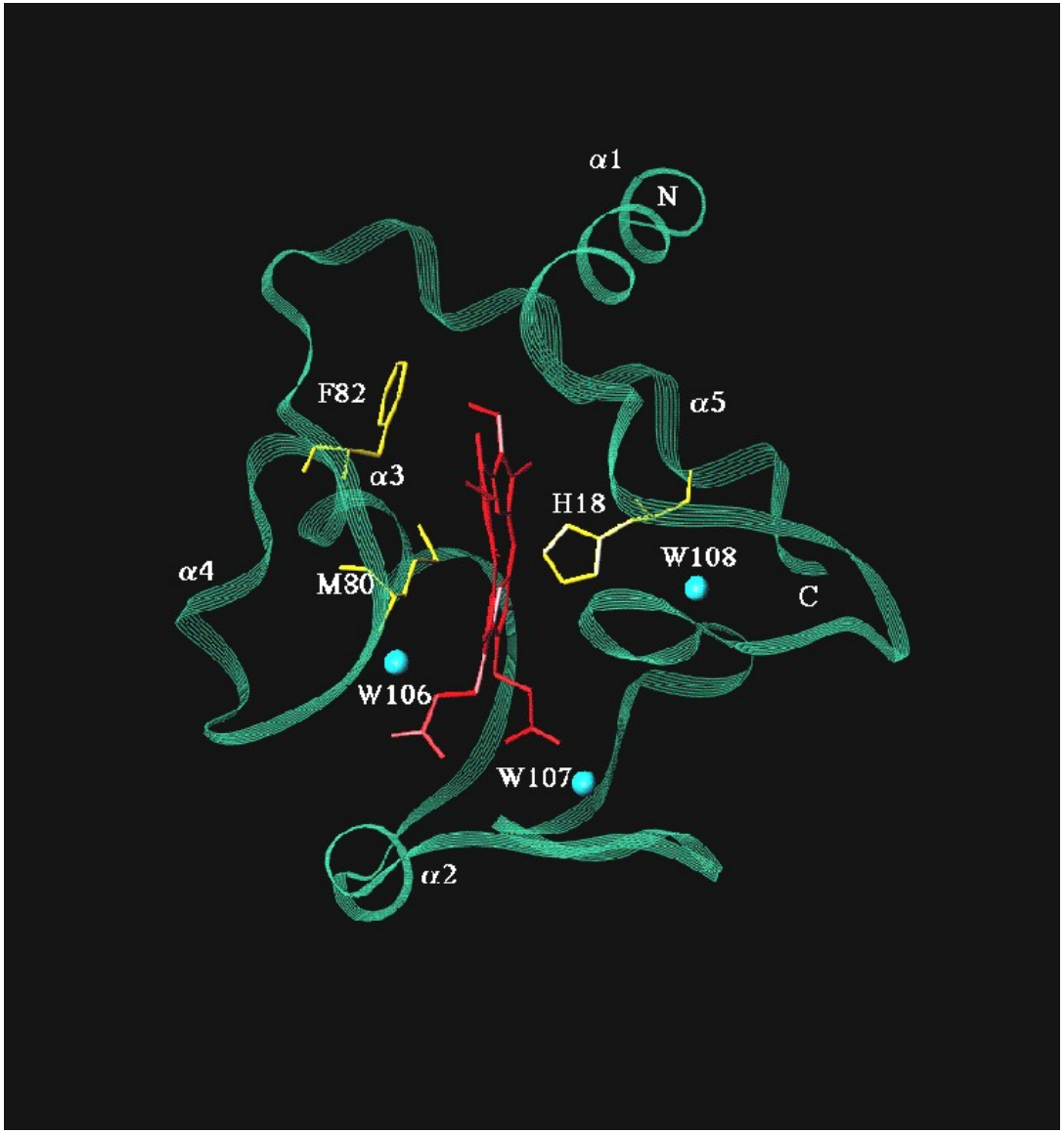




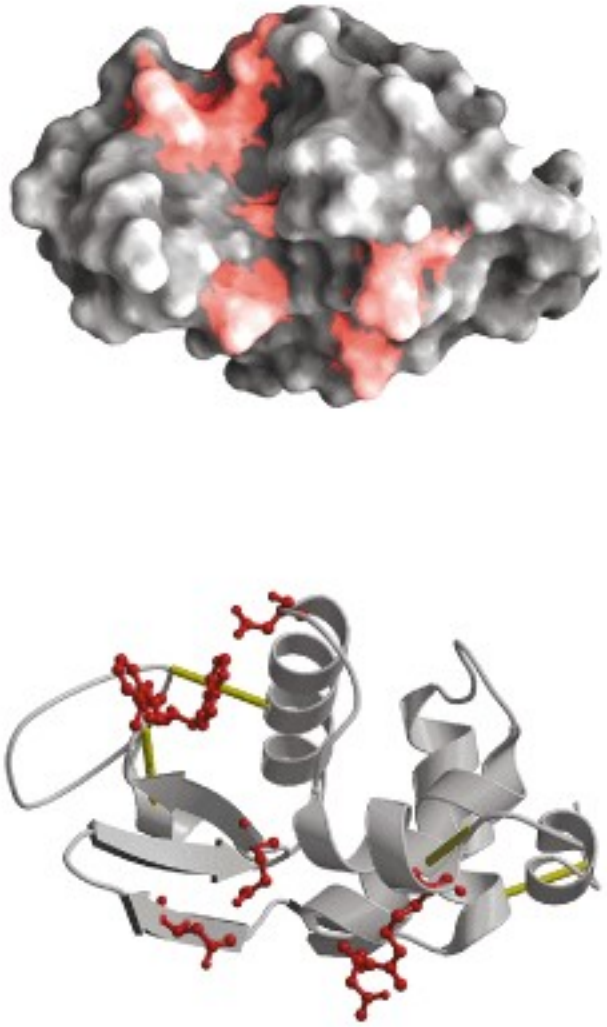






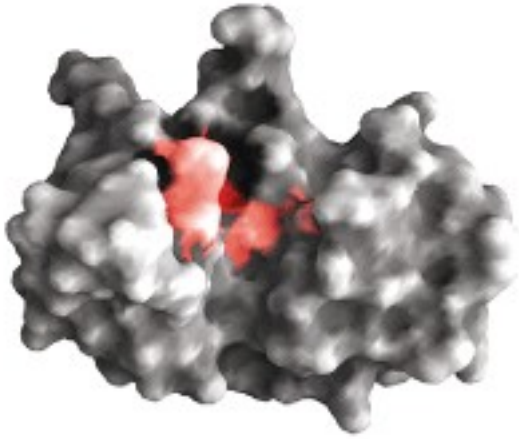


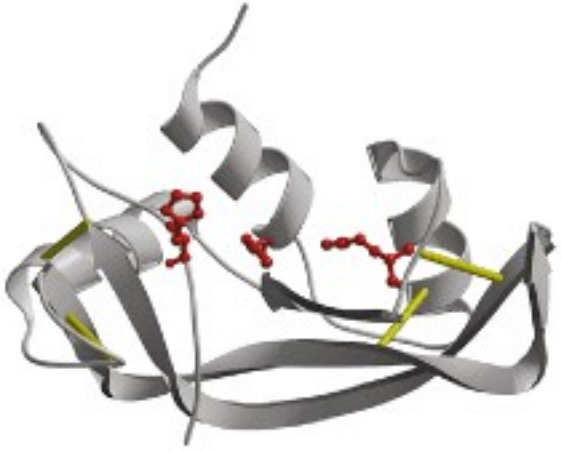
cytochrome c contains only about 40% α-helices, but many irregularly coiled and extended segments



Lysozyme

Lysozyme, an enzyme that catalyzes the hydrolytic cleavage of polysaccharides in some bacterial cell walls, contains about 40% α-helices and about 12% β-sheets.





Ribonuclease

Ribonuclease A contains more β-sheet structures.

Summary of B3

1. Protein structures have conventionally been understood at four different levels.

Primary: N→C ， S-S

secondary ： -helix, β-pleated sheet, β -turn

tertiary, quaternary;

peptide bond, peptide bond is rigid and planar

2. Proteins (ribonuclease A) **denaturation, renaturation;**

2. Protein folding;

4. Determination of 3-D structures of proteins.

B4 Myoglobin and Hemoglobin

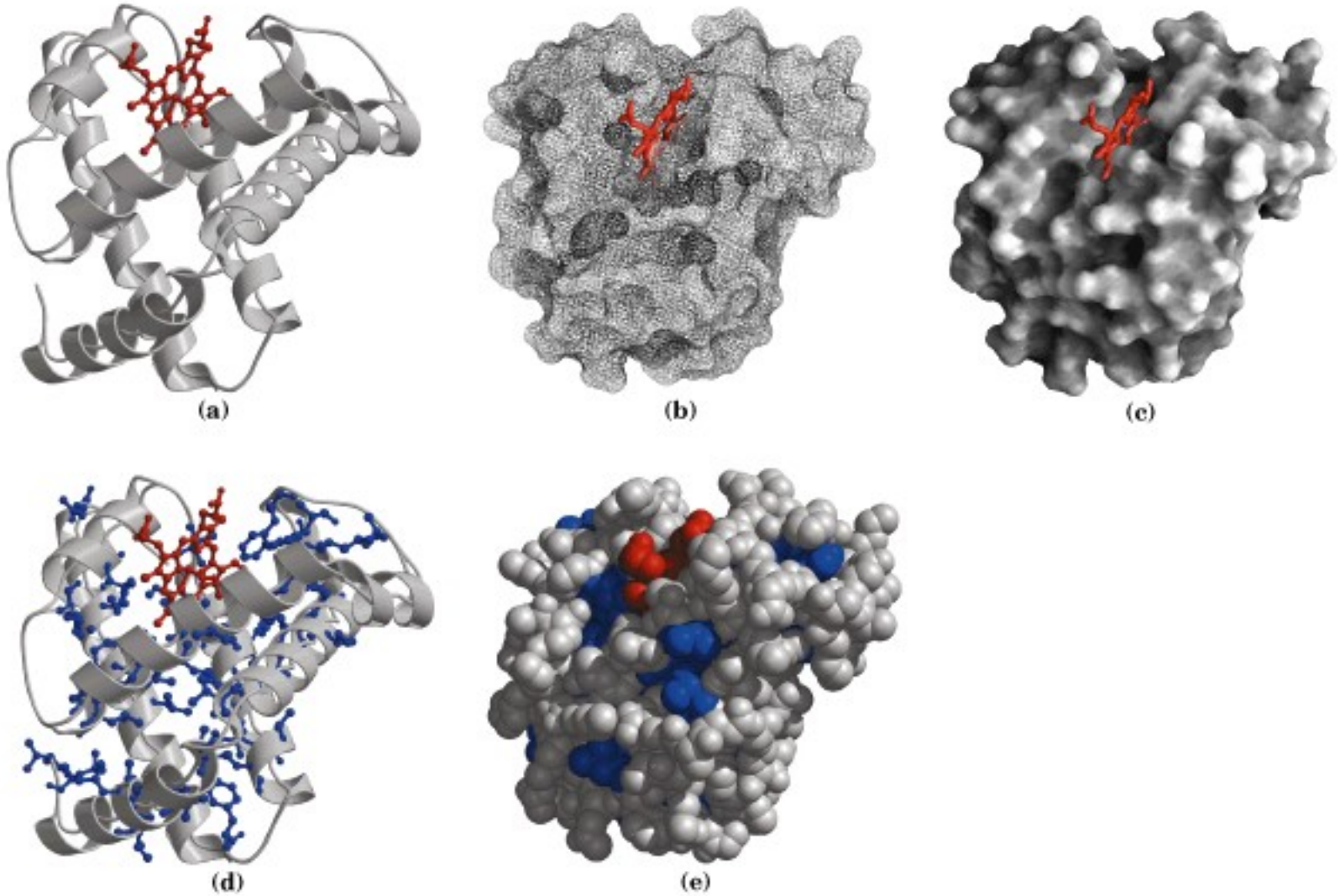
1. Oxygen-binding proteins

Hemoglobin and myoglobin are the two oxygen-binding proteins present in large multicellular organisms.

2. Sperm whale (抹香鲸 , 巨头鲸) myoglobin (肌红蛋白), the oxygen carrier in muscle, was the first protein to be seen in atomic detail by X-ray analysis (John Kendrew, 1950s)



“环保先锋”丹麦围杀1400多只海豚，鲜血染红了海滩 (baidu.com) - 百度新闻



Tertiary structure of sperm whale myoglobin.

a) backbone; b) “mesh” image; c) surface contour image;

d) ribbon; e) space-filling model.

2.1 The existence of α-helices were for the first time directly observed in a protein.

The myoglobin molecule contains eight α-helices.

All the α-helices are right-handed.

All the peptide bonds are in the planar trans configuration.

There is no β-pleated sheets observed in the molecule.

2.2 The myoglobin molecule has a dense hydrophobic core.

Many hydrophobic R groups (e.g., Leu, Val, Met, Phe) are found to be in the interior of the myoglobin molecule.

Hydrophobic interaction is important for the stability of the protein structure.

Only two hydrophilic histidine residues were found in the interior of the protein.

2.3 The polar R groups are located on the outer surface and hydrated.

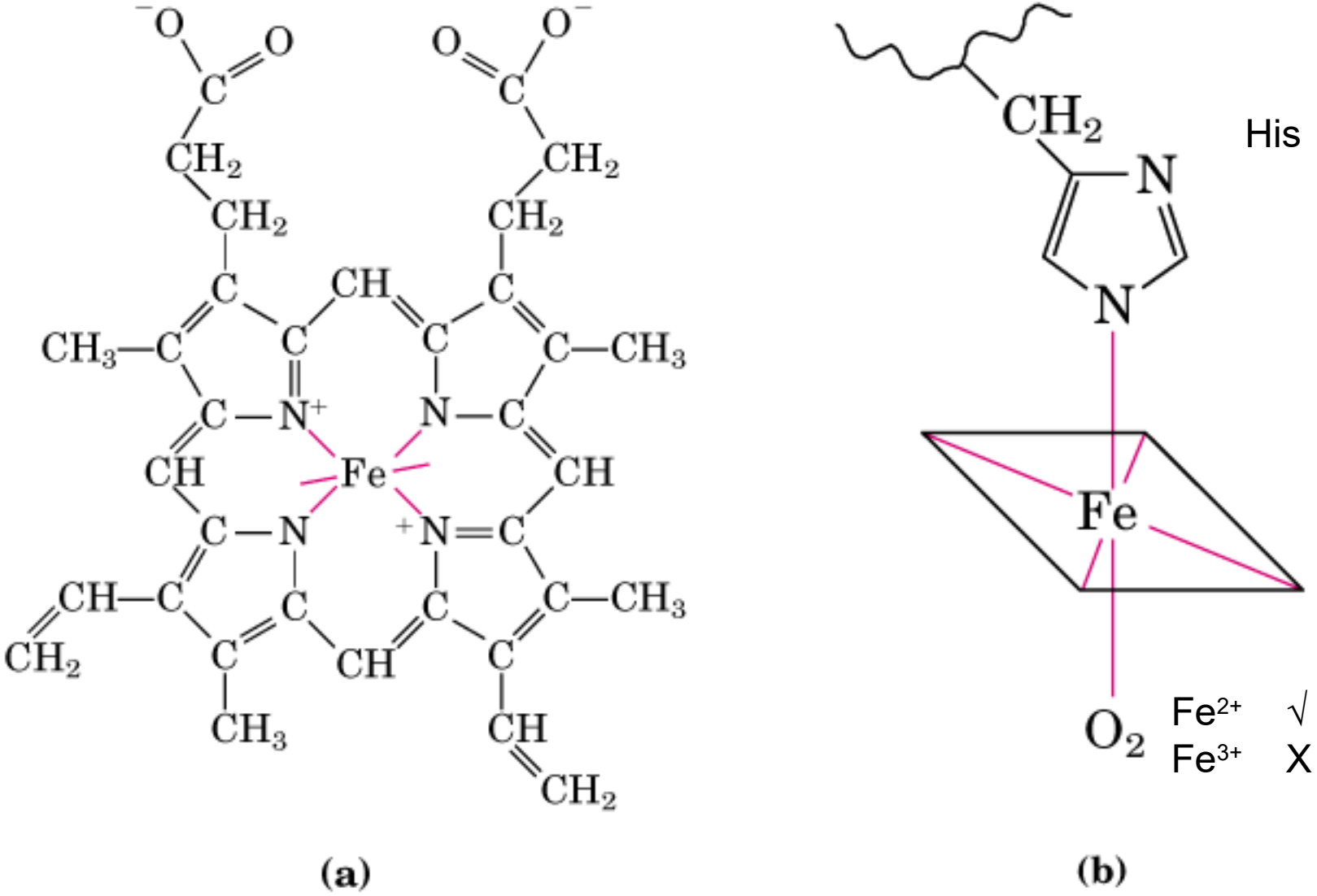
Nonpolar residues are also present on the outer surface!

2.4 All Pro residues are found at bends.

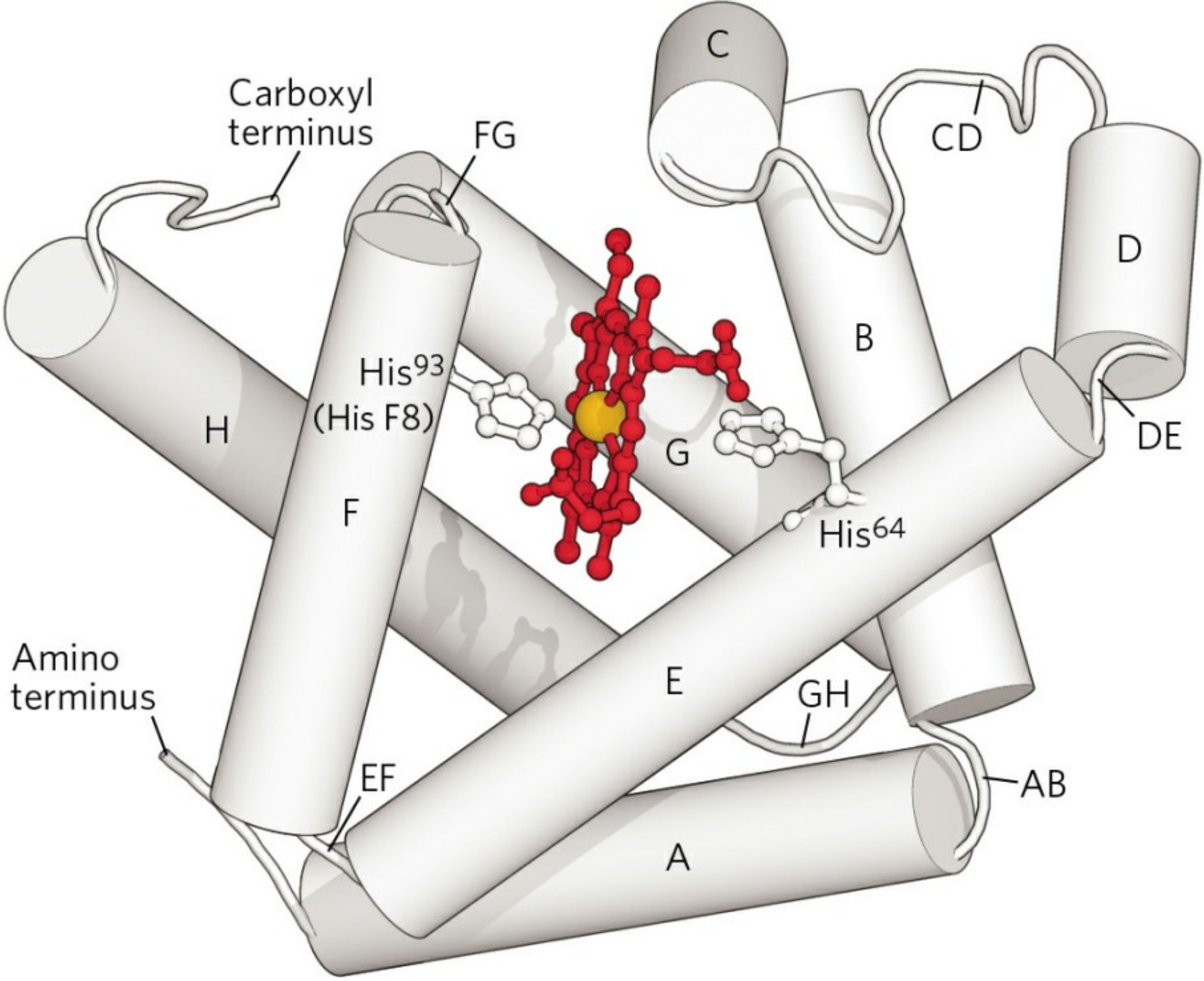
2.5 The flat heme group (the prosthetic group) was revealed to rest in a crevice (pocket).

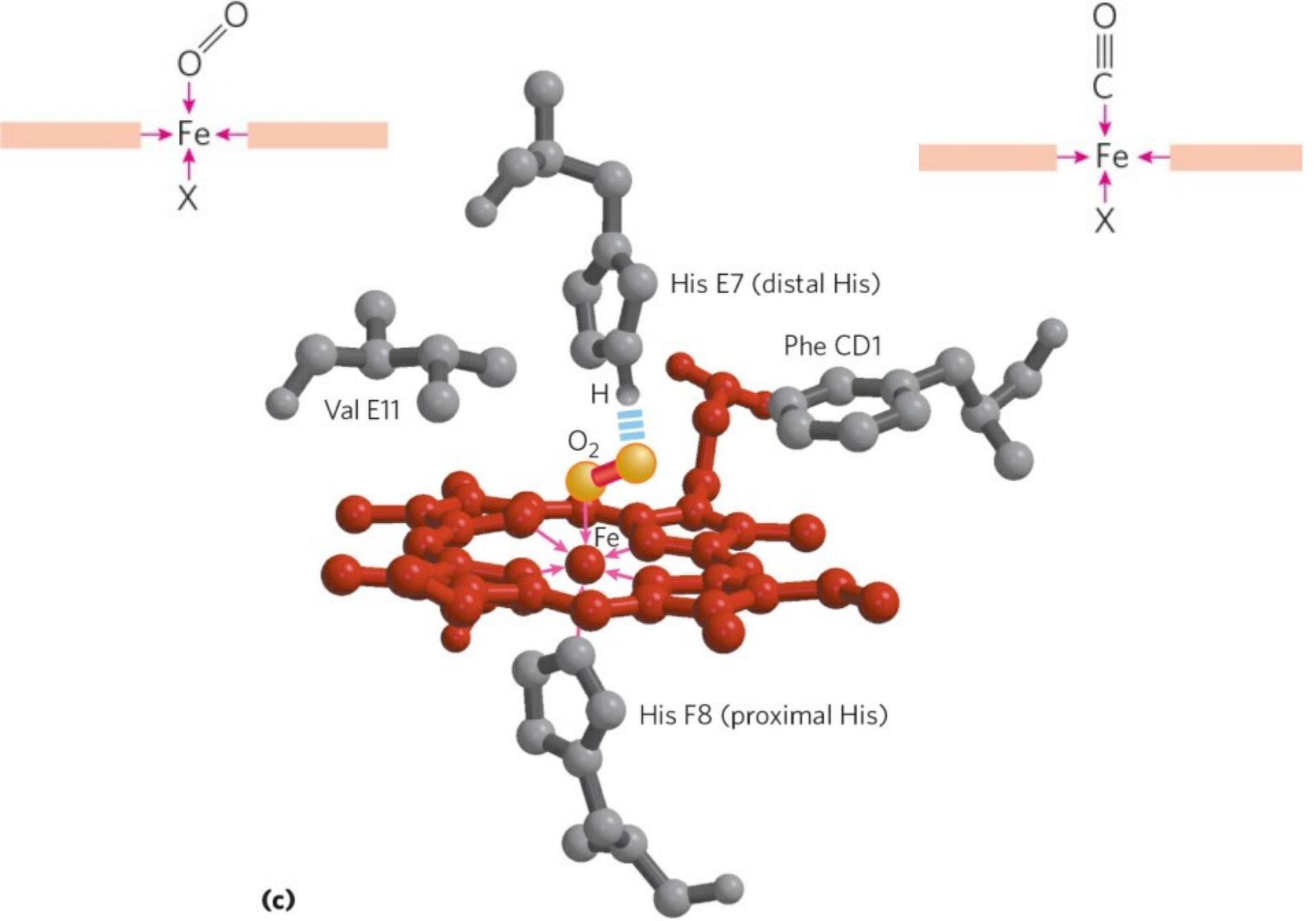
The heme group consists of a complex organic ring structure, protoporphyrin, which is bound to an iron atom in its ferrous (Fe^{2+}) state.
（ferrous/ferric）

The iron atom has six coordination bonds（配位键）, with four in the plane of and bonded to the flat porphyrin molecule and two perpendicular to it.



The heme group.


C
Carboxyl terminus
FG
CD
D
B
His93 (His F8)
H
F
G
DE
His64
Amino terminus
E
GH
EF
AB
A


O
O
Fe
X
O
C
Fe
X
His E7 (distal His)
Phe CD1
Val E11
H
O2
Fe
His F8 (proximal His)
(c)

One of the perpendicular coordination bonds is bound to a nitrogen atom of an interior His residue.

The sixth coordination serves as the binding site for O_2.

The accessibility of the heme group to solvent is highly restricted, thus preventing the oxidation of the ferrous ion (Fe^{2+}) to the ferric ion (Fe^{3+}), which is unable to bind O_2.

Sir John Kendrew and Max Perutz won the Nobel Prize in Chemistry in 1962 for determining the complete atomic structure of myoglobin and hemeglobin.

3. Hemoglobin is a multisubunit allosteric (变构) protein that carries O_2 in erythrocyte.

3.1 Hemoglobin is a well-studied and well-understood protein.

3.1.1 It was one of the first proteins to have its molecular mass accurately determined.

3.1.2 The first protein to be characterized by ultracentrifuge.

3.1.3 The first protein to be associated with a specific physiological function.

3.1.4 The first protein with a single amino acid substitution being related to a genetic disease.

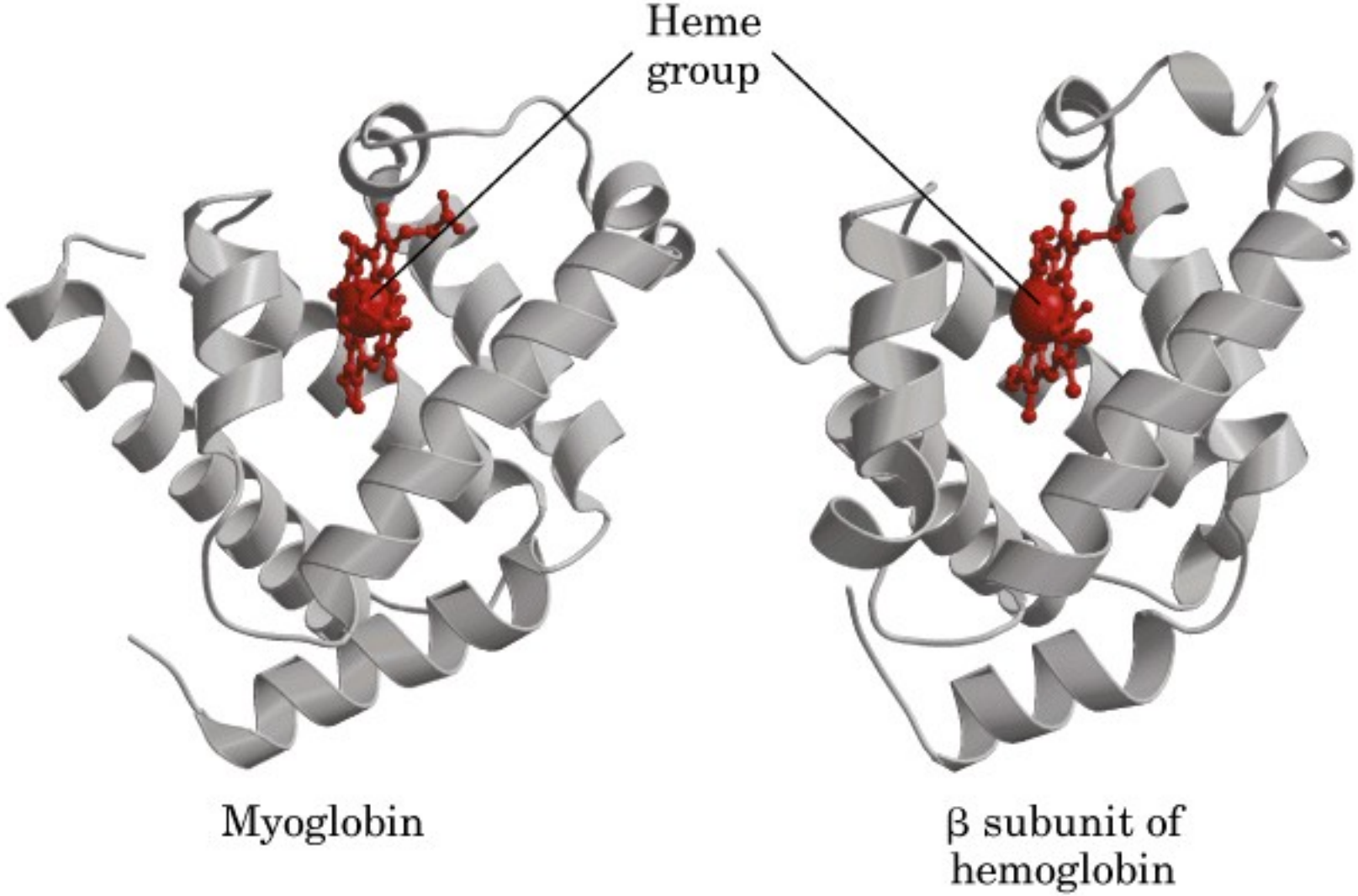
3.1.5 The first multisubunit protein with its detailed atomic structure determined by X-ray crystallography.

3.1.6 The best understood allosteric protein.

3.2 Determination of the atomic structure of hemoglobin A (from normal adult) is very revealing.

3.2.1 The protein molecule exists as a $\alpha_2\beta_2$ tetramer.

3.2.2 Each subunit has a structure strikingly and unexpectedly similar to each other and to that of myoglobin, indicating quite different amino acid sequences can specify very similar 3-D structures.

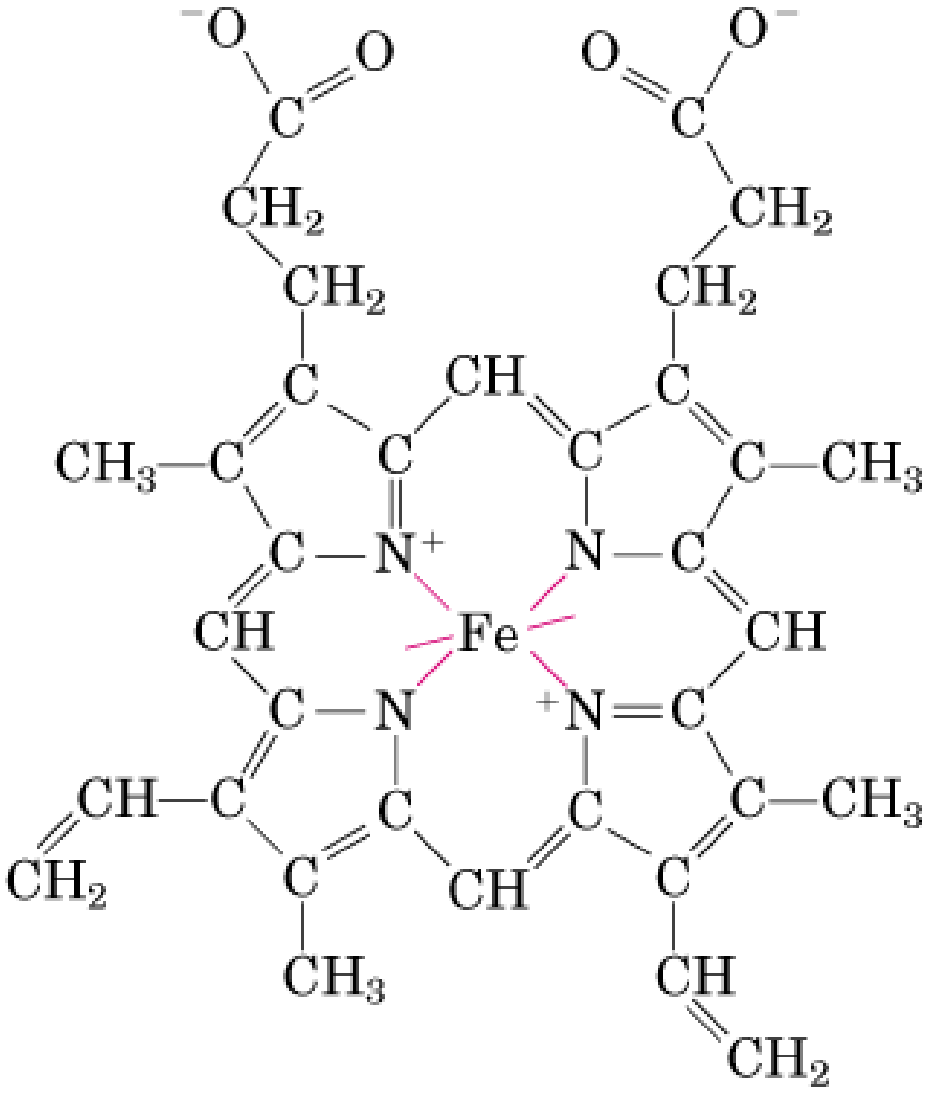

Heme
group
Myoglobin
β subunit of
hemoglobin

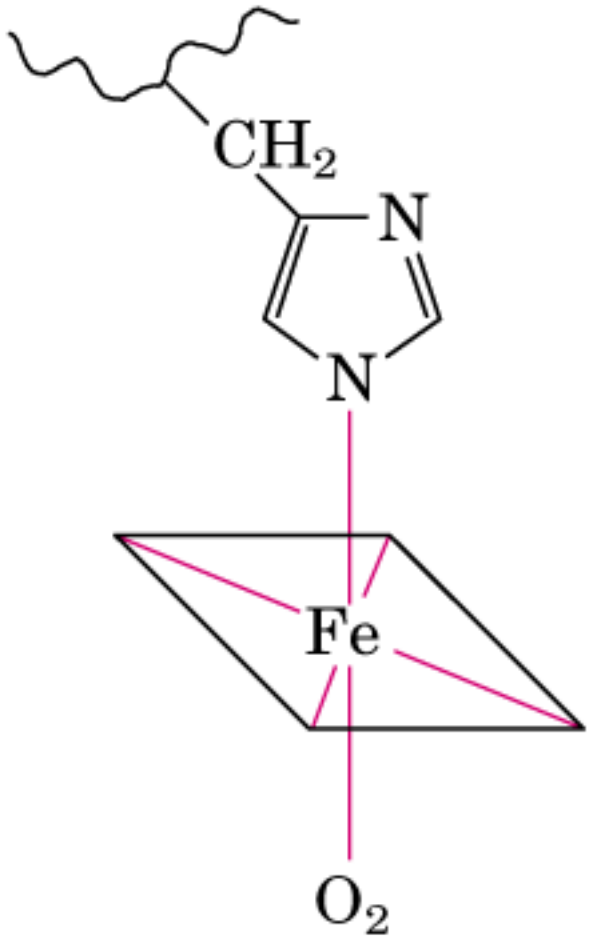
3.2.3 Extensive interactions exist between the unlike subunits through noncovalent interactions.

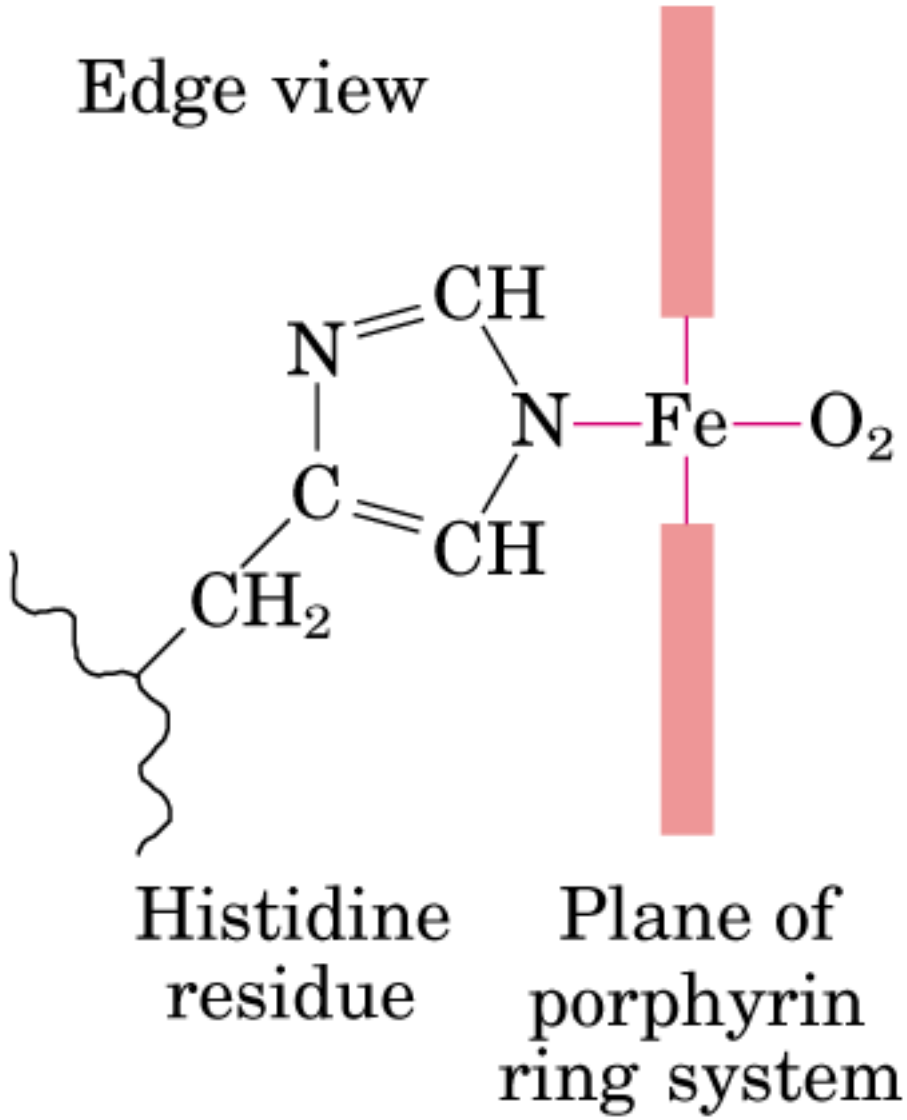
3.2.4 Quaternary structure changes markedly when O_2 binds. Crystals of deoxyhemoglobin shatter (break) when exposed to O_2.

3.2.5 O_2 binds to the sixth coordination position of the ferrous iron (as in myoglobin).

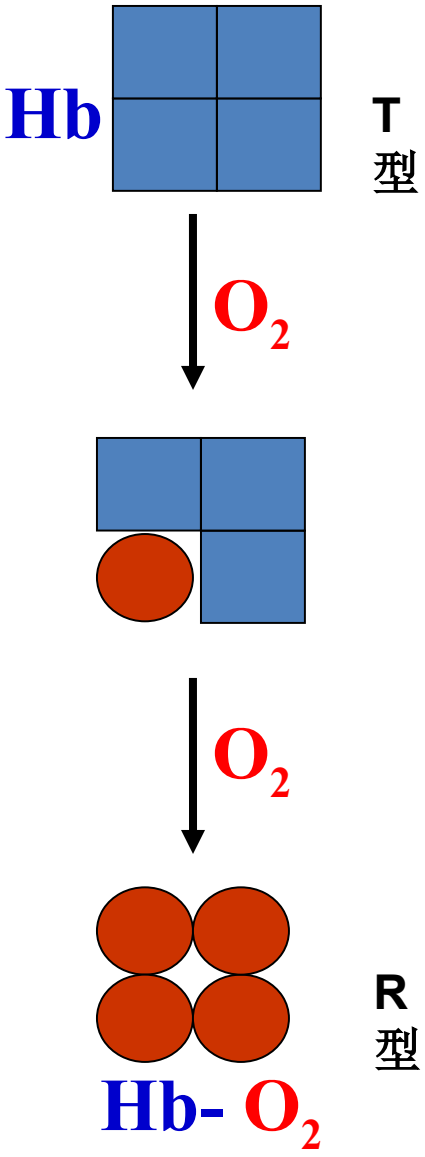
3.2.6 Oxygen binding of hemoglobin shows positive cooperativity. The binding of O_2 (the ligand) at one heme facilitates the binding of O_2 at the other hemes on the same tetramer.

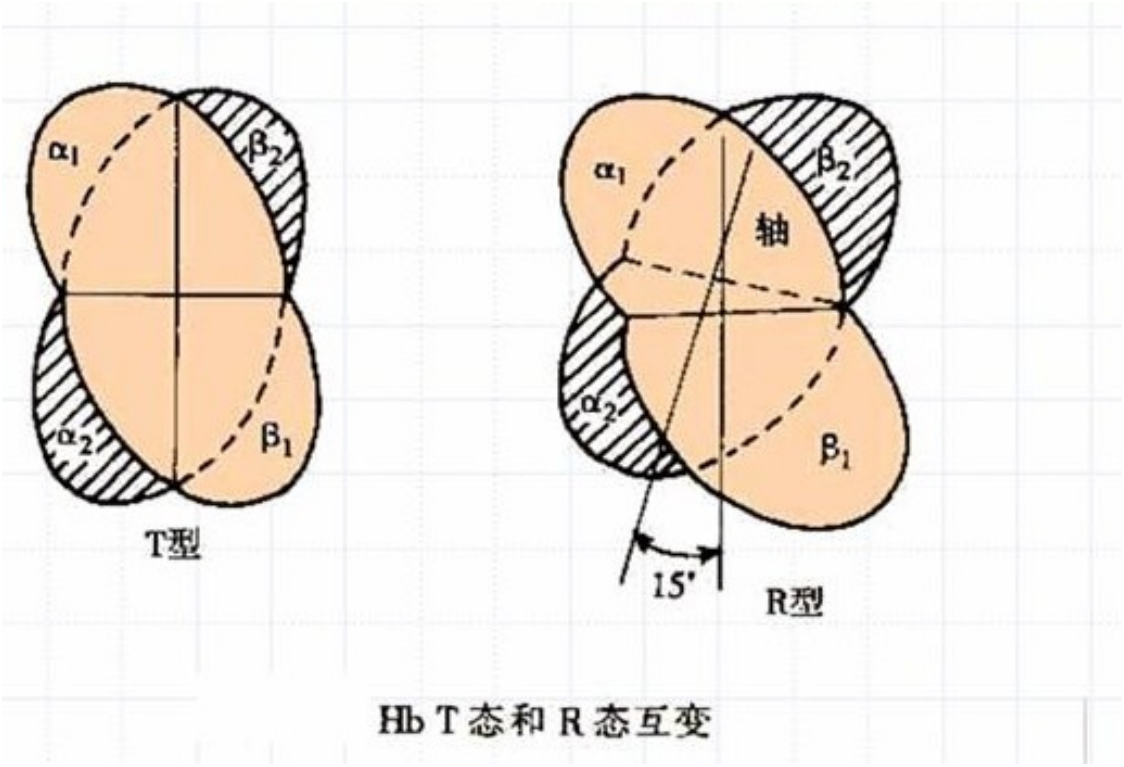

O
O
C
CH2
CH2
C
CH
C
CH3
C
C
C
C
CH3
C
N+
N
C
CH
Fe
CH
C
N
+N
C
CH
C
C
C
C
CH3
CH2
C
CH
C
CH3
CH
CH2


CH2
N
N
Fe
O2

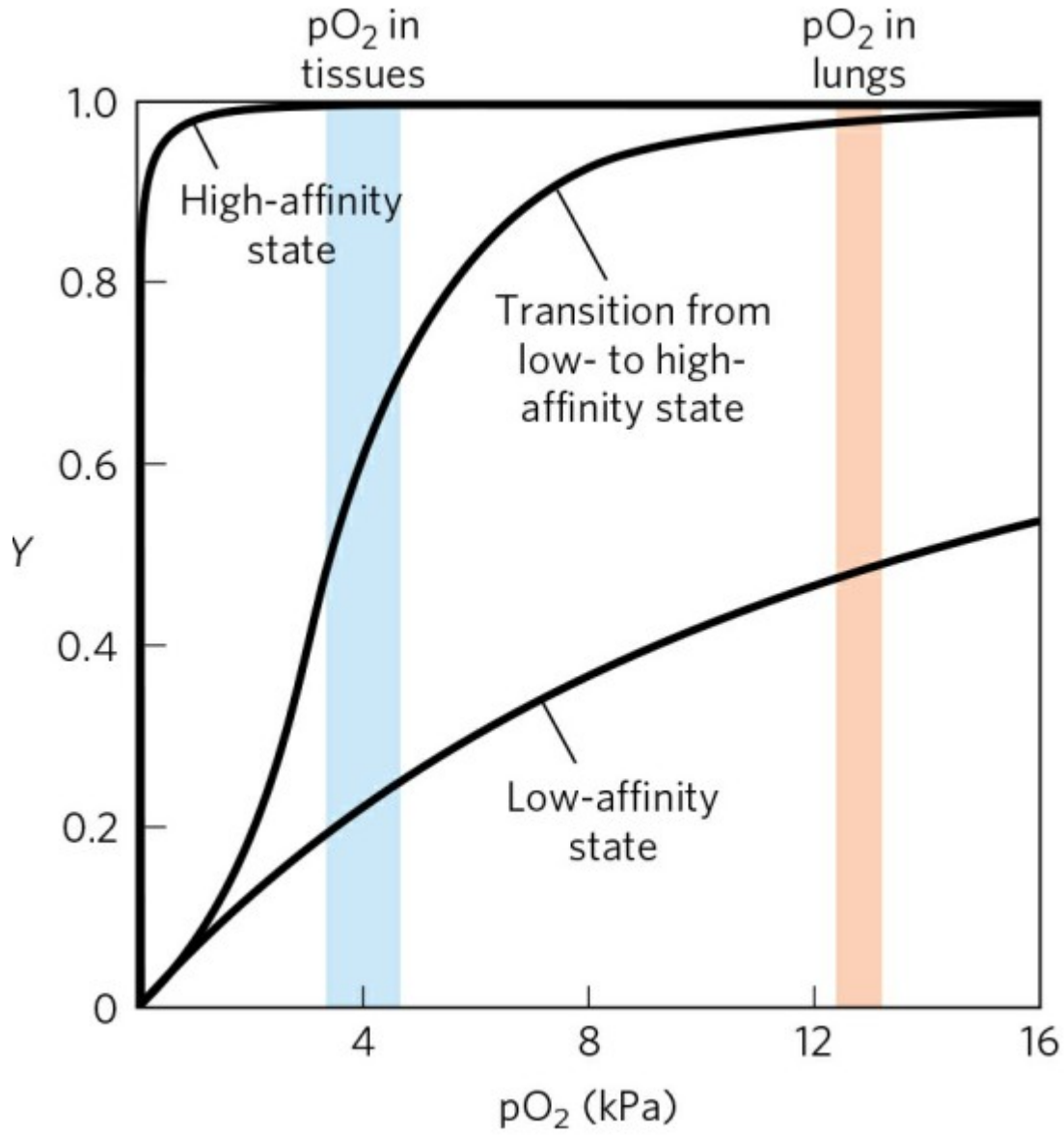

Edge view
CH
N
N
Fe
O2
C
CH
CH2
Histidine residue
Plane of porphyrin ring system

血红蛋白输氧功能和构象变化





Hb T 态和 R 态互变

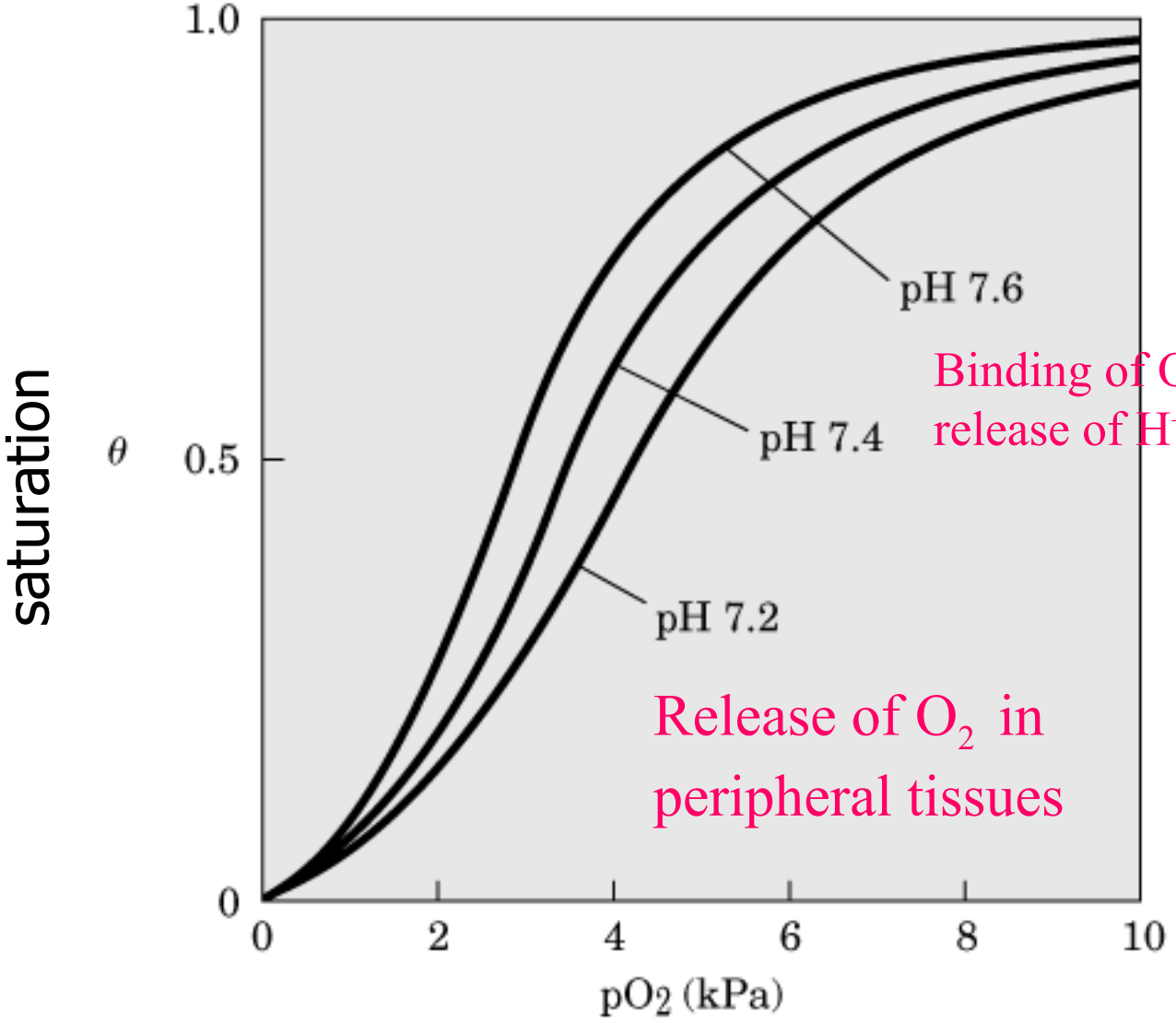

pO_2 in tissues
pO_2 in lungs
1.0
0.8
0.6
0.4
0.2
0
Y
High-affinity state
Transition from low- to high-affinity state
Low-affinity state
4
8
12
16
pO_2 (kPa)

3.3 The Bohr effect

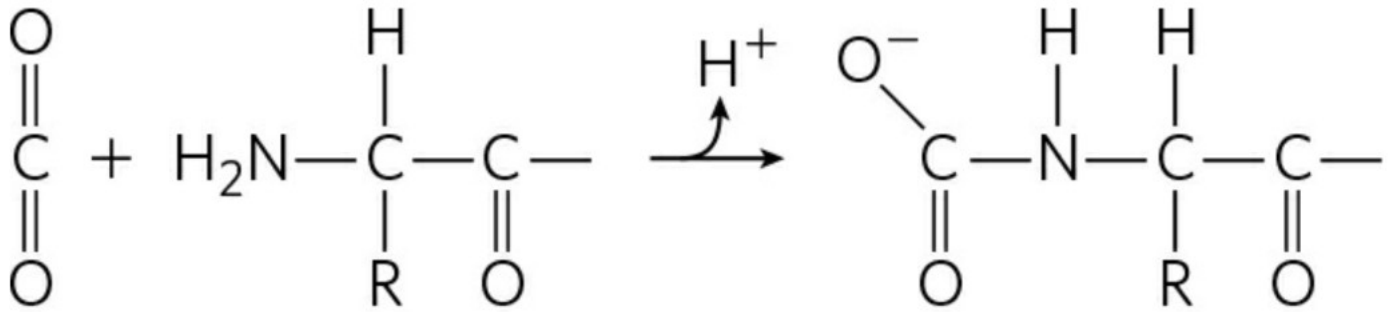
Increasing concentrations of H^+ (with a decrease of pH) or CO_2 lower the O_2 affinity of hemoglobin (H^+ and CO_2 has no effect on O_2 affinity of myoglobin). One molecule of 2,3-diphosphoglycerate (BPG) binds to the central cavity of one tetramer of hemoglobin, which also lowers its O_2 affinity.

This is called Bohr effect, which helps the release of O_2 in the capillaries of actively metabolizing tissues.

Bohr's effect and its molecular mechanism

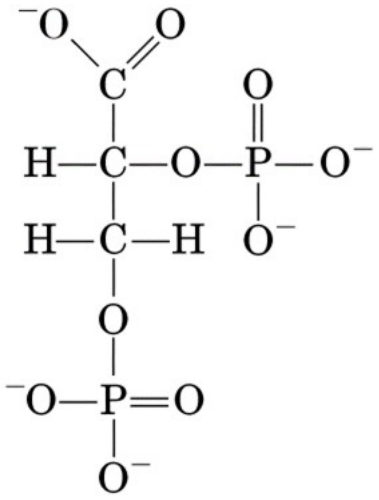


Binding of O2 in lungs with release of H^+ and CO_2.

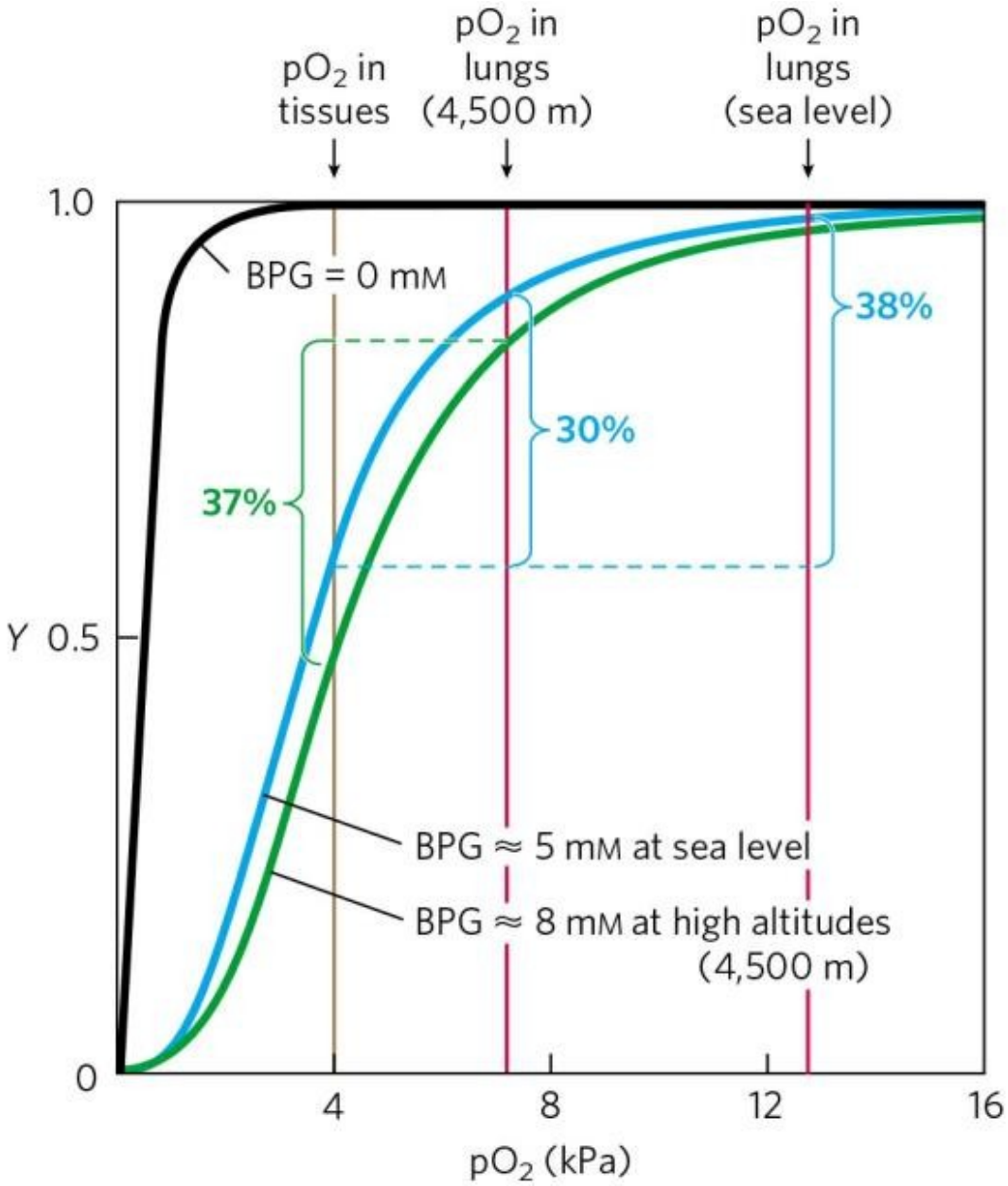


Amino-terminal residue

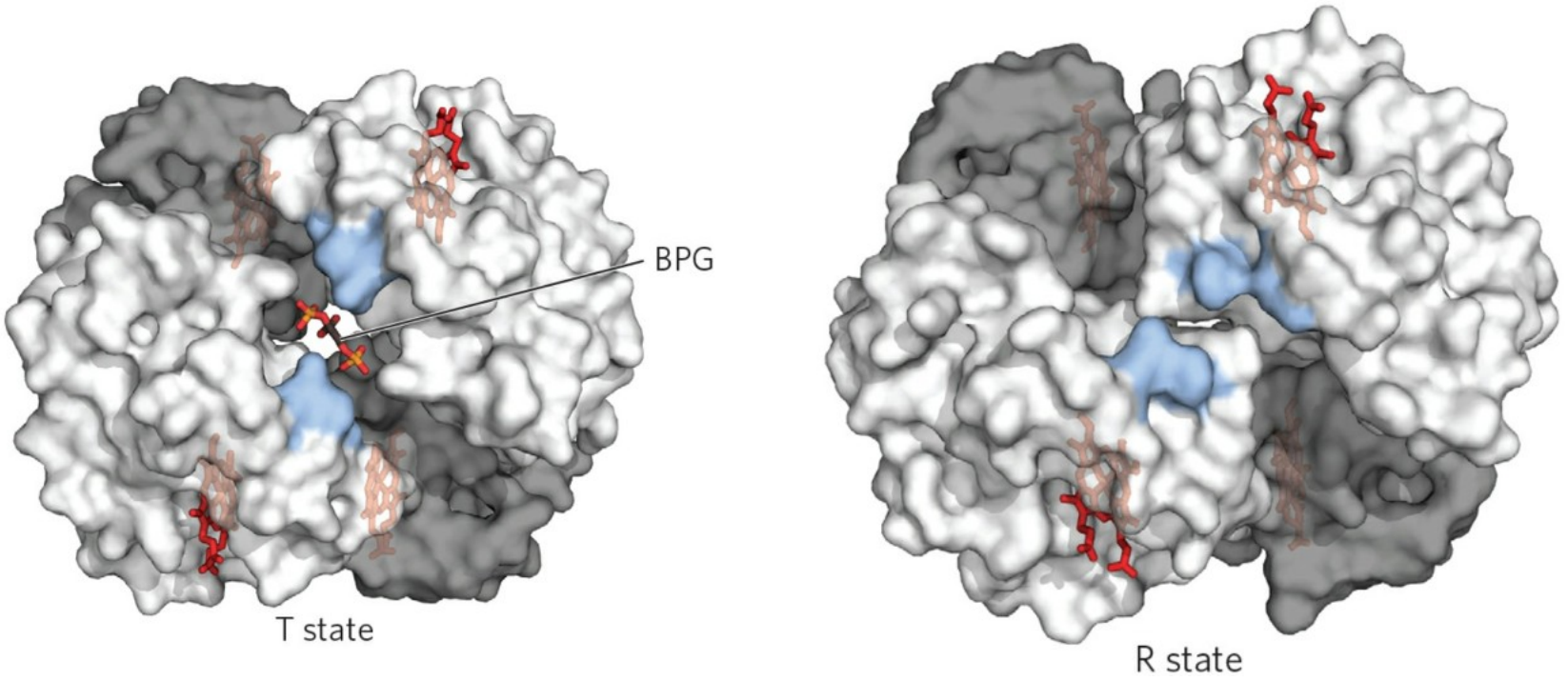
Carbamino-terminal residue



2,3-Bisphosphoglycerate



Effect of 2,3-bisphosphoglycerate on oxygen binding to hemoglobin.



Binding of 2,3-bisphosphoglycerate to deoxyhemoglobin.

3.4 Cooperativity is a particular case of an allosteric effect

3.4.1 Allosteric effect refers to the phenomenon in which a molecule (allosteric effector) bound to one site on a protein causes a conformational change in the protein such that the activity of another site on the protein is altered (increased or decreased).

3.4.2 H^+, CO_2, and BPG all show an allosteric effect for the O_2 binding process of hemoglobin.

3.5 The functional characteristics of an allosteric protein are regulated by specific molecules in its environment. In other words, in the evolutionary transition from myoglobin to hemoglobin, a macromolecule capable of perceiving information from its environment has emerged.

3.6 Sickle-cell anemia was found to be caused by a single amino acid change in the β chain of hemoglobin molecules.

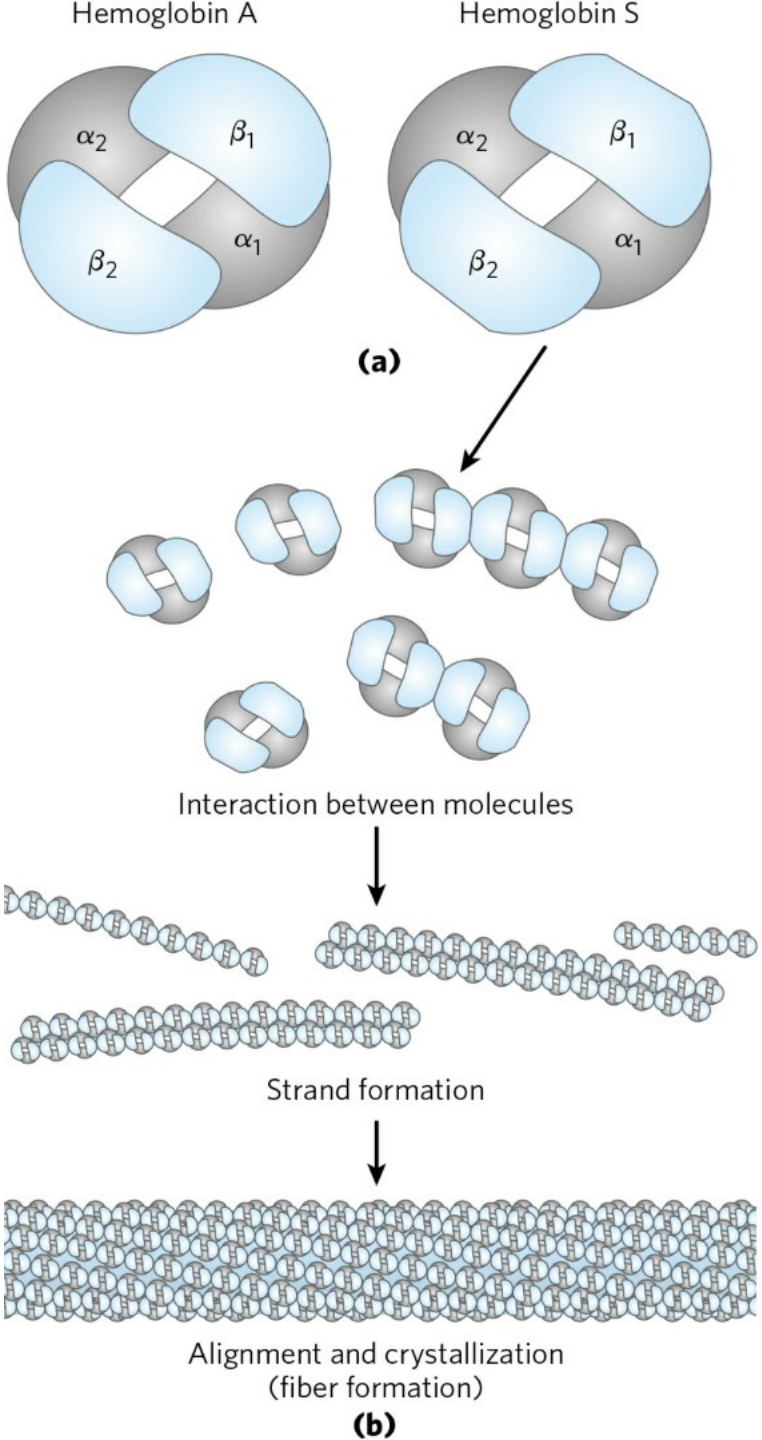
3.6.1 The hemoglobin molecule from sickle-cell anemia patients (HbS) was found to have a higher pI value (having more net positive charges).

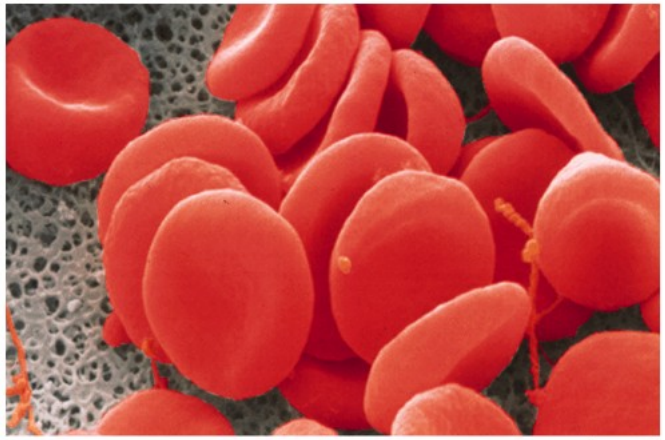
3.6.2 Amino acid sequencing revealed that HbS contains Val instead of Glu at position 6 of the β chain!

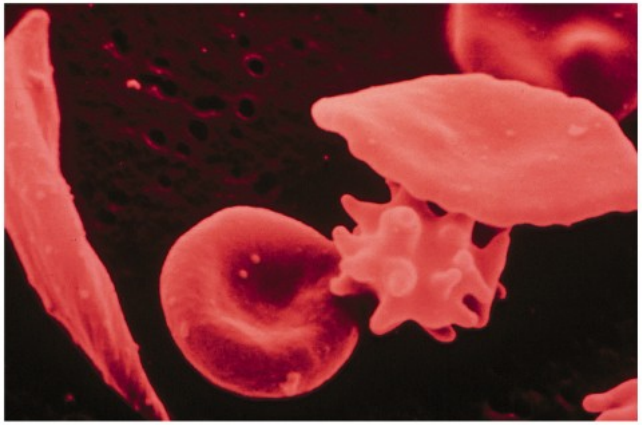
3.6.3 DNA sequencing of the genes further revealed that the codon of β6 changed from GAA(Glu) to GTA(Val).

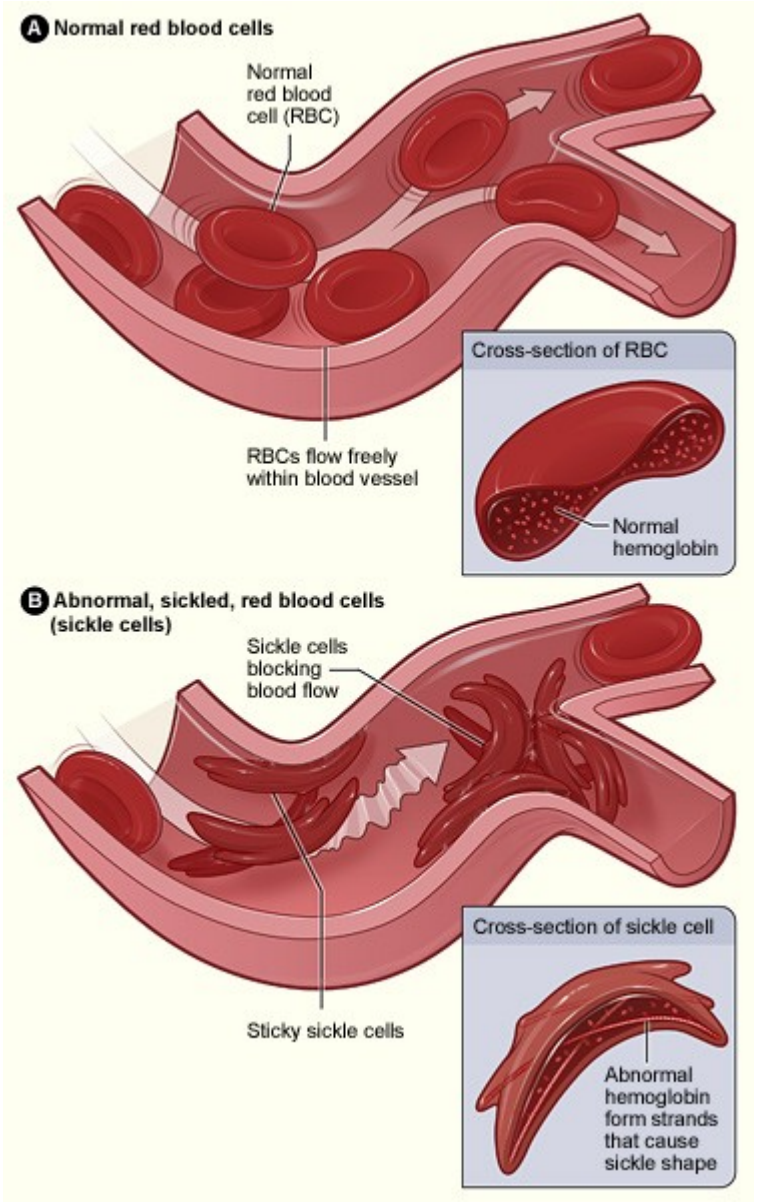
3.6.4 The oxygen binding affinity and allosteric properties of hemoglobin are virtually unaffected by this change (the β6 is located at the surface of the protein).

3.6.5 Presence of Val 6 on the β subunits generates a hydrophobic patch on the surface which complements with another hydrophobic patch formed only in deoxygenated HbS, thus generating the fiber aggregates (a polymer of HbS).


Hemoglobin A
Hemoglobin S
α2
β1
β2
α1
(a)
Interaction between molecules
Strand formation
Alignment and crystallization
(fiber formation)
(b)







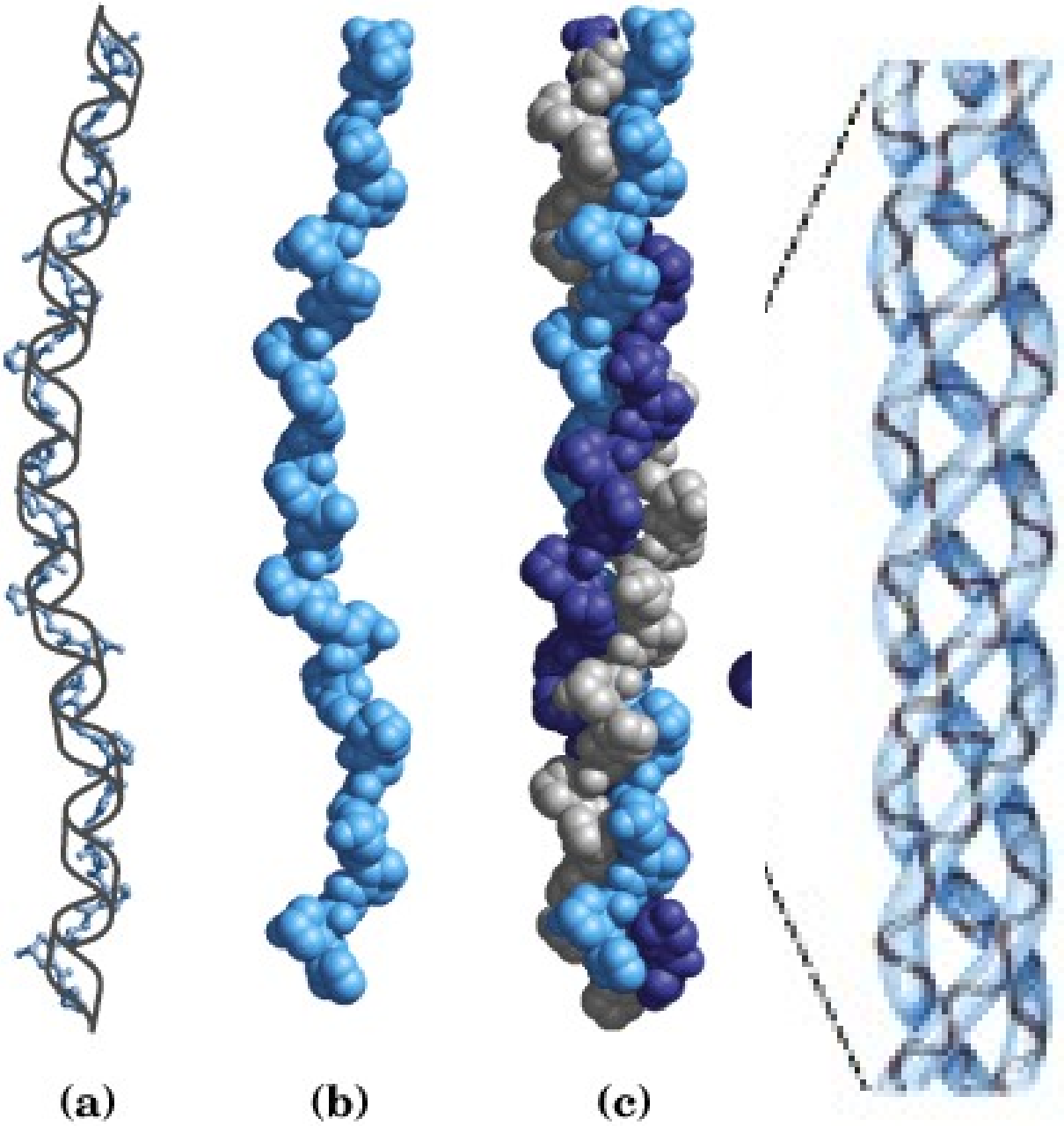
from WIKIPEDIA The Free Encyclopedia

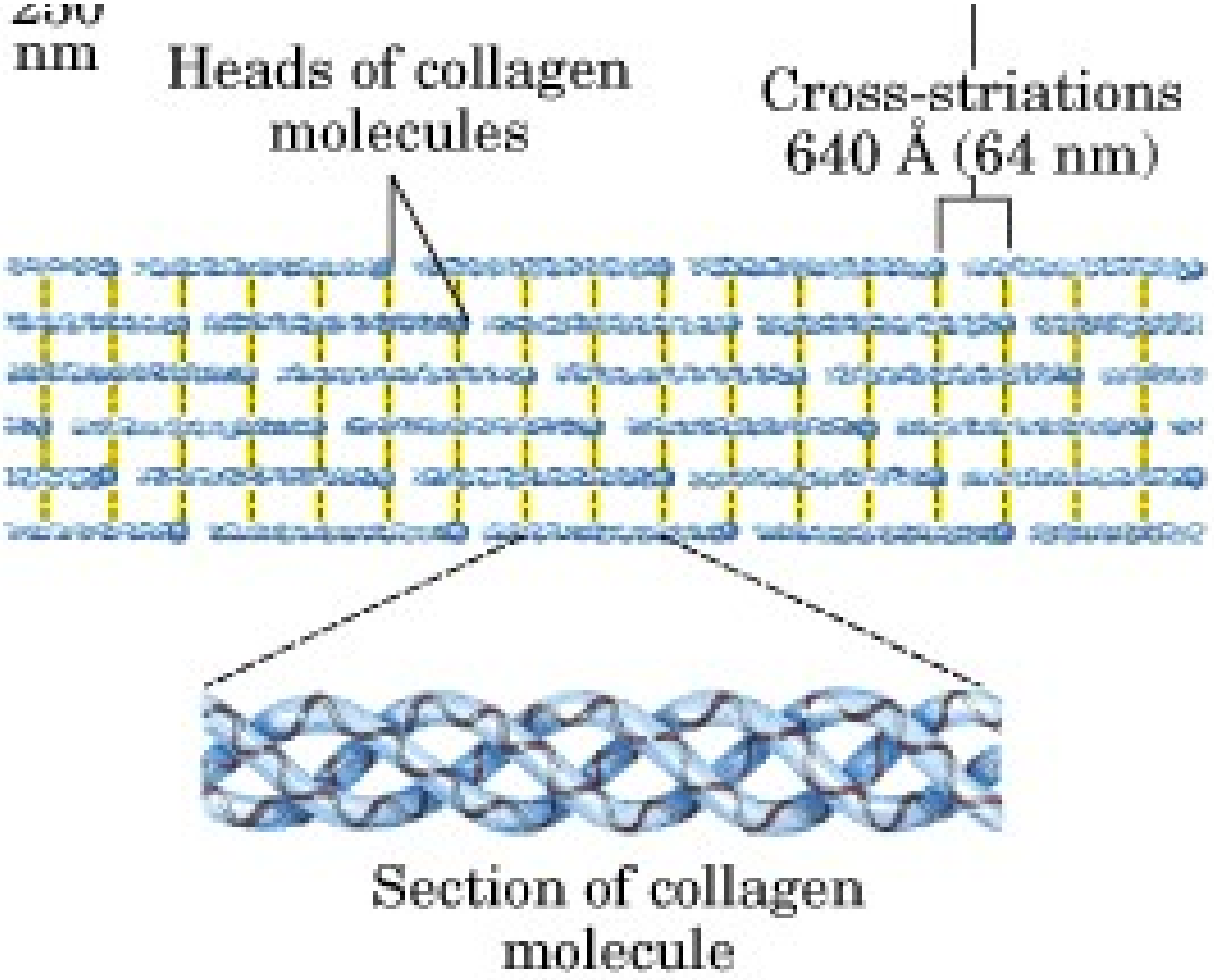
B5 Collagen（胶原蛋白）

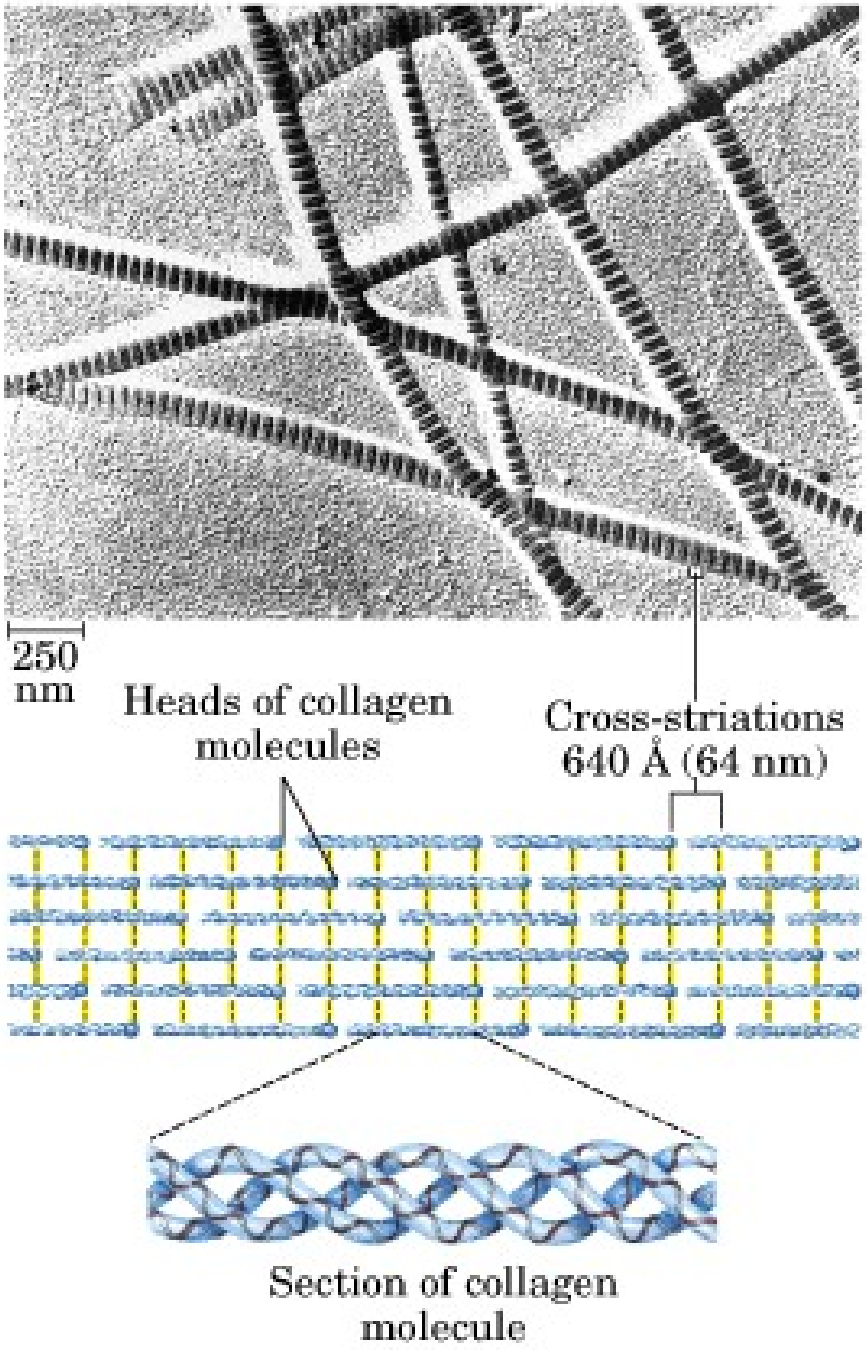
- Fibrous proteins have polypeptide chains arranged in long strands or sheets.
- Globular proteins have polypeptide chains folded into a spherical or globular shape.
- Collegen is the name given to a family of structurally related proteins that form strong insoluble fibers.



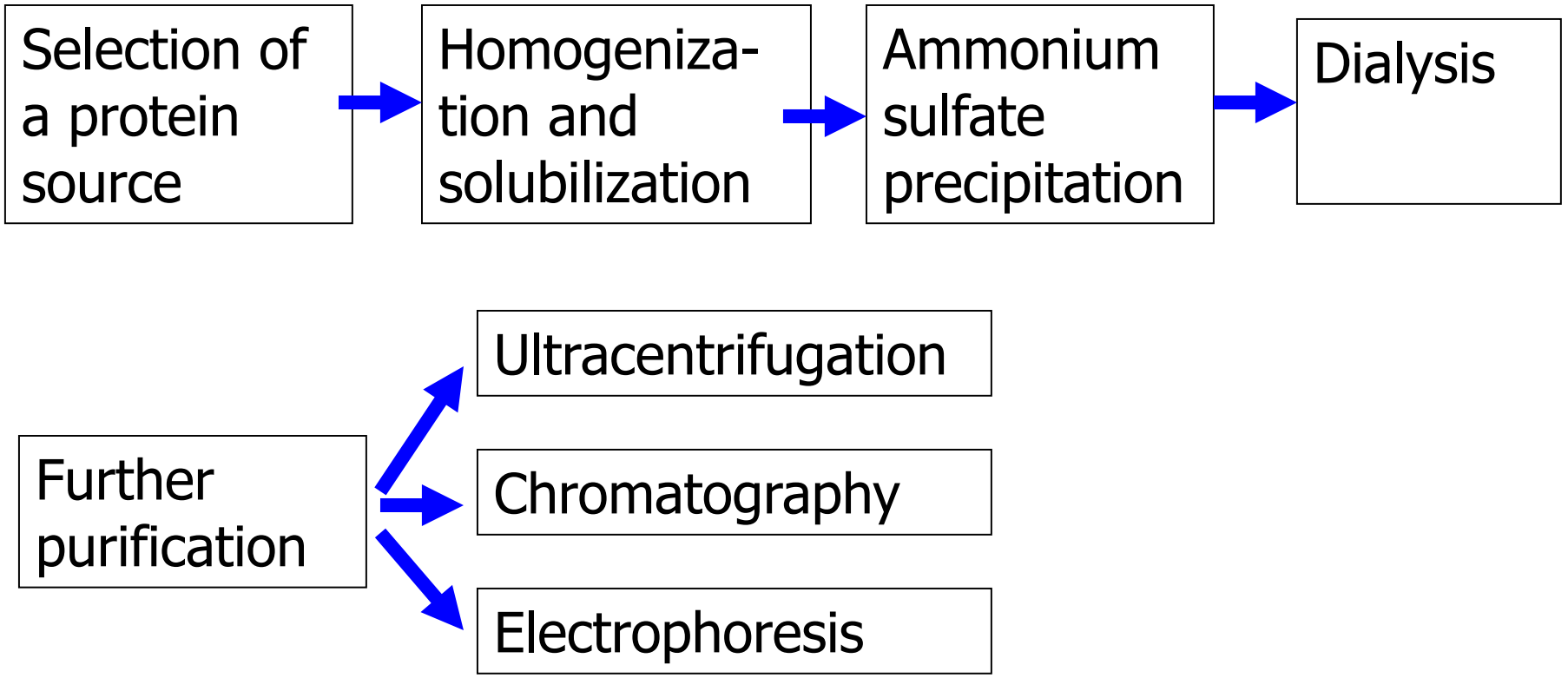
Collagen is the most abundant protein in mammals and is present in most organs of the body, where it serves to hold cells together in discrete units. It is also the major fibrous element of skin, bones, tendons(腱）, cartilage （软骨）, blood vessels and teeth.


(a)
(b)
(c)


nm
Heads of collagen molecules
Cross-striations 640 Å (64 nm)
Section of collagen molecule


250
nm
Heads of collagen
molecules
Cross-striations
640 Å (64 nm)
Section of collagen
molecule

B6 B7 B8 Protein Purification



salting in and salting out

Ionic strength also profoundly influences protein solubility.

Most globular proteins tend to become increasingly soluble as the ionic strength is raised. This phenomenon, called salting in, is attributed to the diminishment by the salt ions of electrostatic attractions between the protein molecules.

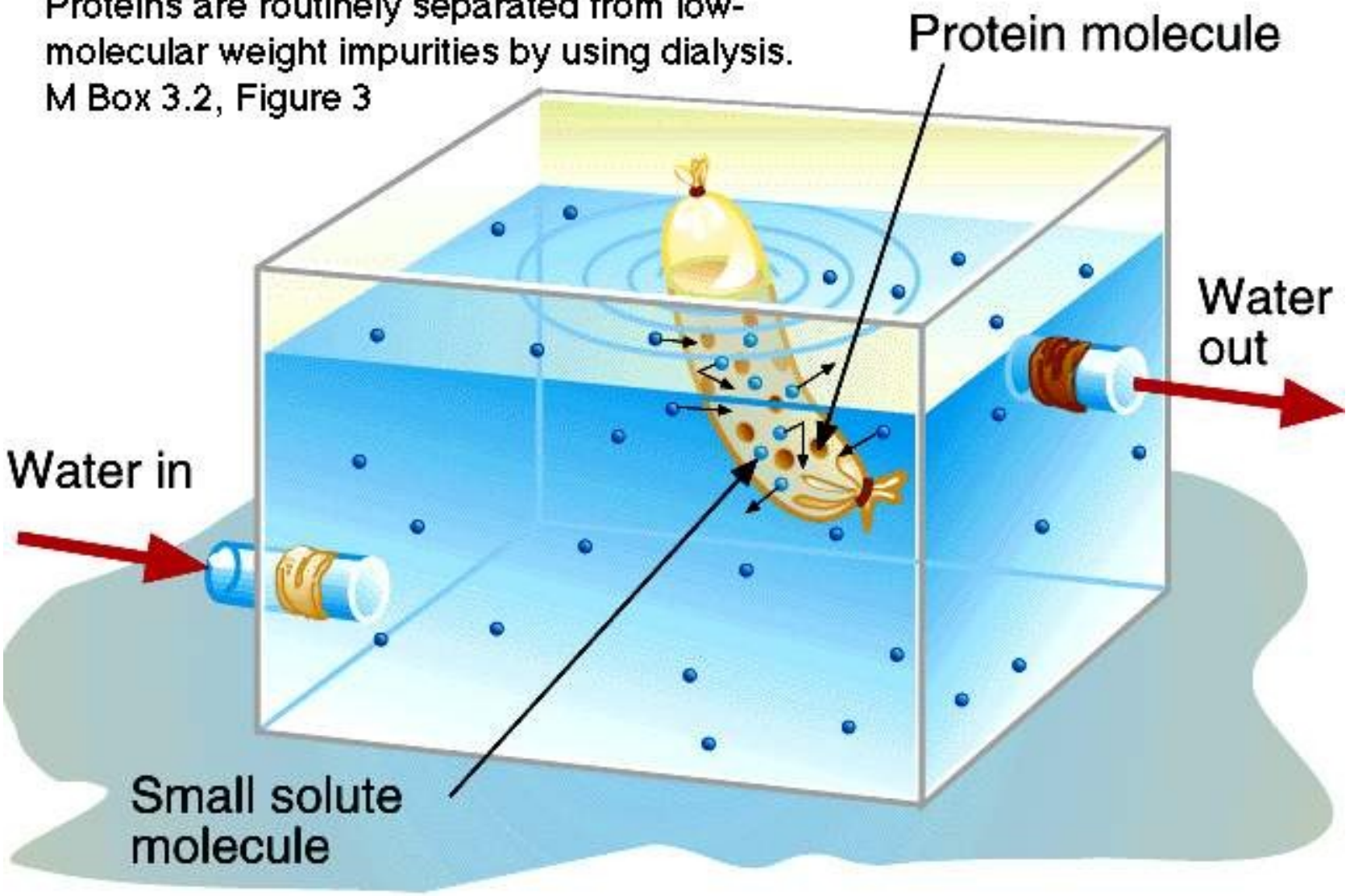
- **However, as the salt concentration reaches high levels (greater than 1 *M*), the effect may reverse so that the solubility of proteins is generally lowered, which called salting out. In such cases, the numerous salt ions begin to compete with the protein for waters of solvation, and, as they win out, the protein becomes insoluble.**

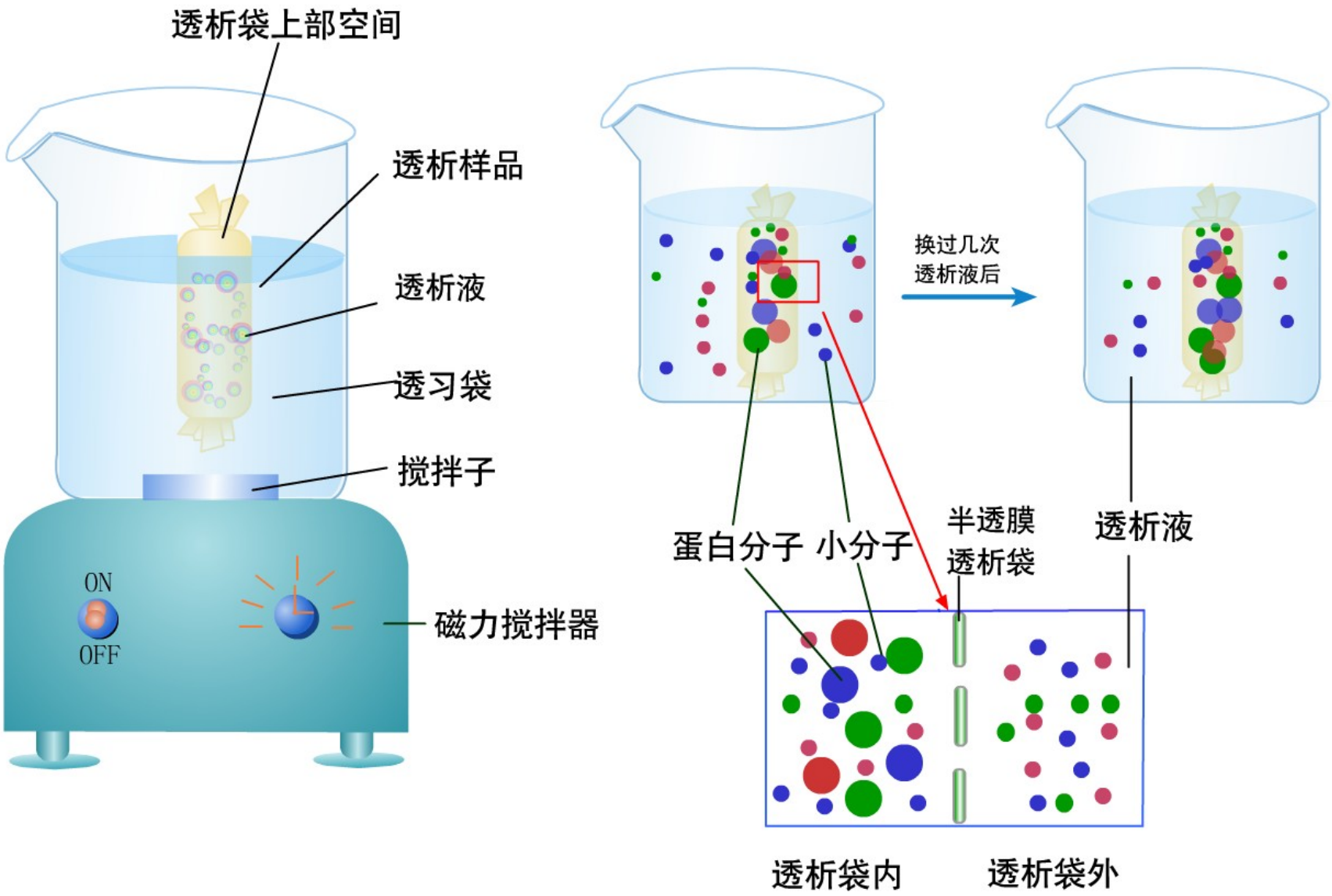
- **Ammonium sulfate is often used for this purpose.**

Dialysis

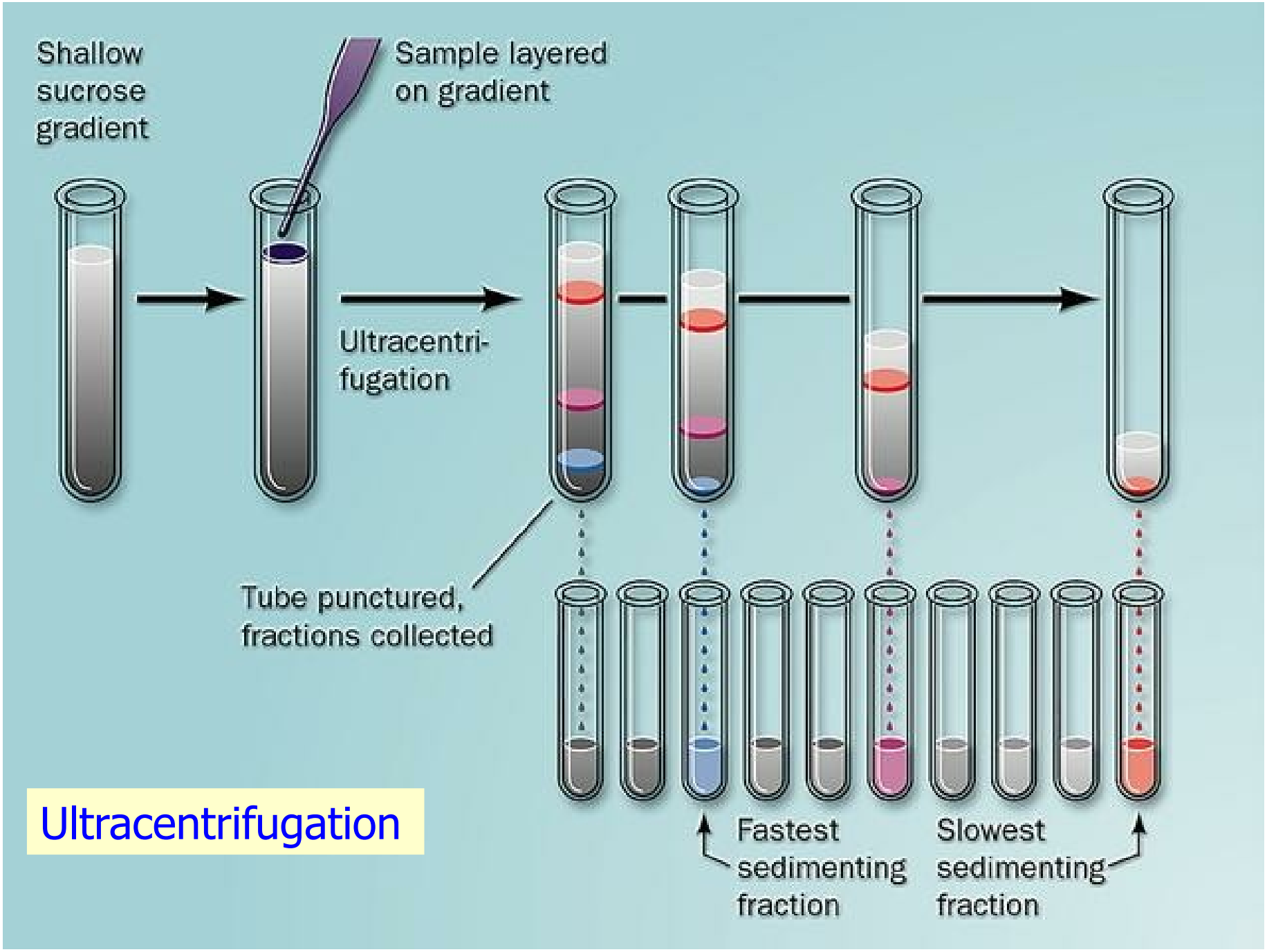
Biochemistry by McKee/McKee, ©1996 McGraw-Hill Higher Education

Proteins are routinely separated from low-molecular weight impurities by using dialysis.
M Box 3.2, Figure 3



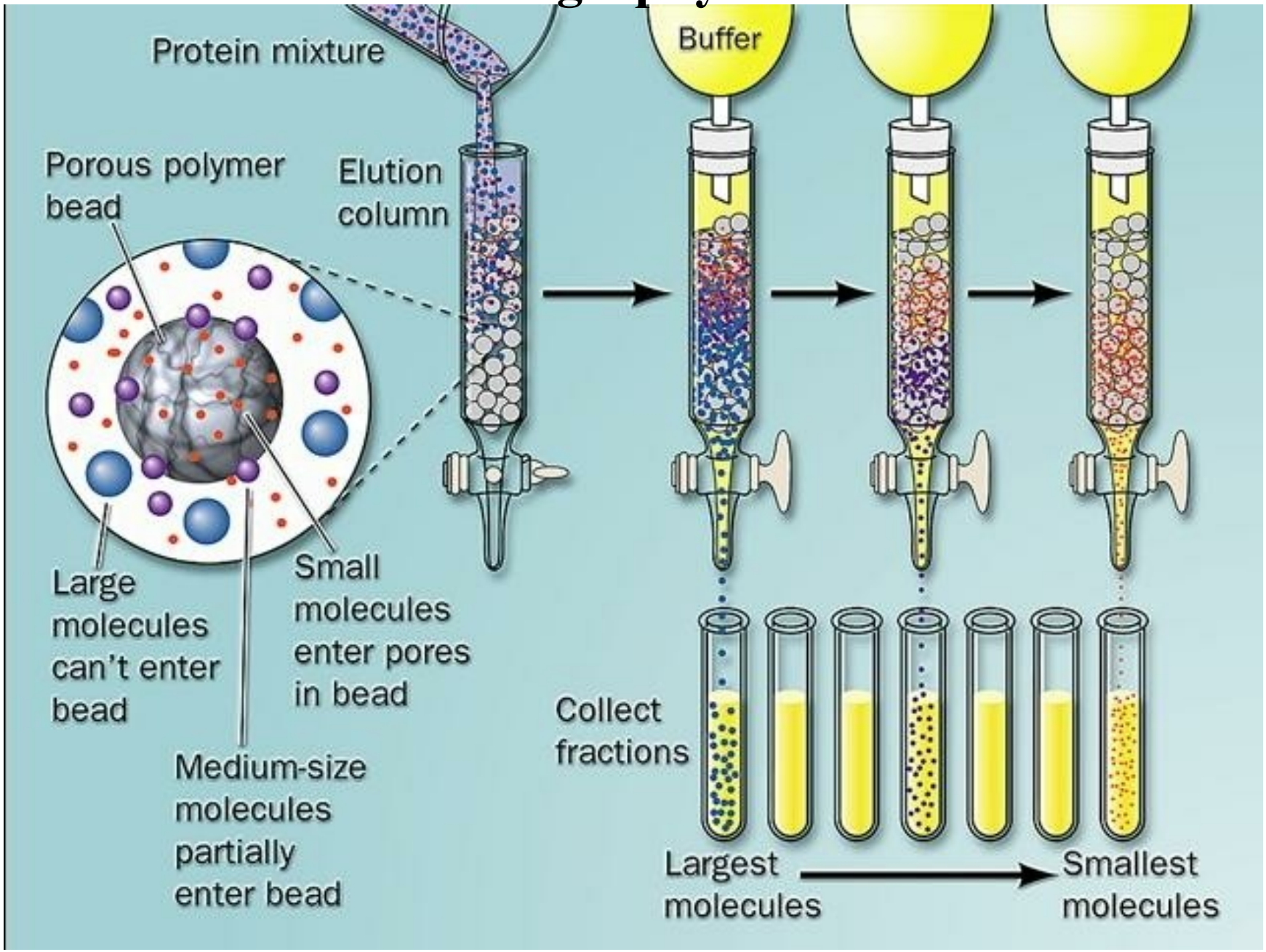


透析装置和透析过程示意图



Ultracentrifugation

Gel filtration chromatography

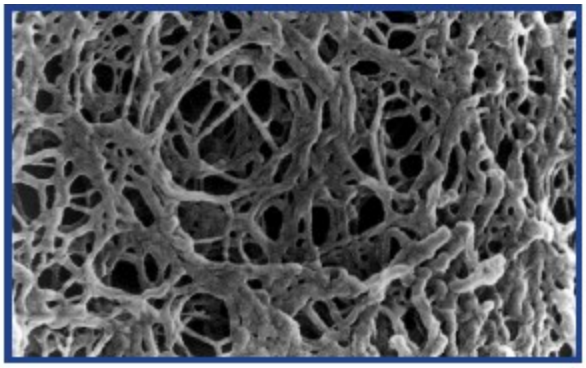


凝胶过滤层析

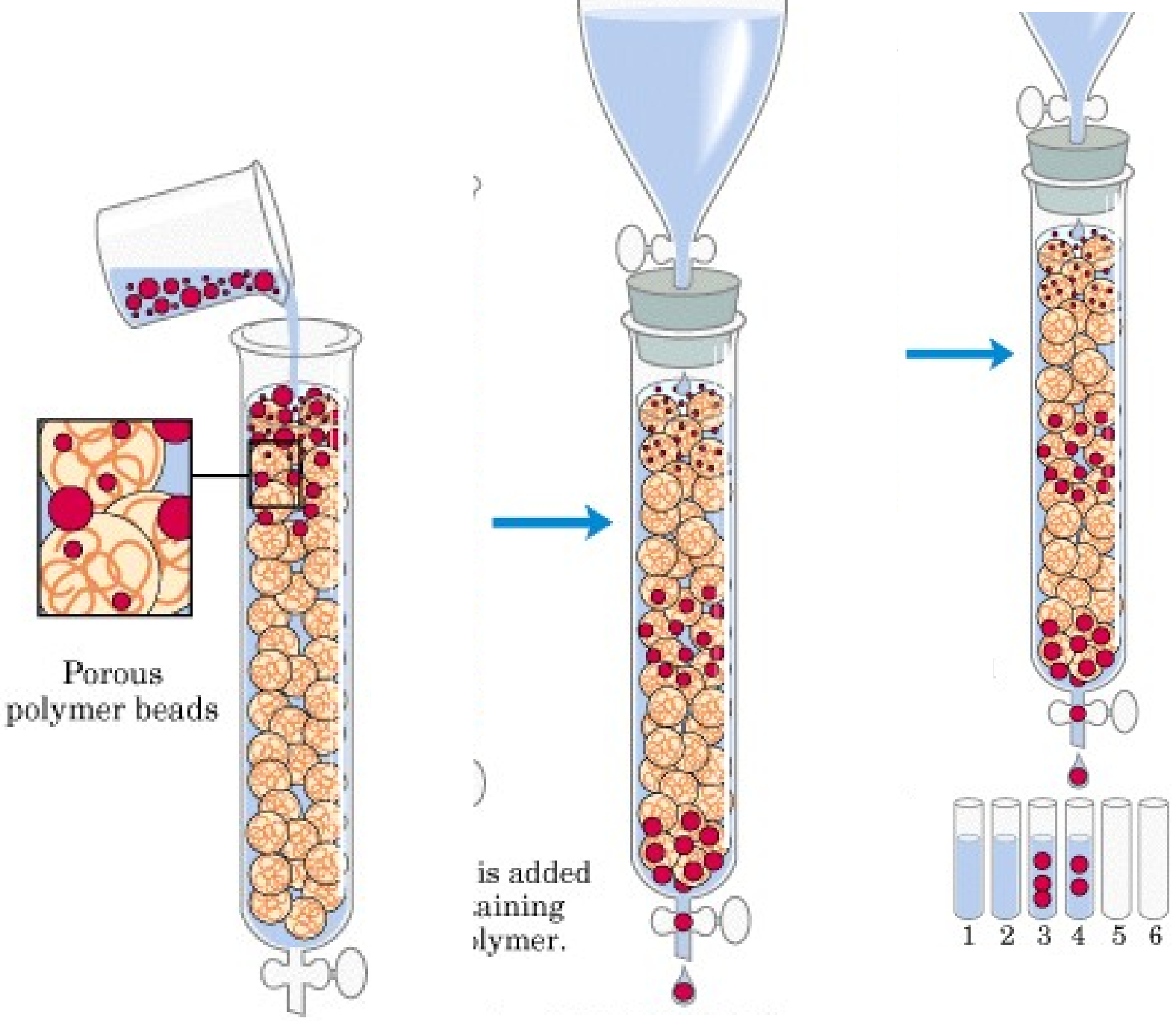
交联的葡聚糖凝胶



凝胶颗粒
网状结构



凝胶颗粒内部的


Porous
polymer beads
is added
aining
lymer.
1 2 3 4 5 6

分子越大越先排出

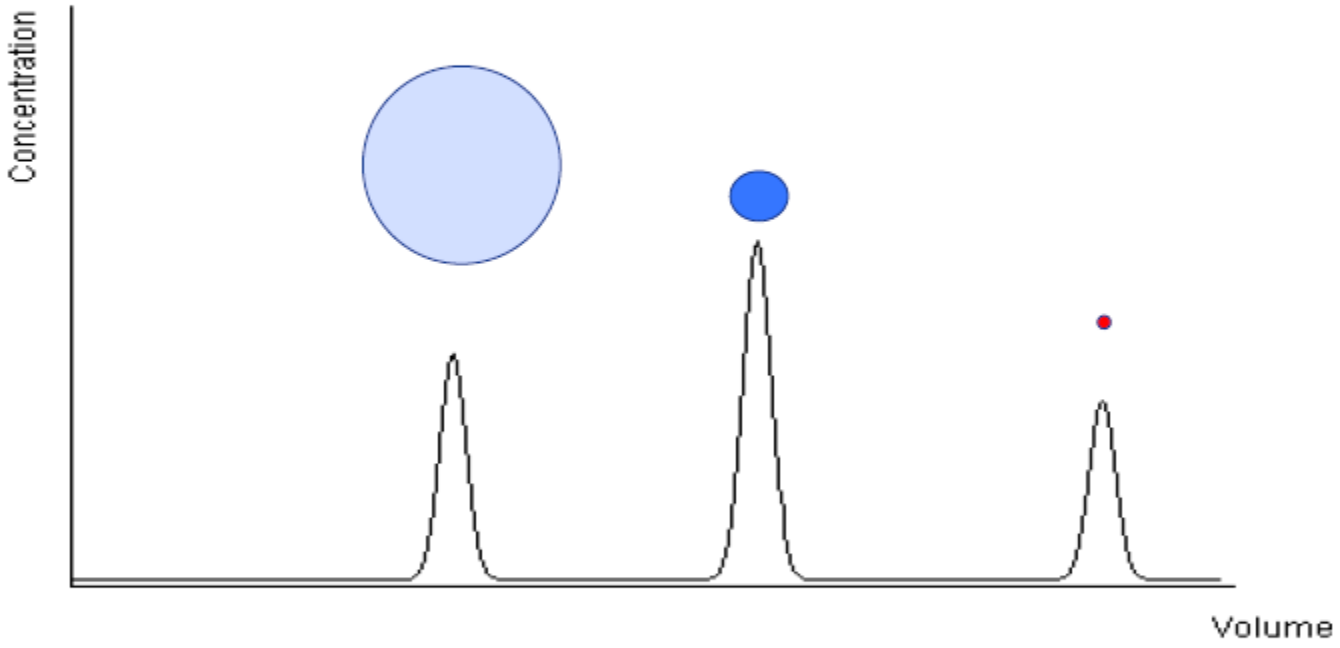
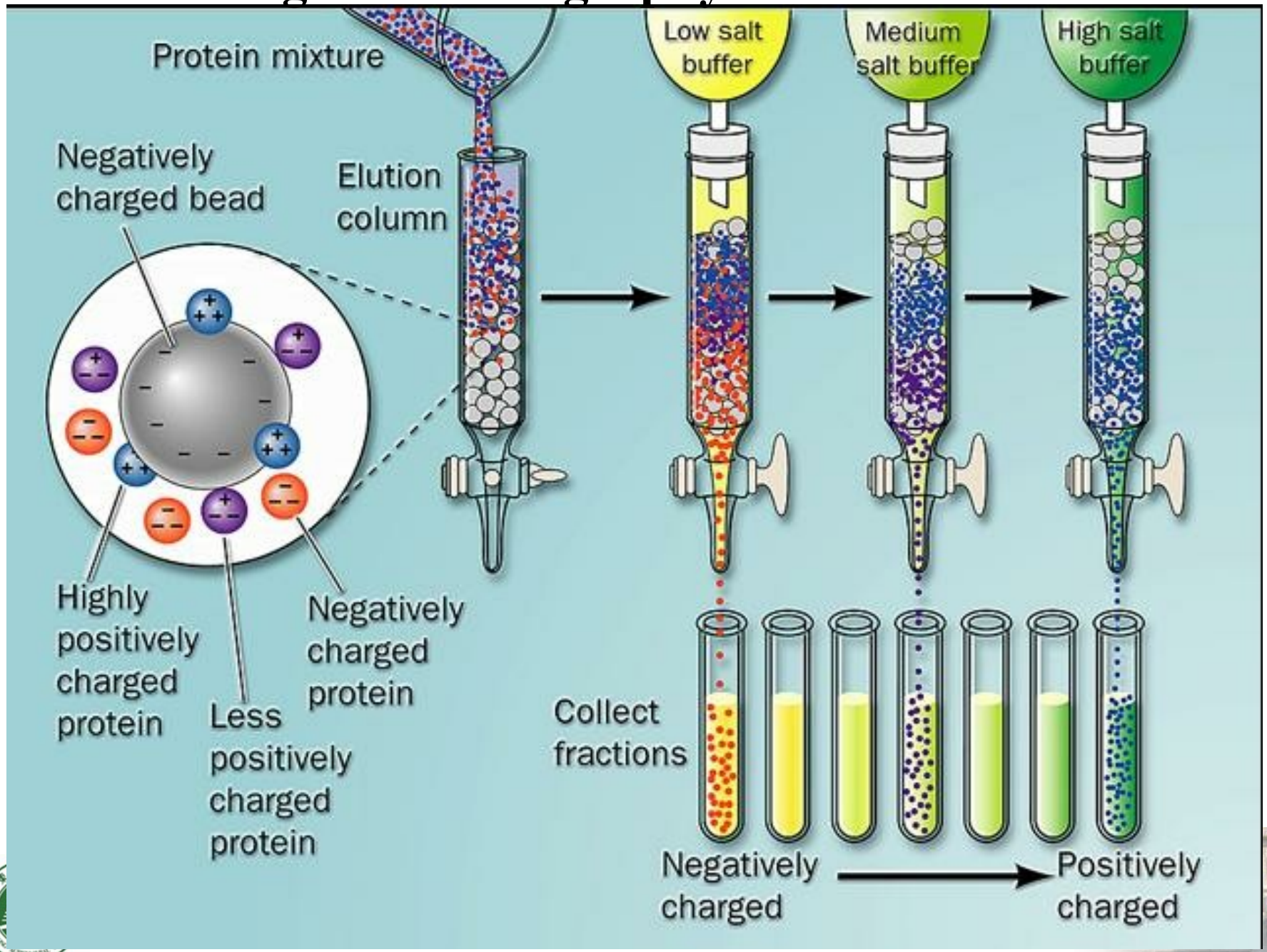


表 1　常用凝胶分离范围

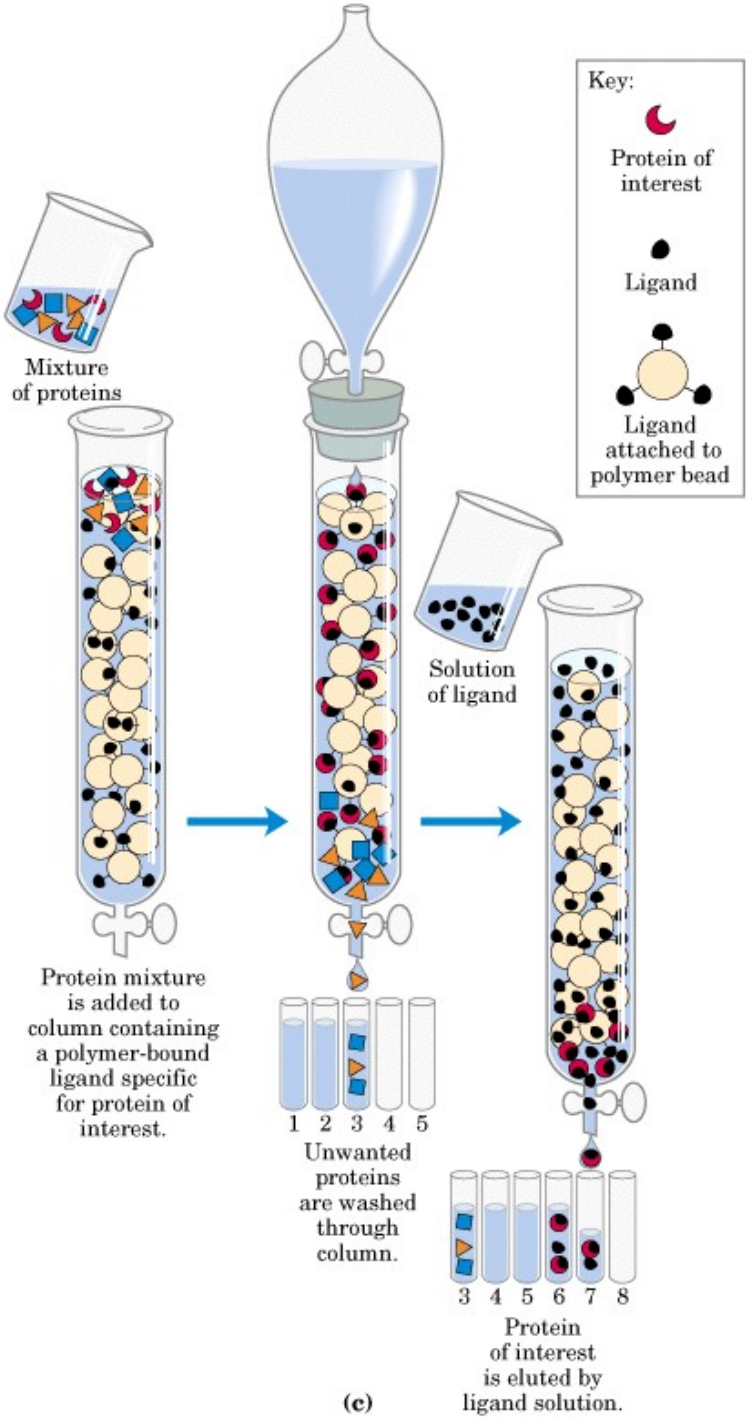
凝胶类型及规格		分离范围
葡聚糖凝胶（sephadex）	G-200	6×10^5
	G-100	1.5×10^5
	G-50	3×10^4
	G-25	5×10^3
聚丙烯酰胺凝胶（Bio-Gel）	P-300	5×10^5
	P-150	1.5×10^5
琼脂糖凝胶（Sepharose）	2B	1.5×10^5
	4B	1.5×10^5
	6B	1.5×10^5

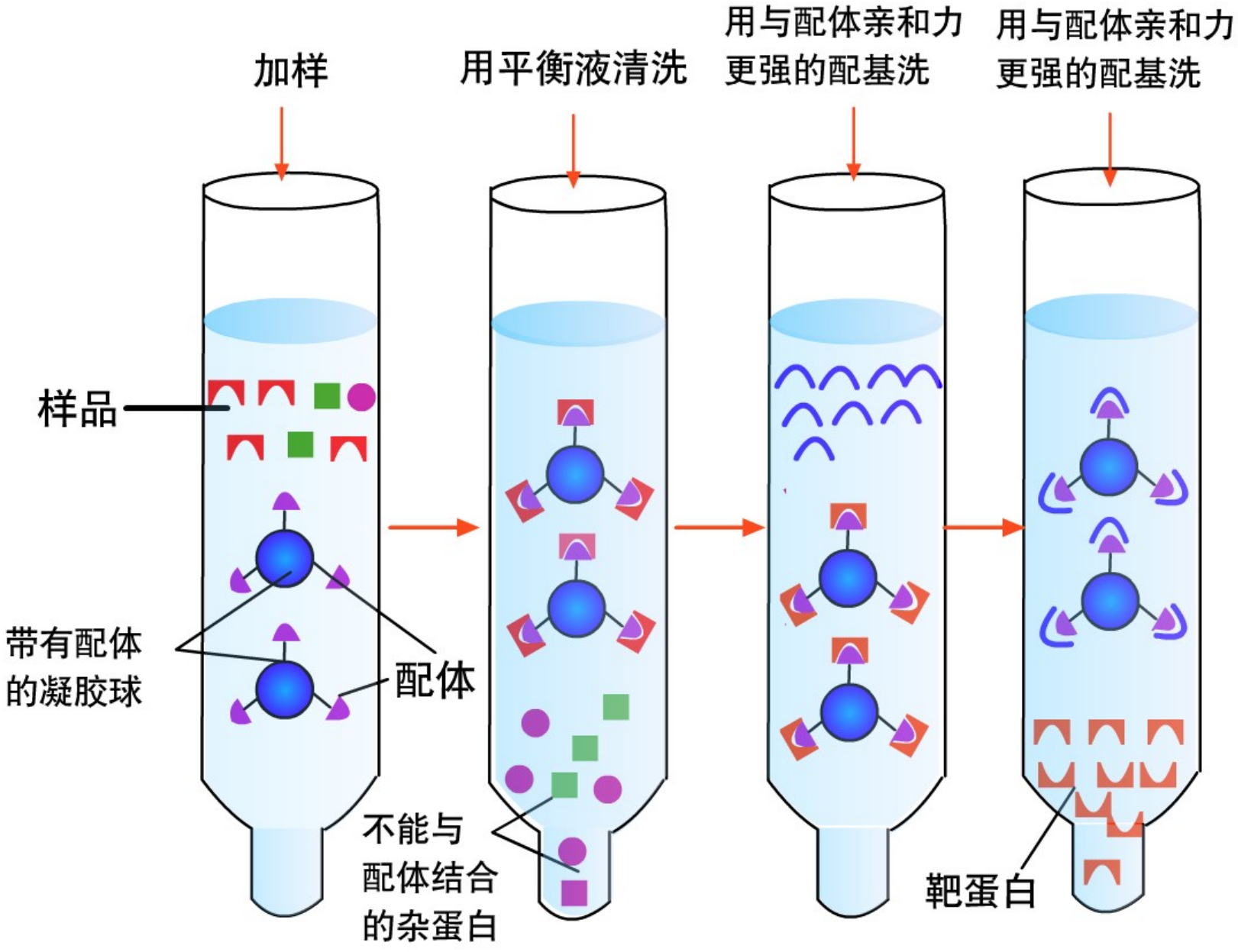


Ion exchange chromatography



Affinity chromatography



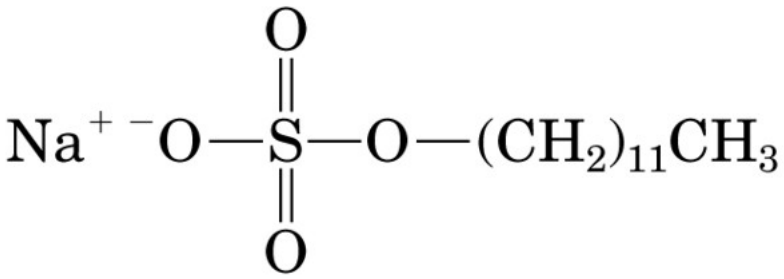


亲和层析
（Affinity chromatography）

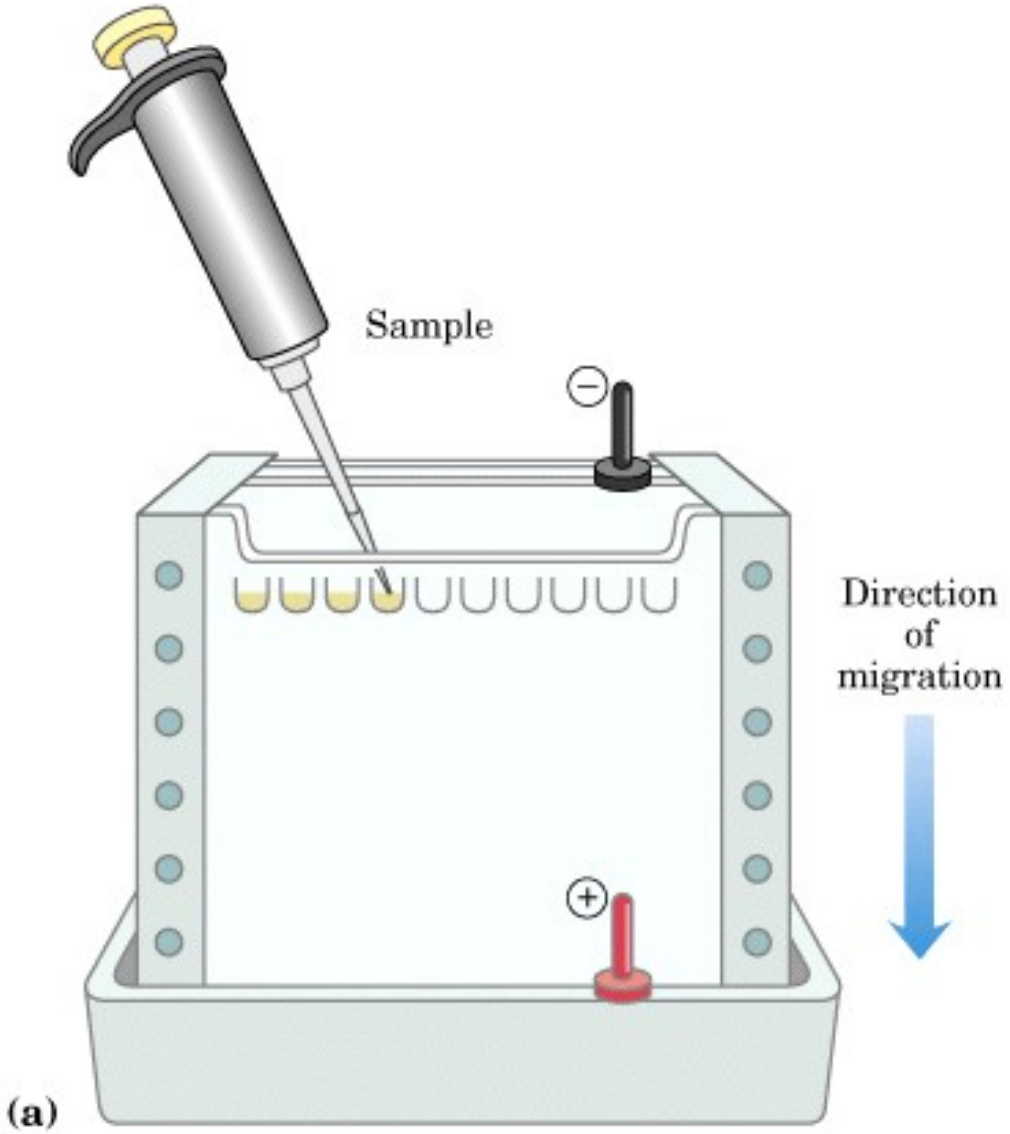
Electrophoresis

Native PAGE

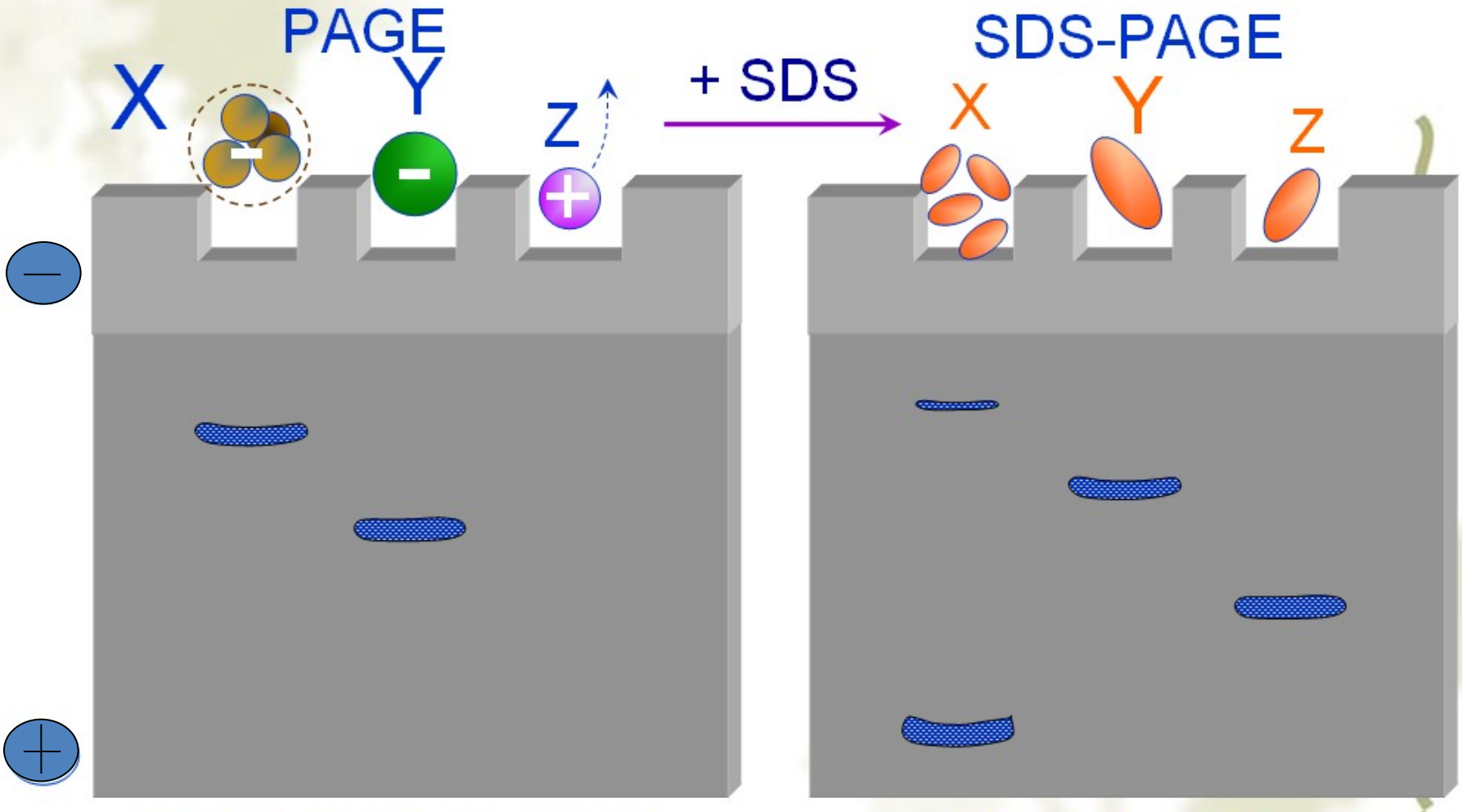
SDS-PAGE

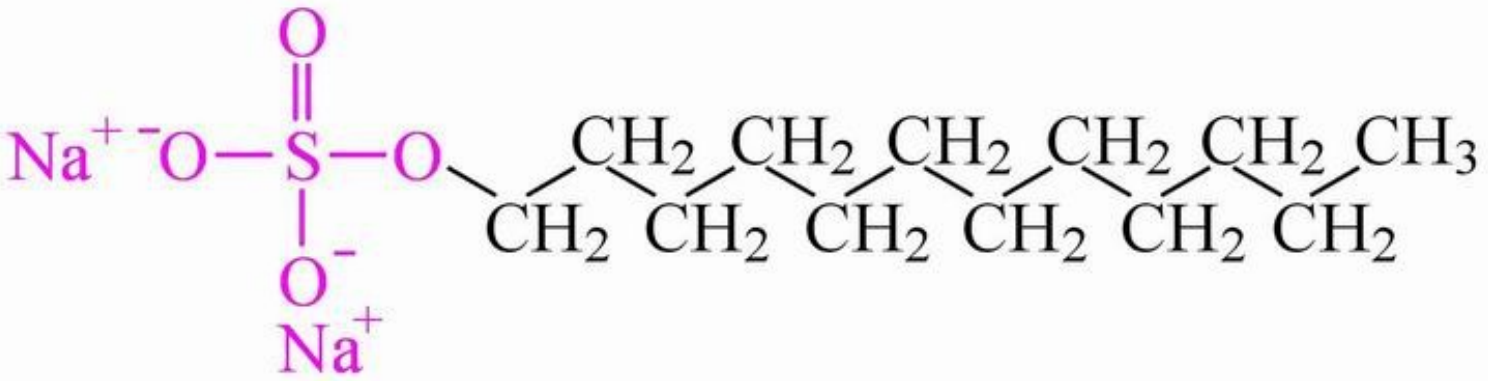


Sodium dodecyl sulfate
(SDS)

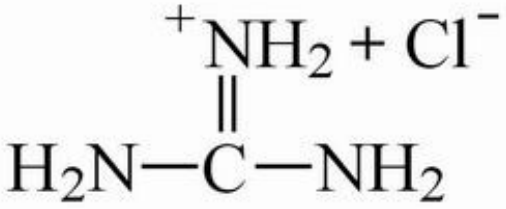


PAGE和SDS-PAGE的比较

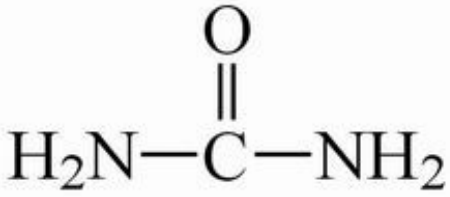




SDS



盐酸胍

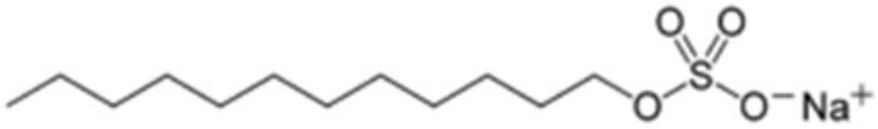


尿素

十二烷基磺酸钠(SDS)

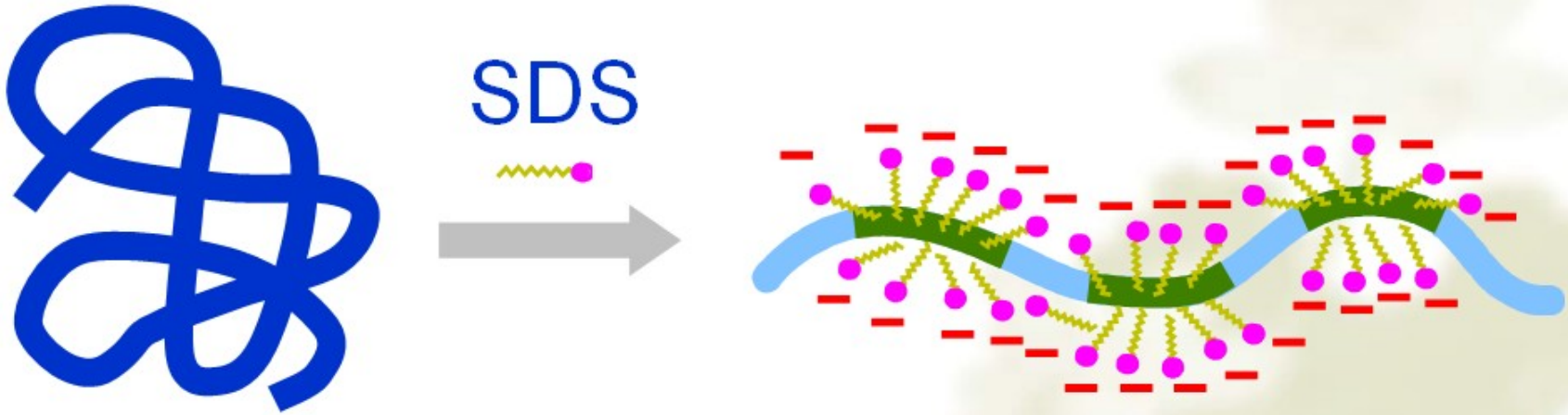
Sodium dodecyl sulphate

$CH_3\text{-}(CH_2)_{10}\text{-}CH_2OSO_3\text{--}Na^+$



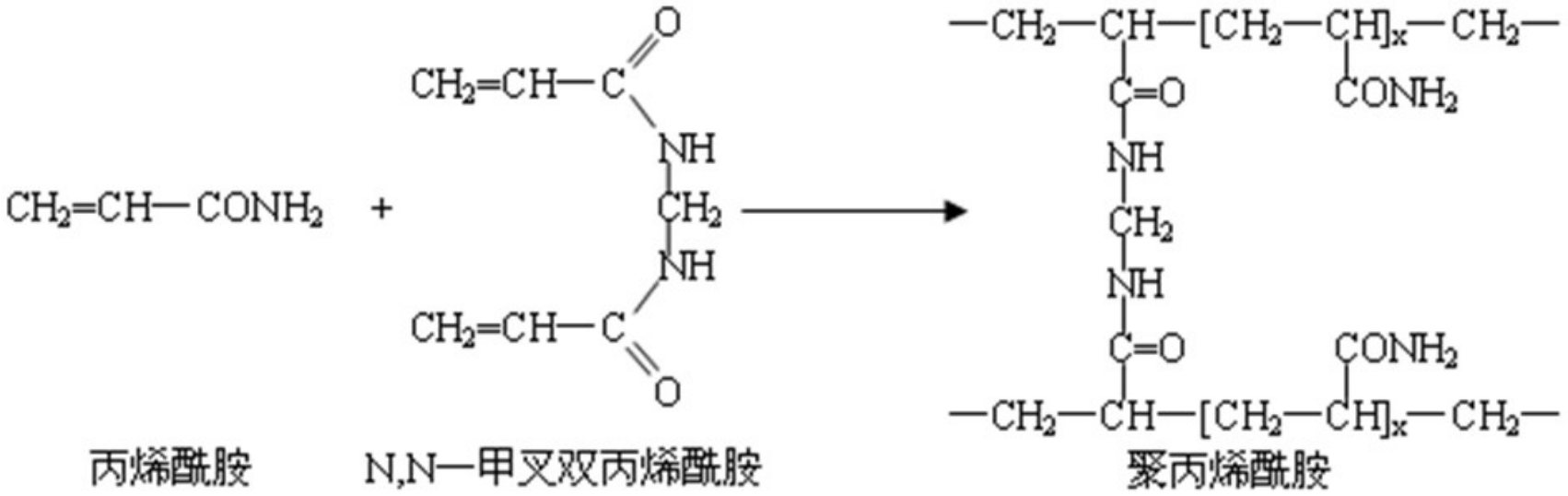
SDS的性质与作用

- ◆ 蛋白质变性
  - ✓ 0.1-1% SDS
  - ✓ 0.1 M 2-巯基乙醇
- ◆蛋白质变与SDS分子按比例结合
  - ✓ 1 ： 1.4

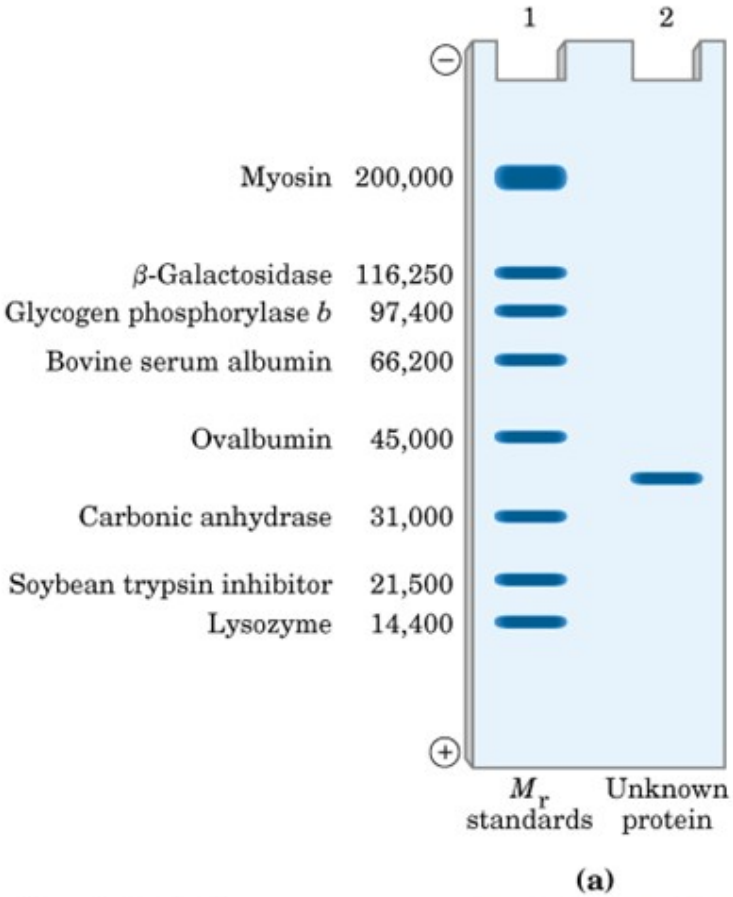


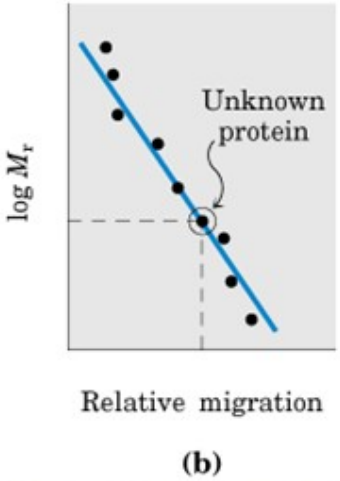
聚丙烯酰胺凝胶性质

- 聚丙烯酰胺凝胶(PAG)是由丙烯酰胺（Acr）和甲叉双丙烯酰胺（Bis）在催化剂的作用下，聚合交联而成的。



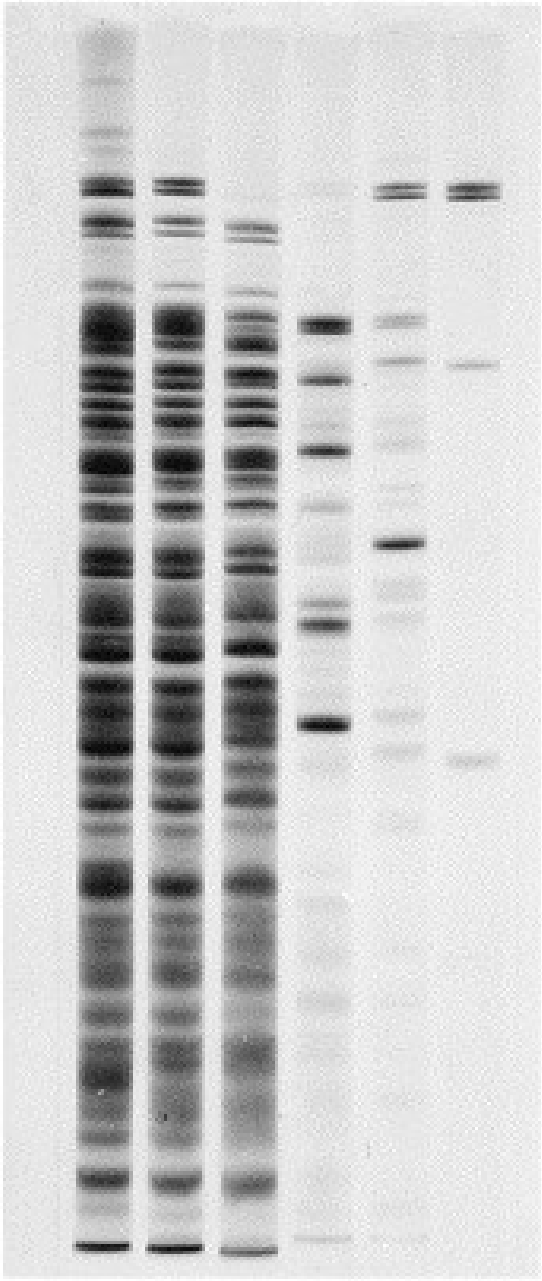
$$\text{相对迁移率} = \frac{\text{蛋白样品距加样端迁移距离（cm）}}{\text{溴酚蓝区带中心距加样端距离（cm）}}$$





以每个蛋白标准的分子量对数对它的相对迁移率作图得标准曲线，量出未知蛋白的迁移率即可测出其分于量。

标难曲线只对同一块凝胶上的样品的分子量测定才具有可靠性。

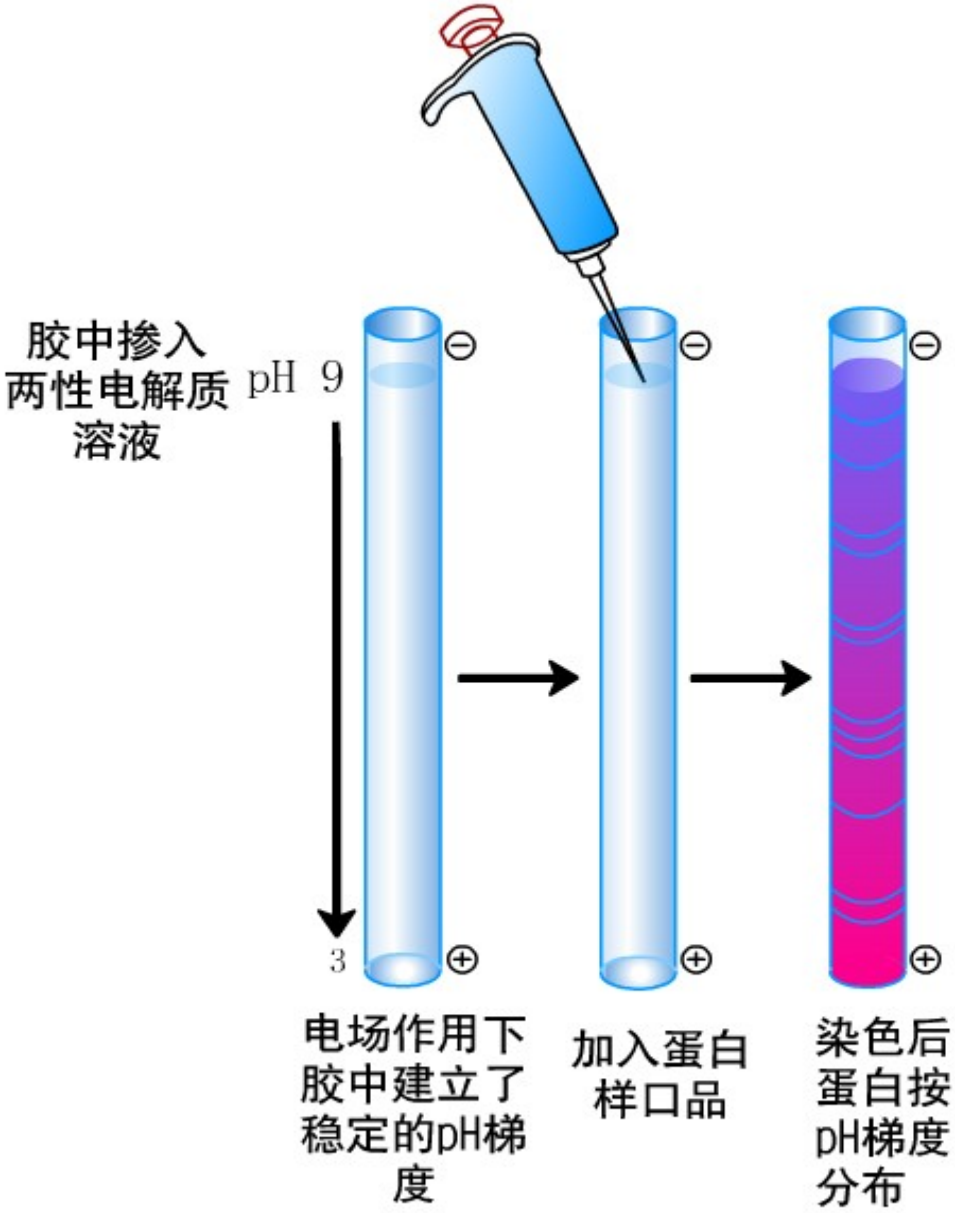


(b)

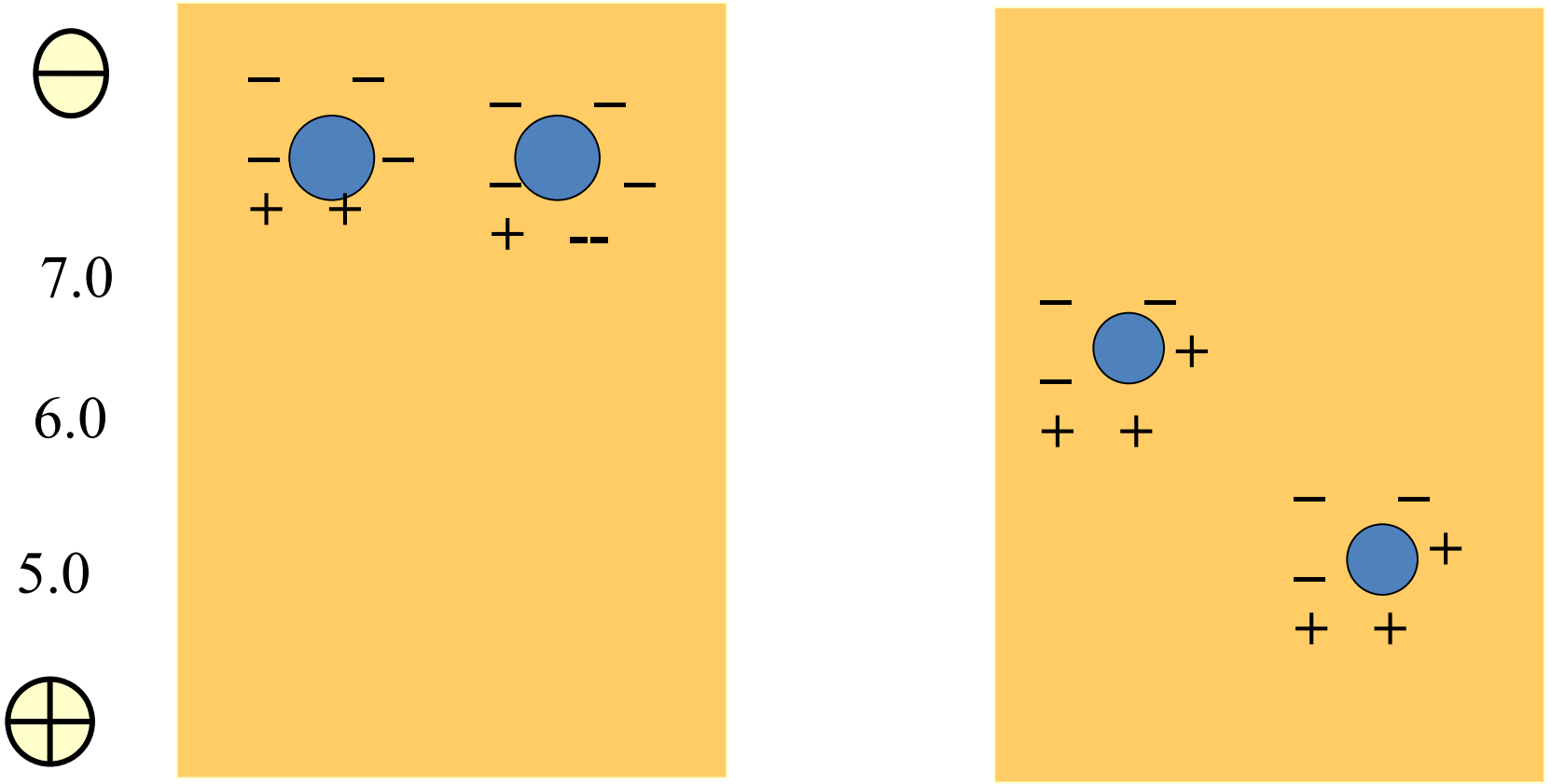
Isoelectric Focusing (IEF)

IEF can be described as electrophoresis in a pH gradient set up between a cathode and an anode with the cathode at higher pH than the anode. Proteins, being amphoteric species, are positively charged at pH values below their pI and negatively charged above their pI. It means that whenever a protein is in the gradient, it will migrate towards its pI.

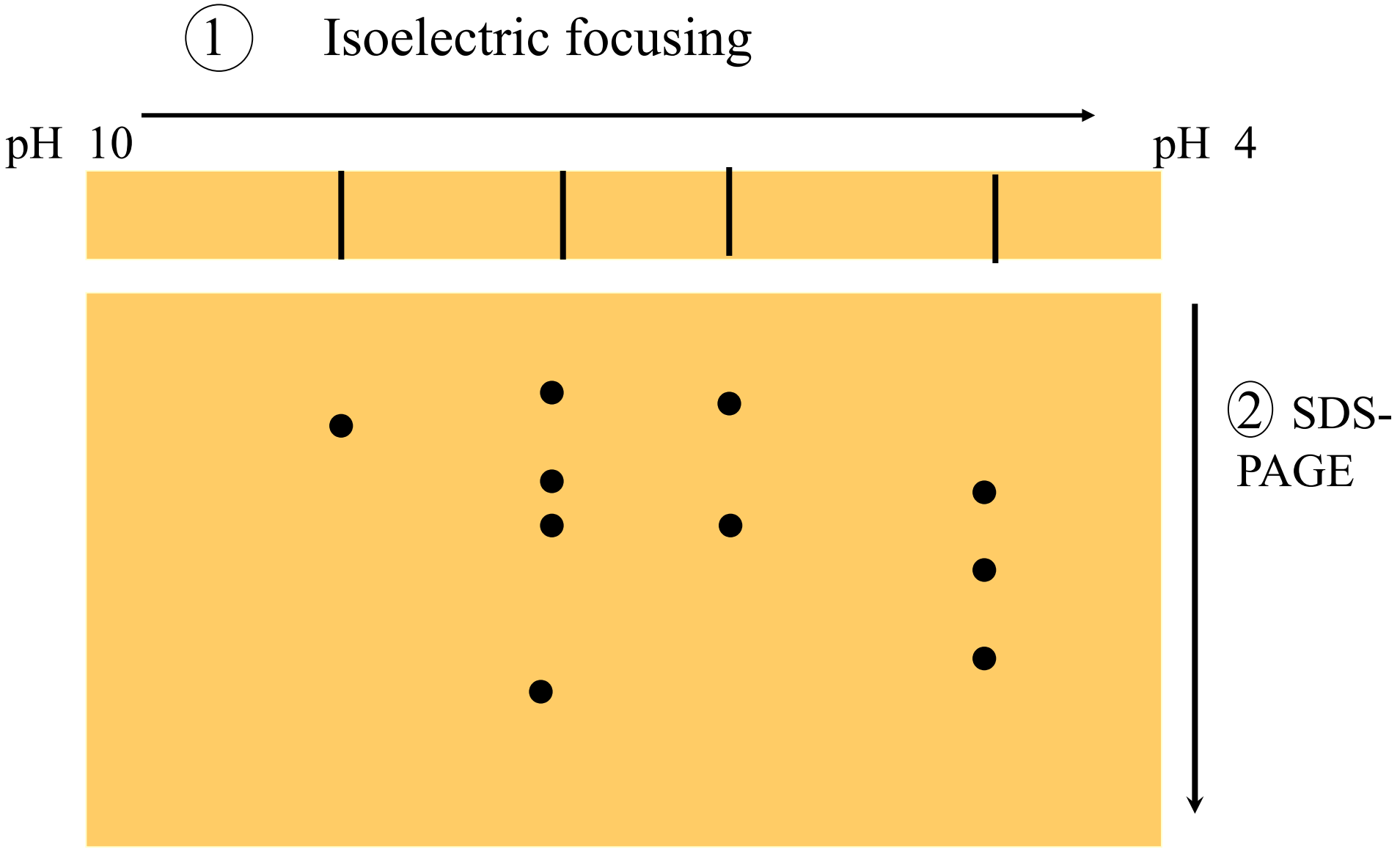



胶中掺入
两性电解质
溶液
pH 9
3
电场作用下
胶中建立了
稳定的pH梯
度
加入蛋白
样口品
染色后
蛋白按
pH梯度
分布

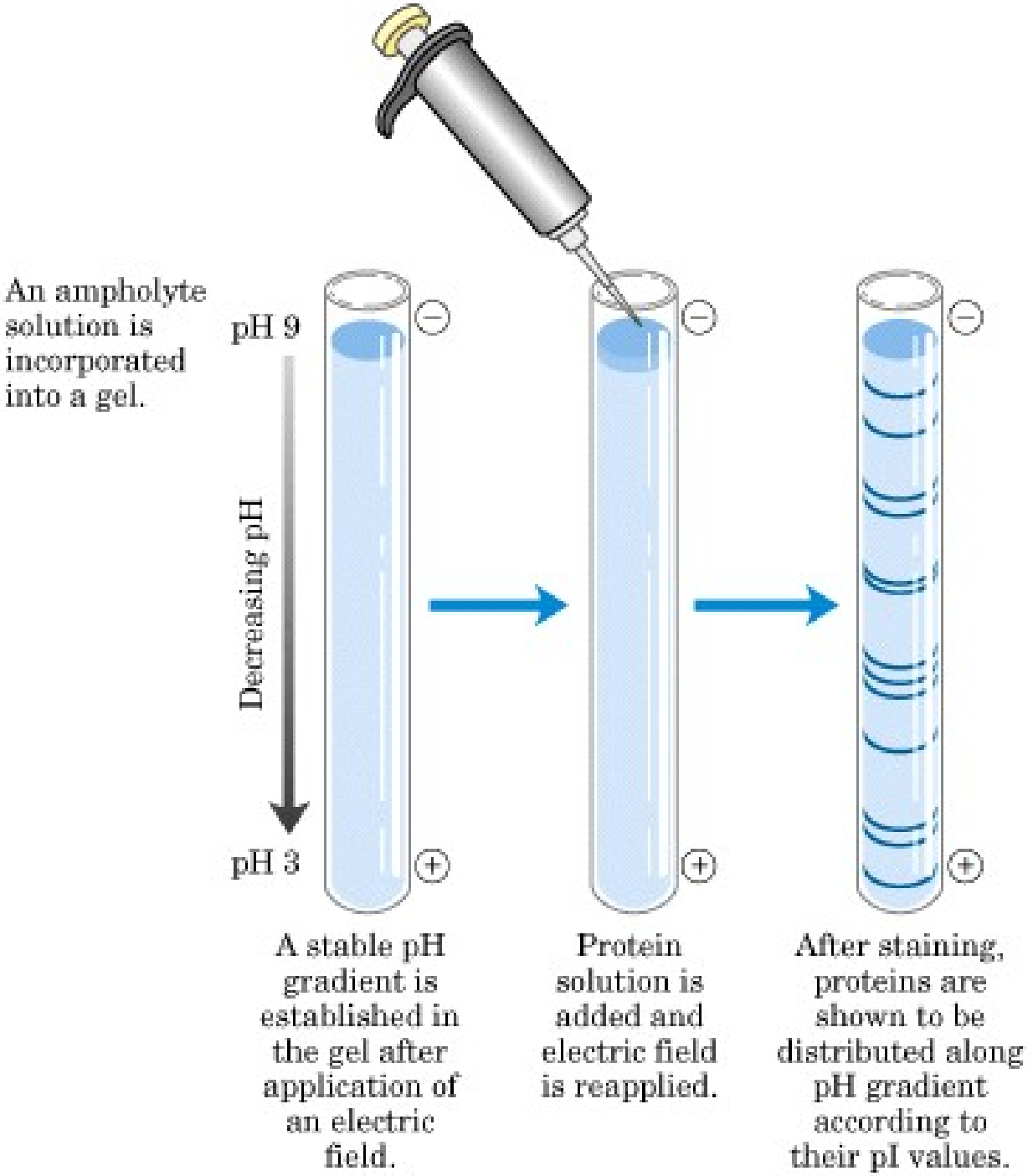
等电聚焦电泳

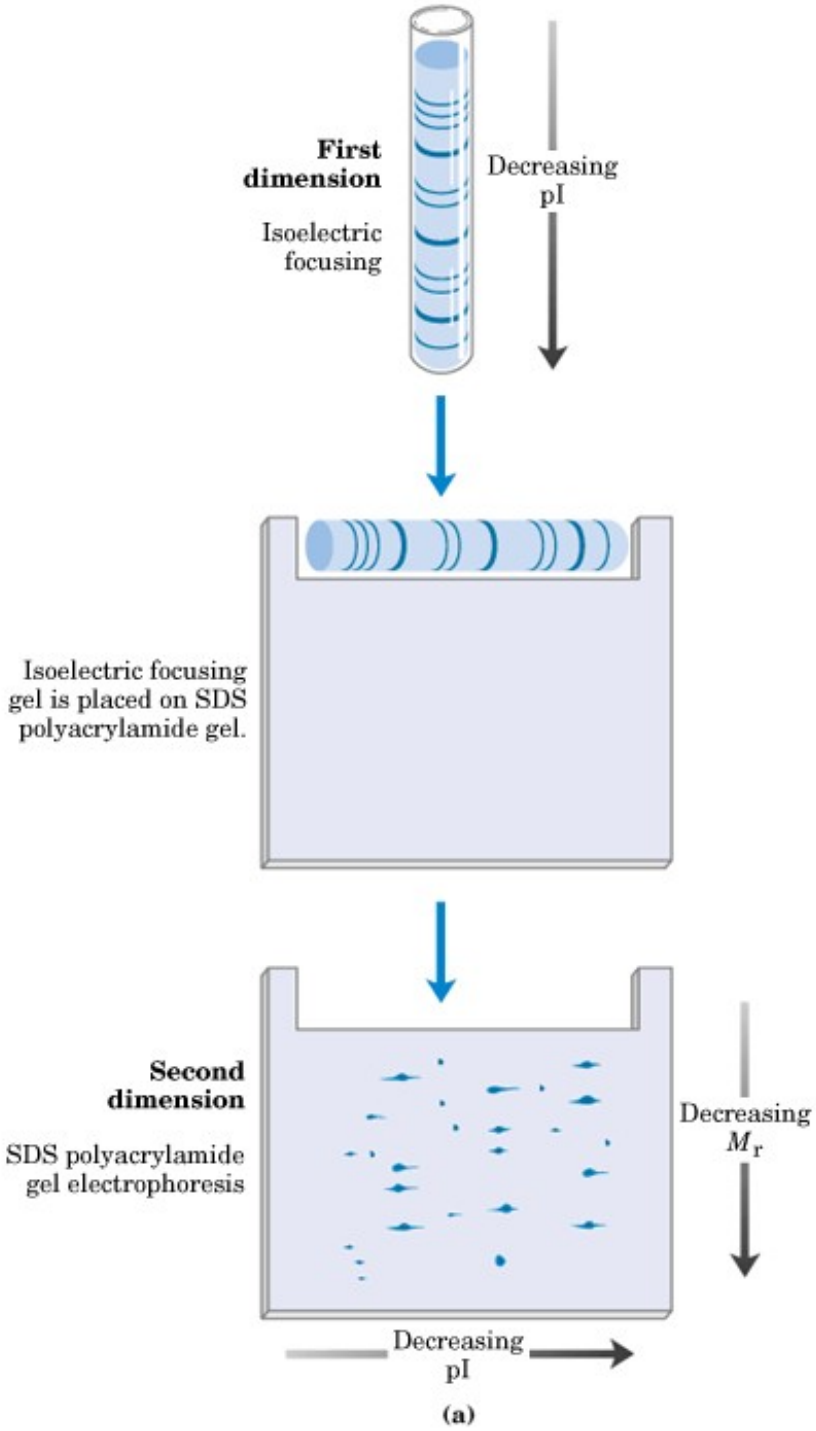


等电聚焦



Two-dimensional gel electrophoresis


An ampholyte solution is incorporated into a gel.
pH 9
Decreasing pH
pH 3
A stable pH gradient is established in the gel after application of an electric field.
Protein solution is added and electric field is reapplied.
After staining, proteins are shown to be distributed along pH gradient according to their pI values.


First dimension
Isoelectric focusing
Decreasing pI
Isoelectric focusing gel is placed on SDS polyacrylamide gel.
Second dimension
SDS polyacrylamide gel electrophoresis
Decreasing M_r
Decreasing pI

(a)



(b)

B9 Protein sequencing and peptide synthesis

- Sequencing strategy
- Amino acid composition analysis
- Edman degradation and Sanger reaction



Three steps

- Determine types and amounts of amino acids in polypeptide.
- Identify terminal residues of polypeptide.
- Identify amino-terminal residue; purify and recycle remaining peptide fragment through Edman process.





Polypeptide

Procedure

hydrolyze; separate amino acids

Result

A	5	H	2	R	1
C	2	I	3	S	2
D	4	K	2	T	1
E	2	L	2	V	1
F	1	M	2	Y	2
G	3	P	3		

Conclusion

Polypeptide has 38 amino acid residues. Trypsin will cleave three times (at one R (Arg) and two K (Lys)) to give four fragments. Cyanogen bromide will cleave at two M (Met) to give three fragments.

react with FDNB; hydrolyze; separate amino acids

Result

2,4-Dinitrophenylglutamate detected

Conclusion

E (Glu) is amino-terminal residue.

reduce disulfide bonds (if present)



cleave with trypsin separate fragments; sequence by Edman degradation

Result

T-1 GASMALIK

T-2 EGAAYHDFEPIDPR

T-3 DCVHSD

T-4 YLIACGPMTK

Conclusion

T-2 placed at amino terminus because it begins with E (Glu).

T-3 placed at carboxyl terminus because it does not end with R (Arg) or K (Lys).

cleave with cyanogen bromide; separate fragments; sequence by Edman degradation

Result

C-1 EGAAYHDFEPIDPRGASM

C-2 TKDCVHSD

C-3 ALIKYLIACGPM

Conclusion

C-3 overlaps with T-1 and T-4, allowing them to be ordered.

establish sequence



Amino terminus

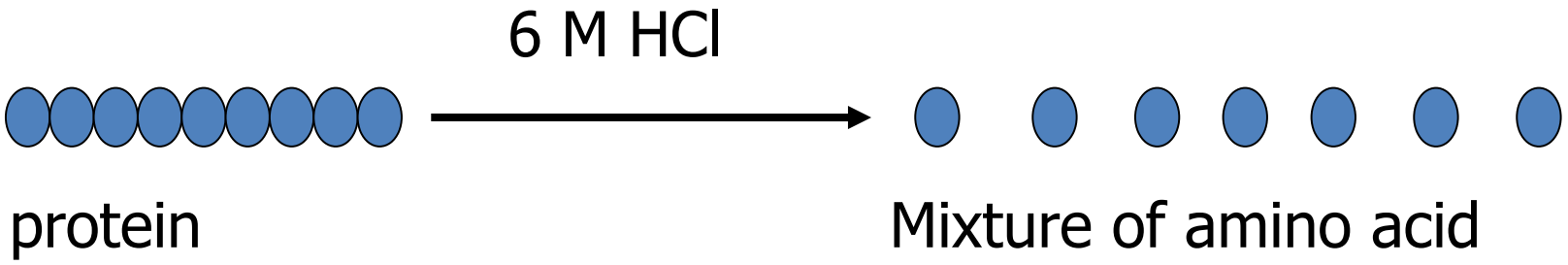
Carboxyl terminus

Cut the peptide into fragments

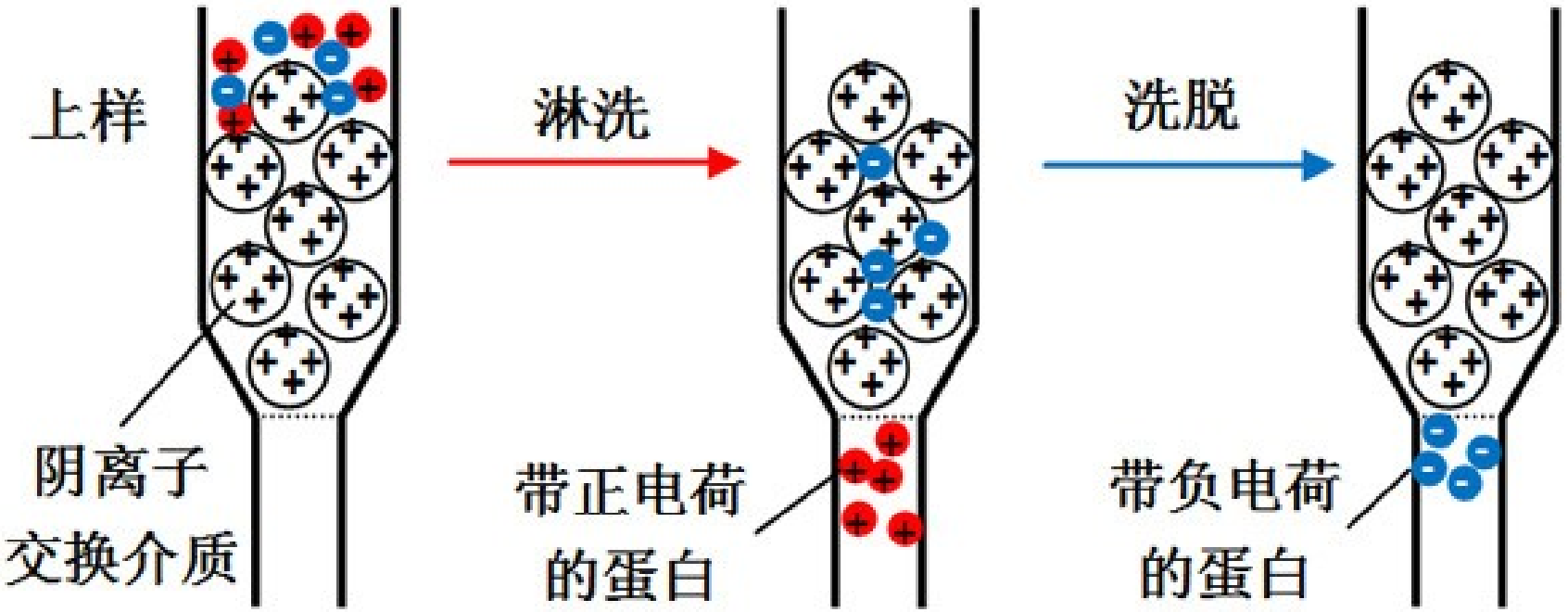
- 胃蛋白酶 ***Pepsin*** ： 芳香族 aa － NH_2 形成的肽键
- 胰蛋白酶 ***trypsin*** ： Lys 或 Arg － COOH （碱性，高度专一）形成的肽键； Pro 抑制水解
- 胰凝乳蛋白酶 ***Chymotrypsin*** ： 芳香族 aa － COOH 形成的肽键 , Leu 、 Met 、 His 水解速度次之 ,Pro 抑制水解。
- 溴化氰 ***cyanogen bromide*** ： Met （高度专一）

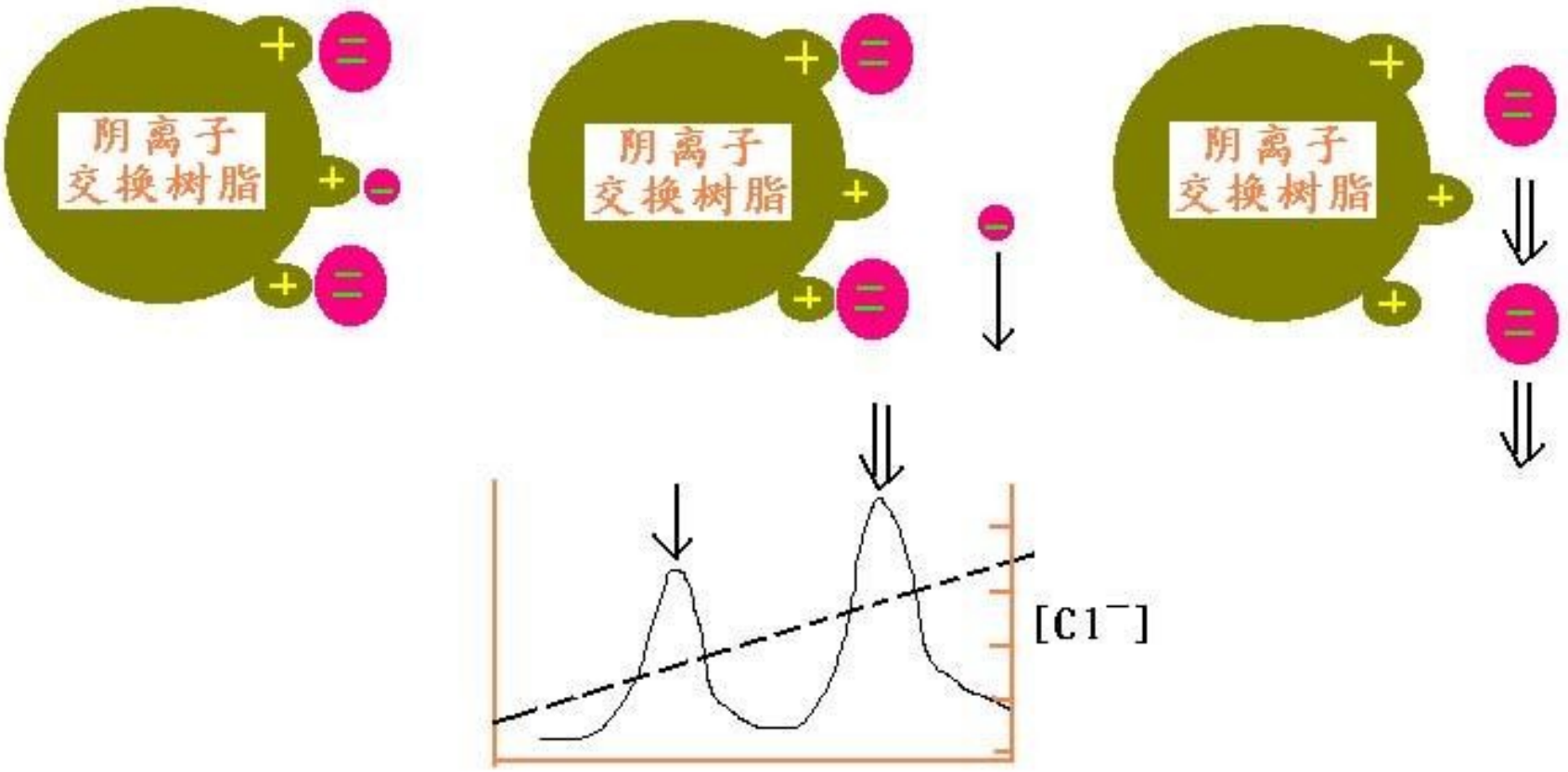
Step 1

Amino acid composition analysis



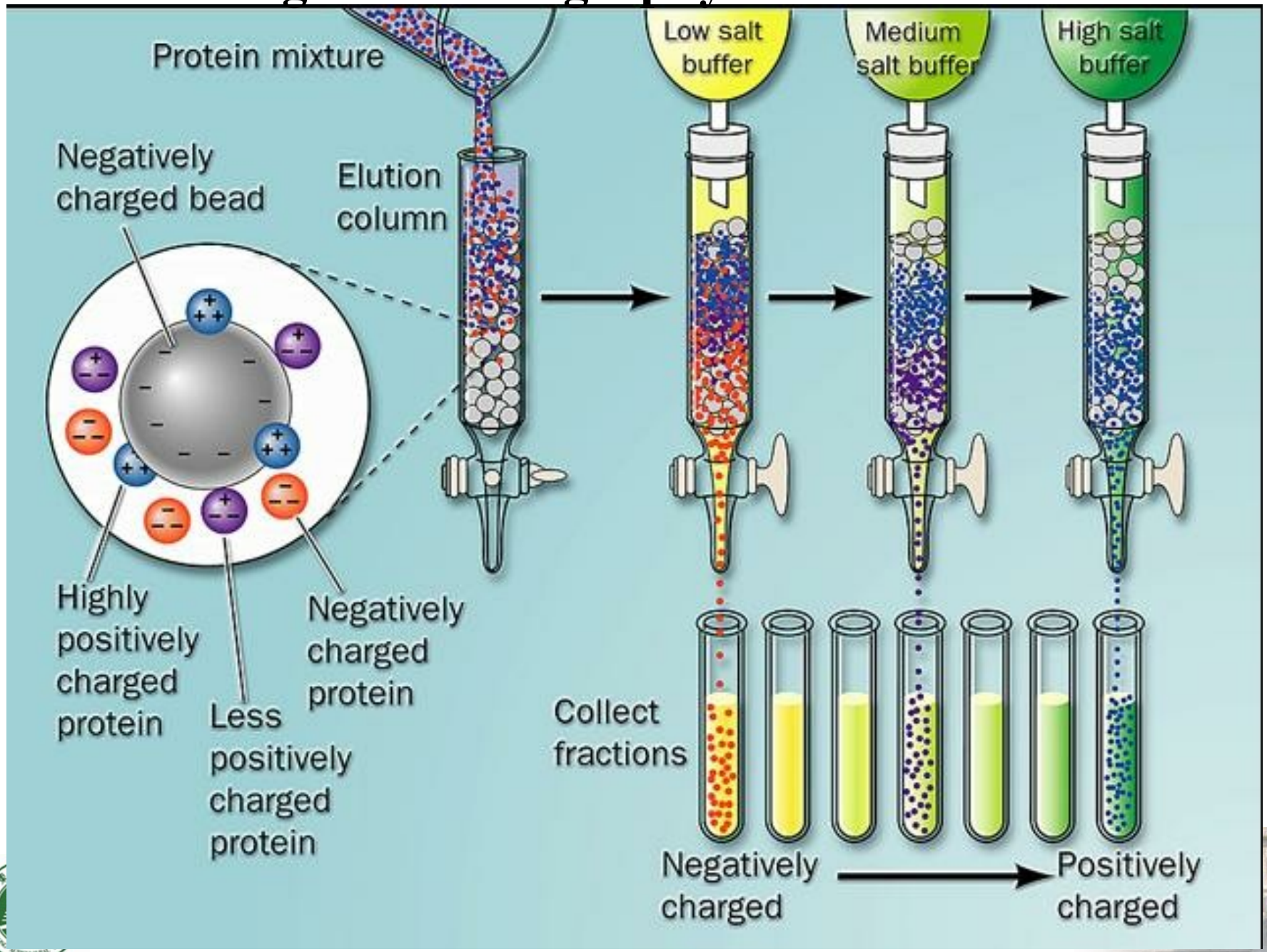
The purified protein sample is hydrolyzed into amino acid by heating it in 6 M HCl at 110 degree for 24 h. The resulting mixture of amino acid is subjected to ion exchange chromatography on a column to separate out the 20 standard amino acids. The separated amino acids are then detected and quantified by reacting them with ninhydrin.

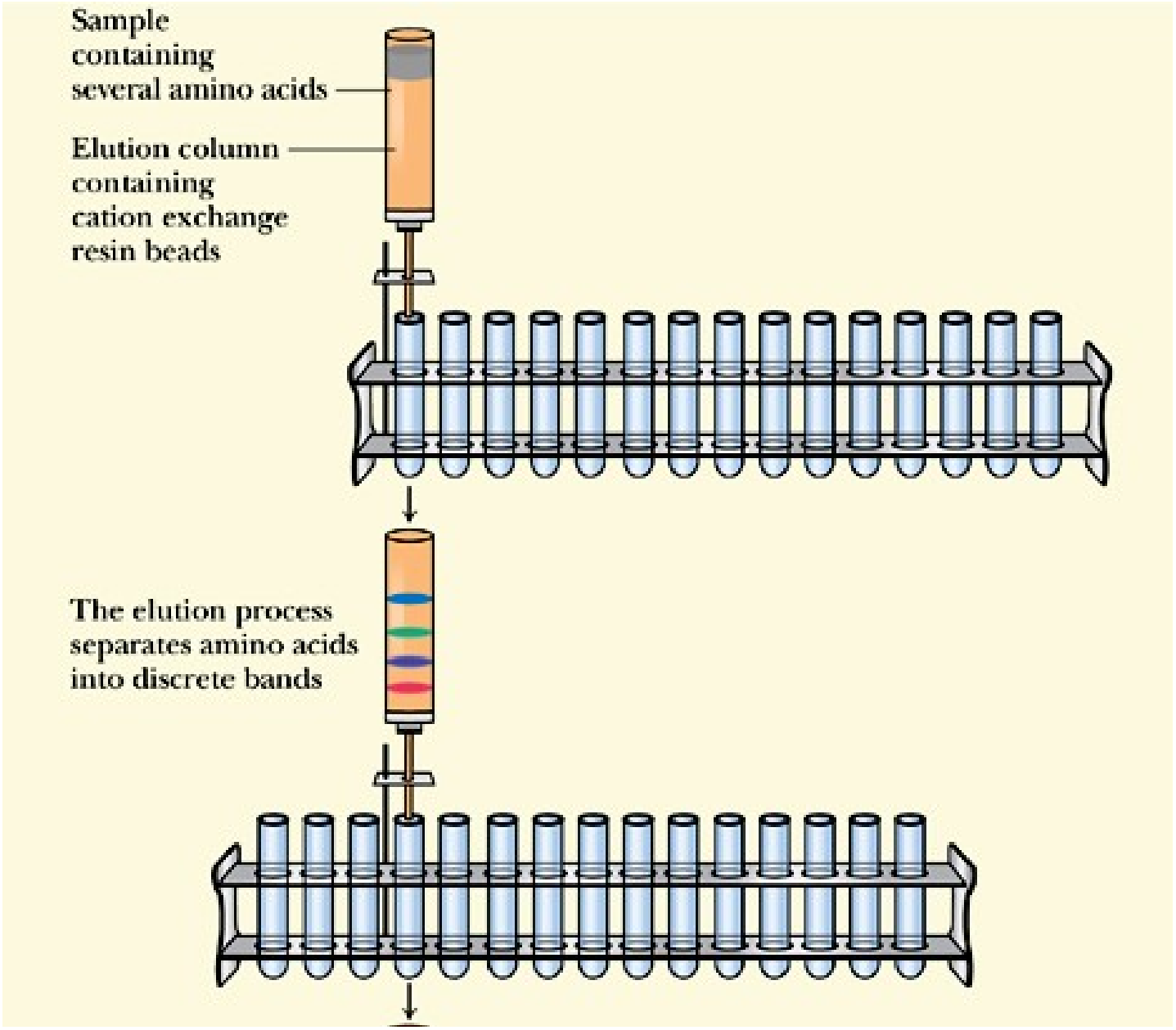

上样
淋洗
洗脱
阴离子
交换介质
带正电荷
的蛋白
带负电荷
的蛋白

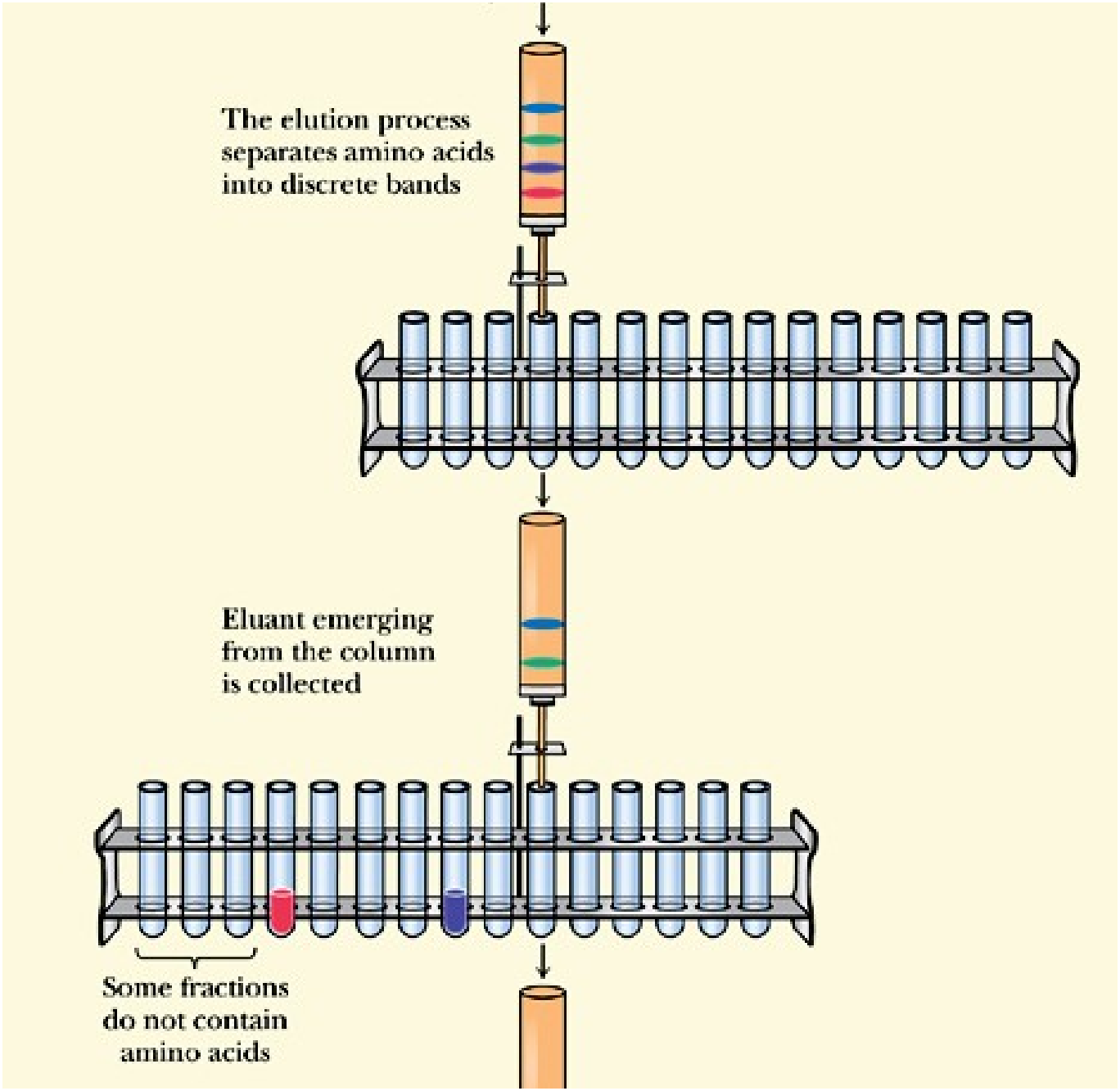


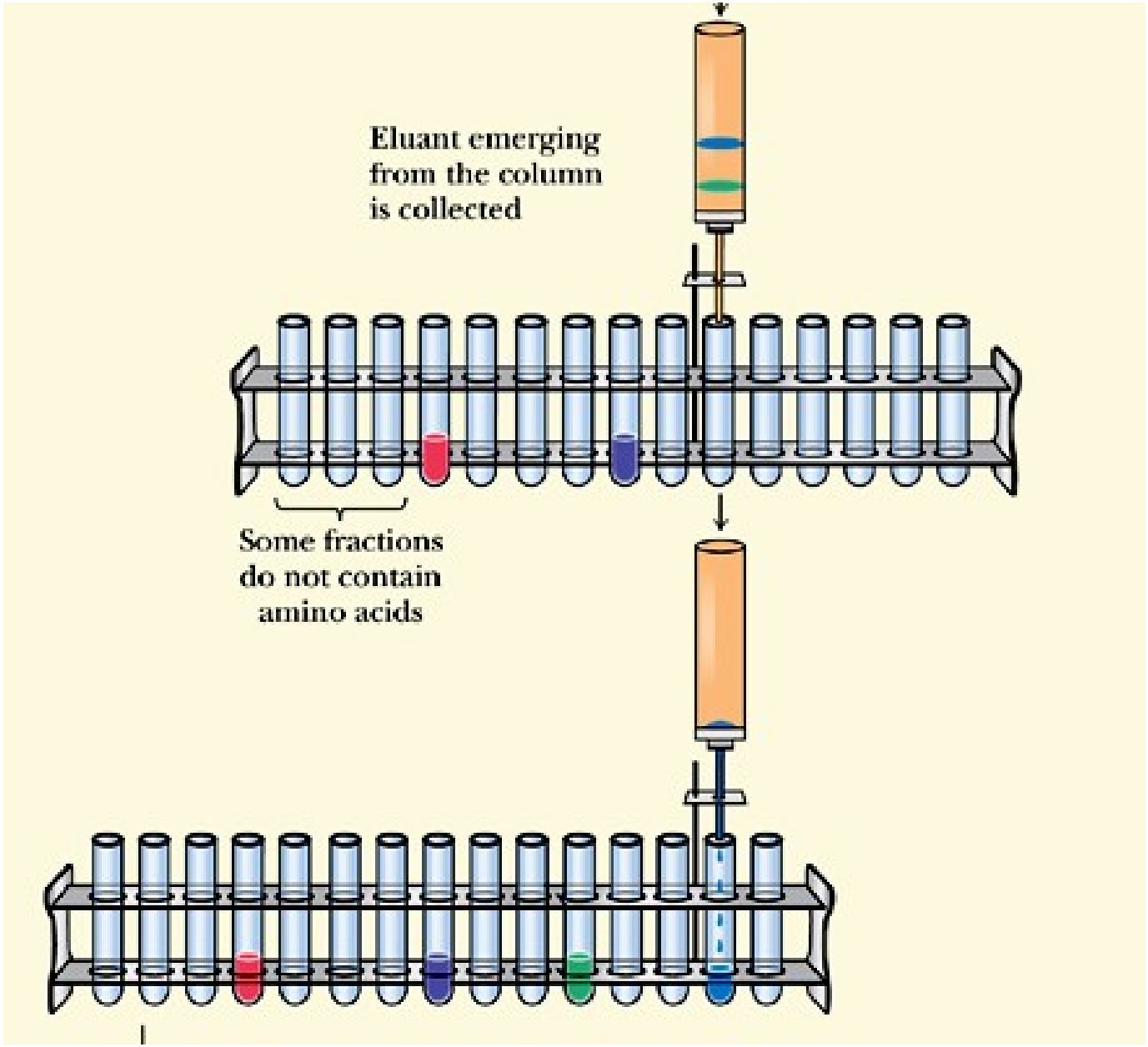
离子交换层析分离蛋白质示意图

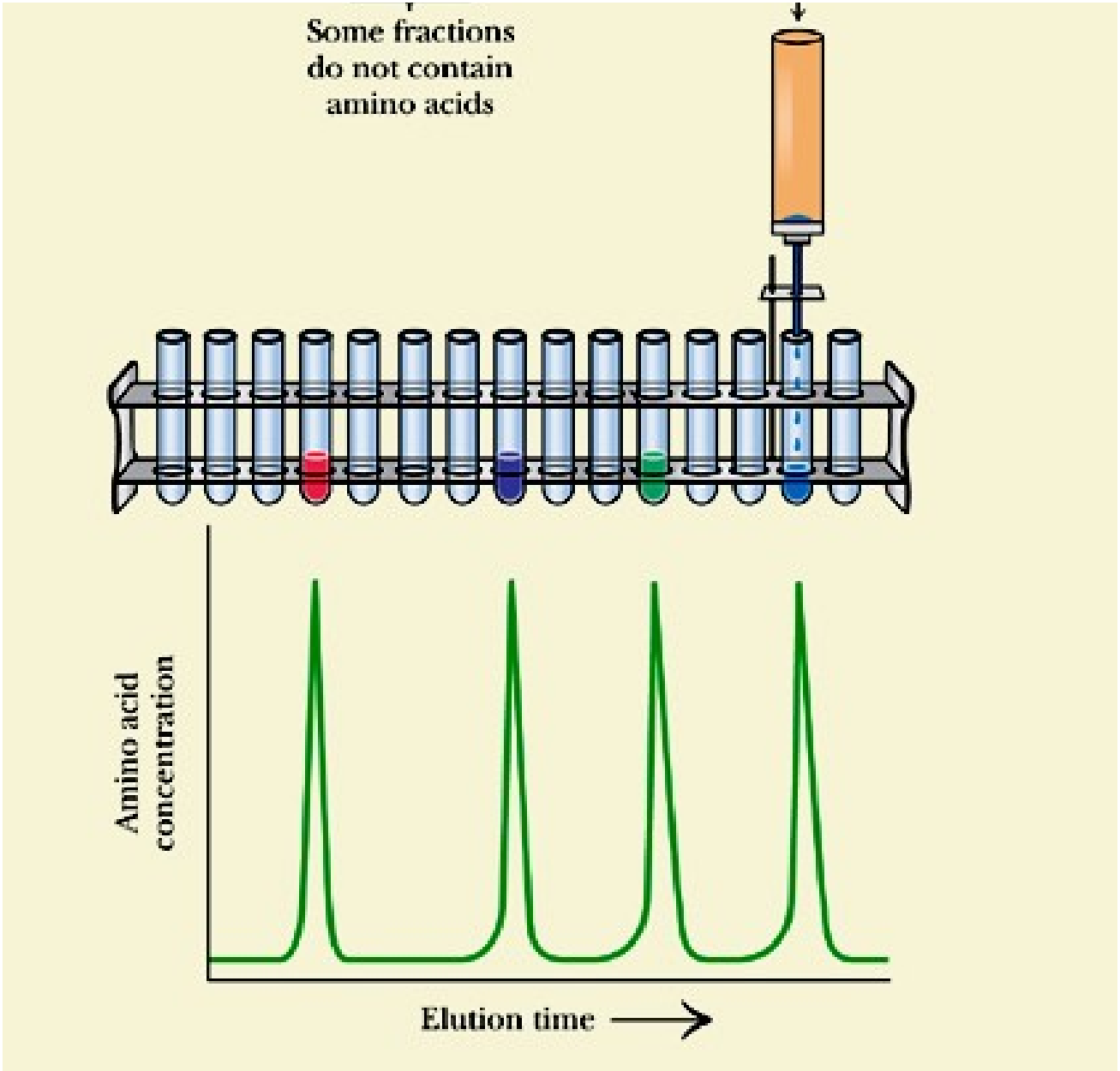
Ion exchange chromatography




Sample containing several amino acids
Elution column containing cation exchange resin beads
The elution process separates amino acids into discrete bands

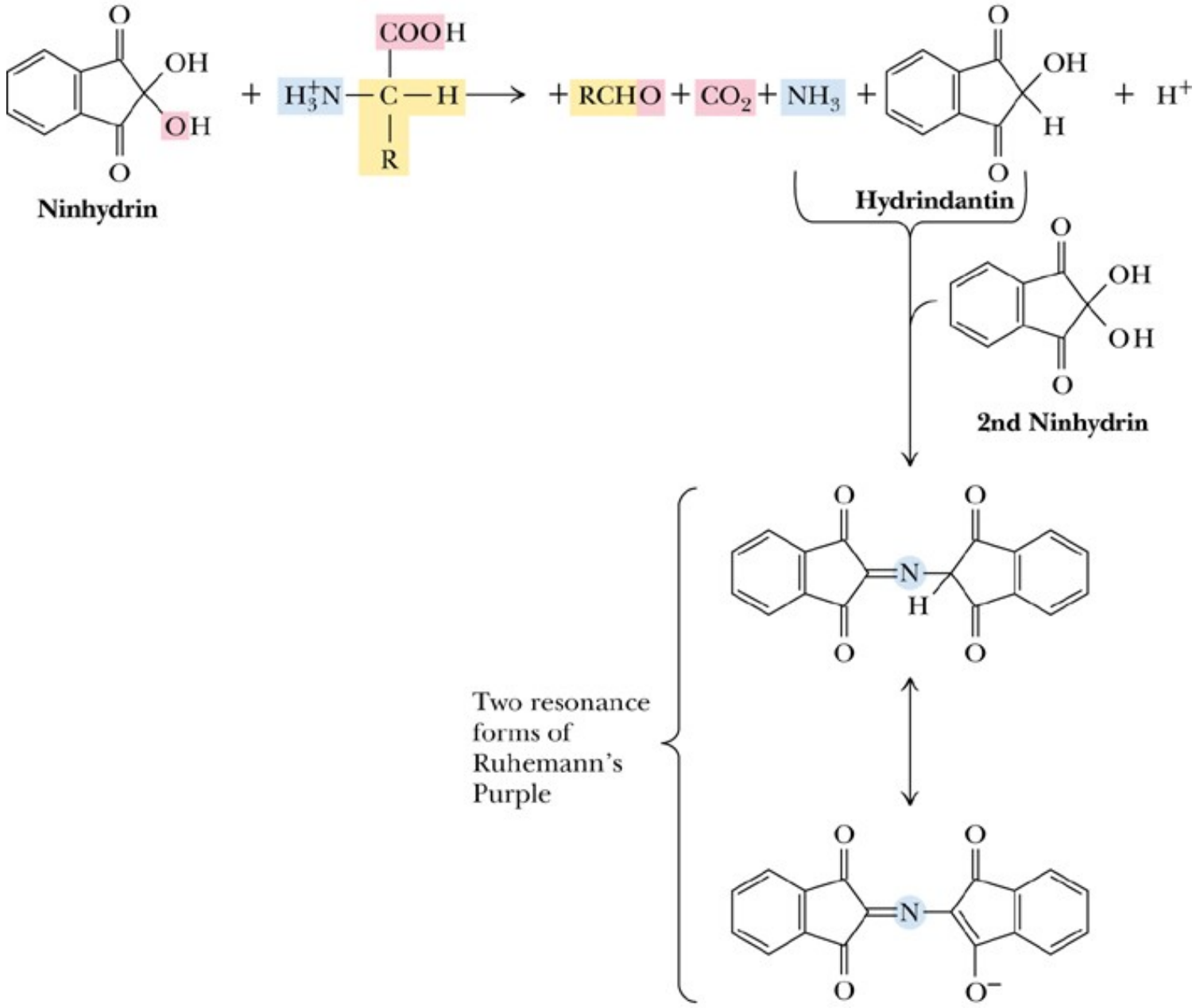

The elution process separates amino acids into discrete bands
Eluant emerging from the column is collected
Some fractions do not contain amino acids


Eluant emerging
from the column
is collected
Some fractions
do not contain
amino acids


Some fractions
do not contain
amino acids
Amino acid
concentration
Elution time



The amino acids are detected by colorimetric reaction with ninhydrin (茚三酮） .


Ninhydrin
COOH
H3N—C—H
R
RCHO
CO2
NH3
H+
Hydrindantin
2nd Ninhydrin
Two resonance forms of Ruhemann's Purple

Step 2

- Identify terminal residues of polypeptide

- Sanger Reaction for N-terminal residue
- Other Reaction



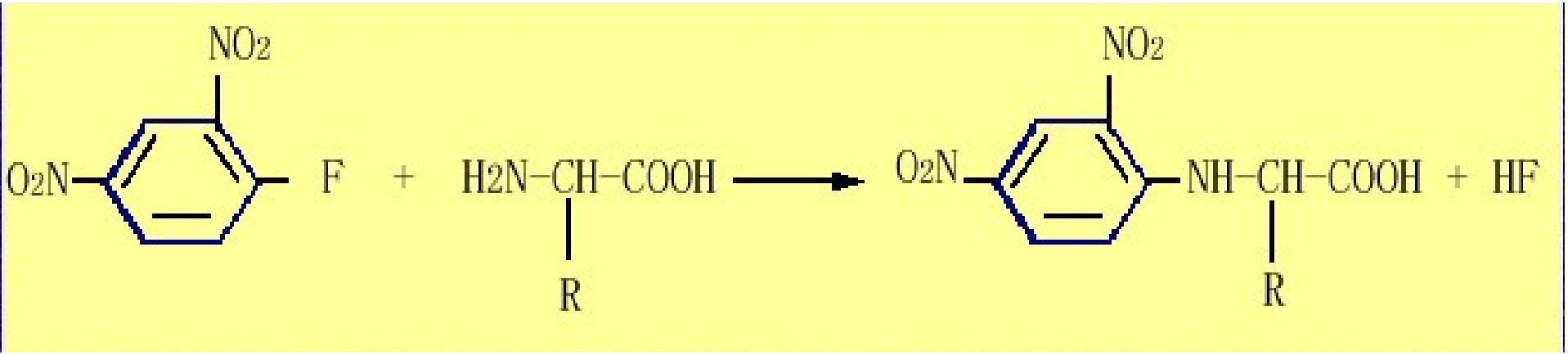
Step 3

- Identify amino-terminal residue; purify and recycle remaining peptide fragment through Edman process.



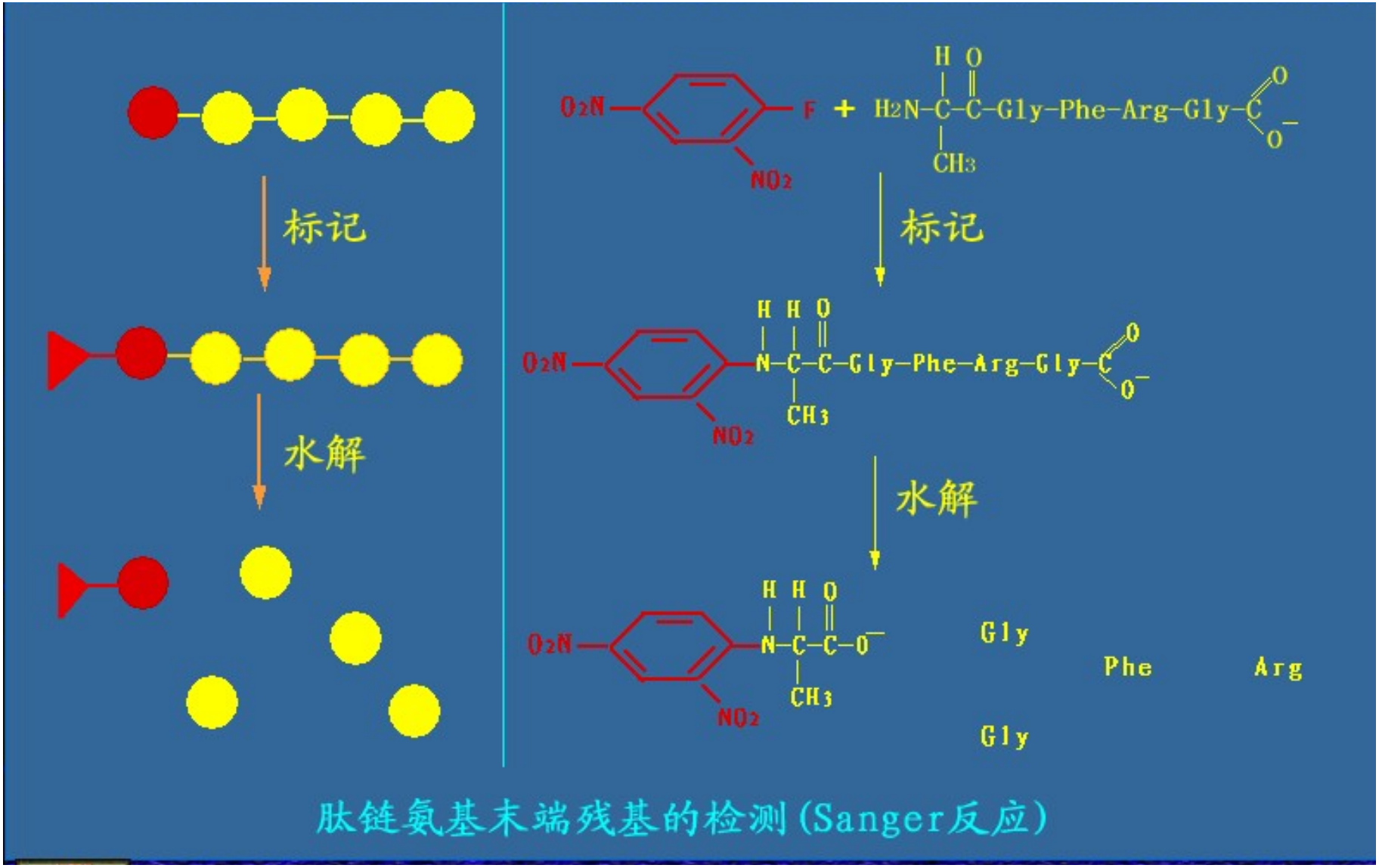
- Edman degradation and Sanger reaction

Sanger reaction



DNFB or FDNB

二硝基氟苯



肽链氨基末端残基的检测(Sanger反应)



Polypeptide

Procedure	Result	Conclusion
hydrolyze; separate amino acids	A 5 C 2 D 4 E 2 F 1 G 3 H 2 I 3 K 2 L 2 M 2 P 3 R 1 S 2 T 1 V 1 Y 2	Polypeptide has 38 amino acid residues. Trypsin will cleave three times (at one R (Arg) and two K (Lys)) to give four fragments. Cyanogen bromide will cleave at two M (Met) to give three fragments.
react with FDNB; hydrolyze; separate amino acids	2,4-Dinitrophenylglutamate detected	E (Glu) is amino-terminal residue.
reduce disulfide bonds (if present)		



cleave with trypsin separate fragments; sequence by Edman degradation	T-1 GASMALIK T-2 EGAAYHDFEPIDPR T-3 DCVHSD T-4 YLIACGPMTK	T-2 placed at amino terminus because it begins with E (Glu). T-3 placed at carboxyl terminus because it does not end with R (Arg) or K (Lys).
cleave with cyanogen bromide; separate fragments; sequence by Edman degradation	C-1 EGAAYHDFEPIDPRGASM C-2 TKDCVHSD C-3 ALIKYLIACGPM	C-3 overlaps with T-1 and T-4, allowing them to be ordered.

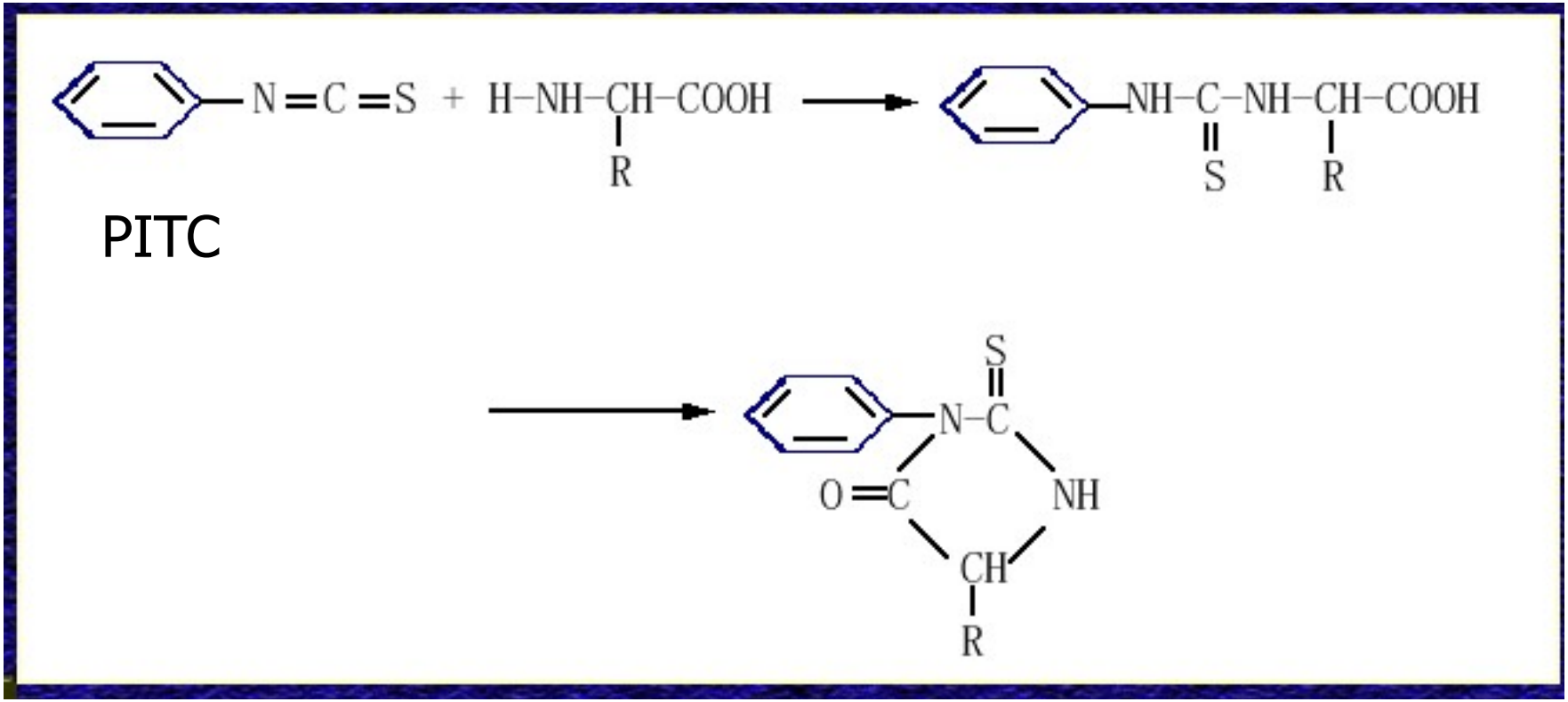
establish sequence

Amino terminus

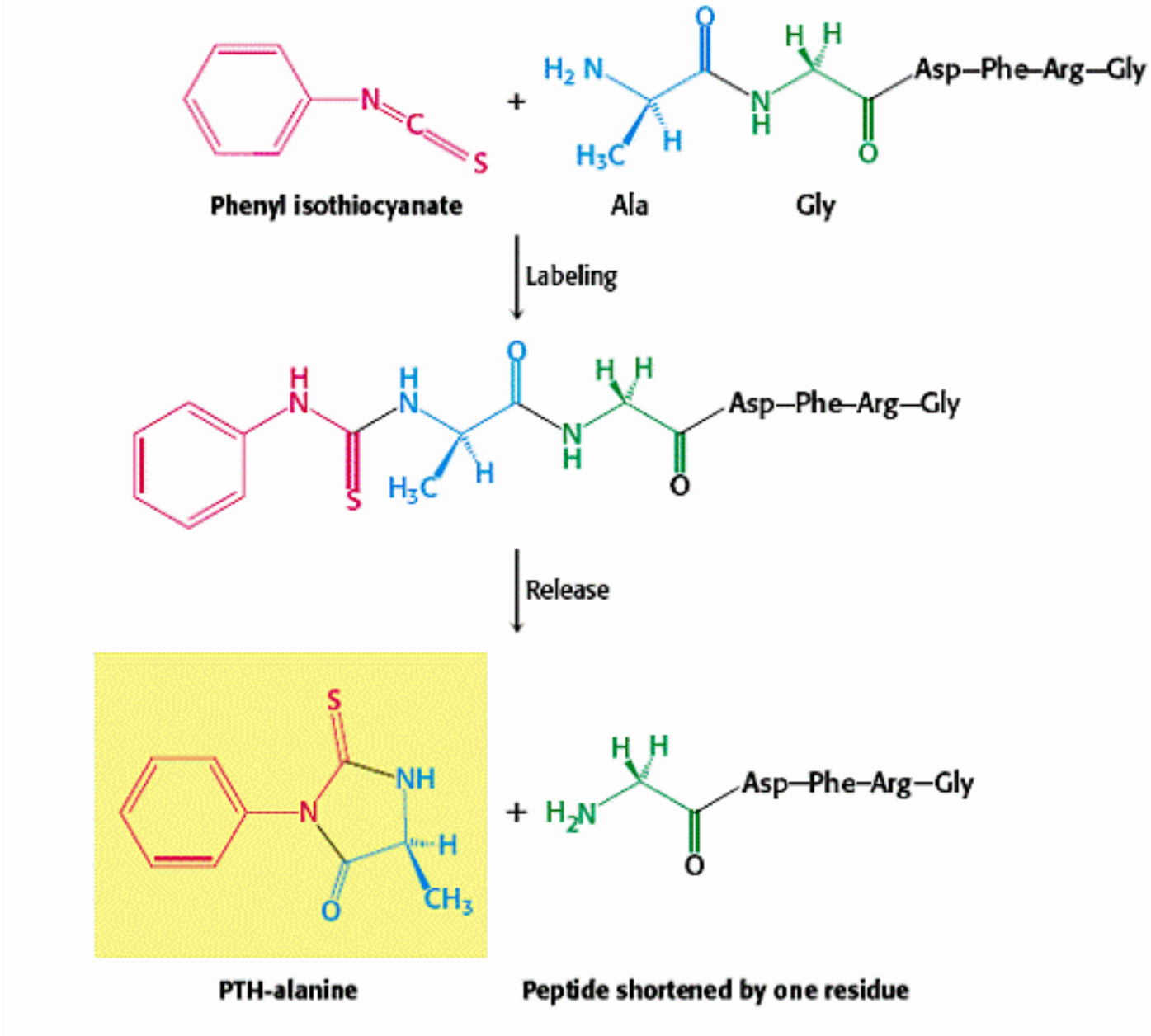


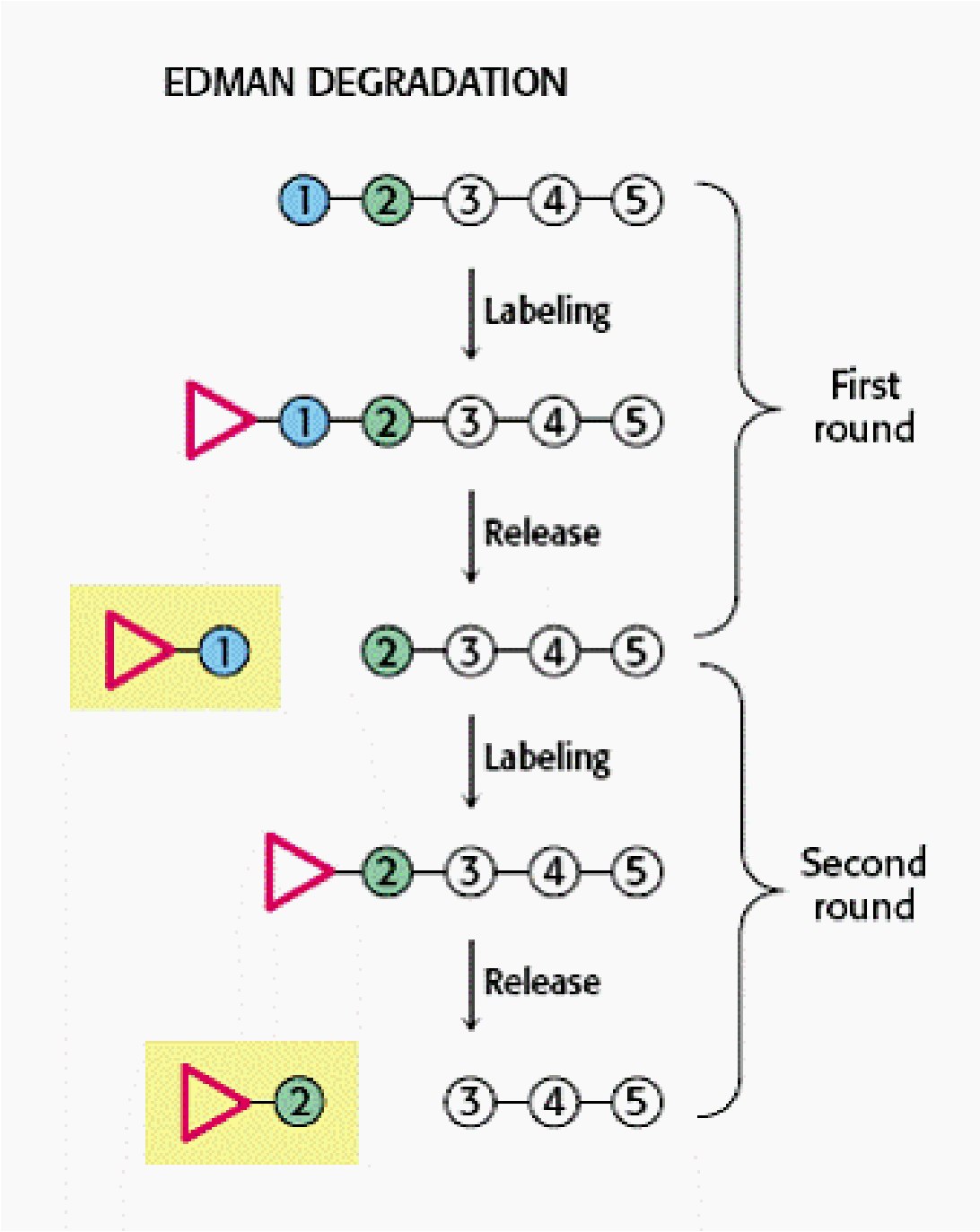
Carboxyl terminus

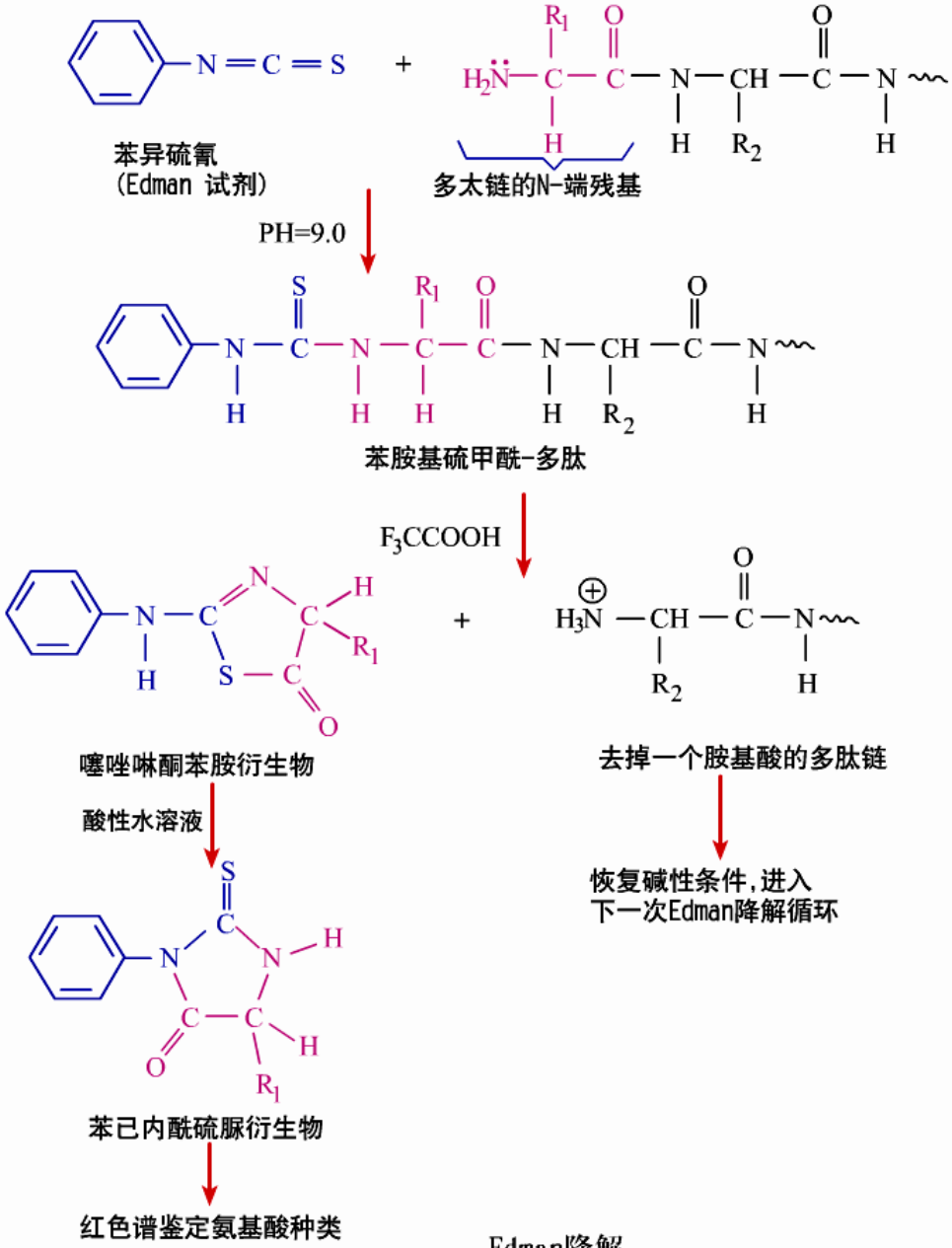
Edman reaction



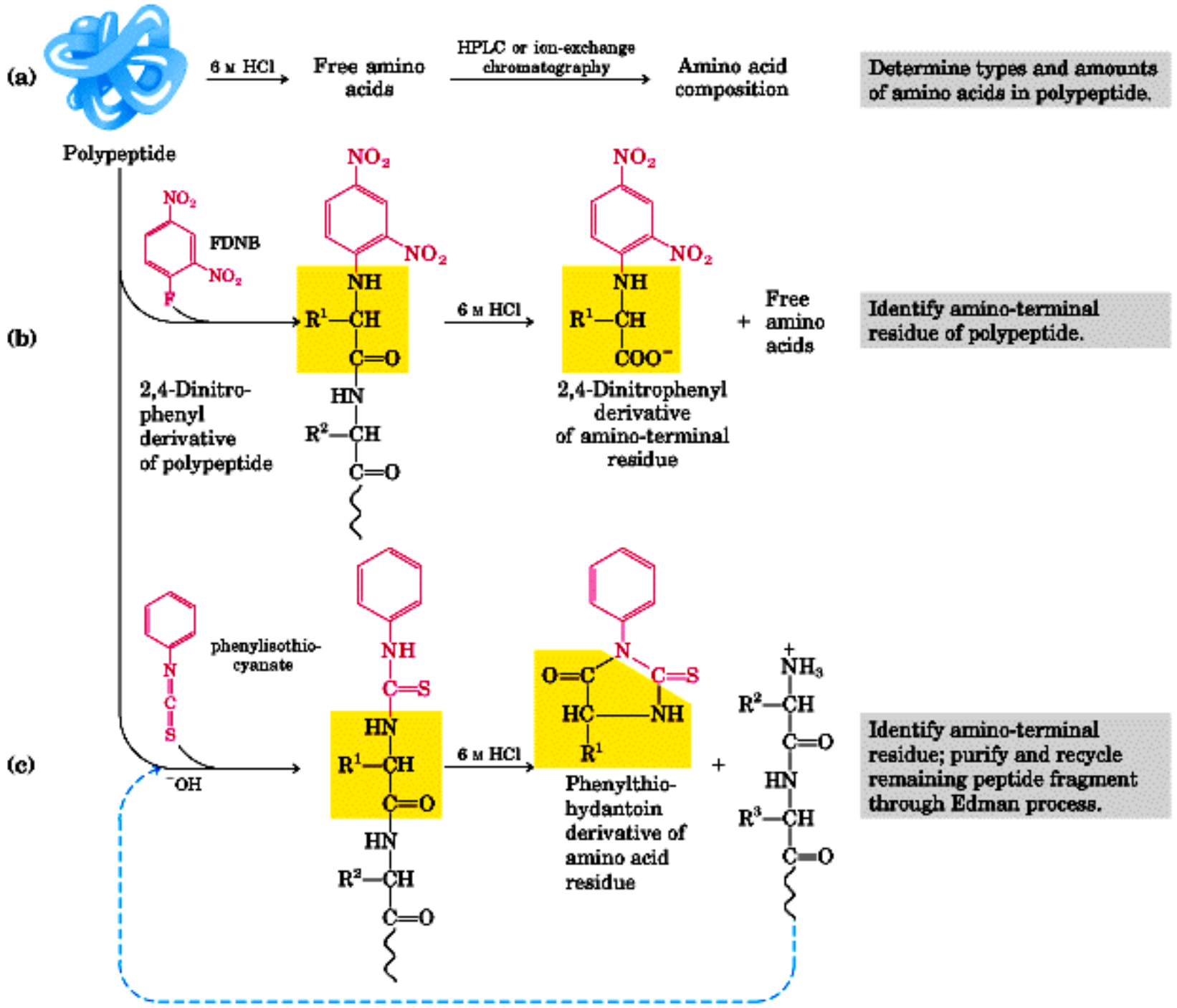
The amino acid sequence of a protein can be determined by Edman degradation which sequentially removes one residue at a time from the N terminus.


Phenyl isothiocyanate
Ala
Gly
Asp–Phe–Arg–Gly
Labeling
Asp–Phe–Arg–Gly
Release
PTH-alanine
Asp–Phe–Arg–Gly
Peptide shortened by one residue


EDMAN DEGRADATION
1 2 3 4 5
Labeling
1 2 3 4 5
First round
Release
1
2 3 4 5
Labeling
2 3 4 5
Second round
Release
2
3 4 5


苯异硫氰
(Edman 试剂)
多太链的N-端残基
PH=9.0
苯胺基硫甲酰-多肽
F3CCOOH
噻唑啉酮苯胺衍生物
去掉一个胺基酸的多肽链
酸性水溶液
恢复碱性条件，进入
下一次Edman降解循环
苯已内酰硫脲衍生物
红色谱鉴定氨基酸种类

Edman降解


(a)
6 M HCl
Free amino acids
HPLC or ion-exchange chromatography
Amino acid composition
Determine types and amounts of amino acids in polypeptide.
Polypeptide
(b)
FDNB
2,4-Dinitrophenyl derivative of polypeptide
6 M HCl
2,4-Dinitrophenyl derivative of amino-terminal residue
+ Free amino acids
Identify amino-terminal residue of polypeptide.
(c)
phenylisothiocyanate
6 M HCl
Phenylthiohydantoin derivative of amino acid residue
+
Identify amino-terminal residue; purify and recycle remaining peptide fragment through Edman process.



Polypeptide

Procedure	Result	Conclusion
hydrolyze; separate amino acids	A 5 C 2 D 4 E 2 F 1 G 3 H 2 I 3 K 2 L 2 M 2 P 3 R 1 S 2 T 1 V 1 Y 2	Polypeptide has 38 amino acid residues. Trypsin will cleave three times (at one R (Arg) and two K (Lys)) to give four fragments. Cyanogen bromide will cleave at two M (Met) to give three fragments.
react with FDNB; hydrolyze; separate amino acids	2,4-Dinitrophenylglutamate detected	E (Glu) is amino-terminal residue.
reduce disulfide bonds (if present)		



cleave with trypsin separate fragments; sequence by Edman degradation	T-1 GASMALIK T-2 EGAAYHDFEPIDPR T-3 DCVHSD T-4 YLIACGPMTK	T-2 placed at amino terminus because it begins with E (Glu). T-3 placed at carboxyl terminus because it does not end with R (Arg) or K (Lys).
cleave with cyanogen bromide; separate fragments; sequence by Edman degradation	C-1 EGAAYHDFEPIDPRGASM C-2 TKDCVHSD C-3 ALIKYLIACGPM	C-3 overlaps with T-1 and T-4, allowing them to be ordered.

establish sequence

Amino terminus



Carboxyl terminus

Thank you!

Isoelectric points (pI)

Supersecondary structure

Protein tertiary structure

Allosteric effect

Protein denaturation

有一个七肽，组成为： Gly 、 Ala 、 Ser 、 Arg 、 Val （ 2 个）、 Phe 。与 DNFB 反应产生 DNP-Ala ，羧肽酶水解可产生游离的 Gly ，用胰蛋白酶水解产生一个三肽（ Ala 、 Arg 、 Ser ）和一个四肽。用胰凝乳蛋白酶处理可得一个五肽和一个二肽。试推测这个多肽的结构。

两次获得诺贝尔奖的科学家

- 诺贝尔奖二次得主
  居里夫人（Marie Curie）
  约翰·巴丁（John Bardeen）
  莱纳斯·鲍林（Linus Pauling）
  弗雷德·桑格尔（Fred Sanger）



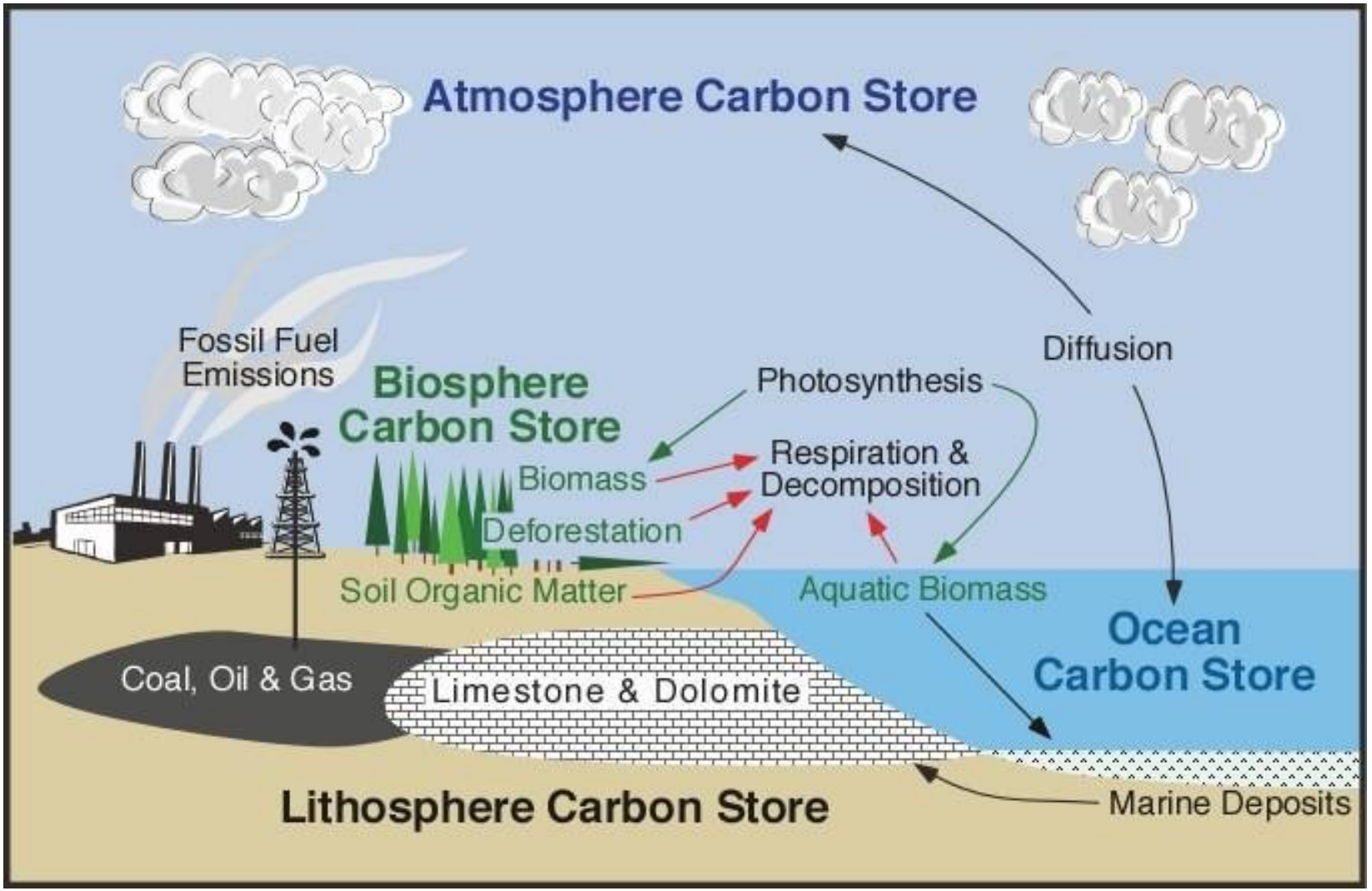
弗雷德·桑格尔（ Fred Sanger ）

- 英国生物化学家 Sanger，(1918 年 8 月 13 日—)，分别获得 1958 年和 1980 年诺贝尔化学奖。他是同一领域内两次获奖的第二人，更关键的是，两次获奖理由都可归结为：测序。并且，他是目前唯一在世的两次获得诺贝尔奖的人。2013 年去世。
- 1958：弗雷德·桑格尔发明酶法测定人胰岛素序列，从而确定胰岛素的分子结构，开创了蛋白质测序的领域。
- 1980：弗雷德·桑格尔、沃尔特·吉尔伯特共同荣获诺贝尔化学奖。他们的贡献在于：分别使用不同的方法测定 DNA 的序列。Sanger 法后来成为主流，并用于人类基因组计划（ HGP ）的测序。



TABLE 1–2 Molecular Components of an *E. coli* Cell

	Percentage of total weight of cell	*Approximate number of different molecular species*
Water	70	1
Proteins	15	3,000
Nucleic acids		
DNA	1	1
RNA	6	>3,000
Polysaccharides	3	5
Lipids	2	20
Monomeric subunits and intermediates	2	500
Inorganic ions	1	20


Atmosphere Carbon Store
Fossil Fuel Emissions
Biosphere Carbon Store
Photosynthesis
Diffusion
Respiration & Decomposition
Biomass
Deforestation
Soil Organic Matter
Aquatic Biomass
Ocean Carbon Store
Coal, Oil & Gas
Limestone & Dolomite
Marine Deposits
Lithosphere Carbon Store


CO_2
的大气库
(占空气的0.032%)
倡导低碳生活，唤起绿色希望
请完成碳循环图。
帮助
水中释放
溶于水中
风化
水草
腐鱼
珊瑚
草
牛
动尸
家庭和工业
汽车
形成化石燃料

Globular protein structures are compact and varied

